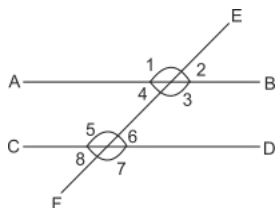


Geometry

LINES

1). Parallel Lines : Two or more lines which will never meet just like railway track.

2). Transversal Lines: A line which cuts parallel lines as shown in the figure below.



- EF is the transverse line. AB and CD are parallel lines. Symbol of parallel lines is "||". To denote a line symbols like $\overline{AB}$ or simply AB are used.
- Let's take a look at various angles made by the transverse line and the parallel lines :

(i) Corresponding Angles:- $\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 4$ and $\angle 8$, $\angle 3$ and $\angle 7$ are pairs of corresponding angles.

- Corresponding angles are equal. E.g. $\angle 1 = \angle 5$

(ii) Alternate Angles:- E.g. ($\angle 1 = \angle 7$), ($\angle 2 = \angle 8$), ($\angle 4 = \angle 6$), ($\angle 3 = \angle 5$).

(iii) Vertically opposite angles :-

E.g. ($\angle 1 = \angle 3$), ($\angle 2 = \angle 4$), ($\angle 6 = \angle 8$), ($\angle 5 = \angle 7$)

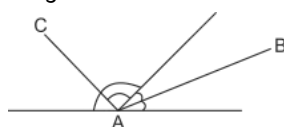
(iv) Adjacent angles:- $\angle 1 + \angle 2 = 180^\circ$ in this case as these are linear pairs. It is not necessary that their sum should be 180° .

(v) Sum of Interior angles on same side = 180° . The angle, made by bisectors of interior angle will be 90° .

(vi) Sum of Exterior angles on same side = 180°

Note: From a point, an infinite number of lines can be drawn.

3). Internal angle bisector and External angle bisector. In the given figure AB is the internal angle bisector and AC is the external angle bisector.



ANGLES

Various types of angles are as follows:

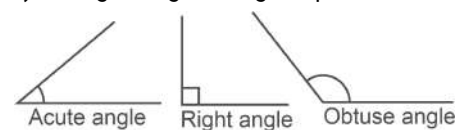
1). Acute Angle: Angles which are less

than 90° .

2). Right Angle: 90° angle is called right angle.

3). Obtuse Angle: Angles greater than 90° and less than 180° .

4). Straight Angles: Angle equal to 180° .



5). Reflex angle: Angles greater than 180° and less than 360° .

6). Complete Angle: 360° angle is called Complete angle or whole angle.



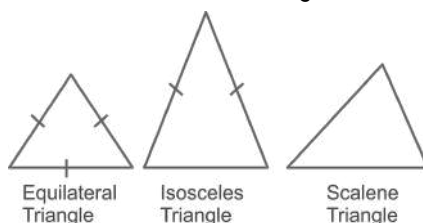
7). Complementary Angles: Sum of two angles is equal to 90° .

8). Supplementary Angles: Sum of the two angles is equal to 180° .

TYPES OF TRIANGLES

1). Based on sides:-

- Equilateral Triangles:** All three sides are equal to each other.
- Isosceles Triangles:** Any two sides are equal to each other.
- Scalene Triangle:** All three sides are different in length.

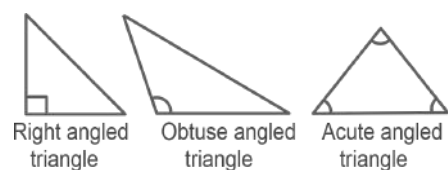


2). Based on angles :-

$\Rightarrow$ **Right angle triangle** :- One of the angle is 90° .

$\Rightarrow$ **Obtuse angled triangle** :- One angle is more than 90° .

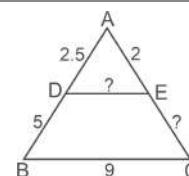
$\Rightarrow$ **Acute angled triangle** :- All three angles are less than 90° .



Properties of Triangles

[1.] If a line DE intersects two sides of a triangle AB and AC at D and E respectively, then : $\frac{AD}{DB} = \frac{AE}{EC}$ and $\frac{AD}{AB} = \frac{DE}{BC}$. Also if D and E are mid points of AB and AC respectively, then:

$DE = \frac{BC}{2}$ and DE will be parallel to BC.

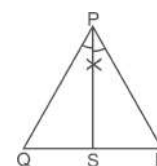


E.g. Let AD = 2.5 units; DB = 5, AE = 2 and BC = 9, then EC = ? and DE = ?

$$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow EC = AE \times \frac{DB}{AD} = 2 \times \frac{5}{2.5} = 4 \text{ units. Similarly DE} = 3 \text{ units.}$$

[2.] In case of Internal angle bisector:

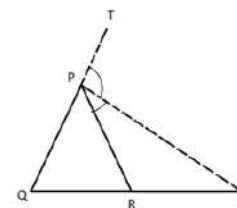
$$\frac{PQ}{QS} = \frac{PR}{SR} \text{ or } \frac{QS}{SR} = \frac{PQ}{PR}$$



Here PS is the internal angle bisector and a common side of the triangles ΔPQS and ΔPSR .

[3.] In case of Exterior angle bisector :

$$\frac{PQ}{PR} = \frac{QS}{RS}$$



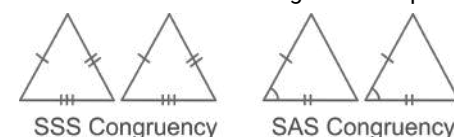
PS is the external angle bisector.

Congruency of Triangles

Two triangles will be congruent if:

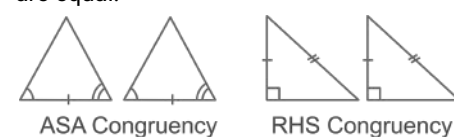
1. SSS: (Side – Side – Side rule): When all three sides are equal.

2. SAS: (Side – Angle – Side rule): When two sides and one angle are equal.



3. ASA: (Angle – Side – Angle rule): When two angles and one side are equal.

4. RHS: (Right angle – Hypotenuse – Side rule): When one side and hypotenuse of the right angled triangle are equal.

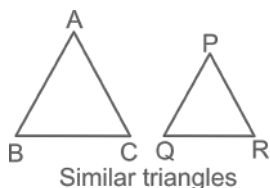


Similarity of Triangles

1). Similar triangles are triangles that have the same shape, but their sizes may

vary.

2). In congruency the triangles are mirror images of each other. We can say that all congruent triangles are similar but the vice versa is not true.



3). Important properties of similar triangles:

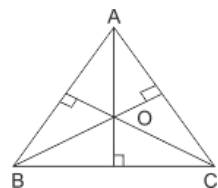
- $\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle PQR} = \frac{AB^2}{PQ^2}$
 $= \frac{AC^2}{PR^2} = \frac{BC^2}{QR^2}$
- $\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR} = \frac{AB}{PQ} = \frac{AC}{PR} = \frac{BC}{QR}$

Altitude :

- It is also known as height. The line segment drawn from the vertex of a triangle perpendicular to its opposite side is called an Altitude of a triangle.

Orthocenter:

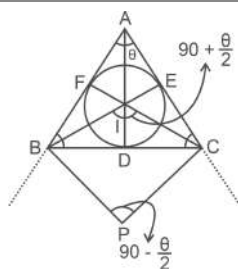
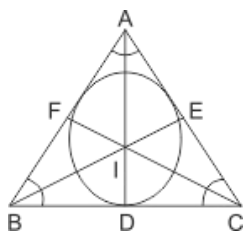
- The point at which the altitudes of a triangle intersect is called an Orthocenter.
- Generally Orthocenter is denoted by 'O'.



- Here, $\angle AOC = 180^\circ - \angle B$
- In the case of a right angled triangle, the orthocenter lies at the vertex of right angle.
- In case of an obtuse angled triangle, the orthocenter lies outside the triangle.

In-center:

The point at which the *internal angle bisectors* of a triangle meet. It is generally denoted by 'I'. From the in-center the in-circle of a triangle is drawn. The radius of the in-circle is equal to $ID = IE = IF$.



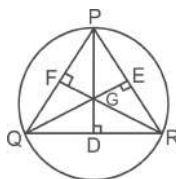
Important Results:

- $\angle BIC = 90^\circ + \frac{\angle A}{2}$ (for Internal angle bisectors)
- $\angle BPC = 90^\circ - \frac{\angle A}{2}$ (for external angle bisectors)
- $\frac{AB}{AC} = \frac{BD}{DC}$
- In-radius $r = \frac{\text{Area of Triangle}}{\text{Semi-perimeter of Triangle}}$
- $AI : ID = AB + AC : BC$
 $BI : IE = AB + BC : AC$
 $CI : IF = AC + BC : AB$
- Length of angle bisector(AD)
 $= \frac{2 \times AB \times AC \cos \theta}{AB + AC}$

Circum-center :-

The point at which the *perpendicular bisectors* of the sides of the triangle meet.

- From the circum-center the circumcircle of a triangle is drawn. The radius of the circum-circle is equal to $PG = RG = QG$.



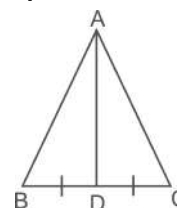
- Since the angle formed at the center of a circle is double of the angle formed at the circumference, we have:
 $\angle QGR = 2\angle P$, $\angle QGP = 2\angle R$ and $\angle PGR = 2\angle Q$.
- For an obtuse angle triangle, the circumcenter lies outside the triangle.

Note: A line which is perpendicular to another line and also bisects it into two equal parts is called a perpendicular bisector.

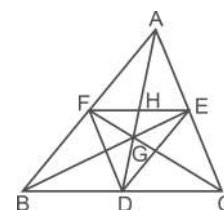
Centroid: It is the point of intersection of the *medians* of a triangle. It is also called the center of mass.

- The median is a line drawn from the vertex of a triangle to the mid-point of the opposite side.

- Each median is divided in the ratio of 2 : 1 at the centroid.
- Median divides the triangle into two equal areas.
- **Apollonius theorem:**



- $AB^2 + AC^2 = 2(AD^2 + BD^2)$
 Other key points for medians:
 All the medians of a triangle divide it into six parts with equal areas.
 Therefore, $\text{Area of } \triangle ABC = 6 \times \text{Area of any of the smaller parts}$.
- $\text{Area of } \triangle FEG = \frac{1}{12} \times \text{Area of } \triangle ABC$



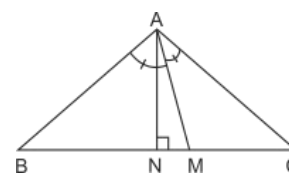
- FE divides the line AG in the ratio of 3 : 1 i.e. $\frac{AH}{HG} = \frac{3}{1}$.
- $\text{Area of } \triangle ABG = \text{Area of } \triangle BCG = \text{Area of } \triangle ACG$
- $\text{Area of } \triangle DEF = \frac{1}{4} \triangle ABC$

Note : In case of Equilateral triangle, orthocenter, in-center, circum-center and centroid lie at the same point.

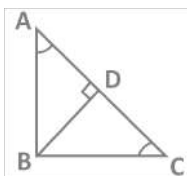
Other Facts about triangles:

- 1). Sum of two sides is always greater than the third side.
 E.g. $AB + BC > CA$, $AB + AC > BC$, $BC + CA > AB$.
- 2). Difference of two sides is always less than the third side.
 E.g. $AB - BC < CA$, $AB - AC < BC$, $BC - CA < AB$.
- 3). In $\triangle ABC$, $\angle B > \angle C$, if AM is the bisector of $\angle BAC$ and $AN \perp BC$, then

$$\angle MAN = \frac{1}{2}(\angle B - \angle C)$$

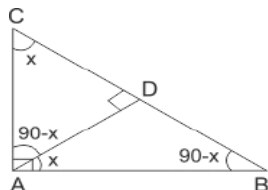


- 4). In right angled $\triangle ABC$, where BD is an altitude to hypotenuse AC, then Altitude $BD = \frac{AB \times BC}{AC}$.



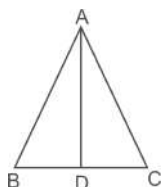
Similarly, if BE is a median drawn to hypotenuse AC, then $BE = \frac{1}{2}AC$. (Length of median is half the length of hypotenuse)

5). In right angled $\triangle ABC$, AD is perpendicular to BC, then



- $AD^2 = BD \times DC$
- $AB^2 = BD \times BC$
- $AC^2 = CD \times BC$

Formulas of triangle :-



In case of equilateral triangle,

- $\text{Area}(A) = \frac{\sqrt{3}}{4} (\text{side})^2$
- $\text{Inradius}(r) = \frac{\text{side}}{2\sqrt{3}}$
- $\text{circumradius}(R) = \frac{\text{side}}{\sqrt{3}}$
- $\text{height}(h) = \frac{\sqrt{3}}{2} \times \text{side}$

In case of isosceles triangle,

$$\bullet \text{Area}(A) = \frac{BC}{4} \sqrt{4AB^2 - BC^2}$$

Where $AB = AC$

- $\text{Inradius}(r) = \frac{\text{area of triangle}}{\text{Semi-perimeter}}$
- $\text{circumradius}(R) = \frac{AB \times BC \times AC}{4 \times \text{Area of triangle}}$
- $\text{height}(h) = \sqrt{AB^2 - \frac{BD^2}{4}}$

In case of scalene triangle,

$$\bullet \text{Area}(A) = \sqrt{s(s-a)(s-b)(s-c)}$$

Where a,b,c are sides of triangle

$$s = \frac{a+b+c}{2}$$

In case of right angle triangle ,

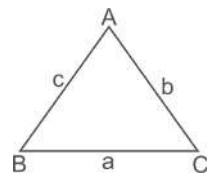
- $\text{Area}(A) = \frac{1}{2} \times \text{Base} \times \text{height}$
- $\text{Inradius}(r) = \frac{\text{Perpendicular} + \text{base} - \text{hypotenuse}}{2}$

$$\bullet \text{circumradius}(R) = \frac{\text{hypotenuse}}{2}$$

Note :- Distance between circumcentre and incentre of any triangle =

$$\sqrt{R^2 - 2Rr}$$

Sine rule :-



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Where R = circumradius of triangle

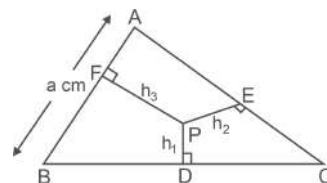
Cosine rule :-

$$\cos A = \frac{c^2 + b^2 - a^2}{2bc}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

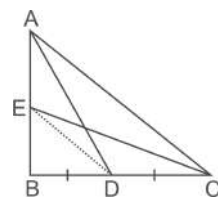
• If P is any point inside the equilateral triangle, the sum of altitude from the point P to the sides AB, BC and AC equal to the sides of the triangle.



$$h_1 + h_2 + h_3 = \frac{\sqrt{3}}{2} a$$

Where a = side of equilateral triangle

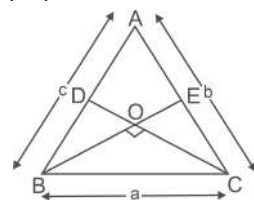
• if $\triangle ABC$ is right angle triangle and AD, CE are medians then:



$$(i) 4(AD^2 + CE^2) = 5 AC^2$$

$$(ii) AD^2 + CE^2 = AC^2 + ED^2$$

• If BE and CD are medians and perpendicular to each other then:



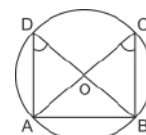
$$AB^2 + AC^2 = 5 BC^2$$

• If length of median of triangles are given then area of triangle = $\frac{4}{3} \times$ area of triangle formed by medians.

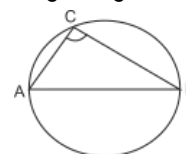
CIRCLE

Properties of circles :-

1). Angles in the same segment of a circle from the same base are always equal.

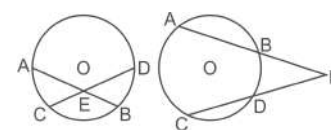


2). Angle dropped on the circumference of a circle with the diameter as base is always a right angle.



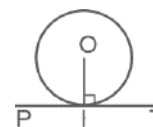
3). Two chords AB and CD of a circle intersect, internally or externally, at E then:

$$AE \times EB = CE \times ED$$

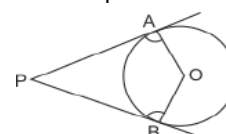


4). The tangent to a circle at a point on its circumference is perpendicular to the radius at that point.

$$OI \perp PT$$

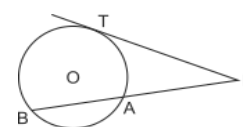


5). Two tangents PA and PB are drawn from an external point P on a circle with center O is equal. $PA = PB$



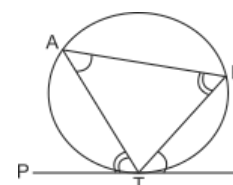
6). For a tangent, PT, and a secant, PB, drawn to a circle from a point P, we have.

$$PT^2 = PB \times PA$$



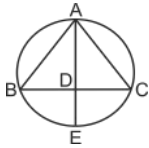
7). Angle made by a chord with a tangent is always equal to the angle dropped on any point of circumference taking the chord as base.

$$\angle PTA = \angle ABT$$

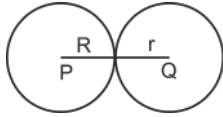


8). In the following case:

$$AB \times AC + AE \times DE = AE^2$$

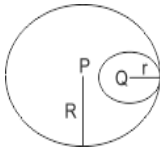


9). If two circles touch each other externally, then distance between their centres is the sum of their radius.



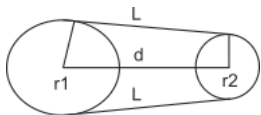
Here, $PQ = R + r$

10). If two circles touch each other internally, then distance between their centres is the difference of their radius.



Here, $PQ = R - r$

Direct common Tangents:

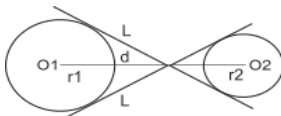


Length of direct common tangent (L) =

$$\sqrt{d^2 - (r_1 - r_2)^2}$$

Where, d = distance between two centers and r_1, r_2 are radii of the circles.

Transverse common Tangents:-

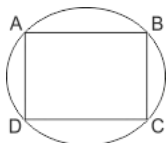


Length of transverse tangent (L) =

$$\sqrt{d^2 - (r_1 + r_2)^2}$$

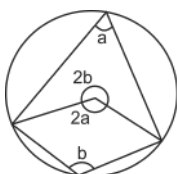
Cyclic Quadrilateral :-

A quadrilateral drawn inside a circle with its vertices lying on the circumference.



Sum of opposite angles = 180°
i.e. $\angle A + \angle C = \angle B + \angle D = 180^\circ$

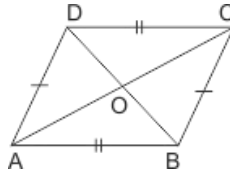
Important result: In the following type of cyclic quadrilateral remember the relationship between the angles, as shown:



QUADRILATERAL

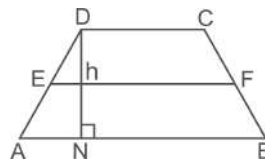
Any closed figure that has four sides is called a quadrilateral. Depending on length of sides, orientation, they can be classified as followed:

1). **Parallelogram** : Opposite sides are equal and parallel.



- $AB = DC$ and $AD = BC$
- $\angle A + \angle B = \angle B + \angle C = \angle C + \angle D = \angle D + \angle A = 180^\circ$
- $\angle A = \angle C$ and $\angle B = \angle D$
- The diagonals bisect each other i.e. $AO = OC$ and $OB = OD$
- Area of parallelogram = Base $\times$ Altitude
- Sum of squares of all four sides = Sum of squares of diagonals, So,
 $AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + AD^2$
- Square, Rectangle and Rhombus are parallelogram.
- All rectangles are parallelogram but all parallelogram are not rectangles.

2). **Trapezium** : Only one pair of opposite sides are parallel.



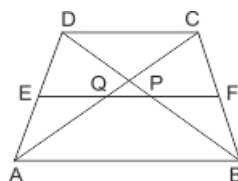
Area of trapezium ABCD

$$= \frac{1}{2} \times \text{Sum of parallel side} \times \text{height}$$

$$= \frac{1}{2} \times (AB + DC) \times h$$

- If E is the midpoint of $\overline{AD}$ and F is the midpoint of $\overline{BC}$, then

$$EF = \frac{AB + DC}{2}$$



- If P is the midpoint of diagonal $\overline{BD}$ and Q is the midpoint of diagonal $\overline{AC}$, then

$$PQ = \frac{AB - DC}{2}$$

3). **Rhombus** : All sides are equal and opposite sides are parallel to each other.

$$\angle A + \angle B = \angle B + \angle C = \angle C + \angle D = \angle D + \angle A = 180^\circ$$

- $\angle A = \angle C$ and $\angle B = \angle D$

$$4a^2 = d_1^2 + d_2^2$$

i.e. Sum of squares of sides = Sum of

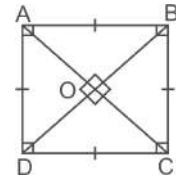
squares of diagonals.

- Diagonals bisect each other at right angles and form four right angled triangles with equal areas.

$$\begin{aligned} \text{Area of } \triangle AOB &= \triangle BOC = \triangle COD \\ &= \triangle DOA = \frac{1}{4} \times \text{area of rhombus } ABCD \end{aligned}$$

- The diagonals of Rhombus are not of equal length.

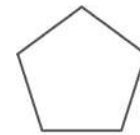
4). **Square**: All sides are equal in length and adjacent sides are perpendicular to each other.



- $AB = BC = CD = AD$ and $\angle A = \angle B = \angle C = \angle D = 90^\circ$
- Diagonals bisect each other at right angles and form four right angled isosceles triangles.
- Diagonals are of equal length: i.e. $AC = BD$.
- In a rectangle the diagonals do not intersect at right angles.

5). **Polygons:**

- Convex polygon: All its interior angles are less than 180° .
- Concave polygon: At least one angle is more than 180° .

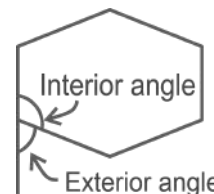


Convex Pentagon



Concave Pentagon

- Regular Polygons: All sides and angles are equal.



* Sum of all exterior angles = 360° .

* Each exterior angle = $\frac{360^\circ}{n}$, where n = number of sides.

* Exterior angle + Interior angle = 180°

* In case of convex polygon, sum of all interior angles = $(2n - 4) \times 90^\circ$

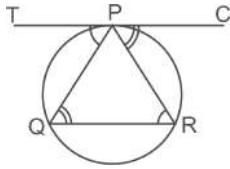
* Number of diagonals = $\frac{n(n-3)}{2}$

Where, n = number of sides.

ALTERNATE SEGMENT THEOREM

The Alternate Segment Theorem states that : An angle between a tangent and a chord through a point of contact is equal

to the angle in the alternate segment.

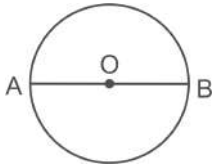


Here, in above diagram, $\overline{TC}$ is tangent. $\overline{PQ}$ and $\overline{PR}$ are chord which touches tangent $\overline{TC}$ at point P. Hence, $\angle TPQ = \angle PRQ$ and $\angle CPR = \angle PQR$

Important formulas :-

Circle :-

• If r be the radius and O be the center of the circle.

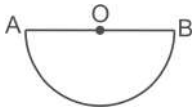


Diameter = $2 \times \text{radius}$

Area = πr^2

Circumference = $2\pi r$

Semi - Circle :-

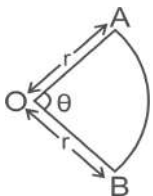


Diameter = $2 \times \text{radius}$

Perimeter = $(\pi + 2)r$

Area = $\frac{\pi r^2}{2}$

Sector :-

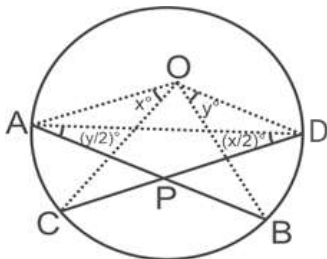


If $\angle AOB = \theta$

Perimeter = $[2r + \frac{\theta}{360} \times (2\pi r)]$

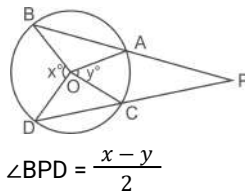
Area = $\frac{\theta}{360} \times \pi r^2$

• Two Chords AB and CD of a circle with center O, intersect each other at P. If $\angle AOC = x^\circ$ and $\angle BOD = y^\circ$ then the value of $\angle BPD$ is :-



$\angle BPD = \frac{x + y}{2}$

• Chords are intersecting at external points.



$\angle BPD = \frac{x - y}{2}$

Variety Questions

Q.1. The length of a direct common tangent to two circles is 24 cm. Their centers are 25 cm apart. If the larger circle has a radius of 10 cm, what is the radius of the smaller circle?

SSC CGL 14/10/2025 (Shift 2)

(a) 2 cm (b) 3 cm (c) 4 cm (d) 5 cm

Q.2. In $\triangle KLM$, $NO \parallel LM$ and intersects KL and KM at N and O . If $KN = 2$, $NL = 4$, and $KO = 3$, find OM .

SSC CGL 14/10/2025 (Shift 2)

(a) 6 (b) 4.5 (c) 3 (d) 9

Q.3. Which of the following statements is always true for a right-angled triangle?

SSC CGL 14/10/2025 (Shift 2)

- (a) The Centroid is outside the triangle.
(b) The Circumcenter is at the midpoint of the hypotenuse.
(c) The Orthocenter, Centroid, Incenter, and Circumcenter all coincide.
(d) The Incenter is one of the vertices of the triangle.

Q.4. Two lines intersect, forming four angles. One of the angles measures $4x - 5$ degrees and its adjacent angle measures $6x + 25$ degrees. What is the measure of the angle vertically opposite to the $4x - 5$ degree angle?

SSC CGL 14/10/2025 (Shift 2)

(a) 59° (b) 55° (c) 45° (d) 60°

Q.5. If two complementary angles are in the ratio 2 : 7, find the angles

SSC CGL 14/10/2025 (Shift 2)

- (a) $10^\circ, 80^\circ$ (b) $20^\circ, 70^\circ$
(c) $40^\circ, 50^\circ$ (d) $35^\circ, 55^\circ$

Q.6. What is the radian measure of 135° ?

SSC CGL 13/09/2025 (Shift 1)

- (a) $\frac{5\pi}{6}$ (b) $\frac{2\pi}{3}$ (c) $\frac{3\pi}{4}$ (d) $\frac{\pi}{3}$

Q.7. From an external point X, two tangents XY and XZ touch a circle at points Y and Z respectively. What is the relationship between the lengths XY and XZ?

Graduate Level 02/08/2025 (Shift - 3)

- (a) $XY > XZ$ (b) $XY < XZ$
(c) $XY = XZ$ (d) They form curved paths

Q.8. A quadrilateral PQRS is inscribed around a circle. If $PQ = 12$ cm, $QR = 14$ cm,

and the perimeter of the quadrilateral is 42 cm, what is the length of side RS?

Graduate Level 31/07/2025 (Shift - 3)

- (a) 8 cm (b) 9 cm (c) 10 cm (d) 14 cm

Q.9. Tangents AB and AC are drawn to a circle from a point A, such that, $\angle BAC = 40^\circ$. A chord CP is drawn parallel to BA. The measure of $\angle CBP$ is:

Graduate Level 31/07/2025 (Shift - 3)

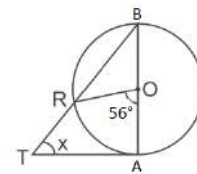
- (a) 55° (b) 45° (c) 35° (d) 40°

Q.10. There are two circles with radii of 7 cm and 4 cm are positioned so that they touch each other internally. What is the distance between their centers?

Graduate Level 25/07/2025 (Shift - 3)

- (a) 1 cm (b) 3 cm (c) 6 cm (d) 7 cm

Q.11. In a given circle, if AB is the diameter, the value of $3x$ is:



Graduate Level 25/07/2025 (Shift - 3)

- (a) 194° (b) 191° (c) 186° (d) 148°

Q.12. Two chords PQ and RS of a circle with centre O intersect each other at X. $\angle PXR = 85^\circ$ and $\angle POS = 120^\circ$. $\angle QOR$ is:

Higher Secondary 31/07/2025 (Shift - 2)

- (a) 70° (b) 95° (c) 110° (d) 125°

Q.13. A circle has two chords, PQ and RS, which are of equal length, subtend angles of x° and y° respectively at the center. Which of the following is true?

Higher Secondary 26/07/2025 (Shift - 2)

- (a) $x > y$ (b) $x < y$ (c) $x = y$ (d) $x + y = 180^\circ$

Q.14. Two circles intersect at points M and N. A line through M meets the first circle again at A and the second circle again at B. Another line through N meets the first circle again at C and the second circle again at D. If $\angle CAM = 55^\circ$ and $\angle MBD = 75^\circ$, find the value of $\angle MNC + \angle MND$.

Higher Secondary 24/07/2025 (Shift - 2)

- (a) 50° (b) 65° (c) 130° (d) 20°

Q.15. A chord of length 24 cm is 7 cm away from the center. Find the radius.

Matriculation Level 30/07/2025 (Shift - 1)

- (a) 14 cm (b) $\sqrt{193}$ cm
(c) $19\sqrt{17}$ cm (d) 17 cm

Q.16. What is the longest chord of a circle called?

Matriculation Level 25/07/2025 (Shift - 1)

- (a) Radius (b) Center
(c) Tangent (d) Diameter

Q.17. From a point T, a tangent TP at

point P, is drawn to a circle with centre O. A secant TQR (point Q is near to point T) is drawn from the point T. ΔPQR is inscribed into the circle by joining the points P, Q and R. Draw lines OQ and OR. If $\angle PTQ = 27^\circ$ and $\angle TPQ = 55^\circ$, what is the degree measure of $\angle ROQ$?

- (a) 86 (b) 82 (c) 98 (d) 94

Q.18. AB is the chord of the circle with centre O. A line segment DOC originating from a point D on the circumference of the circle in the major segment meets AB produced at C such that $BC = OD$. If angle $BCO = 30^\circ$, then angle AOD is:

- (a) 30° (b) 90° (c) 60° (d) 80°

Q.19. A triangle ABC is made on a circle, where AB is the diameter of the circle. If BC is equal to the radius of the circle and $\angle ABC = x \angle BAC$, then x is :

- (a) 2.5 (b) 2 (c) 1 (d) 1.5

Q.20. AB is a chord of length 32 cm of a circle of radius 20 cm, the tangents at A and B intersect at a point T. Find length TA (rounded off to two digits after decimal).

- (a) 36.50 cm (b) 21.33 cm
(c) 26.67 cm (d) 19.93 cm

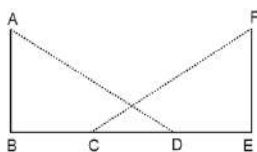
Q.21. WXYZ is a quadrilateral in which XQ and ZP are perpendicular to WY such that $XQ = ZP$. Diagonals ZX and WY intersect each other at point O. If $OX = 12$ cm, then find the value of ZX (in cm).

- (a) 12 (b) 24 (c) 36 (d) 18

Q.22. In triangle ABC, D is the mid-point of BC. If DL perpendicular to AB and DM perpendicular to AC such that $DL = DM$, then the triangle will be:

- (a) isosceles triangle
(b) right angled triangle
(c) obtuse angle triangle
(d) equilateral triangle

Q.23. In ΔABD and ΔFEC , $\angle BAD = 60^\circ$, $I(BD) = I(EC)$, $\angle ABD = \angle FEC = 90^\circ$, and $I(AB) = I(FC)$. Find the ratio of $\frac{\angle BAD}{\angle FCE}$.



- (a) 2 : 3 (b) 1 : 2 (c) 2 : 5 (d) 2 : 1

Q.24. If the length of a common external tangent to two circles is 9 and that of a common internal tangent is 5, then the product of the radii of the two circles is:

- (a) 12 (b) 8 (c) 14 (d) 10

Q.25. X, Y, and Z are three points on a plane with $XY = 11$ cm, $YZ = 13$ cm, and $XZ = 24$ cm. The number of circles that

travel through points X, Y, and Z is:

- (a) 0 (b) 2 (c) 1 (d) 3

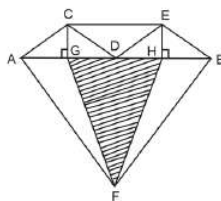
Q.26. In an obtuse-angled triangle ABC, the length of its longest side AB is 50 cm and one of the other two sides are 42 cm. If the area of the triangle is 294 cm^2 , what is the length (in cm) of its third side?

- (a) $12\sqrt{43}$ (b) $8\sqrt{35}$ (c) $15\sqrt{21}$ (d) $2\sqrt{58}$

Q.27. In a triangle PQR, RS intersects PQ at point S. The sides of the triangle $QR = 36$ cm, $SQ = 27$ cm, $RS = 18$ cm and $\angle QRS = \angle QPR$. What is the ratio of the perimeter of ΔPRS to that of ΔQSR ?

- (a) $\frac{5}{8}$ (b) $\frac{7}{9}$ (c) $\frac{8}{6}$ (d) $\frac{9}{12}$

Q.28. In the diagram, if $\overline{AB} \parallel \overline{CE}$, $AG = GD = DH = HB$, then which of the following is correct?



- I) $\Delta ACG \cong \Delta BEH$
II) $\Delta CDE \cong \Delta DEB \cong \Delta DCA$
III) $\Delta HFB \cong \Delta AFG$
(a) I, II and III (b) I and III
(c) Only I (d) I and II

Q.29. ABC is a right-angled triangle, right angled at A in such a way that $AB = AC$ and the bisector of angle B intersects the side AC at D. DM is perpendicular to BC, intersecting BC at M. If $AB = 12$ cm and $BC = 12\sqrt{2}$ cm, then find the length of side AD.

- (a) $12(\sqrt{2} + 2)$ cm (b) $12(\sqrt{2} - 1)$ cm
(c) $12(\sqrt{2} - 2)$ cm (d) $12(\sqrt{2} + 1)$ cm

Q.30. Two circles with centres M and N have radii 5 cm and 8 cm, respectively. The circles touch each other externally at point T. A line PR is drawn such that the points M, T and N lie on PR, P being closer to M. From P, a tangent $PQ = 12$ cm is drawn to the circle with centre M touching at Q, and from R, another tangent $RS = 15$ cm is drawn to the circle with centre N touching at S. What is the length (in cm) of PR?

- (a) 53 (b) 43 (c) 26 (d) 37

Q.31. Two circles of radius 7 units each, intersect in such a way that the common chord is of length 7 units. What is the common area in square units of the intersection?

- (a) $98(\frac{\pi}{6} - \frac{\sqrt{3}}{3})$ (b) $98(\frac{\pi}{3} - \frac{\sqrt{3}}{4})$

- (c) $98(\frac{\pi}{6} - \frac{\sqrt{3}}{6})$ (d) $98(\frac{\pi}{6} - \frac{\sqrt{3}}{4})$

Q.32. In a circle, ABCD is a cyclic quadrilateral in which AE is drawn parallel to CD, and BA is produced to F. If $\angle ABC = 85^\circ$ and $\angle FAE = 24^\circ$, find the value of $\angle BCD$.

- (a) 119° (b) 124° (c) 115° (d) 125°

Q.33. What is the length (in cm) of chord PQ in a circle with a radius of 7 cm, where a diameter AB and non-diameter chord PQ intersect perpendicularly at point C, and the ratio of AC to BC is 4 : 3?

- (a) $4\sqrt{3}$ (b) $8\sqrt{3}$ (c) $2\sqrt{3}$ (d) $6\sqrt{3}$

Q.34. The angles of a triangle are in the ratio 1 : 2 : 3, then find the ratio of the corresponding sides.

- (a) 1 : 2 : $\sqrt{3}$ (b) 1 : 2 : 3
(c) 1 : $\sqrt{2}$: 3 (d) 1 : $\sqrt{3}$: 2

Q.35. In ΔABC , P is a point on AB such that $PB : AP = 3 : 4$ and PQ is parallel to AC. If AR and QS are perpendicular to PC and $QS = 9$ cm, what is the length (in cm) of AR?

- (a) 28 (b) 35 (c) 21 (d) 14

Q.36. The lengths of the three sides of a triangle are given as 3 cm, 4 cm and 5 cm. The ratio of their corresponding altitudes to the opposite vertices will be:

- (a) 3 : 4 : 5 (b) 9 : 8 : 7
(c) 20 : 15 : 12 (d) 5 : 4 : 3

Q.37. The median AD of a triangle ABC is produced and a perpendicular CF is dropped on it. BE is perpendicular to AD. If $BC = 34$ cm and $DF = 8$ cm, what is the length (in cm) of BE?

- (a) 17 (b) 9 (c) 15 (d) 19

Q.38. How many times does the hour hand and minute hand of a clock coincide in half a day?

- (a) 55 (b) 22 (c) 11 (d) 44

Q.39. If a transversal intersects two parallel lines, and the difference between two interior angles on the same side of the transversal is 40° , then find the smallest angle among the interior angles.

- (a) 70° (b) 50° (c) 40° (d) 60°

Q.40. Which of the following statements is NOT true?

- (a) If the sides of a triangle are in the ratio 13 : 5 : 12, then two of its angles are acute angles.
(b) The triangle formed by connecting the midpoints of the sides of a triangle has one-fourth area of the

bigger triangle.

- (c) The area of the triangle with the largest side l always has an area less than $\frac{l^2}{2}$.

- (d) The area of an isosceles triangle with sides $2a$, $2a$ and $3a$ is $\frac{3\sqrt{3}a^2}{2}$.

Q.41. ΔPQR is right angled at Q such that $PQ = (x - y)$, $QR = x$ and $PR = (x + y)$. S is a point on QR such that $QS = PQ$. The ratio $QS : SR$ for any values of x and y is:

- (a) $1 : 3$ (b) $1 : 2$ (c) $2 : 1$ (d) $3 : 1$

Q.42. Two circles with centres P and Q of radii 6 cm and 4 cm, respectively, touch each other internally. If the perpendicular bisector of PQ meets the bigger circle in A and B , then the value of AB is:

- (a) $\sqrt{35}$ cm (b) $\sqrt{5}$ cm
(c) $2\sqrt{5}$ cm (d) $2\sqrt{35}$ cm

Q.43. Two circles of radius 13 cm and 15 cm intersect each other at points A and B . If the length of the common chord is 12 cm, then what is the distance between their centers?

- (a) $\sqrt{145} + \sqrt{184}$ (b) $\sqrt{131} + \sqrt{181}$
(c) $\sqrt{145} + \sqrt{169}$ (d) $\sqrt{133} + \sqrt{189}$

Q.44. M is the mid-point of side QR of a parallelogram $PQRS$ (P being on the top left hand, followed by other points going clockwise). The line SM is drawn intersecting PQ produced at T . What is the length (in terms of the length of SR) of PT ?

- (a) $\frac{3SR}{2}$ (b) $\frac{SR}{2}$ (c) $2SR$ (d) SR

Q.45. Two circles with diameters 50 cm and 58 cm, respectively, intersect each other at points A and B , such that the length of the common chord is 40 cm. Find the distance (in cm) between the centres of these two circles.

- (a) 34 (b) 37 (c) 36 (d) 35

Q.46. PQ is a chord of a circle. The tangent XR at X on the circle cuts PQ produced at R . If $XR = 12$ cm, $PQ = x$ cm, $QR = x - 2$ cm, then x (in cm) is:

- (a) 7 (b) 6 (c) 10 (d) 14

Q.47. A triangle PQR has three sides equal in measurement and if PM is perpendicular to QR , then which of following equality holds?

- (a) $3PM^2 = 2PQ^2$ (b) $3PQ^2 = 4PM^2$
(c) $3PM^2 = 4PQ^2$ (d) $3PQ^2 = 2PM^2$

Q.48. $AB = 8$ cm and $CD = 6$ cm are two parallel chords on the same side of the centre of a circle. If the distance between

them is 2 cm, then the radius (in cm) of the circle is:

- (a) $\frac{\sqrt{265}}{4}$ (b) $\frac{\sqrt{256}}{4}$ (c) $\frac{\sqrt{156}}{4}$ (d) $\frac{\sqrt{198}}{4}$

Q.49. If the distance between the centres of two circles is 12 cm and the radii are 5 cm and 4 cm, then the length (in cm) of the transverse common tangent is:

- (a) 9 (b) $\sqrt{143}$ (c) $\sqrt{63}$ (d) 7

Q.50. In ΔABC , the straight line parallel to the side BC meets AB and AC at the points P and Q , respectively. If $AP = QC$, the length of AB is 16 cm and the length of AQ is 4 cm, then the length (in cm) CQ is:

- (a) $(2\sqrt{21} + 2)$ (b) $(2\sqrt{18} - 2)$
(c) $(2\sqrt{17} - 2)$ (d) $(2\sqrt{19} + 2)$

Q.51. The three sides of a triangle are 9 , 24 and S units. Which of the following is correct?

- (a) $15 < S < 33$ (b) $15 \leq S < 33$
(c) $15 < S \leq 33$ (d) $15 \leq S \leq 33$

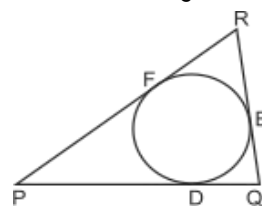
Q.52. If ΔABC and ΔDEF are congruent triangles, then which of the following is FALSE?

- (a) The ratio of AC to DF is $2 : 1$.
(b) The perimeter of both the triangles is equal.
(c) $AB = DE$, and $BC = EF$.
(d) The ratio of the angles in both the triangles is the same.

Q.53. The length of the common chord of two circles of radii 15 cm and 13 cm, whose centres are 14 cm apart, is:

- (a) 14 cm (b) 12 cm (c) 15 cm (d) 24 cm

Q.54. In the given figure, a circle is inscribed in ΔPQR , such that it touches the sides PQ , QR and RP at points D , E , F , respectively. If the lengths of the sides $PQ = 18$ cm, $QR = 13$ cm and $RP = 15$ cm, then find the length of PD .

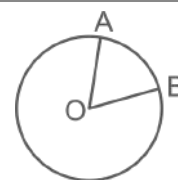


- (a) 10 cm (b) 8 cm (c) 15 cm (d) 12 cm

Q.55. In a ΔABC , $DE \parallel BC$, where D is a point on AB and E is a point on AC . If DE divides the area of ΔABC into two equal parts, then $DB : AB$ is equal to:

- (a) $\sqrt{2} : \sqrt{3}$ (b) $\sqrt{2} : \sqrt{2} + 1$
(c) $\sqrt{2} + 1 : \sqrt{2}$ (d) $\sqrt{2} - 1 : \sqrt{2}$

Q.56. In the given figure, the length of arc AB is equal to 3 times radius r of the circle.

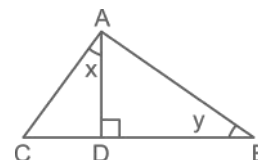


Find the area of sector OAB in terms of radius r .

- (a) $1.5r^2$ (b) $2.5r^2$ (c) $\frac{1}{2}r^2$ (d) $3r^2$

Q.57. In the given figure, if $AD \perp BC$, $AC = 26$ units, $CD = 10$ units, $BC = 42$ units, $\angle DAC = x$ and $\angle B = y$, then the value of

$$\frac{6}{\cos x} - \frac{5}{\cos y} + 8 \tan y$$
 is: (in unit)



- (a) $\frac{16}{9}$ (b) $\frac{13}{6}$ (c) $\frac{25}{4}$ (d) $\frac{15}{7}$

Q.58. AB is the diameter of a circle with centre O . P be a point on it. If $\angle AOP = 95^\circ$, then $\angle OBP =$ _____.

- (a) 57.5° (b) 45.5° (c) 47.5° (d) 55.5°

Q.59. In a parallelogram, one of the parallel sides is 16 cm and the other side is 12 cm. If the perpendicular distance between the two parallel sides of dimension 16 cm is 24 cm, then the perpendicular distance between its other two parallel sides is:

- (a) 16 cm (b) 12 cm (c) 32 cm (d) 24 cm

Q.60. Two medians DM and EN of ΔDEF intersect each other at G at right angles. If $EF = 20$ cm and $EN = 12$ cm, then what is the length of DM ? (in cm)

- (a) 20 (b) 12 (c) 18 (d) 15

Q.61. In a triangle ABC , $AB = 6$ cm, $BC = 11$ cm. How many integral values of AC can be possible?

- (a) 10 (b) 8 (c) 12 (d) 11

Q.62. Two circles of radius of 6 cm each. They intersect each other such that each passes through the centre of the other. What is the length of the common chord?

- (a) $9\sqrt{3}$ cm (b) 12 cm (c) $6\sqrt{3}$ cm (d) 8 cm

Q.63. In a trapezium $ABCD$, AB and DC are parallel to each other with a perpendicular distance of 8 m between them. Also, $(AD) = (BC) = 10$ m, and $(AB) = 15$ m $<$ (DC) . What is the perimeter (in m) of the trapezium $ABCD$?

- (a) 50 (b) 66 (c) 62 (d) 58

Q.64. 'O' is a point in the interior of an equilateral triangle. The perpendicular distance from 'O' to the sides are $\sqrt{3}$ cm,

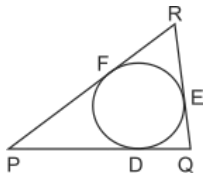
$2\sqrt{3}$ cm, $5\sqrt{3}$ cm. The perimeter of the triangle is:

- (a) 48 cm (b) 32 cm (c) 24 cm (d) 64 cm

Q.65. Three circles of radius 6 cm are kept touching each other. The string is tightly tied around these three circles. What is the length of the string?

- (a) $36 + 12\pi$ cm (b) $36 + 18\pi$ cm
(c) $24 + 36\pi$ cm (d) $36 + 20\pi$ cm

Q.66. In the given figure, a circle is inscribed in $\triangle PQR$, such that it touches the sides PQ, QR and RP at points D, E, F, respectively. If the lengths of the sides PQ = 15 cm, QR = 11 cm and RP = 13 cm, then find the length of PD.

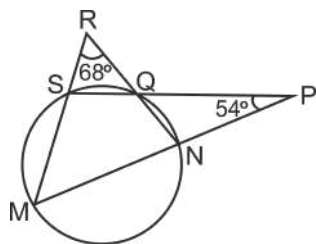


- (a) 9 cm (b) 8 cm (c) 7.5 cm (d) 8.5 cm

Q.67. Two circles touch each other externally at P. AB is a direct common tangent to the two circles, A and B are points of contact, and $\angle PAB = 40^\circ$. The measure of $\angle ABP$ is:

- (a) 45° (b) 55° (c) 50° (d) 40°

Q.68. In the given figure, MNP, SQP, NQR and MSR are straight lines. $\angle NPQ = 54^\circ$ and $\angle QRS = 68^\circ$. What is the degree measure of $\angle SMN$?



- (a) 29° (b) 38° (c) 54° (d) 68°

Q.69. PQ and RS are two parallel chords of a circle such that PQ is 48 cm and RS is 40 cm. If the chords are on the opposite sides of the centre and the distance between them is 22 cm, what is the radius (in cm) of the circle?

- (a) 25 (b) 24 (c) 35 (d) 22

Q.70. An arc on a circle that is 15 cm long subtends a 24° angle at the centre. What is the circumference of the circle?

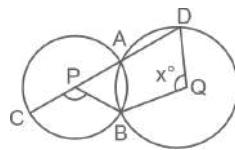
- (a) 240 cm (b) 220 cm
(c) 236 cm (d) 225 cm

Q.71. Calculate the area of the quadrilateral formed with the vertices (-3, 2), (5, 4), (7, -6) and (-5, -4).

- (a) 80 square units (b) 160 square units
(c) 0 square units (d) 150 square units

Q.72. In the following figure, P and Q are

centers of two circles. The circles are intersecting at points A and B. PA produced on both the sides meets the circles at C and D. If $\angle CPB = 100^\circ$, then find the value of x.



- (a) 115 (b) 120 (c) 110 (d) 100

Q.73. An isosceles $\triangle MNP$ is inscribed in a circle. If $MN = MP = 16\sqrt{5}$ cm, and $NP = 32$ cm what is the radius (in cm) of the circle?

- (a) 20 (b) $18\sqrt{5}$ (c) 18 (d) $20\sqrt{5}$

Q.74. In triangle ABC, D and E are points on BC such that $AD = AE$ and $\angle BAD = \angle CAE$. If $AB = (2p + 3)$, $BD = 2p$, $AC = (3q - 1)$ and $CE = q$, then find the value of $(p + q)$.

- (a) 3 (b) 4.5 (c) 3.6 (d) 2

Q.75. In a circle with centre O, AC and BD are two chords. AC and BD meet at E, when produced. If AB is a diameter and $\angle AEB = 36^\circ$, then the measure of $\angle DOC$ is:

- (a) 112° (b) 124° (c) 136° (d) 108°

Q.76. AB is a chord in the minor segment of a circle with center O. C is a point on the minor arc (between A and B). The tangents to the circle at A and B meet at a point P. If $\angle ACB = 108^\circ$, then $\angle APB$ is equal to:

- (a) 36° (b) 54° (c) 27° (d) 18°

Q.77. In $\triangle ABC$, M is the midpoint of the side AB. N is a point in the interior of $\triangle ABC$ such that CN is the bisector of $\angle C$ and $CN \perp NB$. What is the length (in cm) of MN, if $BC = 10$ cm and $AC = 15$ cm?

- (a) 5 (b) 2.5 (c) 4 (d) 2

Q.78. The side of an equilateral $\triangle ABC$ is $3\sqrt{7}$ cm. P is a point on side BC such that $BP : PC = 1 : 2$. The length (in cm) of AP is:

- (a) $6\sqrt{3}$ (b) $7\sqrt{3}$ (c) 6 (d) 7

Q.79. How many isosceles triangles with integer sides are possible such that the sum of two of the sides is 16 cm?

- (a) 18 (b) 15 (c) 9 (d) 24

Q.80. If a and b are the length of two sides of a triangle such that the product $ab = 24$, where a and b are integers, then how many such triangles are possible?

- (a) 15 (b) 12 (c) 18 (d) 16

Q.81. In $\triangle ABC$, AD is a median. If points E, F and G midpoints of AD, AE and DE, respectively, then what will be the area Δ

BFG?

- (a) $\frac{1}{2}(\text{Area } \triangle ABC)$ (b) $\frac{1}{8}(\text{Area } \triangle ABC)$
(c) $\frac{1}{4}(\text{Area } \triangle ABC)$ (d) $\frac{1}{2}(\text{Area } \triangle BGC)$

Q.82. D, E and F are the feet of the perpendiculars from the vertices A, B and C respectively, of a triangle ABC. If angle BED and angle BFE (in degrees) are 24 and 110, respectively, what is the measure of angle EBF? (in degrees)

- (a) 55 (b) 67 (c) 86 (d) 46

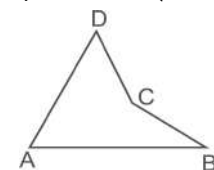
Q.83. In a $\triangle ABC$, $\angle A = 90^\circ$, if BM and CN are two medians, $\frac{BM^2 + CN^2}{BC^2}$ is equal

- (a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{5}{4}$ (d) $\frac{3}{4}$

Q.84. The side MN of $\triangle LMN$ is produced by X. If $\angle LNX = 117^\circ$ and $\angle M = \frac{1}{2}\angle L$, then $\angle L$ is:

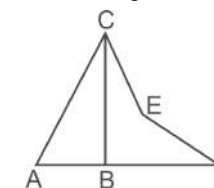
- (a) 77° (b) 78° (c) 76° (d) 75°

Q.85. In the following figure (not to scale), $\angle DAB + \angle CBA = 90^\circ$, $BC = AD$, $AB = 20$ cm, $CD = 10$ cm, then the area of the quadrilateral (in cm^2) ABCD is:



- (a) 120 (b) 150 (c) 100 (d) 75

Q.86. If in the given figure, $\angle ACB + \angle BAC = 80^\circ$; $\angle BDE = 35^\circ$; $\angle BCE = 45^\circ$, then the marked angle $\angle CED$ is:



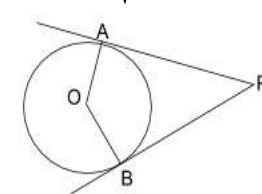
- (a) 150° (b) 120° (c) 160° (d) 135°

Q.87. XYZ is a triangle. If the medians ZL and YM intersect each other at G, then (Area of $\triangle GLM$: Area of $\triangle XYZ$)

- (a) 1 : 14 (b) 1 : 12 (c) 1 : 11 (d) 1 : 10

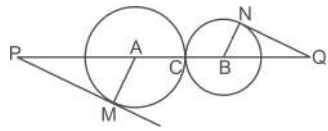
Q.88. PA and PB are tangents to the circle and O is the centre of the circle. The radius is 5 cm and PO is 13 cm. If the area of the triangle PAB is M, then the

value of $\sqrt{\frac{M}{15}}$ is:



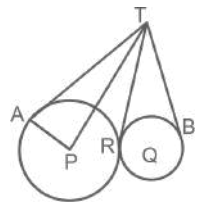
(a) $\sqrt{\frac{24}{13}}$ (b) $\frac{24}{13}$ (c) $\frac{12}{13}$ (d) $\sqrt{\frac{12}{13}}$

Q.89. In the given figure, MP is the tangent to a circle with centre A and NQ is a tangent to a circle with centre B. If MP = 15 cm, NQ = 8 cm, PA = 17 cm and BQ = 10 cm, then AB is:



(a) 13.5 cm (b) 23 cm (c) 14 cm (d) 28 cm

Q.90. In the figure, two circles with centres P and Q touch externally at R. Tangents AT and BT meet the common tangent TR at T. If AP = 6cm and PT = 10 cm, then BT = ?



(a) 6 cm (b) 12 cm (c) 8 cm (d) 10 cm

Q.91. PRT is a tangent to a circle with centre O, at the point R on it. Diameter SQ of the circle is produced to meet the tangent at P and QR is joined. If $\angle QRP = 28^\circ$, then the measure of $\angle SPR$ is :
(a) 29° (b) 34° (c) 62° (d) 32°

Q.92. PQRS is a cyclic quadrilateral in which PQ = x cm, QR = 16.8 cm, RS = 14 cm and PS = 25.2 cm and PR bisects QS. What is the value of x ?
(a) 18 (b) 21 (c) 28 (d) 24

Q.93. In $\triangle ABC$, $\angle C = 90^\circ$, M and N are the mid-points of the sides AB and AC respectively. CM and BN intersect each other at D and $\angle BDC = 90^\circ$. If BC = 8 cm, then the length of BN is:

(a) $6\sqrt{3}$ cm (b) $6\sqrt{6}$ cm
(c) $4\sqrt{6}$ cm (d) $8\sqrt{3}$ cm

Q.94. Two equilateral triangle of side $10\sqrt{3}$ cm are joined to form a quadrilateral. What is the altitude of the quadrilateral ?
(a) 12 cm (b) 14 cm (c) 16 cm (d) 15 cm

Q.95. The lengths of the parallel sides of a trapezium are 51 cm, and 21 cm, and that of each of the other two sides is 39 cm. What is the area (in cm^2) of the trapezium?
(a) 1206 (b) 1296 (c) 1152 (d) 1260

Q.96. ABCD is a cyclic quadrilateral. Sides AB and DC, when produced, meet at E, and sides BC and AD, when produced, meet at F. If $\angle BFA = 60^\circ$ and

$\angle AED = 30^\circ$, then the measure of $\angle ABC$ is :

(a) 65° (b) 75° (c) 70° (d) 80°

Q.97. PT is a tangent at the point R on a circle with centre O. SQ is a diameter. Which when produced meets the tangent PT at P. If $\angle SPT = 32^\circ$, then what will be the measure of $\angle QRP$?

(a) 32° (b) 58° (c) 30° (d) 29°

Q.98. PQRS is a rectangle. T is a point on PQ such that RTQ is an isosceles triangle and $PT = 5QT$. If the area of the triangle RTQ is $12\sqrt{3} \text{ cm}^2$, then the area of rectangle PQRS is :

(a) $142\sqrt{3} \text{ cm}^2$ (b) $134\sqrt{3} \text{ cm}^2$
(c) $144\sqrt{3} \text{ cm}^2$ (d) 142 cm^2

Q.99. If in $\triangle PQR$, $\angle P = 120^\circ$, $PS \perp QR$ at S and $PQ + QS = SR$, then the measure of $\angle Q$ is :

(a) 20° (b) 50° (c) 40° (d) 30°

Q.100. A circle is inscribed in a quadrilateral ABCD touching AB, BC, CD and AD at the points P, Q, R and S, respectively, and $\angle B = 90^\circ$. If AD = 24 cm, AB = 27 cm and DR = 6cm, then what is the circumference of the circle ?

(a) 20π (b) 18π (c) 15π (d) 12π

Q.101. The sum of the interior angles of a regular polygon is 1260° . What is the difference between an exterior angle and an interior angle of the polygon?

(a) 105° (b) 100° (c) 120° (d) 90°

Q.102. In a trapezium ABCD, $DC \parallel AB$, AB = 12 cm and DC = 7.2 cm. What is the length of the line segment joining the mid-points of its diagonals?

(a) 2.6 (b) 4.8 (c) 2.4 (d) 3.6

Q.103. If each interior angle of a regular polygon is $(128\frac{4}{7})^\circ$ then what is the sum of the number of its diagonal and the number of its sides?

(a) 15 (b) 19 (c) 17 (d) 21

Q.104. A quadrilateral ABCD is inscribed in a circle with centre O. If $\angle BOC = 92^\circ$ and $\angle ADC = 112^\circ$, then $\angle ABO$ is equal to :

(a) 22° (b) 24° (c) 28° (d) 26°

Q.105. The perimeter of $\triangle ABC$ is 24 cm and its side, BC = 9 cm. AD is the bisector of $\angle BAC$, while I is the incentre AI : ID is equal to :

(a) 7 : 5 (b) 5 : 2 (c) 3 : 2 (d) 5 : 3

Q.106. The area of a sector of a circle with central angle 60° is A. The circumference of the circle is C. Then A is equal to :

(a) $\frac{c^2}{6\pi}$ (b) $\frac{c^2}{18\pi}$ (c) $\frac{c^2}{24\pi}$ (d) $\frac{c^2}{4\pi}$

Q.107. It is given that the area of a triangle is A. The values of its perimeter, inradius, circumradius and the average of the lengths of the medians are respectively p, r, R and d. The ratio A : p is equal to

(a) $(R + r)^2 : d$ (b) $r : 2$

(c) $r : 1$ (d) $(R - r)^2 : r$

Q.108. A regular hexagon is inscribed in a circle. What is the ratio of the area of the circle to that of its portion not covered by the hexagon?

(a) $\frac{2\pi}{2\pi - 3\sqrt{3}}$ (b) $\frac{\pi}{\pi - 3\sqrt{3}}$

(c) $\frac{2\pi}{\sqrt{3}}$ (d) $\frac{\pi}{\sqrt{3}}$

Q.109. In a stadium an athlete is running on a circular path with uniform speed during a practice session. The angle covered by him during one second is found to be 10° by a coach observing him from the centre of the circular track. What would be the measure of angle (in degrees) described by the athlete by an observer standing on the circle?

(a) 5

(b) It depends on the exact position of the observer on the circle

(c) 10

(d) 20

Q.110. The circumcentre, incentre, orthocentre and the centroid of a triangle are one and the same point. The triangle must be:

(a) Isosceles (b) right-angled

(c) right-angled isosceles (d) equilateral

Q.111. There is a polygon of 11 sides. How many triangles can be drawn by only using the vertices of the polygon?

(a) 165 (b) 150 (c) 175 (d) 180

Q.112. In a circle with centre O, AB is a diameter. Points C, D and E are on the circle on one side of AB such that ABEDC is a pentagon. The sum of angles ACD and DEB is :

(a) 240° (b) 225° (c) 270° (d) 180°

Q.113. In $\triangle ABC$, D is a point on BC such that $\angle BAD = \frac{1}{2}\angle ADC$, $\angle BAC = 87^\circ$ and $\angle C = 42^\circ$. What is the measure of $\angle ADB$?
(a) 94° (b) 68° (c) 102° (d) 78°

Q.114. In a circle with centre O, A diameter AB and a chord CD intersect each other at E, AC and AD are joined. If $\angle BOC = 48^\circ$ and $\angle AOD = 100^\circ$, then what is the measure of $\angle CEB$?



(a) 72° (b) 74° (c) 78° (d) 82°

Q.115. In $\triangle ABC$, AD, the bisector of $\angle A$, meets BC at D. If $BC = a$, $AC = b$ and $AB = c$, then $BD - DC =$

- (a) $\frac{ac}{b+c}$ (b) $\frac{a(c+b)}{c-b}$
(c) $\frac{a(c-b)}{c+b}$ (d) $\frac{ab}{b+c}$

Q.116. In a circle with centre O, ABDC is a cyclic quadrilateral with AB as a diameter of the circle. AC and BD produced meet at E. If $\angle CED = 70^\circ$, then what is the measure of $\angle COD$?

- (a) 45° (b) 60° (c) 30° (d) 40°

Q.117. In $\triangle ABC$, BD is perpendicular to AC. E is a point on BC such that $\angle BEA = x^\circ$. If $\angle EAC = 38^\circ$ and $\angle EBD = 40^\circ$, then the value of x is:

- (a) 88° (b) 68° (c) 78° (d) 72°

Q.118. A circle is inscribed in a quadrilateral ABCD, touching sides AB, BC, CD and DA at P, Q, R, and S respectively. If $AS = 8$ cm, $BC = 11$ cm and $CR = 5$ cm, then the length AB (in cm) is equal to:

- (a) 12 (b) 13 (c) 16 (d) 14

Q.119. In $\triangle ABC$, $\angle A = 90^\circ$. If BL and CM are medians, then:

- (a) $4(BL^2 + CM^2) = 3BC^2$
(b) $4(BL^2 + CM^2) = 5BC^2$
(c) $3(BL^2 + CM^2) = 4BC^2$
(d) $5(BL^2 + CM^2) = 4BC^2$

Q.120. Two concentric circles are of radius 15 cm and 9 cm. What is the length of the chord of the larger circle which is tangent to the smaller circle (in cm)?

- (a) 24 (b) 18 (c) 20 (d) 25

Q.121. Tangents AB and AC are drawn to a circle from a point A, such that, $\angle BAC = 40^\circ$. A chord CP is drawn parallel to BA. The measure of $\angle CBP$ is:

- (a) 55° (b) 45° (c) 35° (d) 40°

Q.122. In a circle with centre O, AD is a diameter and AC is a chord. B is a point on AC, such that $OB = 5$ cm and $\angle OBA = 60^\circ$. If $\angle DOC = 60^\circ$, then what is the length of BC (in cm)?

- (a) 4 (b) $3\sqrt{5}$ (c) $5\sqrt{3}$ (d) 5

Q.123. Chords AB and CD of a circle, when produced, meet at point P outside the circle. If $AB = 6$ cm, $CD = 3$ cm and $PD = 5$ cm, then PB is equal to:

- (a) 6 cm (b) 6.25 cm (c) 5 cm (d) 4 cm

Q.124. In a $\triangle ABC$, the sides are $AB =$

16cm, $AC = 63$ cm, $BC = 65$ cm. From A, a straight line AM is drawn up to the midpoint M of side BC. Then the length of AM is equal to in (cm):

- (a) 32.5 (b) 24.5 (c) 23.5 (d) 31.5

Q.125. In $\triangle ABC$, AD is the median and G is a point on AD such that $AG : GD = 2 : 1$, then $\text{ar}(\triangle BDG) : (\triangle ABC)$ is equal to:

- (a) 1 : 4 (b) 1 : 9 (c) 1 : 6 (d) 1 : 3

Q.126. From a point P, outside a circle, PAB is a secant and PT is a tangent to the circle, where A, B and T are points on the circle. If $PT = 5$ cm, $PA = 4$ cm and $AB = x$, then x is equal to:

- (a) 2.25 (b) 2.75 (c) 2.45 (d) 1.75

Q.127. Let O be the centre of a circle and AC be its diameter. BD is a chord intersecting AC at E. Point A is joined at B and D. If $\angle BOC = 50^\circ$ and $\angle AOD = 110^\circ$, then $\angle BEC = ?$

- (a) 80° (b) 70° (c) 55° (d) 90°

Q.128. In a $\triangle ABC$, right angled at B, $AB = 7$ cm and $(AC - BC) = 1$ cm. The value of $(\sec C + \cot A)$ is:

- (a) $\frac{19}{24}$ (b) $\frac{4}{3}$ (c) $\frac{3}{4}$ (d) 1

Q.129. In a circle with centre O, PQR is a tangent at the point Q on it. AB is a chord in the circle parallel to the tangent such that $\angle BQR = 70^\circ$. What is the measure of $\angle AQB$?

- (a) 40° (b) 60° (c) 55° (d) 35°

Q.130. The base and altitude of an isosceles triangle are 10 cm and 12 cm respectively. Then the length of each equal side (in cm) is:

- (a) 10 (b) 7.5 (c) 8.5 (d) 13

Q.131. ABCD is a quadrilateral whose side AB is the diameter of a circle through A, B, C and D. If $\angle ADC = 130^\circ$, then the measure of $\angle BAC$ is:

- (a) 40° (b) 45° (c) 35° (d) 50°

Q.132. In $\triangle ABC$, $\angle A$ is a right angle. The lengths of AC and BC are 6 cm and 10 cm respectively. Point D is on AB such that $BD = 4$ cm. What is the length of CD (in cm)?

- (a) $2\sqrt{13}$ (b) $3\sqrt{10}$ (c) $2\sqrt{10}$ (d) $3\sqrt{13}$

Q.133. AB is a diameter of a circle with centre O. CB is a tangent to the circle at B. AC intersects the circle at G. If the radius of the circle is 6 cm and $AG = 8$ cm, then the length of BC is in (cm):

- (a) $2\sqrt{5}$ (b) $6\sqrt{6}$ (c) $2\sqrt{6}$ (d) $6\sqrt{5}$

Q.134. In a circle with centre O, ACBO is a parallelogram where C is a point on the minor arc AB. What is the measure of

$\angle AOB$?

- (a) 150° (b) 100° (c) 110° (d) 120°

Q.135. The vertices of $\triangle PQR$ lie on a circle with centre O. SR is a tangent to the circle at the point R. If QR bisects the $\angle ORS$, then what is the measure of $\angle RPQ$?

- (a) 60° (b) 45° (c) 40° (d) 30°

Q.136. In $\triangle ABC$ with sides 6 cm, 7 cm and 8 cm, the angle bisector of the largest angle divides the opposite side into two segments. What is the length of the shorter segment?

- (a) $\frac{24}{5}$ (b) $\frac{21}{5}$ (c) $\frac{48}{13}$ (d) $\frac{56}{13}$

Q.137. ABCD is a cyclic quadrilateral whose diagonals intersect at P. If $AB = BC$, $\angle DBC = 70^\circ$ and $\angle BAC = 30^\circ$, then the measure of $\angle PCD$ is:

- (a) 35° (b) 50° (c) 55° (d) 30°

Q.138. ABCD is a cyclic quadrilateral in which $\angle A = 67^\circ$ and $\angle B = 92^\circ$. What is the difference between the measure of $\angle C$ and $\angle D$?

- (a) 29° (b) 27° (c) 19° (d) 25°

Q.139. In $\triangle ABC$, $AD \perp BC$ and $BE \perp AC$, AD and BE intersect each other at F. If $BF = AC$, then the measure of $\angle ABC$ is:

- (a) 45° (b) 60° (c) 70° (d) 50°

Q.140. A triangle ABC is inscribed in a circle with centre O. AO is produced to meet the circle at K and $AD \perp BC$. If $\angle B = 80^\circ$ and $\angle C = 64^\circ$, then the measure of $\angle DAK$ is:

- (a) 10° (b) 16° (c) 12° (d) 20°

Q.141. In a circle of radius 10 cm, with centre O, PQ and PR are two chords each of length 12 cm. PO bisects chord QR at the points S. The length of OS (in cm):

- (a) 2.8 (b) 2.5 (c) 3.2 (d) 3

Q.142. In $\triangle ABC$, F and E are the points on sides AB and AC respectively, such that $FE \parallel BC$ and FE divide the triangle in two parts of equal area. If $AD \perp BC$ and AD intersect FE at G, then $GD : AG = ?$

- (a) $\sqrt{2} : 1$ (b) $(\sqrt{2} - 1) : 1$
(c) $2\sqrt{2} : 1$ (d) $(\sqrt{2} + 1) : 1$

Q.143. A circle is inscribed in a triangle ABC. It touches the sides AB, BC and AC at the points R, P and Q respectively. If $AQ = 4.5$ cm, $PC = 5.5$ cm and $BR = 6$ cm, then the perimeter of the triangle ABC is:

- (a) 30.5 (b) 28 (c) 32 (d) 26.5

Q.144. A unique circle can always be drawn through x number of given non-collinear points, then x must be:

- (a) 2 (b) 3 (c) 4 (d) 1



Q.145. Side AB of a triangle ABC is 80 cm long, whose perimeter is 170 cm. If angle ABC = 60° , the shortest side of triangle ABC measures _____ cm.
(a) 17 (b) 15 (c) 25 (d) 21

Q.146. Which of the following solids has the least number of faces?
(a) Cube (b) Cone
(c) Triangular prism (d) Square pyramid

Q.147. Which of the following has the maximum number of vertices?
(a) Cuboid (b) Triangular Prism
(c) Hexagonal pyramid (d) Tetrahedron

Examination wise Questions

SSC CGL 2025 Tier - I

Q.148. What is $\frac{7\pi}{6}$ radians in degrees ?
SSC CGL 12/09/2025 (Shift 2)
(a) 225° (b) 210° (c) 240° (d) 360°

Q.149. What is the central angle of a sector with an arc length of 20 cm in a circle of radius 10 cm ?
SSC CGL 12/09/2025 (Shift 2)
(a) 6 radians (b) 3 radians
(c) 2 radians (d) 4 radians

Q.150. Two triangles are similar with sides in the ratio 2:5. What is the ratio of their areas?
SSC CGL 12/09/2025 (Shift 2)
(a) 16 : 25 (b) 25 : 8 (c) 4 : 25 (d) 25 : 9

Q.151. The angles of a cyclic quadrilateral are in the ratio 1 : 2 : 4 : 5. What is the measure of the smallest angle?
SSC CGL 12/09/2025 (Shift 2)
(a) 30° (b) 36° (c) 108° (d) 150°

Q.152. In triangles ABC and PQR, if the lengths of the sides are such that AB equals PQ, BC equals QR, and CA equals RP, which congruence rule demonstrates that triangle ABC is congruent to triangle PQR ?
SSC CGL 13/09/2025 (Shift 1)
(a) SSS (b) SAS (c) ASA (d) AAS

Q.153. In a cyclic quadrilateral PQRS, if angle P is 110° , what is the measure of angle R ?
SSC CGL 13/09/2025 (Shift 1)
(a) 60° (b) 70° (c) 90° (d) 120°

Q.154. There are two parallel chords measuring 14 cm and 10 cm, both situated on the same side of the center of a circle. The space between the two chords is 4 cm. What is the radius of the circle?

SSC CGL 13/09/2025 (Shift 3)
(a) 7.07 cm (b) 8.07 cm
(c) 6.05 cm (d) 5.05 cm

Q.155. Convert 3.5 radians to degrees.
SSC CGL 14/09/2025 (Shift 3)
(a) 130° (b) 143.24° (c) 180° (d) 200.55°

Q.156. In similar triangles, the ratio of corresponding altitudes is 3 : 4. What is the ratio of their areas?
SSC CGL 14/09/2025 (Shift 3)
(a) 2 : 5 (b) 9 : 16 (c) 2 : 25 (d) 5 : 2

Q.157. The diameter of a circle measures 15 cm. What is the maximum length of a chord in this circle?
SSC CGL 14/09/2025 (Shift 3)
(a) 15 cm (b) 35 cm (c) 65 cm (d) 90 cm

Q.158. A circular disc of diameter 14 cm is inscribed inside an equilateral triangle. What is the approximate area of the remaining portion of the triangle?
SSC CGL 15/09/2025 (Shift 1)
(a) 155.6 cm^2 (b) 168.2 cm^2
(c) 250.3 cm^2 (d) 100.00 cm^2

Q.159. In triangle PQR, medians PS, QT, and RU intersect at the centroid G. What is the ratio of the area of triangle GPQ to the area of triangle PQR ?
SSC CGL 15/09/2025 (Shift 1)
(a) 2 : 1 (b) 1 : 3 (c) 3 : 1 (d) 3 : 4

Q.160. In ΔPQR , a line segment DE is parallel to QR, with D on PQ and E on PR. If the ratio of the area of ΔPDE to the area of the trapezoid DERQ is 4 : 21, what is the ratio of PD to DQ ?
SSC CGL 15/09/2025 (Shift 1)
(a) 2 : 3 (b) 2 : 7 (c) 2 : 5 (d) 4 : 7

Q.161. Two circles have radii of 12 cm and 5 cm. If the length of a direct common tangent is 24 cm, what is the distance between their centers?
SSC CGL 15/09/2025 (Shift 1)
(a) 16 cm (b) 22 cm (c) 19 cm (d) 25 cm

Q.162. The distance between the centers of two circles is D. The lengths of their direct and transverse common tangents are P and Q, respectively. If $P^2 + Q^2 = 400$ and the sum of the squares of their radii is 200, what is the value of D ?
SSC CGL 15/09/2025 (Shift 1)
(a) 20 (b) $20\sqrt{2}$ (c) $5\sqrt{2}$ (d) 30

Q.163. Two parallel chords of a circle are 30 cm and 16 cm long. Both chords lie on the same side of the circle's center, and the distance between them is 7 cm. Find the diameter of the circle.
SSC CGL 15/09/2025 (Shift 1)
(a) 28 cm (b) 30 cm (c) 24 cm (d) 34 cm

Q.164. If two circles are touching externally how many common tangents do they have?
SSC CGL 15/09/2025 (Shift 2)
(a) 1 (b) 2 (c) 3 (d) 0

Q.165. Two equal chords, AB and CD, are at a distance of 9 cm from the center of a circle. If the radius is 41 cm, what is the length of AB ?
SSC CGL 15/09/2025 (Shift 2)
(a) 24 cm (b) 48 cm (c) 72 cm (d) 80 cm

Q.166. In a triangle PQR, medians PS and QT intersect at O. If the length of median PS is 15 cm, what is the length of the segment PO?
SSC CGL 16/09/2025 (Shift 1)
(a) 10 cm (b) 6 cm (c) 8 cm (d) 9 cm

Q.167. Two right-angled triangular blocks, PQR and XYZ, have $\angle Q = \angle Y = 90^\circ$. If the lengths of the hypotenuses PR and XZ are equal, and the sides PQ and XY are equal, are the triangles congruent? If so, by what rule?
SSC CGL 16/09/2025 (Shift 1)
(a) Yes, by RHS (b) Yes, by SAS
(c) Yes, by SSS (d) Yes, by ASA

Q.168. A right triangle ABC is inscribed in a circle with a diameter of 26 cm. An altitude BD is drawn from the vertex B to the hypotenuse AC. If the length of the side AB is 10 cm, what is the length of the segment AD?
SSC CGL 16/09/2025 (Shift 1)
(a) 3.8 cm (b) 3 cm (c) 5.2 cm (d) 4 cm

Q.169. Two circles touch each other externally. Which of the following statements is true?
SSC CGL 16/09/2025 (Shift 1)
(a) There are three common tangents to the circles
(b) There are two common tangents to the circles
(c) There are four common tangents to the circles
(d) There is only one common tangent to the circles

Q.170. Two circles with radii R and r touch each other externally. If the length of their direct common tangent is D, which of the following is the correct relationship between D, R, and r?
SSC CGL 16/09/2025 (Shift 1)
(a) $D = R + r$ (b) $D = 2(R + r)$
(c) $D = 2\sqrt{Rr}$ (d) $D = R^2 + r^2$

Q.171. A chord of a circle has a length of 10 cm. The angle subtended by the chord at a point on the circumference is 30° . What is the distance from the center of the circle to the chord?



SSC CGL 16/09/2025 (Shift 1)

(a) $5\sqrt{3}$ cm (b) $3\sqrt{3}$ cm (c) 5 cm (d) 3 cm

Q.172. What is the area of the segment formed by a chord in a circle of radius 4 cm, if the angle subtended at the center is 60° ?

SSC CGL 16/09/2025 (Shift 1)

- (a) $(\frac{8\pi}{3}) - 4\sqrt{3}$ (b) $(\frac{16\pi}{3}) - 4\sqrt{3}$
 (c) $(\frac{8\pi}{3}) - 8\sqrt{3}$ (d) $(\frac{16\pi}{3}) - 8\sqrt{3}$

Q.173. Four non-overlapping, non-touching circles are drawn in a plane. What is the maximum number of distinct common tangents that can be drawn to them in total?

SSC CGL 16/09/2025 (Shift 2)

- (a) 20 (b) 24 (c) 28 (d) 32

Q.174. In a circle, chord AB and chord CD intersect at E such that AE : EB = 3 : 4 and CE : ED = 5 : 2. If AB = x and CD = y, which of the following is true?

SSC CGL 16/09/2025 (Shift 2)

- (a) $\frac{x^2}{y^2} = \frac{5}{6}$ (b) $\frac{x}{y} = \frac{6}{15}$
 (c) $\frac{x}{y} = \frac{10}{9}$ (d) $\frac{x^2}{y^2} = \frac{144}{100}$

Q.175. The angle bisectors of triangle PQR converge at the in-center 'O'. Given that the measure of $\angle QOR$ is 125° , what is the measure of $\angle QPR$?

SSC CGL 16/09/2025 (Shift 3)

- (a) 40° (b) 70° (c) 90° (d) 60°

Q.176. In a quadrilateral PQRS, PQ is parallel to RS, and PR is a diagonal. If PQ = RS, which of the following is true?

SSC CGL 16/09/2025 (Shift 3)

- (a) $\triangle PQR \cong \triangle RSP$ by SSS
 (b) $\triangle PQR \cong \triangle RSP$ by SAS
 (c) $\triangle PQR \cong \triangle RSP$ by ASA
 (d) The triangles are similar, but not necessarily congruent

Q.177. In triangle PQR, a line segment ST is drawn through the centroid O, parallel to side QR. S lies on PQ and T lies on PR. What is the ratio of the area of the smaller triangle PST to the area of the trapezoid STRQ?

SSC CGL 16/09/2025 (Shift 3)

- (a) 3 : 1 (b) 1 : 2 (c) 4 : 5 (d) 4 : 9

Q.178. Two circles of radii 9 cm and 5 cm have their centers 15 cm apart. What is the length of the internal common tangent?

SSC CGL 16/09/2025 (Shift 3)

- (a) 12 cm (b) $\sqrt{26}$ cm
 (c) $\sqrt{29}$ cm (d) $\sqrt{25}$ cm

Q.179. Two circles with radii $r_1 = 14$ cm and $r_2 = 6$ cm have their centers 40 cm apart. Find the length of the transverse common tangent between them.

SSC CGL 16/09/2025 (Shift 3)

- (a) $\sqrt{1200}$ cm (b) 1200 cm
 (c) $\sqrt{1275}$ cm (d) 13 cm

Q.180. In a circle, two chords are 8 cm and 10 cm long. If the 10 cm chord is 6 cm away from the center, what can be said about the distance of the 8 cm chord from the center?

SSC CGL 16/09/2025 (Shift 3)

- (a) It is less than 3 cm.
 (b) It is equal to 3 cm.
 (c) It is greater than 3 cm.
 (d) It cannot be determined.

Q.181. A circle is circumscribed about a triangle. If one of the angles of the triangle is 150° , and the radius of the circumscribed circle is R, what is the length of the side opposite the 150° angle?

SSC CGL 16/09/2025 (Shift 3)

- (a) $R\sqrt{3}$ (b) 2R (c) R (d) $R\sqrt{2}$

Q.182. A triangle can have:

SSC CGL 17/09/2025 (Shift 1)

- (a) three right angles
 (b) One acute angle and one right angle
 (c) Only one right or one obtuse angle
 (d) Three obtuse angles each measuring Less than 90°

Q.183. The area of a triangle PQR is 8 cm². If a similar triangle DEF has sides that are twice the length of $\triangle PQR$'s sides, what is the area of $\triangle DEF$?

SSC CGL 17/09/2025 (Shift 1)

- (a) 32 cm² (b) 48 cm²
 (c) 64 cm² (d) 80 cm²

Q.184. From an outside point T, two tangents, TP and TQ, are drawn to a circle. Given that the length of TP is 15 cm, what is the length of TQ?

SSC CGL 17/09/2025 (Shift 1)

- (a) 16 cm (b) 10 cm (c) 15 cm (d) 24 cm

Q.185. A circle is inscribed within a right triangle. Considering that the lengths of the two legs measure 12 cm and 16 cm, what is the radius of the inscribed circle?

SSC CGL 17/09/2025 (Shift 1)

- (a) 6 cm (b) 9 cm (c) 4 cm (d) 5 cm

Q.186. A triangle ABC is inscribed in a circle. A tangent is drawn to the circle at point A, intersecting the extension of side BC at D. If AD = 12 cm and CD = 8 cm, what is the length of side BC?

SSC CGL 17/09/2025 (Shift 2)

- (a) 10 cm (b) 8 cm (c) 12 cm (d) 18 cm

Q.187. A circle with radius $2x$ touches another circle with radius $4x$ externally. What is the length of a direct common tangent?

SSC CGL 17/09/2025 (Shift 2)

- (a) $2x$ (b) $3x$ (c) $4x\sqrt{2}$ (d) $3x\sqrt{2}$

Q.188. Two circles have radii 5 cm and 3 cm, and their centres are 8 cm apart. How many common tangents can be drawn between them, and what type(s) of tangents are they?

SSC CGL 17/09/2025 (Shift 2)

- (a) 4 tangents: 2 direct and 2 transverse
 (b) 3 tangents: 2 direct and 1 transverse
 (c) 2 tangents: only direct tangents
 (d) 0 tangents: one circle lies completely inside the other

Q.189. Two chords PQ and RS intersect at a point T inside a circle. If PT = 8 cm, TQ = 10 cm, and RT = 6 cm, what is the length of TS?

SSC CGL 17/09/2025 (Shift 2)

- (a) $\frac{40}{3}$ cm (b) $\frac{20}{6}$ cm
 (c) $\frac{40}{6}$ cm (d) $\frac{50}{3}$ cm

Q.190. In a circle, a chord AB is 16 cm long and is at a distance of 6 cm from the center. What is the radius of the circle?

SSC CGL 17/09/2025 (Shift 2)

- (a) 12 cm (b) 10 cm
 (c) 8 cm (d) None of these

Q.191. What is the total measure of all interior angles in an 8-sided polygon?

SSC CGL 17/09/2025 (Shift 3)

- (a) 1540° (b) 1080° (c) 1220° (d) 900°

Q.192. A tangent line PA is drawn from an external point P to a circle. If the radius is 7 cm and the distance from P to the center is 25 cm, what is the length of the tangent PA?

SSC CGL 17/09/2025 (Shift 3)

- (a) 19 cm (b) 26 cm (c) 24 cm (d) 28 cm

Q.193. When two tangents, PA and PB, are drawn to a circle from a point P outside the circle, and the angle between them is 90° , what is the distance from P to the center (O) of the circle, if the radius is 6 cm?

SSC CGL 17/09/2025 (Shift 3)

- (a) 6 cm (b) $6\sqrt{2}$ cm (c) 12 cm (d) $6\sqrt{3}$ cm

Q.194. Each side of a regular octagon is $\frac{3}{4}$ of the side of a regular hexagon. If the perimeter of the hexagon is 24 cm, what will be the perimeter of the octagon?



SSC CGL 18/09/2025 (Shift 1)

(a) 18 cm (b) 20 cm (c) 24 cm (d) 16 cm

Q.195. How many diagonals are there in a decagon?

SSC CGL 18/09/2025 (Shift 1)

(a) 35 (b) 45 (c) 54 (d) 65

Q.196. The distance between the centers of two circles is 20 cm. The length of the internal common tangent joining both circles is 12 cm. If the radius of the larger circle is 10 cm, what will be the radius of the smaller circle?

SSC CGL 18/09/2025 (Shift 1)

(a) 6 cm (b) 8 cm (c) 5 cm (d) 7 cm

Q.197. In a cyclic quadrilateral PQRS, PR is the diameter of the circle. If $\angle PRS = 55^\circ$, then what is the measure of $\angle RPS$?

SSC CGL 18/09/2025 (Shift 1)

(a) 40° (b) 35° (c) 45° (d) 90° **Q.198.** What is the measure of each interior angle in a regular twelve-sided polygon?

SSC CGL 18/09/2025 (Shift 2)

(a) 120° (b) 150° (c) 180° (d) 90° **Q.199.** In a trapezoid, the two parallel sides are in the ratio 2 : 3. If the height of the trapezoid is 12 cm and its total area is 360 cm^2 , determine the lengths of the parallel sides.

SSC CGL 18/09/2025 (Shift 2)

(a) 26 cm and 38 cm
(b) 12 cm and 24 cm
(c) 24 cm and 36 cm
(d) 17 cm and 23 cm**Q.200.** ΔPQR has sides $PQ = 8 \text{ cm}$, $QR = 9 \text{ cm}$, $RP = 10 \text{ cm}$. ΔMNO is congruent to ΔPQR . What is the perimeter of ΔMNO ?

SSC CGL 18/09/2025 (Shift 2)

(a) 27 cm (b) 32 cm (c) 25 cm (d) 21 cm

Q.201. A circle has radius 9 cm. A tangent is drawn from an external point A. If the length of the tangent is 12 cm, what is the distance from A to the center of the circle?

SSC CGL 18/09/2025 (Shift 2)

(a) 16 cm (b) 35 cm (c) 41 cm (d) 15 cm

Q.202. From a point outside a circle, two tangents are drawn to the circle. If one of the tangents measures 15 cm, what is the length of the other tangent?

SSC CGL 18/09/2025 (Shift 2)

(a) 15 cm (b) 13 cm (c) 12 cm (d) 14 cm

Q.203. What is the total sum of the interior angles of a convex nonagon (a polygon with 9 sides)?

SSC CGL 18/09/2025 (Shift 3)

(a) 1080° (b) 1260° (c) 1360° (d) 1440° **Q.204.** Two circles have radii in the ratio 2 : 1. The length of a direct common tangent is 4 cm, and the distance between their centers is 5 cm. What is the radius of the smaller circle?

SSC CGL 18/09/2025 (Shift 3)

(a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm

Q.205. In a circle having a diameter 20 cm, a chord is 6 cm away from the center. Find the length of the chord.

SSC CGL 18/09/2025 (Shift 3)

(a) 16 cm (b) 15 cm (c) 13 cm (d) 17 cm

Q.206. In a parallelogram:

SSC CGL 19/09/2025 (Shift 1)

(a) Only one diagonal is equal.
(b) The diagonals are equal.
(c) The diagonals bisect each other.
(d) All angles are equal.**Q.207.** A right-angled triangle has a hypotenuse of 17 cm and one leg of 8 cm. A second right-angled triangle is similar to the first, and its hypotenuse is 51 cm. What is the area of the second triangle?

SSC CGL 19/09/2025 (Shift 1)

(a) 390 cm^2 (b) 460 cm^2
(c) 260 cm^2 (d) 540 cm^2 **Q.208.** "In triangles ABC and PQR, if $AB = PQ$, $AC = PR$, and $BC = QR$, by which congruence rule are they congruent?"

SSC CGL 19/09/2025 (Shift 1)

(a) Yes, by SSS
(b) Yes, by SAS
(c) Yes, by ASA
(d) No, they are not congruent**Q.209.** The length of each of two tangents drawn from an external point to a circle is 15 cm. What is the approximate distance from the external point to the center of the circle, if the radius is 11 cm?

SSC CGL 19/09/2025 (Shift 1)

(a) 24 cm (b) 15 cm (c) 19 cm (d) 16 cm

Q.210. A circle is inscribed in a right triangle with legs of length 9 and 12. What is the radius of the circle?

SSC CGL 19/09/2025 (Shift 1)

(a) 2 (b) 3 (c) 4 (d) 5

Q.211. If every exterior angle of a regular polygon is 72° , how many sides does the polygon have?

SSC CGL 19/09/2025 (Shift 2)

(a) 4 (b) 5 (c) 6 (d) 8

Q.212. Two circles, with radii of r_1 and r_2 respectively, have their centers separated by a distance of d . If the length of a transverse common tangent is equal to the distance between the centers, which

of the following is true?

SSC CGL 19/09/2025 (Shift 2)

(a) $r_1 \times r_2 = 0$ (b) $(r_1 - r_2) = 0$
(c) $(r_1 + r_2) = 0$ (d) $r_1 = 2r_2$ **Q.213.** A chord of a circle has a length of $2\sqrt{17}$ cm. The distance of the chord from the center is 8 cm. What is the radius of the circle?

SSC CGL 19/09/2025 (Shift 2)

(a) 6.89 cm (b) 6 cm (c) 9.52 cm (d) 9 cm

Q.214. A regular polygon has each interior angle measuring 120° . Find the number of its sides

SSC CGL 19/09/2025 (Shift 3)

(a) 6 (b) 1 (c) 13 (d) 8

Q.215. A point T is 17 cm away from the center of a circle. A tangent is drawn from T to the circle, and its length is 15 cm. What is the area of the circle?

SSC CGL 19/09/2025 (Shift 3)

(a) $121\pi \text{ cm}^2$ (b) $512\pi \text{ cm}^2$
(c) $64\pi \text{ cm}^2$ (d) $16\pi \text{ cm}^2$ **Q.216.** In ΔPQR , an angle bisector from P meets QR at S. If PS bisects $\angle QPR$, and $PQ = PR$, are ΔPQS and ΔPRS congruent? If so, by what rule?

SSC CGL 19/09/2025 (Shift 3)

(a) Yes, by SSS
(b) Yes, by SAS
(c) Yes, by ASA
(d) No, they are not congruent**Q.217.** The angle between two tangents drawn from an external point to a circle is 70° . What is the angle subtended by the chord connecting their points of contact at the center?

SSC CGL 19/09/2025 (Shift 3)

(a) 80° (b) 110° (c) 115° (d) 60° **Q.218.** From a point L outside a circle, a tangent LK and a secant LMN are drawn. If $LK = 10 \text{ cm}$ and $MN = 15 \text{ cm}$, what is the length of the segment LM?

SSC CGL 19/09/2025 (Shift 3)

(a) 6 cm (b) 8 cm (c) 5 cm (d) 11 cm

Q.219. When a transversal intersects two parallel lines, one of the alternate interior angles is 135° . What is the measure of the angle that lies opposite to it on the alternate interior side?

SSC CGL 20/09/2025 (Shift 1)

(a) 75° (b) 80° (c) 160° (d) 135° **Q.220.** If $\Delta PQR \sim \Delta MNO$ and $PQ = 8 \text{ cm}$, $QR = 10 \text{ cm}$, $MN = 16 \text{ cm}$, what is the length of NO?

SSC CGL 20/09/2025 (Shift 1)

(a) 20 cm (b) 8 cm (c) 4 cm (d) 12 cm

Q.221. A circle that has a radius of 17

centimeters, and there is a chord measuring 30 centimeters in length. What is the distance from the center of the circle to the chord?

SSC CGL 20/09/2025 (Shift 1)

- (a) 6 centimeters (b) 8 centimeters
(c) 10 centimeters (d) 12 centimeters

Q.222. Two parallel lines are intersected by a transversal. If a pair of corresponding angles is formed, and one of the angles measures 100° , what is the measure of the other angle?

SSC CGL 20/09/2025 (Shift 3)

- (a) 60° (b) 90° (c) 100° (d) 150°

Q.223. If $\triangle PQR \sim \triangle MNO$, and $\angle P = 70^\circ$, $\angle Q = 80^\circ$, then $\angle N = ?$

SSC CGL 20/09/2025 (Shift 3)

- (a) 70° (b) 80°
(c) 50° (d) Cannot be determined

Q.224. A circle has two chords, each 8 cm long. If one chord is 2 cm away from the center, what is the distance of the other chord from the center?

SSC CGL 20/09/2025 (Shift 3)

- (a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm

Q.225. By an external point X, two tangents are drawn to a circle, touching the circle at points M and N. The chord of contact MN subtends an angle α at the center of the circle O. What is the angle $\angle MXN$ between the tangents in terms of α ?

SSC CGL 20/09/2025 (Shift 3)

- (a) α (b) 2α (c) $180^\circ - \alpha$ (d) $90^\circ - \alpha$

Q.226. If the measure of an exterior angle of a regular polygon is 30° determine the total number of its sides.

SSC CGL 21/09/2025 (Shift 1)

- (a) 10 (b) 12 (c) 8 (d) 6

Q.227. In $\triangle PQR$, a median PS is drawn to the side QR. What is the ratio of the area of $\triangle PQS$ to the area of $\triangle PRS$?

SSC CGL 21/09/2025 (Shift 1)

- (a) 1 : 1 (b) 3 : 2 (c) 2 : 1 (d) 1 : 2

Q.228. If in $\triangle ABC$ and $\triangle DEF$, $\angle A = \angle D$, $AB = DE$, and $\angle B = \angle E$, which rule proves $\triangle ABC \cong \triangle DEF$?

SSC CGL 21/09/2025 (Shift 2)

- (a) SSS (b) SAS (c) ASA (d) AAS

Q.229. The line segment from the center of a circle to the midpoint of a chord is 8 cm long. If the radius is 17 cm, what is the length of the chord?

SSC CGL 21/09/2025 (Shift 2)

- (a) 12 cm (b) 24 cm (c) 36 cm (d) 30 cm

Q.230. A chord of a circle subtends an angle of 80° at the center. What is the

angle subtended at a point on the circle in the same segment?

SSC CGL 21/09/2025 (Shift 2)

- (a) 60° (b) 40° (c) 30° (d) 90°

Q.231. The sum of all interior angles of a 18-sided polygon is:

SSC CGL 21/09/2025 (Shift 3)

- (a) 2880° (b) 3600°
(c) 3060° (d) 3420°

Q.232. A right-angled triangle ABC has legs $AB = 15$ cm and $BC = 20$ cm. An altitude BD is drawn from the vertex B to the hypotenuse AC. What is the length of the altitude BD?

SSC CGL 21/09/2025 (Shift 3)

- (a) 12 cm (b) 16 cm (c) 20 cm (d) 15 cm

Q.233. In a triangle ABC, side BC is extended to a point D. The exterior angle at C is 112° . The interior angles at A and B are in the ratio 3:4. What is the measure of the largest interior angle of the triangle?

SSC CGL 22/09/2025 (Shift 1)

- (a) 60° (b) 68° (c) 105° (d) 15°

Q.234. Consider two triangles, $\triangle ABC$ and $\triangle DEF$. It is stated that these two triangles are congruent to each other, which is denoted as $\triangle ABC \cong \triangle DEF$ and $\angle A = 40^\circ$, what is the measure of $\angle D$?

SSC CGL 22/09/2025 (Shift 1)

- (a) 40° (b) 50° (c) 60° (d) 130°

Q.235. A chord of length 12 cm lies 4 cm away from the center of a circle. What is the approximate radius of the circle?

SSC CGL 22/09/2025 (Shift 1)

- (a) 10 cm (b) 14 cm (c) 7 cm (d) 25 cm

Q.236. A chord AB of a circle has a length of 20 cm. The angle subtended by the chord at a point C on the major arc is 30° . Find the radius of the circle.

SSC CGL 22/09/2025 (Shift 1)

- (a) 17 cm (b) 20 cm (c) 12 cm (d) 18 cm

Q.237. Two angles are supplementary. The measure of one angle is 30° less than twice the measure of the other. What is the measure of the larger angle?

SSC CGL 22/09/2025 (Shift 2)

- (a) 110° (b) 125° (c) 170° (d) 105°

Q.238. In a triangle ABC, if the exterior angle bisectors of $\angle B$ and $\angle C$ meet at point E (the excenter opposite A), and $\angle BEC = 50^\circ$, what is the measure of $\angle A$?

SSC CGL 22/09/2025 (Shift 2)

- (a) 110° (b) 100° (c) 80° (d) 70°

Q.239. In triangle LMN, a line segment OP is drawn parallel to the side MN, with O on LM and P on LN. If $LO = 4$ cm, $LM =$

12 cm, and $LP = 5$ cm, find the length of PN.

SSC CGL 22/09/2025 (Shift 2)

- (a) 11 cm (b) 12 cm (c) 15 cm (d) 10 cm

Q.240. The sum of the measures of 8 interior angles of a 9-sided polygon is $1,120^\circ$. What is the measure of the 9th exterior angle?

SSC CGL 22/09/2025 (Shift 3)

- (a) 60° (b) 20° (c) 40° (d) 80°

Q.241. Two triangles, $\triangle ABC$ and $\triangle XYZ$, have $AB = XY$, $AC = XZ$, and $\angle BAC = \angle YXZ$. By what rule are they congruent?

SSC CGL 22/09/2025 (Shift 3)

- (a) SSS (b) SAS (c) ASA (d) AAS

Q.242. A chord subtends an angle of 140° at the center of a circle. What is the measure of the angle subtended by the same chord at any point on the remaining part of the circle?

SSC CGL 22/09/2025 (Shift 3)

- (a) 35° (b) 70° (c) 140° (d) 105°

Q.243. A circle is inscribed in a regular hexagon. What is the angle subtended by one side of the hexagon at the center of the circle?

SSC CGL 22/09/2025 (Shift 3)

- (a) 36° (b) 54° (c) 60° (d) 108°

Q.244. What is the minimum distance between centers of two circles having radii 7 cm and 4 cm such that exactly three common tangents exist?

SSC CGL 23/09/2025 (Shift 1)

- (a) 11 cm (b) 3 cm (c) 7 cm (d) 14 cm

Q.245. In a circle, angle subtended by chord AB at the center is 100° . What is the measure of $\angle ACB$, where C lies on the circle?

SSC CGL 23/09/2025 (Shift 1)

- (a) 40° (b) 45° (c) 50° (d) 60°

Q.246. In a circle with center O, two chords, AB and CD, cross each other at right angles. If the distance from the center O to chord AB is 6 cm, while the distance from O to chord CD is 8 cm. What is the radius of the circle?

SSC CGL 23/09/2025 (Shift 1)

- (a) 11 cm (b) 7 cm (c) 9 cm (d) 10 cm

Q.247. If the angle subtended by a chord at the center is 120° , what is the angle subtended at a point on the circle in the same segment?

SSC CGL 23/09/2025 (Shift 1)

- (a) 60° (b) 75° (c) 90° (d) 80°

Q.248. In a triangle PQR, the sides are p, q, and r, and the area of the triangle is T. What is the radius of the inscribed circle (inradius)?



SSC CGL 23/09/2025 (Shift 1)

- (a) $\frac{T}{p+q+r}$ (b) $\frac{2T}{p+q+r}$
 (c) $\frac{p+q+r}{2T}$ (d) $\frac{T}{2(p+q+r)}$

Q.249. If the difference between the interior and exterior angles of a regular polygon is 60° , find the number of sides of the polygon.

SSC CGL 23/09/2025 (Shift 2)

- (a) 6 (b) 9 (c) 13 (d) 11

Q.250. Convert 0.6 radians to degrees.

SSC CGL 23/09/2025 (Shift 2)

- (a) 35.7° (b) 34.39° (c) 25.24° (d) 67.3°

Q.251. In an acute-angled triangle, the orthocenter lies at which position ?

SSC CGL 23/09/2025 (Shift 2)

- (a) On one of the sides
 (b) Outside the triangle
 (c) Inside the triangle
 (d) At a vertex

Q.252. A chord is drawn in a circle with a radius of 17 cm. If the distance of the chord from the center is 15 cm, what is the length of the chord?

SSC CGL 23/09/2025 (Shift 2)

- (a) 13 cm (b) 14 cm (c) 16 cm (d) 31 cm

Q.253. The tangent to a circle from a point P is 9 cm and the distance from P to the centre is 15 cm. Find the radius.

SSC CGL 23/09/2025 (Shift 2)

- (a) 12 cm (b) 11 cm
 (c) $\sqrt{3}$ cm (d) $\sqrt{16}$ cm

Q.254. Two circles have radii of 16 cm and 6 cm. The distance between their centers is 30 cm. What is the length of a direct common tangent ?

SSC CGL 23/09/2025 (Shift 3)

- (a) 20 cm (b) 22 cm
 (c) $20\sqrt{2}$ cm (d) $12\sqrt{2}$ cm

Q.255. A line drawn from the center of a circle is perpendicular to a chord. What does it do to the chord ?

SSC CGL 23/09/2025 (Shift 3)

- (a) Divides the chord into two equal parts
 (b) Doubles the length of the chord
 (c) Makes the chord tangent to the circle
 (d) Reduces the chord length to half

Q.256. A chord divides a circle into two arcs. The angle subtended by the major arc at the circumference is 90° . What is the central angle subtended by the major arc ?

SSC CGL 23/09/2025 (Shift 3)

- (a) 180° (b) 110° (c) 270° (d) 150°

Q.257. In a circle, chord AB subtends an angle of 85° at a point C on the circle.

What is the angle subtended by the same chord AB at another point D on the same arc as C ?

SSC CGL 23/09/2025 (Shift 3)

- (a) 55° (b) 50° (c) 85° (d) 40°

Q.258. From an outside point P, two tangents, PA and PB, are drawn to a circle. A third tangent to the circle at a point Q intersects PA and PB at R and S respectively. If PA = 20 cm, what is the perimeter of ΔPRS ?

SSC CGL 23/09/2025 (Shift 3)

- (a) 40 cm (b) 25 cm (c) 20 cm (d) 10 cm

Q.259. What is the ratio of the interior angle of a regular pentagon to the interior angle of a regular decagon ?

SSC CGL 24/09/2025 (Shift 1)

- (a) 3 : 4 (b) 6 : 5 (c) 4 : 3 (d) 2 : 9

Q.260. In a quadrilateral PQRS, angles P, Q, R are in the ratio 2 : 3 : 4. If angle S = 90° , what is angle Q ?

SSC CGL 24/09/2025 (Shift 1)

- (a) 90° (b) 72° (c) 45° (d) 120°

Q.261. Convert $\frac{5\pi}{6}$ radians to degrees.

SSC CGL 24/09/2025 (Shift 1)

- (a) 135° (b) 115° (c) 45° (d) 150°

Q.262. In a right-angled triangle, the point of intersection of the altitudes is found at which of the following points ?

SSC CGL 24/09/2025 (Shift 1)

- (a) At the midpoint of the hypotenuse
 (b) At the vertex of the angle where the right angle is formed
 (c) Somewhere on the hypotenuse, but not exactly in the middle
 (d) At the midpoint of any side

Q.263. A circle has a radius of 29 cm. A chord is drawn at a distance of 20 cm from the center. What is the length of the chord ?

SSC CGL 24/09/2025 (Shift 1)

- (a) 34 cm (b) 42 cm (c) 28 cm (d) 24 cm

Q.264. A tangent is drawn from a point 20 cm from the center of a circle of radius 12 cm. Find the length of the tangent.

SSC CGL 24/09/2025 (Shift 1)

- (a) 20 cm (b) 17 cm (c) 11 cm (d) 16 cm

Q.265. The radii of two circles are 8 cm and 3 cm. The centers of the two circles are 12 cm apart. What is the length of a transverse common tangent ?

SSC CGL 24/09/2025 (Shift 2)

- (a) $\sqrt{21}$ cm (b) $\sqrt{22}$ cm
 (c) $\sqrt{23}$ cm (d) $\sqrt{35}$ cm

Q.266. In a circle, a chord AB subtends

an angle of 40° at point C on the circumference. What is the angle it subtends at point D on the circumference, on the same side as C ?

SSC CGL 24/09/2025 (Shift 2)

- (a) 40° (b) 80° (c) 140° (d) 150°

Q.267. A chord of length 24 cm is drawn on a circle with a radius of 13 cm. What is the distance of the chord from the center of the circle ?

SSC CGL 24/09/2025 (Shift 2)

- (a) 5 cm (b) 10 cm (c) 8 cm (d) 11 cm

Q.268. In a circle, the chord PQ subtends an angle of 100° at the center O of the circle. What is the measure of the angle PRQ subtended by the same chord PQ at any point R on the minor arc of the circle?

SSC CGL 24/09/2025 (Shift 2)

- (a) 130° (b) 60° (c) 90° (d) 120°

Q.269. A line is drawn from a point P, 15 cm away from the center O, of a circle with a radius of 12 cm. If this line is a tangent to the circle, what is the length of the tangent from point P to the point of contact with the circle?

SSC CGL 24/09/2025 (Shift 2)

- (a) 3 cm (b) 9 cm (c) 12 cm (d) 10 cm

Q.270. If half the interior angle of a regular polygon is k times its exterior angle, express the number of sides (n) of the polygon in terms of k.

SSC CGL 24/09/2025 (Shift 3)

- (a) $2k + 4$ (b) $2k - 4$ (c) $4k + 2$ (d) $4k - 3$

Q.271. What is $\frac{5\pi}{4}$ radians in degrees?

SSC CGL 24/09/2025 (Shift 3)

- (a) 175° (b) 120° (c) 225° (d) 135°

Q.272. If a triangle has a right angle, where is its Orthocenter located?

SSC CGL 24/09/2025 (Shift 3)

- (a) Outside the triangle.
 (b) On the midpoint of the hypotenuse.
 (c) At the vertex of the right angle.
 (d) Inside the triangle.

Q.273. In a circle with center O, chords AB and CD are equal in length. If the distance from O to AB is 16 cm, what is the distance from O to CD ?

SSC CGL 24/09/2025 (Shift 3)

- (a) 4 cm (b) 8 cm (c) 16 cm (d) 12 cm

Q.274. A circle that has a diameter of 18 centimeters. A line is tangent to the circle. What is the angle between the tangent and the diameter at the point of contact?

SSC CGL 24/09/2025 (Shift 3)

- (a) 30° (b) 45° (c) 90° (d) 180°



Q.275. An angle of $\frac{2\pi}{3}$ radians

represent what percentage of a full circle?

SSC CGL 25/09/2025 (Shift 1)

(a) 33.33% (b) 50% (c) 43.33% (d) 35%

Q.276. If two circles of radius 8 cm and 3 cm have their centers 13 cm apart, what is the length of the direct common tangent?

SSC CGL 25/09/2025 (Shift 1)

(a) $\sqrt{124}$ cm (b) 114 cm
(c) 78 cm (d) 12 cm

Q.277. A chord separates a circle into two segments. If the angle formed by the chord at the center is 100° , what is the angle formed at a point located in the major segment?

SSC CGL 25/09/2025 (Shift 1)

(a) 90° (b) 75° (c) 50° (d) 120°

Q.278. In a circle, the chords PQ and RS are equal. The angle subtended by PQ at the circumference is 75° . What is the angle subtended by RS at the circumference?

SSC CGL 25/09/2025 (Shift 1)

(a) 75° (b) 37.5° (c) 120° (d) 95°

Q.279. A chord CD is drawn in a circle with center O. The tangent at C and the tangent at D meet at a point Q. If $\angle CDQ = 60^\circ$, measure the angle subtended by the chord CD at the circumference of the circle?

SSC CGL 25/09/2025 (Shift 1)

(a) 30° (b) 60° (c) 90° (d) 120°

Q.280. A line is tangent to a circle at point P. If a chord PQ has a length of 10 cm and makes an angle of 45° with the tangent, what is the radius of the circle?

SSC CGL 25/09/2025 (Shift 1)

(a) $4\sqrt{4}$ cm (b) $3\sqrt{3}$ cm
(c) $5\sqrt{3}$ cm (d) $5\sqrt{2}$ cm

Q.281. The radii of three concentric circles are in arithmetic progression. If the innermost radius is 5 cm and the outermost is 15 cm, what is the radius of the middle circle?

SSC CGL 25/09/2025 (Shift 2)

(a) 12 cm (b) 11 cm (c) 10 cm (d) 14 cm

Q.282. Convert 5 radians to degrees.

SSC CGL 25/09/2025 (Shift 2)

(a) 171.89° (b) 240° (c) 286.62° (d) 540°

Q.283. The measures of the interior angles of a pentagon are $y, y + 5, y + 10, y + 15$, and $y + 20$. What is the measure of the smallest angle?

SSC CGL 25/09/2025 (Shift 2)

(a) 88° (b) 98° (c) 108° (d) 118°

Q.284. If a chord subtends an angle of 100° at the center, the angle it subtends at the circumference on the same side is:

SSC CGL 25/09/2025 (Shift 2)

(a) 50° (b) 37.5° (c) 45° (d) 60°

Q.285. If an angle $\theta = 3$ radians, what percentage of the full circle does it represent?

SSC CGL 25/09/2025 (Shift 3)

(a) 45.74% (b) 47.74%
(c) 50.8% (d) 46.83%

Q.286. Two circles of radii 5 cm and 3 cm are such that the distance between their centers is 8 cm. How many common tangents can be drawn?

SSC CGL 25/09/2025 (Shift 3)

(a) 1 (b) 0 (c) 2 (d) 3

Q.287. A circle with a radius of 12 cm, and a chord makes a 60° angle at the center. What is the length of that chord?

SSC CGL 25/09/2025 (Shift 3)

(a) 6 cm (b) 12 cm
(c) $12\sqrt{3}$ cm (d) $12\sqrt{2}$ cm

Q.288. Four points P, Q, R, and S are located on the circumference of a circle. If $\angle RPS = 50^\circ$ and $\angle PQS = 70^\circ$, what is the measure of $\angle PRS$?

SSC CGL 25/09/2025 (Shift 3)

(a) 60° (b) 90° (c) 50° (d) 70°

Q.289. In a circle, points P, Q, R, and S lie on the circumference such that chords PR and QS intersect at the center, and both $\angle RPQ$ and $\angle RSQ$ are subtended by the same arc RQ. If $\angle RPQ = 70^\circ$, what is the measure of $\angle RSQ$?

SSC CGL 25/09/2025 (Shift 3)

(a) 80° (b) 70° (c) 100° (d) 110°

Q.290. A circle with radius r has a tangent line at a point P on the circle. The distance of the tangent from point P to the external point Q is 13 cm. The distance from the center of the circle to point Q is 15 cm. Find the radius of the circle.

SSC CGL 25/09/2025 (Shift 3)

(a) $\sqrt{56}$ (b) $\sqrt{26}$ (c) $\sqrt{36}$ (d) $\sqrt{59}$

Q.291. A sector of radius 7 cm has an area of 38.5 cm^2 . Find the angle of the sector.

SSC CGL 26/09/2025 (Shift 1)

(a) 90° (b) 60° (c) 30° (d) 75°

Q.292. Convert 3.5 radians to degrees.

SSC CGL 26/09/2025 (Shift 1)

(a) 190.56° (b) 120° (c) 200.63° (d) 80°

Q.293. In a right triangle, one of the acute angles is 60° . What is the measure

of the exterior angle at the right-angled vertex?

SSC CGL 26/09/2025 (Shift 1)

(a) 45° (b) 30° (c) 60° (d) 90°

Q.294. A triangle has one angle measuring 105° has its Orthocenter located where?

SSC CGL 26/09/2025 (Shift 1)

(a) Inside the triangle.
(b) Outside the triangle.
(c) On one of the sides of the triangle.
(d) On one of the vertices of the triangle.

Q.295. In triangle PQR, ST is drawn parallel to QR, intersecting PQ and PR at S and T respectively. If $PS = 5$ cm, $SQ = 15$ cm, and $PT = 4$ cm, find TR.

SSC CGL 26/09/2025 (Shift 1)

(a) 14 cm (b) 16 cm (c) 18 cm (d) 12 cm

Q.296. If the angle subtended by arc AB at the center is 105° , what is the reflex angle made by this arc at the center?

SSC CGL 26/09/2025 (Shift 1)

(a) 255° (b) 210° (c) 105° (d) 245°

Q.297. In a circle of diameter d , a chord subtends an angle θ (in radians) at the center. Which expression represents the length of the minor arc?

SSC CGL 26/09/2025 (Shift 1)

(a) $\frac{d\theta}{2}$ (b) $\frac{1}{d2\theta}$ (c) $d\sin\frac{\theta}{2}$ (d) πd

Q.298. P, Q, and R together can complete a task in 6 days. If P alone takes 12 days and Q alone takes 18 days, how long will R alone take?

SSC CGL 26/09/2025 (Shift 1)

(a) 15 days (b) 20 days
(c) 18 days (d) 36 days

Q.299. The sum of the interior angles of a regular polygon is five times the sum of its exterior angles. Determine the number of diagonals in the polygon.

SSC CGL 26/09/2025 (Shift 2)

(a) 63 (b) 72 (c) 54 (d) 97

Q.300. A triangle with sides 8 cm, 15 cm, and 17 cm has an inradius of 2 cm. What is its area?

SSC CGL 26/09/2025 (Shift 2)

(a) 10 cm^2 (b) 20 cm^2
(c) 30 cm^2 (d) 40 cm^2

Q.301. In a circle, there is a chord measuring 18 cm that is located 13 cm away from the center. What is the approximate radius of the circle?

SSC CGL 26/09/2025 (Shift 2)

(a) 10 cm (b) 18 cm (c) 16 cm (d) 25 cm

Q.302. A tangent is drawn to a circle with a radius of 8 cm. At the point of



tangency, what is the angle between the tangent and the radius ?

SSC CGL 26/09/2025 (Shift 2)

- (a) 45° (b) 60° (c) 90° (d) 180°

Q.303. The angle subtended by an arc at the center is 36°, and its length is 11 cm. Find the radius of the circle.

SSC CGL 26/09/2025 (Shift 3)

- (a) 22 cm (b) 17.5 cm
(c) 30 cm (d) 8.75 cm

Q.304. Convert 6° to radians.

SSC CGL 26/09/2025 (Shift 3)

- (a) $\frac{5\pi}{180}$ (b) $\frac{5\pi}{60}$ (c) $\frac{\pi}{120}$ (d) $\frac{\pi}{30}$

Q.305. At an intersection of two lines, two angles are vertically opposite. If one of the angles measures 55°, what is the measure of the other angle ?

SSC CGL 26/09/2025 (Shift 3)

- (a) 45° (b) 55° (c) 125° (d) 180°

Q.306. If in triangle PQR, DE || QR and

$\frac{PD}{DQ} = \frac{1}{2}$, and PE = 6 cm, then ER is:

SSC CGL 26/09/2025 (Shift 3)

- (a) 2.5 cm (b) 12 cm
(c) 10 cm (d) 14 cm

Q.307. Chord AB and chord CD are equal and subtend angles of 35° and x° at the center respectively. Find X

SSC CGL 26/09/2025 (Shift 3)

- (a) 25° (b) 30° (c) 90° (d) 35°

Q.308. The area of an equilateral triangle increases by 44%. By what percentage did the side length increase ?

SSC CGL 14/10/2025 (Shift 1)

- (a) 30% (b) 20% (c) 34% (d) 16%

Q.309. In a quadrilateral ABCD, angles A, B, C are in the ratio 5 : 4 : 5. If angle D = 80°, what is angle B ?

SSC CGL 14/10/2025 (Shift 1)

- (a) 80° (b) 72° (c) 45° (d) 120°

Q.310. Two circles of radii 8 cm and 5 cm are such that the distance between their centers is 13 cm. How many common tangents can be drawn ?

SSC CGL 14/10/2025 (Shift 1)

- (a) 0 (b) 1 (c) 2 (d) 3

Q.311. What is $\frac{3\pi}{2}$ radians in degrees ?

SSC CGL 14/10/2025 (Shift 1)

- (a) 180° (b) 360° (c) 270° (d) 150°

Q.312. In a circle, the chords EF and GH are equal. The angle subtended by EF at the circumference is 95°. What is the angle subtended by GH at the circumference ?

SSC CGL 14/10/2025 (Shift 1)

- (a) 95° (b) 47.5° (c) 120° (d) 85°

Q.313. In a triangle PQR, medians PS and QT intersect at O. If the length of median PS is 21 cm, what is the length of the segment PO ?

SSC CGL 14/10/2025 (Shift 3)

- (a) 14 cm (b) 16 cm (c) 10 cm (d) 12 cm

Q.314. A regular decagon can be divided into how many isosceles triangles by joining its vertices to its center ?

SSC CGL 14/10/2025 (Shift 3)

- (a) 8 (b) 10 (c) 12 (d) 5

Q.315. A point P is 17 cm away from the center of a circle. A tangent is drawn from P to the circle, and its length is 15 cm. What is the perimeter of the circle ?

SSC CGL 14/10/2025 (Shift 3)

- (a) 12π cm (b) 15π cm
(c) 16π cm (d) 18π cm

SSC Selection Post (XIII)

Q.316. A triangle has sides 5 cm, 12 cm, and 13 cm. What type of triangle is this?

Matriculation Level 24/07/2025 (Shift - 1)

- (a) Right-angled (b) Obtuse
(c) Acute (d) Equilateral

Q.317. From a point P, a tangent is drawn to a circle with center O. The radius is 5 cm, and the distance from O to P is 13 cm. What is the length of the tangent?

Matriculation Level 24/07/2025 (Shift - 1)

- (a) 46 cm (b) 12 cm (c) 40 cm (d) 44 cm

Q.318. A triangle has angles 40°, 60°, and 80°. It is a/an _____.

Matriculation Level 25/07/2025 (Shift - 1)

- (a) Equilateral (b) Right-angled
(c) Isosceles (d) Scalene

Q.319. How many tangents can be drawn at a specific point on a circle?

Matriculation Level 26/07/2025 (Shift - 1)

- (a) infinitely many (b) 2 (c) 1 (d) 3

Q.320. A tangent BC at point B of a circle with radius 8 cm meets a line through the center A at point C such that AC = 17 cm. What is the length of the tangent BC ?

Matriculation Level 28/07/2025 (Shift - 1)

- (a) 12 (b) 15 (c) 13 (d) 9

Q.321. In a triangle, two angles are 35° and 65°. What is the measure of the third angle?

Matriculation Level 29/07/2025 (Shift - 1)

- (a) 90° (b) 80° (c) 100° (d) 70°

Q.322. A line segment joining any two points on the circumference of a circle is

called a _____.

Matriculation Level 29/07/2025 (Shift - 1)

- (a) Tangent (b) Radius
(c) Chord (d) Diameter

Q.323. What is the number of sides of a regular polygon if the sum of its interior angles is 1440 degrees?

Matriculation Level 29/08/2025 (Shift - 3)

- (a) 9 (b) 10 (c) 11 (d) 12

Q.324. In a triangle, if one exterior angle is 140°, and the opposite interior angle is 60°, what is the other interior angle adjacent to it?

Matriculation Level 31/07/2025 (Shift - 1)

- (a) 100° (b) 130° (c) 80° (d) 180°

Q.325. In a circle with center X, a chord PQ is such that $\angle PXQ = 140^\circ$. What is the measure of $\angle PRQ$, where R lies on the arc PQ ?

Matriculation Level 01/08/2025 (Shift - 1)

- (a) 30° (b) 70° (c) 65° (d) 60°

Q.326. A triangle has angles in the ratio 3 : 1 : 2. What are the angles?

Matriculation Level 01/08/2025 (Shift - 1)

- (a) 40°, 60°, 80° (b) 30°, 45°, 105°
(c) 36°, 54°, 90° (d) 90°, 30°, 60°

Q.327. A triangle has two angles measuring 55° and 75°. What is the measure of the third angle?

Matriculation Level 01/08/2025 (Shift - 2)

- (a) 65° (b) 70° (c) 60° (d) 50°

Q.328. In a triangle PQR, if angle P = 60° and angle Q = 50°, then what is the measure of angle R?

Matriculation Level 01/08/2025 (Shift - 3)

- (a) 80° (b) 50° (c) 70° (d) 90°

Q.329. Two circles have the same centre. Their radii are 200 m and 220 m respectively. A small wheel of radius 60 cm rolls once around the inner circle. Another wheel rolls once around the outer circle. After completing one full round each, both wheels are found to have completed the same number of rotations about their own axes. What is the radius of the second wheel?

Matriculation Level 02/08/2025 (Shift - 1)

- (a) 23 (b) 66 (c) 50 (d) 121

Q.330. How many diagonals are there in a regular decagon (a polygon with 10 sides)?

Matriculation Level 29/08/2025 (Shift - 3)

- (a) 35 (b) 40 (c) 45 (d) 50

Q.331. A right prism is cut vertically into two equal halves. How does its volume change?

Matriculation Level 29/08/2025 (Shift - 3)



- (a) Becomes double (b) Becomes half
(c) Stays the same (d) Increases by 25%

Q.332. Consider the following two statements, labeled Assertion (A) and Reason (R).

Assertion (A): Two circles touching internally have only one common tangent.

Reason (R): One circle lies entirely inside the other, touching at a single point, which prevents the existence of direct common tangents.

Higher Secondary 24/07/2025 (Shift - 2)

- (a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

Q.333. A quadrilateral PQRS is cyclic, $\angle P = 60^\circ$, $\angle R = 120^\circ$ and diagonals intersect at point T. What is the value of $\angle Q + \angle S$?

Higher Secondary 25/07/2025 (Shift - 2)

- (a) 360° (b) 180° (c) 90° (d) 120°

Q.334. Which of the following properties regarding circles is CORRECT?

Higher Secondary 25/07/2025 (Shift - 2)

- (a) The angle in a semicircle is a right angle (90°)
(b) The angle in a semicircle is 180°
(c) Angles in the same segment of a circle are always unequal.
(d) The angle subtended at the circumference is twice the angle at the center.

Q.335. In right-angled triangle ABC with $\angle B = 90^\circ$, let D be the midpoint of AB. From D drop a perpendicular DE to the hypotenuse AC, meeting AC at E. if AB = 40 cm and BC = 30 cm, find the area of $\triangle AED$ (in cm^2).

Higher Secondary 25/07/2025 (Shift - 2)

- (a) 84 cm^2 (b) 96 cm^2
(c) 72 cm^2 (d) 108 cm^2

Q.336. Consider a triangle ABC where the angle at vertex B measures 90° degrees and P is the midpoint of hypotenuse AC. What can be said about point P?

Higher Secondary 26/07/2025 (Shift - 2)

- (a) P is the centroid of the triangle
(b) P is the incenter of the triangle
(c) P is the circumcenter of the triangle
(d) P lies on side BC

Q.337. $\triangle ABC$ and $\triangle PQR$ are similar if their corresponding sides are proportional. This is known as the:

Higher Secondary 26/07/2025 (Shift - 2)

- (a) AA (Angle-Angle) Similarity
(b) SAS (Side-Angle-Side) Similarity
(c) SSS (Side-Side-Side) Similarity

(d) ASA (Angle-Side-Angle) Similarity

Q.338. Two tangents are drawn from an external point P to a circle, touching at points A and B. If the distance from P to the center O is 10 cm, and radius is 8 cm, what is the length of tangent PA?

Higher Secondary 28/07/2025 (Shift - 2)

- (a) 5 cm (b) 6 cm (c) 4 cm (d) 7 cm

Q.339. If triangle PQR is congruent to triangle XYZ, which of the following is true?

Higher Secondary 28/07/2025 (Shift - 2)

- (a) $PQ = ZY$ (b) $QR = YX$
(c) $RP = XZ$ (d) $\angle P = \angle Y$

Q.340. If a chord of a circle is equal in length to the radius, what is the angle subtended by the chord at the center of the circle?

Higher Secondary 28/07/2025 (Shift - 2)

- (a) 30° (b) 90° (c) 45° (d) 60°

Q.341. The radii of two concentric circles with centre 'O' are 6 cm and 10 cm. AB is a chord of the larger circle which is tangent to the smaller circle at point 'C'. If AD is a diameter of the larger circle, then find the length of CD.

Higher Secondary 29/07/2025 (Shift - 2)

- (a) $\sqrt{136}$ cm (b) $\sqrt{208}$ cm
(c) $\sqrt{244}$ cm (d) $\sqrt{164}$ cm

Q.342. The distance between the centres of the two circles is 58 cm, and the radii of the circles are 28 cm and 14 cm. Find the length of the transverse common tangent drawn to the two circles.

Higher Secondary 29/07/2025 (Shift - 2)

- (a) 36 cm (b) 40 cm (c) 42 cm (d) 48 cm

Q.343. Which of the following sets of conditions is NOT sufficient to prove that two triangles are similar?

Higher Secondary 30/07/2025 (Shift - 2)

- (a) AAA (b) SAS (c) SSS (d) SSA

Q.344. In a circle, a chord subtends an angle of 80° at the center. What is the angle subtended by the same chord at any point on the remaining part of the circle?

Higher Secondary 30/07/2025 (Shift - 2)

- (a) 90° (b) 110° (c) 40° (d) 60°

Q.345. In an acute angled triangle ABC, AP is an altitude drawn on side BC. O is a point on AP such that AB = 15 cm, OB = 6 cm, and OC = 10 cm, then find the length of AC.

Higher Secondary 30/07/2025 (Shift - 2)

- (a) $15\sqrt{3}$ cm (b) 25 cm
(c) $15\sqrt{2}$ cm (d) 17 cm

Q.346. Comparing the measure of an internal angle to the measure of an external angle of a regular polygon is 4 : 1. Find the number of sides of the polygon.

Higher Secondary 31/07/2025 (Shift - 2)

- (a) 15 (b) 18 (c) 10 (d) 9

Q.347. PQR is an equilateral triangle inscribed in a circle. 'S' is any point on the minor arc QR and $\angle RQS = 40^\circ$. Find the measure of $\angle QRS$.

Higher Secondary 31/07/2025 (Shift - 2)

- (a) 30° (b) 20° (c) 45° (d) 60°

Q.348. In $\triangle PQR$, EF is parallel to QR. If PE = 16 cm, PQ = 30 cm and PR = 90 cm, then find the length of PF.

Higher Secondary 31/07/2025 (Shift - 2)

- (a) 48 cm (b) 52 cm (c) 36 cm (d) 44 cm

Q.349. A circle is completely enclosed within a parallelogram in such a way that all four sides of the parallelogram are tangents to the circle. The specific type of parallelogram formed in this manner will have its angles as ?

Higher Secondary 02/08/2025 (Shift - 2)

- (a) All equal (b) All unequal
(c) Two equal (d) Three equal

Q.350. Match the concepts in Column I with their corresponding properties in Column II:

Column I (Concept)	Column II (Property)
A. Distance from the center to the circumference	1. 90° angle with the tangent
B. Equal chords	2. Subtend equal angles at center
C. Distance from center to chord	3. Helps compare chord lengths
D. Perpendicular from the center to the tangent	4. Equal to the radius

Higher Secondary 02/08/2025 (Shift - 2)

- (a) A-3, B-2, C-4, D-1 (b) A-4, B-2, C-3, D-1
(c) A-3, B-4, C-1, D-2 (d) A-2, B-1, C-4, D-3

Q.351. Two circles touch each other externally. How many common tangents can be drawn?

Higher Secondary 02/08/2025 (Shift - 2)

- (a) 3 (b) 2 (c) 1 (d) 0

Q.352. In a triangle, which of the following centers is always located inside the triangle?

Higher Secondary 29/08/2025 (Shift - 2)

- (a) Circumcenter (b) Orthocenter
(c) Incenter (d) None of these

Q.353. Within a circle, two parallel chords are positioned such that the center lies between them if the distance



between these chords is 28 cm, and the length of one chord is 24 cm having radius of 20 cm, what is the length of the other chord?

Higher Secondary 29/08/2025 (Shift - 2)
(a) 36 cm (b) 35 cm (c) 32 cm (d) 40 cm

Q.354. Two complementary angles differ by 24° . What are the angles?

Graduate Level 24/07/2025 (Shift - 3)

- (a) $33^\circ, 57^\circ$ (b) $30^\circ, 60^\circ$
(c) $43^\circ, 47^\circ$ (d) $23^\circ, 67^\circ$

Q.355. Two tangents PA and PB are drawn from an external point P to a circle with center O. If the angle between the tangents is 50° , what is the angle subtended by the chord of contact AB at a point on the major arc?

Graduate Level 24/07/2025 (Shift - 3)

- (a) 50° (b) 65° (c) 80° (d) 130°

Q.356. The length of a tangent drawn from an external point to a circle is 16 cm. If the shortest distance from the external point to the circle is 8 cm, what is the radius of the circle?

Graduate Level 24/07/2025 (Shift - 3)

- (a) 10 cm (b) 12 cm (c) 20 cm (d) 24 cm

Q.357. Complement of an angle is 9° , more than twice the angle. Find the angle.

Graduate Level 25/07/2025 (Shift - 3)

- (a) 28° (b) 32° (c) 34° (d) 27°

Q.358. Find the angle if 2 times its complement equals the angle itself.

Graduate Level 26/07/2025 (Shift - 3)

- (a) 22.5° (b) 60° (c) 30° (d) 65.5°

Q.359. The angle between two tangents drawn from an external point to a circle is 80° . What is the angle subtended by the chord connecting their points of contact at the center?

Graduate Level 26/07/2025 (Shift - 3)

- (a) 100° (b) 110° (c) 115° (d) 120°

Q.360. Two circles have radii of 10 cm and 5 cm. The distance between their centers is 25 cm. What is the length of a direct common tangent?

Graduate Level 26/07/2025 (Shift - 3)

- (a) 10 cm (b) 20 cm
(c) $10\sqrt{6}$ cm (d) $12\sqrt{6}$ cm

Q.361. There are two circles with radii of 7 cm and 5 cm are positioned so that they touch each other internally. What is the distance between their centers?

Graduate Level 26/07/2025 (Shift - 3)

- (a) 2 cm (b) 3 cm (c) 4 cm (d) 8 cm

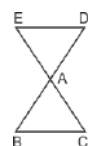
Q.362. If the difference between an angle and its complement is 24° , what is the

angle?

Graduate Level 28/07/2025 (Shift - 3)

- (a) 38° (b) 57° (c) 62° (d) 70°

Q.363. In the given figure, AB = AD = 10 cm and AC = AE = 15 cm and BC = 18 cm. Find ED



Graduate Level 28/07/2025 (Shift - 3)

- (a) 18 cm (b) 16 cm (c) 14 cm (d) 15 cm

Q.364. Two circles with radii 8 cm and 7 cm have their centers 25 cm apart. What is the length of their transversal common tangent?

Graduate Level 28/07/2025 (Shift - 3)

- (a) 12 cm (b) 18 cm (c) 20 cm (d) 22 cm

Q.365. Complement of angle y is equal to $(y + 20)$. Find the angle.

Graduate Level 29/07/2025 (Shift - 3)

- (a) 35° (b) 40° (c) 45° (d) 50°

Q.366. In a triangle, the distances from the incenter to the vertices are 4 cm, 5 cm, and 6 cm. Find the inradius.

Graduate Level 29/07/2025 (Shift - 3)

- (a) 2 cm (b) 3 cm
(c) 4 cm (d) Cannot be determined

Q.367. If ABC is an equilateral triangle inscribed in a circle and D is any point on arc BC. What is the measure of $\frac{1}{2}\angle ADB$.

Graduate Level 29/07/2025 (Shift - 3)

- (a) 20° (b) 30° (c) 60° (d) 15°

Q.368. A quadrilateral PQRS has a circle inscribed within it. Given that the lengths of sides PQ, QR, and RS are 5 cm, 6 cm, and 7 cm respectively, what is the length of side PS?

Graduate Level 29/07/2025 (Shift - 3)

- (a) 4 cm (b) 6 cm (c) 5 cm (d) 8 cm

Q.369. A circular disc of radius 15 cm is divided into 4 equal sectors. What is the angle of each sector?

Graduate Level 31/07/2025 (Shift - 3)

- (a) 15° (b) 60° (c) 30° (d) 90°

Q.370. The complement of an angle is one-fifth the angle. Find the angle

Graduate Level 31/07/2025 (Shift - 3)

- (a) 75° (b) 7° (c) 42° (d) 85°

Q.371. The inradius of a triangle with sides 9 cm, 40 cm, 41 cm is:

Graduate Level 31/07/2025 (Shift - 3)

- (a) 2 cm (b) 4 cm (c) 6 cm (d) 7 cm

Q.372. Chords AB and CD intersect inside a circle at point O. If AO = 16 cm,

OB = 7 cm, and CO = 8 cm, find OD. (O is the origin)

Graduate Level 02/08/2025 (Shift - 3)

- (a) 14 cm (b) 15 cm (c) 17 cm (d) 12 cm

Q.373. An arc of a circle with radius 12 cm subtends an angle of 3 radians at the center. Find the length of the arc.

Graduate Level 02/08/2025 (Shift - 3)

- (a) 36 cm (b) 14 cm (c) 27 cm (d) 25 cm

Q.374. If the circumradius of triangle ABC is 21 cm, and its area is 84cm^2 , find the product of its sides.

Graduate Level 02/08/2025 (Shift - 3)

- (a) 7048cm^3 (b) 7192cm^3
(c) 7204cm^3 (d) 7056cm^3

Q.375. A circular garden is divided into 4 equal parts. If the total area is 300m^2 , what is the central angle of one part and its area?

Graduate Level 29/08/2025 (Shift - 1)

- (a) $60^\circ, 75\text{m}^2$ (b) $90^\circ, 75\text{m}^2$
(c) $120^\circ, 90\text{m}^2$ (d) $90^\circ, 60\text{m}^2$

Q.376. A sector having a radius of 8 cm and has a central angle of 90° . What is the area of the sector?

Graduate Level 29/08/2025 (Shift - 1)

- (a) $16\pi\text{cm}^2$ (b) $13\pi\text{cm}^2$
(c) $8\pi\text{cm}^2$ (d) $\frac{8\pi}{6}\text{cm}^2$

eduquity-based pattern (ebp) Questions

Q.377. In a triangle PQR, incentre is A and $\angle QAR = 116^\circ$. Find the measure of $\angle QPR$.

- (a) 52° (b) 54° (c) 27° (d) 26°

Q.378. The numbers of Commerce, Science and Arts postgraduates working in a company are 40, 80 and 60, respectively. If these figures are represented by a pie chart, then what is the angle corresponding to Commerce postgraduates?

- (a) 60° (b) 50° (c) 70° (d) 80°

Q.379. A quadrilateral having equal diagonals and equal opposite sides is always a _____.

- (a) trapezium (b) kite
(c) rhombus (d) rectangle

Q.380. ABCD is a trapezium in which AB is parallel to CD, and E and F are the mid-points of non-parallel sides AD and BC. If EF is parallel to AB, AB = 28 cm and CD = 18 cm, then find EF (in cm).

- (a) 46 (b) 10 (c) 5 (d) 23

Q.381. The centroid of a ΔXYZ is G. The



area of $\triangle XYZ$ is 45 cm². The area of $\triangle GYZ$ is:

- (a) 22.5 cm² (b) 7.5 cm²
(c) 15 cm² (d) 30 cm²

Q.382. A quadrilateral ABCD has sides AB = 5 units, BC = 12 units, CD = DA and $\angle ABC = \angle ADC = 90^\circ$. What is the area of the quadrilateral (in square units) ?

- (a) 72.75 (b) 74.75 (c) 74.25 (d) 72.25

Q.383. If the perimeter of an equilateral triangle is 24 cm, then the length of each median is:

- (a) $3\sqrt{2}$ cm (b) $4\sqrt{2}$ cm
(c) $4\sqrt{3}$ cm (d) $3\sqrt{3}$ cm

Q.384. ABCD is a quadrilateral in which diagonal BD = 62 cm, $AL \perp BD$ and $CM \perp BD$, such that AL = 14.3 cm and CM = 17.7 cm. The area of the quadrilateral ABCD is:

- (a) 982 cm² (b) 902 cm²
(c) 992 cm² (d) 900 cm²

Q.385. Which of the following statements is sufficient to conclude that two triangles are congruent ?

- (a) These have two equal sides and the same perimeter
(b) These have the same area and the same base.
(c) One side and one angle of both triangles are equal.
(d) These have the same base and the same height.

Q.386. If the sum of two sides of an equilateral triangle is 16 cm, then find the third side.

- (a) 1.4 cm (b) 16 cm
(c) Cannot be found (d) 8 cm

Q.387. Let O be the centre of the circle and AB and CD are two parallel chords on the same side of the radius. OP is perpendicular to AB and OQ is perpendicular to CD. If AB = 10 cm, CD = 24 cm and PQ = 7 cm, then the diameter (in cm) of the circle is equal to:

- (a) 26 (b) 13 (c) 12 (d) 24

Q.388. The distance between the centres of two circles of radii 22 cm and 10 cm is 37 cm. If the points of contact of a direct common tangent to these circles are M and Q, then find the length of the line segment MQ.

- (a) 35 cm (b) 39 cm (c) 29 cm (d) 25 cm

Q.389. The points P and S are on the same side of the line segment QR, such that $\angle PQR = 90^\circ$, $\angle SRQ = 90^\circ$ and PQ = SR. Select the correct statement.

- (a) $\triangle PQR \cong \triangle SRQ$ by RHS
(b) $\triangle PQR \cong \triangle SQR$ by SAS

(c) $\triangle PQR \cong \triangle SQR$ by RHS

(d) $\triangle PQR \cong \triangle SRQ$ by SAS

Q.390. The angles of triangle are such that one is average of other two, then the angles are:

- 1) $\frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}$ 2) $\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{2}$
3) $\frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{4}$ 4) $\frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{3}$
(a) 4 (b) 3 (c) 2 (d) 1

Q.391. Let C be a circle with centre O and P be an external point to C. Let PA and PB be two tangents to C with A and B being the points of tangency, respectively. If PA and PB are inclined to each other at an angle of 60° , then find $\angle POA$.

- (a) 40° (b) 60° (c) 80° (d) 30°

Q.392. Two triangles EFG and HIJ are congruent. If the area of $\triangle EFG$ is 124 cm², then the area of $\triangle HIJ$ will be:

- (a) 124 cm² (b) 248 cm²
(c) 62 cm² (d) 31 cm²

Q.393. By applying which of the following criteria can two triangles NOT be proved congruent?

- (a) angle-side-angle (b) angle-angle-angle
(c) side-side-side (d) side-angle-side

Q.394. Two circles of radii 18 cm and 12 cm touch each other externally. Find the length (in cm) of their direct common tangent.

- (a) $15\sqrt{6}$ (b) $10\sqrt{6}$ (c) $12\sqrt{6}$ (d) $18\sqrt{6}$

Q.395. The side of an equilateral triangle is 28 cm. Taking each vertex as the centre, a circle is described with a radius equal to half the length of the side of the triangle. Find the area of that part of the triangle which is not included in the circles (use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$).

- (a) 30.89 cm² (b) 38.08 cm²
(c) 31.08 cm² (d) 39.08 cm²

Q.396. $\triangle EFG$ and $\triangle HIJ$ are similar. Also, $\angle E = \angle H$ and $\angle F = \angle I$. If $3EF = HI$ and $FG = 9$ cm, then IJ is equal to _____.

- (a) 9 cm (b) 27 cm (c) 3 cm (d) 18 cm

Q.397. If 45° and 65° are the angles of a triangle, then find the exterior angle of the third angle (remaining angle).

- (a) 110° (b) 120° (c) 70° (d) 100°

Q.398. In a circle of radius $5\sqrt{13}$ cm, a chord is at a distance of 10 cm from the centre of the circle. Find the length (in cm) of the chord.

- (a) 30 (b) 28 (c) 36 (d) 15

Q.399. If $\triangle PQR \cong \triangle MON$ such that $\angle PQR = 82^\circ$, $\angle QRP = 47^\circ$, $\angle RPQ = 51^\circ$, $\angle MON = (5y - 8)^\circ$, QR = 24 and ON = $3x + y$, then

find $(x + y)$.

- (a) 14 (b) 18 (c) 16 (d) 20

Q.400. In a circle, two chords MN and PQ intersect at O. If MO = 9 cm, ON = 5 cm and OQ = 6 cm, then the value of OP (in cm) is:

- (a) 7.5 (b) 6.5 (c) 6 (d) 7

Q.401. In a $\triangle ABC$ right-angled at C, if $\tan A = \sqrt{3}$, then the value of $\sin A \cos B \cot(A + B)$ is:

- (a) $\sqrt{3}$ (b) 1 (c) 0 (d) 2

Q.402. In a rhombus ABCD, O is any interior point such that OA = OC. What is five-ninths of the degree measure of $\angle DOB$?

- (a) 120° (b) 150° (c) 100° (d) 90°

Q.403. In $\triangle PQR$, M, N, and S are the mid-points of sides PQ, PR and QR, respectively. If the area of the $\triangle PQR$ is 46 cm², what is the area (in cm²) of the quadrilateral MNRQ?

- (a) 39 (b) 27.6 (c) 23 (d) 34.5

Q.404. In a triangle $\triangle ABC$, the $\angle ABC = 90^\circ$.

If $\sin(A) = \frac{1}{2}$, then $\cos(C)$ is equal to:

- (a) $\frac{1}{\sqrt{2}}$ (b) 1 (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$

Q.405. In a triangle $\triangle ABC$, AB = AC and $\angle A = 80^\circ$, then find $\angle B$.

- (a) 50° (b) 80° (c) 40° (d) 45°

Q.406. $\triangle ABC$ and $\triangle RPQ$ are congruent, then the side of $\triangle ABC$ that corresponds to PQ is:

- (a) AB (b) CA (c) AR (d) BC

Q.407. In a $\triangle ABC$, if $\frac{1}{2}\angle A + \angle B + \angle C =$

120° , then the value of $\frac{1}{4}\angle A$ is :

- (a) 20° (b) 40° (c) 60° (d) 30°

Q.408. P and Q are centres of two circles whose radii are 7 cm and 3 cm, respectively. If the direct common tangents to the circles meet PQ extended at A, then A divides PQ _____.

- (a) externally in the ratio 3 : 7
(b) internally in the ratio 3 : 7
(c) externally in the ratio 7 : 3
(d) internally in the ratio 7 : 3

Q.409. Consider a triangle PQR, right angled at R, in which PQ = 29 units, QR = 21 units and $\angle PQR = \theta$. Find the value of $\cos^2 \theta - \sin^2 \theta$.

- (a) $\frac{21}{841}$ (b) 1 (c) $\frac{20}{841}$ (d) $\frac{41}{841}$

Q.410. In $\triangle ABC$, two medians AD and BE intersect at G at right angles. If AD = 12 cm and BE = 9 cm, then the length of AB



is equal to:

- (a) 14 cm (b) 16 cm (c) 12 cm (d) 10 cm

Q.411. In a triangle PQR, S is a point on the side QR such that $PS \perp QR$, then which of the following options is true?

- (a) $PS^2 + PR^2 = PQ^2 + QR^2$
 (b) $PR^2 + QS^2 = PQ^2 + SR^2$
 (c) $PQ^2 + PR^2 = QS^2 + SR^2$
 (d) $PS^2 + QS^2 = PQ^2 + PR^2$

Q.412. In triangles ABC and DEF, $AB = FD$ and $\angle A = \angle D$. The two triangles are congruent by SAS criterion if :

- (a) $BC = DE$ (b) $AC = EF$
 (c) $BC = EF$ (d) $AC = DE$

Q.413. The measures of the three angles of a triangle are in the ratio 17 : 13 : 15. Find the positive difference between the greatest and the smallest of these three angles.

- (a) 16° (b) 24° (c) 20° (d) 12°

Q.414. In triangle ABC, the length of side BC is 2 cm less than twice the length of side AB. The length of side AC exceeds the length of side AB by 10 cm. The perimeter of the triangle is 40 cm. Determine the length (in cm) of the smallest side of the triangle.

- (a) 12 (b) 14 (c) 10 (d) 8

Q.415. In $\triangle DEF$, the bisector of $\angle D$ intersects side EF at point N. If $DE = 36$ cm, $DF = 40$ cm and $EF = 38$ cm, then the length (in cm) of NF is ____.

- (a) 16 (b) 20 (c) 18 (d) 12

Q.416. From a point T, the length of the tangent to a circle is 32 cm and the distance of T from the centre is 40 cm. The diameter (in cm) of the circle is:

- (a) 36 (b) 16 (c) 32 (d) 48

Q.417. A pair of straight lines from an external point F intersects a circle at A and B ($FA < FB$), and touches the circle at C. O is the centre of the circle. Given that $\angle ACF = 50^\circ$ and $\angle AFC = 30^\circ$, find $\angle AOB$.

- (a) 80° (b) 90° (c) 100° (d) 40°

Q.418. If $\triangle PQR \cong \triangle ABC$, $AB = 6$ cm, $\angle B = 50^\circ$ and $\angle A = 70^\circ$, then which of the following is true?

- (a) $QR = 6$ cm, $\angle Q = 60^\circ$
 (b) $QR = 6$ cm, $\angle R = 60^\circ$
 (c) $QP = 6$ cm, $\angle P = 60^\circ$
 (d) $QP = 6$ cm, $\angle R = 60^\circ$

Q.419. A circle's centre is connected to its 50 cm long chord by a perpendicular that is 21 cm long. Find the circle's radius.

- (a) $\sqrt{1065}$ cm (b) $\sqrt{1068}$ cm
 (c) $\sqrt{1064}$ cm (d) $\sqrt{1066}$ cm

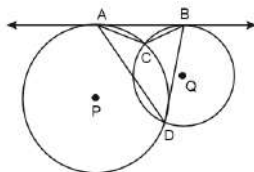
Q.420. ABC is an equilateral triangle. P, Q and R are the mid-points of sides AB, BC and CA, respectively. If the length of the side of the triangle ABC is 11 cm, then the area (in cm^2) of $\triangle PQR$ is:

- (a) $\frac{21\sqrt{3}}{16}$ (b) $\frac{11\sqrt{3}}{16}$
 (c) $\frac{121\sqrt{3}}{16}$ (d) $\frac{111\sqrt{3}}{16}$

Q.421. P and Q are two fixed points 10 cm apart, and R is a point on PQ such that $PR = 6$ cm. By what percentage is the length of QR decreased when the length of PR is increased by 5%?

- (a) 8.5% (b) 7.5% (c) 7% (d) 8%

Q.422. Two circles, centred at P and Q intersect at two points C and D. AB is tangent to the two circles at A and B. If $\angle ADB = 68^\circ$, then $\angle ACB =$ ____.



- (a) 132° (b) 112° (c) 124° (d) 102°

Q.423. The perimeters of two similar triangles $\triangle RST$ and $\triangle ABC$ are 26 cm and 39 cm, respectively. If $AB = 24$ cm, then RS is ____.

- (a) 16 cm (b) 36 cm (c) 18 cm (d) 24 cm

Q.424. In the following sets, which set represents the angles of a triangle?

- (a) $\{30^\circ, 60^\circ, 90^\circ\}$ (b) $\{80^\circ, 50^\circ, 90^\circ\}$
 (c) $\{30^\circ, 50^\circ, 110^\circ\}$ (d) $\{30^\circ, 50^\circ, 90^\circ\}$

Q.425. What is the difference in area (in cm^2) of $\triangle ABC$ having sides of 10 cm, 20 cm and 20 cm, and a right angled triangle $\triangle PQR$ with hypotenuse of 13 cm and one of the perpendiculars of 12 cm?

Note: $\sqrt{2} = 1.41$, $\sqrt{3} = 1.73$, $\sqrt{7} = 2.65$, $\sqrt{13} = 3.61$, $\sqrt{15} = 3.87$, $\sqrt{21} = 4.58$

- (a) 70.05 (b) 36.57 (c) 66.75 (d) 53.58

Q.426. In a triangle, the lengths of sides are 6 units, 12 units and x units. How many possible integer values are there for x?

- (a) 9 (b) 12 (c) 11 (d) 10

Q.427. If a triangle has a perimeter of 52 units, then all its sides have length ____ units.

- (a) < 26 (b) < 20 (c) < 24 (d) < 18

Q.428. A chord of length 32 cm is drawn at a distance of 12 cm from the centre of a circle. Find the radius of the circle.

- (a) 18 cm (b) 22 cm (c) 20 cm (d) 28 cm

Q.429. In a circle with centre O, AOC is the diameter. B is a point on the

circumference of the circle such that arc AB is $\frac{1}{5}$ of the arc BC. What is the degree measure of $\angle BOC$?

- (a) 120° (b) 150° (c) 80° (d) 30°

Q.430. If $\triangle ABC$ and $\triangle QPR$ are congruent triangles such that $AB = 7$ cm, $AC = 8$ cm, $\angle B = 65^\circ$ and $\angle C = 75^\circ$, then which of the following is true?

- (a) $\angle R = 75^\circ$ and $PQ = 8$ cm
 (b) $PQ = 7$ cm and $\angle R = 75^\circ$
 (c) $\angle Q = 40^\circ$ and $RQ = 7$ cm
 (d) $AC = PR$ and $\angle P = 65^\circ$

Q.431. On a circle of radius 7 units, PQ and QR are chords of length 7 units each. What is the length of the chord PR in units?

- (a) $\frac{28}{\sqrt{3}}$ (b) $\frac{7}{\sqrt{3}}$ (c) $\frac{14}{\sqrt{3}}$ (d) $\frac{21}{\sqrt{3}}$

Q.432. If arcs PAQ and RBS of a circle are congruent, then find the ratio of PQ and RS.

- (a) 2 : 3 (b) 1 : 3 (c) 1 : 1 (d) 1 : 2

Q.433. In a $\triangle PQR$, $PQ = PR$ and PT is perpendicular to QR. If $PQ = 17$ cm, $PT = 15$ cm, then what is the measure (in cm) of QR?

- (a) 8 (b) 24 (c) 16 (d) 32

Q.434. Two equal circles pass through each other's centre. If the radius of each of the circle is 13 cm, then what is the length of the common chord?

- (a) $13\sqrt{6}$ (b) $7\sqrt{3}$ (c) $13\sqrt{2}$ (d) $13\sqrt{3}$

Q.435. The side MN of a parallelogram MNOP is produced to Q such that $MN = NQ$. PQ intersects ON at R. The point R divides ON in the ratio:

- (a) 3 : 1 (b) 2 : 1 (c) 1 : 2 (d) 1 : 1

Q.436. If we extend a chord infinitely on both sides, then the chord becomes a ____.

- (a) diameter (b) tangent
 (c) secant (d) normal

Q.437. P, Q and R are the three angles of a triangle. If $P - Q = 20^\circ$ and $Q - R = 26^\circ$, then $\angle P$ is equal to:

- (a) 52° (b) 46° (c) 92° (d) 82°

Q.438. PQ and RS are common tangents to two circles intersecting at A and B. A and B, when produced on both sides, meet the tangents PQ and RS at X and Y, respectively. If $AB = 3$ cm and $XY = 5$ cm, then PQ is ____.

- (a) 5 cm (b) 3 cm (c) 6 cm (d) 4 cm

Q.439. A quadrilateral PQRS is inscribed in a circle of centre O, such that PQ is a diameter and $\angle PSR = 120^\circ$. Find the value of $\angle QPR$.



(a) 30° (b) 60° (c) 40° (d) 50°

Q.440. $\triangle ABC$ and $\triangle DEF$ are two triangles such that $\triangle ABC \cong \triangle FDE$. If $AB = 17$ cm, $\angle B = 52^\circ$ and $\angle A = 95^\circ$, then which of the following options is true?

- (a) $DE = 17$ cm, $\angle F = 33^\circ$
 (b) $DF = 17$ cm, $\angle E = 33^\circ$
 (c) $DE = 17$ cm, $\angle E = 33^\circ$
 (d) $DF = 17$ cm, $\angle D = 33^\circ$

Q.441. In a kite $ABCD$, longer diagonal BD is drawn. If $\angle ABD = 54^\circ$ and $\angle ADB = 50^\circ$, then what is the measure (in degrees) of the $\angle DCB$?

- (a) 108 (b) 76 (c) 100 (d) 54

Q.442. In $\triangle PQR$, if PT is the median, then which of the following is correct?

- (a) $PQ^2 + PR^2 = PT^2 + QR^2$
 (b) $PQ^2 + PR^2 = 2(PT^2 + QT^2)$
 (c) $PQ^2 + PR^2 = 2(PT^2 - QT^2)$
 (d) $PQ^2 + PR^2 = PT^2 + QT^2$

Q.443. In the following sets, which set represents the sides of a right-angled triangle?

- (a) {3, 4, 6} (b) {8, 15, 16}
 (c) {5, 10, 12} (d) {5, 12, 13}

Q.444. Which of the following are the angles of a scalene triangle?

- (a) 60° , 30° and 90° (b) 60° , 60° and 60°
 (c) 110° , 35° and 35° (d) 90° , 45° and 45°

Q.445. The distance between the longest chord and the centre of a circle is:

- (a) Equal to zero unit
 (b) Equal to the diameter
 (c) Equal to half of the radius
 (d) Equal to the radius

Q.446. If the medians of a triangle are equal, then the triangle will be:

- (a) Equilateral triangle
 (b) Obtuse-angle triangle
 (c) Scalene triangle
 (d) Right-angled triangle

Q.447. If in a circle a chord of length 14 cm is at a distance of 24 cm from its centre, then the length of the radius of the circle is:

- (a) 28 cm (b) 31 cm (c) 24 cm (d) 25 cm

Q.448. If each side of an equilateral triangle is $37\sqrt{3}$ cm, then its altitude (in cm) is equal to

- (a) 37.5 (b) 18.5 (c) 60.5 (d) 55.5

Q.449. In $\triangle PQR$, if $4\angle P = 5\angle Q = 20\angle R$, then the value of $\angle Q$ is:

- (a) 72° (b) 36° (c) 90° (d) 45°

Q.450. Triangle XYZ is an isosceles triangle with sides $XY = XZ$. If the measure of angle Y is 80° , what is the measure of angle X ?

(a) 30° (b) 40° (c) 20° (d) 50°

Q.451. PQ is the diameter of a circle with centre A . M is a point on the circumference of the circle. If $m\angle MAP = 110^\circ$, then find the value of $m\angle MQA$.

- (a) 55° (b) 35° (c) 65° (d) 70°

Q.452. Let C be a circle with centre O and PQ be a minor arc on C . Let R and S be two points on the major arc of PQ on C . If PQ subtends an angle 80° at O , find $\angle PRQ$ and $\angle PSQ$, respectively.

- (a) 40° and 60° (b) 50° and 80°
 (c) 40° and 40° (d) 40° and 50°

Q.453. In $\triangle DEF$, $DE = 12$ cm, $EF = 15$ cm, and $\angle DEF = 90^\circ$. $\triangle DEF$ is congruent to $\triangle XYZ$. If $YZ = 15$ cm, then what is the length of XZ ?

- (a) $3\sqrt{41}$ cm (b) $2\sqrt{41}$ cm
 (c) $5\sqrt{41}$ cm (d) $4\sqrt{41}$ cm

Q.454. If two straight lines are coinciding with each other, then the number of points of intersection is/are:

- (a) infinitely many points of intersection
 (b) finite number of point of intersection
 (c) no points of intersection
 (d) unique point of intersection

Q.455. Let ABC be a triangle. If the mid-points of AB , BC and AC are D , E and F , respectively, then $\triangle DEF$ is NOT congruent to ____.

- (a) $\triangle BDE$ (b) $\triangle ECF$ (c) $\triangle ADF$ (d) $\triangle ABC$

Q.456. In $\triangle ABC$, $AB = 6$ cm, $BC = 9$ cm and $AC = 12$ cm. If AD is the angle bisector of $\angle BAC$, where D is a point on BC , then find DC .

- (a) 9 (b) 6 (c) 3 (d) 7

Q.457. A line cuts two concentric circles. The length of chords formed by this line on the circles is 6 cm and 18 cm. Find the difference in the squares of the radii of two circles.

- (a) 90 (b) 120 (c) 60 (d) 72

Q.458. It is always possible to draw a circle passing through _____ non-collinear points in a plane.

- (a) 3 (b) 6 (c) 5 (d) 4

Q.459. If the lengths of two sides of an isosceles triangle are 6 cm and 12 cm, then find the length of the third side.

- (a) 8 cm (b) 6 cm (c) 12 cm (d) 14 cm

Q.460. The centres of two circles are 36 cm apart. If the radii of these two circles are 15 cm and 9 cm, respectively, then what is the sum of the lengths (in cm) of a direct common tangent and a transverse common tangent of these two circles?

- (a) $6\sqrt{5}(\sqrt{7} + 2)$ (b) $6\sqrt{7}(\sqrt{5} + 2)$
 (c) $6\sqrt{5}(\sqrt{5} + 2)$ (d) $6\sqrt{7}(\sqrt{7} + 2)$

Q.461. The side BC of $\triangle ABC$ is produced to a point D . If $AC = BC$ and $\angle BAC = 70^\circ$, then find the value of $\angle ACD - \angle ABC$.

- (a) 70° (b) 60° (c) 75° (d) 45°

Q.462. In a quadrilateral $ABCD$, $AB = BC$, $AD = DC$, $\angle ABD = 68^\circ$, $\angle ADB = (2y - 7)^\circ$, $\angle BDC = 33^\circ$, $\angle DBC = (3x + 2)^\circ$. Then the value of $2x + 3y$ is:

- (a) 108 (b) 144 (c) 104 (d) 118

Q.463. The two circles intersect at two points P and Q . PR and PS are diameters of the two circles. What is $\angle PQR$?

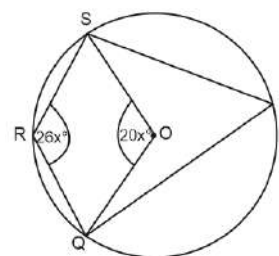
- (a) 90° (b) 0° (c) 45° (d) 60°

Q.464. A circular arc whose radius is 14 cm, makes an angle of 45° at the centre. Find the perimeter (in cm) of the sector.

(Use $\pi = \frac{22}{7}$)

- (a) 38 (b) 28 (c) 39 (d) 34

Q.465.



NOT TO SCALE

In the figure shown above, quadrilateral $PQRS$ has its vertices on the circumference of the circle with centre O . If $m\angle QOS = 20x^\circ$ and $m\angle QRS = 26x^\circ$, then what is the value of ' x '?

- (a) 6 (b) 5 (c) 4 (d) 3

Q.466. For triangles, which of the following statement(s) are true?

- (a) Sum of interior angles is 180° .
 (b) All interior angles can't be equal.
 (c) Can have at the most one obtuse angle.
 (d) Sum of two sides may or may not be equal to the third side.
 (a) (a) and (b) (b) (b), (c) and (d)
 (c) (a), (b) and (d) (d) (a) and (c)

Q.467. If the angles of a triangle are in the ratio $7 : 8 : 3$, then the value of the largest angle is:

- (a) 120° (b) 80° (c) 100° (d) 60°

Q.468. $\triangle ABC$ is a right-angled triangle with $\angle ABC = 90^\circ$. If $m(\overline{AB}) = 28$ cm, and $m(\overline{BC}) = 96$ cm, find the area (in cm^2) of the circumcircle of $\triangle ABC$. (Use $\pi = 3.14$.)

- (a) 7,850 (b) 8,164 (c) 7,693 (d) 7,536



Q.469. The side BC of $\triangle ABC$ is produced to a point D. If $AC = BC$ and $\angle BAC = 70^\circ$, then find the value of $2.5\angle ACD - 1.5\angle ABC$.

(a) 245° (b) 235° (c) 225° (d) 230°

Q.470. In $\triangle ABC$, $DE \parallel BC$ and $5AE = 3EC$. If $AB = 6.4$ units, then the value of DB (in units) is:

(a) 2.4 (b) 5 (c) 3.2 (d) 4

Q.471. In a circle with center O, an arc ABC subtends an angle of 138° at the centre of the circle. The chord AB is produced to a point P. Then, the measure of $\angle CBP$ is:

(a) 108° (b) 42° (c) 111° (d) 69°

Q.472. If the altitude from two vertices of a triangle to the opposite sides are equal, then the triangle will be:

(a) isosceles triangle
(b) equilateral triangle
(c) scalene triangle
(d) obtuse angle triangle

Q.473. In a triangle PQR, $\angle P + \angle Q = 84^\circ$. Find the value of $\angle R$.

(a) 74° (b) 96° (c) 60° (d) 102°

Q.474. In triangle XYZ, A is a point on YZ such that $XA = YA$. If $\angle XYA = 50^\circ$ and $\angle AXZ = 19^\circ$, what is the degree measure of $\angle XZA$.

(a) 61° (b) 53° (c) 49° (d) 41°

Q.475. Given that the perimeters of two similar triangles, $\triangle ABC$ and $\triangle DEF$, are 24 cm and 9 cm, respectively, and $DE = 3$ cm, what is the ratio between the area of $\triangle ABC$ and that of $\triangle DEF$?

(a) 54 : 9 (b) 54 : 7 (c) 64 : 9 (d) 64 : 7

Q.476. Which of the following is a correct statement?

(a) If two circles touch each other, the point of contact lies on the line joining the two centres.
(b) Equal chords are equidistant from the centre of the circle but not always subtend equal angle at centre of the circle
(c) The sum of the angles of a cyclic quadrilateral is always 180° .
(d) Angles subtended by the arc in the same segment of the circle are in ratio of 2 : 1.

Q.477. PQ is a diameter of a circle with centre O. The tangents at R meets PQ produced at A. If $\angle RPQ = 27^\circ$, then measure of $\angle RQP$ is:

(a) 63° (b) 54° (c) 27° (d) 153°

Q.478. Two circles touch each other at a point O. AB is a simple common tangent

to both the circles touching at point A and point B. If radii of the circles are 9 cm and 4 cm, then find AB.

(a) 72 cm (b) 12 cm (c) 24 cm (d) 144 cm

Q.479. In a $\triangle ABC$, two medians AD and BE intersect at G at right angles. If $AD = 18$ cm and $BE = 12$ cm, then the length of BD is equal to:

(a) 15 cm (b) 10 cm (c) 20 cm (d) 8 cm

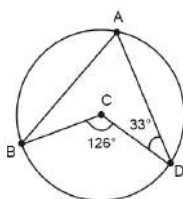
Q.480. CD is a tangent to a circle of circumference of 88 cm with its centre at O. OC cuts the circle at P while OD cuts circle at V such that $\angle COD = 90^\circ$. Length of PC is 6 cm. What is half of CD (in cm) if OD exceeds OC by 1 cm?

(a) 14.5 (b) 15.5 (c) 16.5 (d) 31

Q.481. In a triangle ABC, $AB = 6$ units, $BC = 8$ units and $AC = 10$ units. Let M be a point on AC such that $BM = 5$ units. With a point D, a triangle BMD is formed and the triangle BMD is similar to the triangle ABC with $\frac{BM}{AB} = \frac{BD}{AC}$. What is the length of BD in units?

(a) $\frac{40}{3}$ (b) $\frac{10}{3}$ (c) $\frac{25}{3}$ (d) $\frac{35}{3}$

Q.482. Given is a circle with centre at C. A, B and D are the points on the circumference. Find $\angle ABC$.



(a) 37° (b) 33° (c) 35° (d) 30°

Q.483. Diagonals of a trapezium ABCD with $AB \parallel DC$, intersect each other at the point 'O'. If $AB = 2.5 CD$, find the ratio of the area of triangle AOB to the area of triangle COD.

(a) 16 : 1 (b) 25 : 4 (c) 9 : 2 (d) 5 : 2

Q.484. The areas of two similar triangles $\triangle PQR$ and $\triangle XYZ$ are 12.96 cm^2 and 635.04 cm^2 , respectively. If $QR = 2.9$ cm, then the length (in cm) of YZ equals:

(a) 23.2 (b) 25.2 (c) 30.4 (d) 20.3

Q.485. If $\triangle ABC$ and $\triangle PQR$ are similar. $AB = 8$ cm, $PQ = 12$ cm, $QR = 18$ cm and $RP = 24$ cm, then the perimeter of $\triangle ABC$ is _____ cm.

(a) 36 (b) 42 (c) 54 (d) 27

Q.486. Two circles of diameters 16 cm and 20 cm are such that the distance between their centers is 13 cm. What is the length (in cm) of a common tangent to these circles that does not intersect the line joining the centers?

(a) $\sqrt{167}$ (b) $\sqrt{165}$ (c) $\sqrt{163}$ (d) $\sqrt{173}$

Q.487. If $\triangle ABC \sim \triangle QRP$; area of $\triangle ABC$: area of $\triangle PQR = 9 : 4$; $AB = 18$ cm and $BC = 15$ cm, then find the length of PR.

(a) 15 cm (b) 12 cm (c) 10 cm (d) 14 cm

Q.488. In $\triangle EFG$, $XY \parallel FG$, area of the quadrilateral XFGY = 44 m^2 . If $EX : XF = 2 : 3$, then find the area of $\triangle EXY$ (in m^2).

(a) 7.28 (b) 8.10 (c) 8.38 (d) 9.46

Q.489. Two circles touch each other externally at M. PQ is a direct common tangent to the two circles, P and Q are points of contact and $\angle MPQ = 38^\circ$. $\angle MQP$ is:

(a) 38° (b) 42° (c) 52° (d) 48°

Q.490. In a right-angled triangle ABC such that $\angle BAC = 90^\circ$ and AD is perpendicular to BC. If the area of $\triangle ABC$ is 63 cm^2 , area of $\triangle ACD = 7 \text{ cm}^2$, and $AC = 5$ cm, then the length of BC is equal to:

(a) 10 cm (b) 18 cm (c) 12 cm (d) 15 cm

Q.491. A, B, C are three points such that $AB = 11$ cm, $BC = 13$ cm and $AC = 24$ cm. Number of circles passing through the points A, B, C is:

(a) 0 (b) 1 (c) 2 (d) 3

Q.492. A circle of diameter 26 cm has two equal parallel chords of length 10 cm each, then the distance between the two chords is:

(a) 21 cm (b) 24 cm (c) 26 cm (d) 23 cm

Q.493. PQ is parallel to SR in a trapezium PQRS. It is given that $PQ > SR$ and the diagonals PR and QS intersect at O. If $PO = 3x - 15$, $OQ = x + 9$, $OR = x - 5$ and $OS = 5$ and x has two values x_1 and x_2 , then the value of $(x_1^2 - x_2^2)$ is:

(a) 15 (b) 11 (c) 13 (d) 19

Q.494. If the perimeters of two similar triangles are in the ratio of 4 : 7, and the sum of the areas is 195 cm^2 , then what is $\frac{1}{3}$ of the difference between the areas

(in cm^2) of the two triangles?

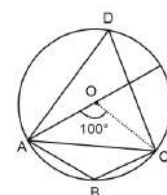
(a) 99 (b) 33 (c) 63 (d) 54

Q.495. Two parallel chords of length 8 units each are 6 units away from each other in a circle.

What is the radius of the circle in units?

(a) 6 (b) 5 (c) 4 (d) 3

Q.496. In the given figure, O is the centre of the circle, AE is the diameter and $\angle AOC = 100^\circ$.



The value of $\angle CDE + \angle CEA$ is:

- (a) 90° (b) 60° (c) 80° (d) 100°

Q.497. There is a circle with a centre T and a radius of 8 cm. Two tangents KE and KI are drawn from point K, which is 17 cm from the centre. The area of the quadrilateral TEKI is:

- (a) 127.5 sq. cm (b) 130 sq. cm
(c) 136 sq. cm (d) 120 sq. cm

Q.498. $\triangle ABC$ is a right-angled triangle where $m\angle ABC = 90^\circ$. If $m(\overline{AB}) = 15$ cm, $m(\overline{BC}) = 20$ cm, and $\overline{BD} \perp \overline{AC}$, intersecting $\overline{AC}$ at the point D, find $m(\overline{BD})$ (in cm).

- (a) 9 (b) 12 (c) 12.5 (d) 10

Q.499. In $\triangle ABC$, P and Q are the middle points of the sides AB and AC, respectively. R is a point on the segment PQ such that $PR : RQ = 1 : 5$. If $PR = 6$ cm, then $BC =$ ____.

- (a) 72 cm (b) 68 cm (c) 66 cm (d) 70 cm

Q.500. The measures of the three angles of a triangle are such that the smallest angle measures 42° less than the greatest angle, while the measure of the remaining angle is 24° more than the measure of the smallest angle. Find the measure of the smallest angle of the triangle.

- (a) 40° (b) 42° (c) 36° (d) 38°

Q.501. In right triangle ABC with right angle at C, M is the mid - point of hypotenuse AB. C is joined to M and produced to a point D, such that $DM = CM$. Point D is joined to B. If $CD = 10$ cm, and $BD = 6$ cm, find the value of CM.

- (a) 11.4 cm (b) 5 cm (c) 9.5 cm (d) 8 cm

Q.502. The centers of two circles of radii 25 cm and 35 cm are 80 cm apart. What is the ratio of the lengths of the transverse common tangent to the direct common tangent to these circles?

- (a) $\sqrt{7} : 3$ (b) $2 : 3$
(c) $2 : \sqrt{7}$ (d) $3\sqrt{7} : 2$

Q.503. A circle of diameter 26 cm has two equal chords of length 10 cm each, separated by a distance h cm. What is the value of h?

- (a) 16 (b) 18 (c) 24 (d) 12

Q.504. In a triangle ABC, $AB = AC$. BC is extended to D such that $CD = AB$ and the angle ADC is 30° . What are the angles of triangle ABC?

- (a) $60^\circ, 60^\circ, 60^\circ$ (b) $50^\circ, 60^\circ, 70^\circ$
(c) $30^\circ, 60^\circ, 90^\circ$ (d) $45^\circ, 60^\circ, 75^\circ$

Q.505. In a circle, O is the centre and AOB is the diameter. AT is a tangent to

the circle. Line TB intersects the circle at Q. Given that $\angle AOQ = 94^\circ$, find $\angle ATQ$.

- (a) 133° (b) 43° (c) 86° (d) 47°

Q.506. AB is parallel to DC in a trapezium ABCD. It is given that $AB > DC$ and the diagonals AC and BD intersect at O. If $AO = 3x - 15$, $OB = x + 9$, $OC = x - 5$ and $OD = 5$, and x has two values x_1 and x_2 , then the value of $(x_1^2 + x_2^2)$ is:

- (a) 45 (b) 56 (c) 61 (d) 73

Q.507. In triangle XYZ, $XY = YZ$ and $m\angle X = 78^\circ$. Find the value of $m\angle Y$.

- (a) 24° (b) 39° (c) 34° (d) 28°

Q.508. Two parallel chords are drawn in a circle of radius 25 cm. The distance between the two chords is 27 cm. If the length of one chord is 48 cm, then length of the other chord is equal to:

- (a) 42 cm (b) 48 cm (c) 30 cm (d) 36 cm

Q.509. In a triangle ABC, BD is perpendicular to AC. E is a point on BC such that angle BEA = x° . If angle EAC = 38° and angle EBD = 40° , then the value of x is:

- (a) 88 (b) 82 (c) 48 (d) 96

Q.510. If the angle of the sector with radius 7 cm is 120° then find the length of arc (in cm). (Use $\pi = \frac{22}{7}$)

- (a) $\frac{67}{3}$ (b) $\frac{22}{3}$ (c) $\frac{47}{3}$ (d) $\frac{44}{3}$

Q.511. Let ABC be a triangle such that $\angle ABC = 70^\circ$ and $\angle ACB = 50^\circ$. Let O be the incentre of the triangle. Find $\angle BOC$.

- (a) 120° (b) 130° (c) 100° (d) 60°

Q.512. Let C be a circle with center O and AB be a chord of C such that the length of AB is equal to the radius of C. Let D be any point on major arc of AB. Find $\angle AOB$ and $\angle ADB$, respectively.

- (a) $60^\circ; 30^\circ$ (b) $90^\circ; 60^\circ$
(c) $120^\circ; 100^\circ$ (d) $100^\circ; 80^\circ$

Q.513. Two circles have radii of 27 cm and 36 cm. The distance between their centres is 45 cm. What is the length (in cm) of their common chord?

- (a) 47.5 cm (b) 37.2 cm
(c) 40.5 cm (d) 43.2 cm

Q.514. Find the altitude (in cm) of side MT of triangle MNT with side $MN = 36$ cm, $MT = 36$ cm and $NT = 48$ cm.

- (a) $18\sqrt{3}$ (b) $16\sqrt{5}$ (c) $12\sqrt{5}$ (d) $24\sqrt{3}$

Q.515. Side BC of a $\triangle ABC$ is extended upto D such that $CD = AC$. If $\angle BAD = 118^\circ$ and $\angle ACB = 80^\circ$, then the value of $\angle ABC$ is equal to:

- (a) 36° (b) 27° (c) 22° (d) 32°

Q.516. Two triangles ABC and DEF are similar. The smallest side of ABC is equal to 15 units. If the sides of ABC are in the ratio $3 : 4 : 5$, and the area of DEF is half of the area of ABC, then what is the largest side of DEF (in units)?

- (a) $\frac{25}{\sqrt{3}}$ (b) $\frac{25}{\sqrt{2}}$ (c) $25\sqrt{2}$ (d) $25\sqrt{3}$

Q.517. The hypotenuse of a right triangle is 6m more than twice the shortest side. If the third side is 2 m less than the hypotenuse, find the area(in m^2) of the triangle.

- (a) 110 (b) 100 (c) 120 (d) 130

Q.518. Find the length of the sides of a triangle, if its angles are in the ratio $2 : 4 : 6$ and its circumradius is 12 cm.

- (a) 6cm, $6\sqrt{3}$ cm, 12 cm
(b) 5 cm, 6 cm, 10 cm
(c) 15 cm, $15\sqrt{3}$ cm, 30 cm
(d) 12 cm, $12\sqrt{3}$ cm, 24 cm

Q.519. PQ is the diameter of a circle with centre O. PT is a tangent touching the circle at P. The secant QT intersects the circle at S. O and S are joined. If $\angle SOP = 104^\circ$, what is the measure (in degrees) of $\angle PTQ$?

- (a) 45 (b) 62 (c) 76 (d) 38

Q.520. $\triangle XYZ$ is similar to $\triangle PQR$. If the ratio of the perimeter of $\triangle XYZ$ to the perimeter of $\triangle PQR$ is $16 : 9$ and $PQ = 3.6$ cm, then what is the length (in cm) of XY?

- (a) $3\frac{4}{5}$ (b) $2\frac{3}{5}$ (c) $6\frac{2}{5}$ (d) $4\frac{3}{5}$

Q.521. Consider AD is a tangent to a circle of radius 6 cm. AC is a secant meeting the circle at B and CD is a diameter. If AB is 7 cm, then the value of AC (in cm) is:

- (a) 16 (b) 20 (c) 9 (d) 18

Q.522. The sides ST and TR and the median SU of the $\triangle STR$ are equal to the sides MN and NJ and the median MV of the $\triangle MNJ$, respectively. If $\angle TSU = 46^\circ$, $\angle NMJ = 79^\circ$ and $\angle MVN = 88^\circ$, what is the degree measure of $\frac{7}{11}\angle SRT$?

- (a) 35° (b) 56° (c) 47° (d) 55°

Q.523. The lengths of the three sides of a triangle are 19 cm, 25 cm and y cm. Which of the following options most accurately gives the possible values of 'y'?

- (a) $6 \leq y \leq 44$ (b) $6 < y < 44$
(c) $6 < y \leq 44$ (d) $6 \leq y < 44$



Q.524. If three sides of a triangle are 6 cm, 8 cm and 10 cm, then the area of the triangle is:

- (a) 24 cm^2 (b) $12\sqrt{3} \text{ cm}^2$
(c) $24\sqrt{3} \text{ cm}^2$ (d) 30 cm^2

Q.525. By decreasing 17° from each angle of a triangle, the ratio of their angles is $3 : 4 : 7$. The radian measure of the greatest angle is:

- (a) $\frac{153\pi}{360}$ (b) $\frac{167\pi}{180}$ (c) $\frac{163\pi}{180}$ (d) $\frac{163\pi}{360}$

Q.526. R wishes to use a stick of length 6 units as a common internal tangent to two circles of radius two units and three units. What can be the maximum distance (in units) between the centres of the circle?

- (a) $\sqrt{67}$ (b) $\sqrt{33}$ (c) $\sqrt{37}$ (d) $\sqrt{61}$

Q.527. A line from point A is drawn that is tangent to the circle at point B. A secant is also drawn from point A to the circle intersecting it at points C and D. If $AB = 42 \text{ cm}$ and $AC = 21 \text{ cm}$, then what is the ratio between AB and CD?

- (a) $3 : 2$ (b) $2 : 5$ (c) $3 : 4$ (d) $2 : 3$

Q.528. The sides AB and AC of $\triangle ABC$ are produced to points D and E, respectively. The bisectors of $\angle CBD$ and $\angle BCE$ meet at P. If $\angle A = 72^\circ$, then the measure of $\angle P$ is:

- (a) 65° (b) 54° (c) 55° (d) 35°

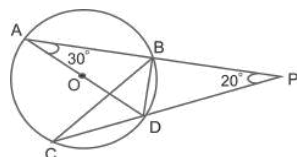
Q.529. Let ABCD be a quadrilateral and AC, BD be diagonals. The length of BD = 20 cm and the heights of the triangles ABD and BCD are, respectively, 6 cm, 8 cm. Find the area of the quadrilateral ABCD (in cm^2).

- (a) 140 (b) 1282 (c) 148 (d) 120

Q.530. In $\triangle ABC$, if $AB = \sqrt{3}$ units. $BC = 1$ unit and $AC = 2$ units, then $\angle ACB$ is equal to:

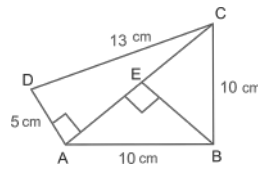
- (a) 30° (b) 90° (c) 60° (d) 45°

Q.531. PBA and PDC are two secants. AD is the diameter of the circle with the centre at O. $\angle A = 30^\circ$, $\angle P = 20^\circ$. Find the measure of $\angle DBC$.



- (a) 45° (b) 30° (c) 50° (d) 40°

Q.532. What is the area of a quadrilateral ABCD. (shown below) in which sides AB and BC are equal, sides AD and CD are of lengths 5 cm and 13 cm, respectively, and side AD is perpendicular to the diagonal AC?



- (a) 78 cm^2 (b) 75 cm^2
(c) 82 cm^2 (d) 80 cm^2

Q.533. Find the altitude (in cm) from the vertex Q to side PR of triangle PQR with side lengths $PQ = 40 \text{ cm}$, $PR = 40 \text{ cm}$ and $QR = 60 \text{ cm}$.

- (a) $25\sqrt{7}$ (b) $20\sqrt{7}$ (c) $15\sqrt{7}$ (d) $30\sqrt{7}$

Q.534. Which of the following is NOT true regarding a circle?

- (a) Given any three points in a plane, there is always a circle passing through these points
(b) The largest chord in a circle has a length twice its radius
(c) The ratio of the perimeter and radius of a circle is always constant
(d) A triangle with vertices on a circle and one side as its diameter is a right-angled triangle

Q.535. $\triangle ABC \sim \triangle EDF$ and $\text{area}(\triangle ABC) : \text{area}(\triangle EDF) = 1 : 4$. If $AB = 7 \text{ cm}$, $BC = 8 \text{ cm}$ and $CA = 9 \text{ cm}$, then DF is equal to:

- (a) 14 cm (b) 12 cm (c) 18 cm (d) 16 cm

Q.536. Let C be a circle with center O and PQ be the diameter of C. Let AB be a chord on C. If $\angle QOB = 40^\circ$, $\angle AOP = 80^\circ$, then find $\angle AOB$.

- (a) 70° (b) 90° (c) 100° (d) 60°

Q.537. If L is the midpoint of side BC of $\triangle ABC$ and the area of $\triangle ABL$ is 15 cm^2 , then the area (in sq cm) of $\triangle ABC$ is:

- (a) 32 (b) 27 (c) 30 (d) 25

Q.538. A right angled triangle of area 600 m^2 is to be made such that the hypotenuse is double the base. For the triangle, the hypotenuse is _____ more than the base.

- (a) 26.32 m (b) 22.32 m
(c) 24.36 m (d) 20.36 m

Q.539. From an external point A, tangents AP and AQ are drawn to a circle with centre O. If $\angle APQ = 40^\circ$, then find $\angle POQ$.

- (a) 110° (b) 100° (c) 60° (d) 80°

Q.540. What is the length of the arc of a circle whose radius is 35 cm and whose arc subtends an angle of 72° at the centre of the circle?

- (a) 28 cm (b) 44 cm (c) 38 cm (d) 56 cm

Q.541. In a $\triangle ABC$, angle $BAC = 30^\circ$ and angle $BCA = 60^\circ$. If $AC = 13 \text{ cm}$ and $AB = 12 \text{ cm}$, then BC is equal to:

- (a) 6 cm (b) 5 cm (c) 4 cm (d) 7 cm

Q.542. Triangle DEF is an equilateral triangle with a side length of 12 cm. If point G is the mid-point of side DE, what is the length (in cm) of side FG?

- (a) 6 (b) $6\sqrt{3}$ (c) 12 (d) $3\sqrt{6}$

Q.543. Which of the following statements is/are correct?

- A. A triangle can have all angles less than 60° .
B. A triangle can have one obtuse angle.
C. A triangle can have two right angles.
D. A triangle can have two acute angles.
(a) B (b) B and D (c) A and C (d) A

Q.544. If one angle of triangle PQR is greater than the sum of the other two, the triangle PQR will be:

- (a) obtuse angle (b) equilateral
(c) right angle (d) acute angle

Q.545. Taking any three of the line segments out of segments of length 2 cm, 5 cm, 7 cm and 8 cm, the number of triangles that can be formed is:

- (a) 2 (b) 4 (c) 1 (d) 3

Q.546. The secant CD intersects the circle at E and CF is tangent to the circle at F. If the length of the secant CD is equal to 20 cm and the length of tangent CF is 15 cm, find the length of the chord DE.

- (a) 3.75 cm (b) 11.25 cm
(c) 9.25 cm (d) 8.75 cm

Q.547. A scalene triangle ABC has two sides whose measures are 3.8 cm and 6 cm, respectively. Which of the following CANNOT be the measure (in cm) of its third side?

- (a) 3.0 (b) 2.7 (c) 2.4 (d) 2.2

Q.548. If the vertical angle of an isosceles triangle is 42° , then the measure of the other two angles will be:

- (a) 59° and 59° (b) 79° and 79°
(c) 89° and 89° (d) 69° and 69°

Q.549. $\triangle ABC$ is inscribed in a circle with centre O. If $AB = 35 \text{ cm}$, $BC = 12 \text{ cm}$ and $AC = 37 \text{ cm}$, then what is the circumradius of a triangle?

- (a) 13.5 cm (b) 16.5 cm
(c) 18.5 cm (d) 14.5 cm

Q.550. In a parallelogram ABCD, if vertices are in respective order and diagonals AC and BD intersect at O, then which of the following is NOT always correct?

- (a) $\triangle AOB \cong \triangle COD$ (b) $\triangle AOD \cong \triangle COB$
(c) $\triangle ABC \cong \triangle ADC$ (d) $\triangle BOC \cong \triangle AOD$

Q.551. In $\triangle ABC$, D is the mid-point of BC and G is the centroid. If $GD = 10 \text{ cm}$, then



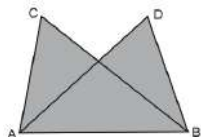
the length of AD is _____.

- (a) 20 cm (b) 10 cm (c) 15 cm (d) 30 cm

Q.552. In a triangle ABC, P is the midpoint of BC. If $AB = (2x + 4)$ cm, $AC = 6$ cm and $AP \perp BC$, then the value of x is:

- (a) 3 (b) 4 (c) 1 (d) 2

Q.553. $\triangle ABC$ and $\triangle ABD$ are on a common base AB and $AC = BD$ and $BC = AD$ as shown in the given figure.



Which of the following options is true ?

- (a) There are no congruent triangles in the given figure
(b) $\triangle ABC \cong \triangle ABD$
(c) $\triangle ABC \cong \triangle BAD$
(d) $\triangle ABC \cong \triangle ADB$

Q.554. In an isosceles $\triangle LMN$, $LM = LN$, and $\angle MLN = 37^\circ$. Find $\angle MNL$.

- (a) 70.0° (b) 65.0° (c) 71.5° (d) 60.5°

Q.555. If $x^2 - 1$, $2x$ and $x^2 + 1$ are the three sides of a right angled triangle, then which of the following can be its hypotenuse ?

- (a) $x^2 + 1$ (b) x^2 (c) $2x$ (d) $x^2 - 1$

Q.556. PQ is a diameter of a circle and PS is a chord. If PQ is 82 cm and PS is 80 cm, the distance (in cm) of PS from the center of the circle is:

- (a) 18 (b) 9 (c) 39 (d) 24

Q.557. If the angles of a triangle are in the ratio of 9 : 11 : 16, then the difference between the greatest angle and the smallest angle is :

- (a) 30° (b) 25° (c) 40° (d) 35°

Q.558. In $\triangle ABC$ and $\triangle PQR$, $AB = 7$ m, $BC = 8$ m, $AC = 9$ m, $PQ = 7$ m, $QR = 8$ m and $PR = 9$ m.

Which of the following is true for these triangles ?

- (a) $\triangle CBA \cong \triangle PQR$ (b) $\triangle ABC \cong \triangle PQR$
(c) $\triangle ABC \cong \triangle RQP$ (d) $\triangle ABC \cong \triangle QRP$

Q.559. Select the correct statement about the properties of a triangle.

- (a) The sum of two sides is always greater than the third side.
(b) The sum of two sides may be equal to the third side.
(c) The sum of two sides is always equal to the third side.
(d) The sum of two sides is always less than the third side.

Q.560. $\triangle PQR$ and $\triangle ABC$ are two congruent triangles. If the area of $\triangle PQR$

is 225 cm^2 , then the area of $\triangle ABC$ is:

- (a) 15 cm^2 (b) 112.5 cm^2
(c) 25 cm^2 (d) 225 cm^2

Q.561. If three angles of a triangle are $(18x + 6^\circ)$, $(10x + 30^\circ)$ and $(15x + 15^\circ)$, then the triangle is:

- (a) scalene (b) equilateral
(c) isosceles (d) right angled

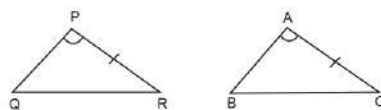
Q.562. $\triangle XYZ$ is an isosceles triangle such that $XY = XZ$. It is given that $YS \perp XZ$ and $ZT \perp XY$. What is the relation between YS and ZT ?

- (a) $YS > ZT$ (b) $YS = 2 ZT$
(c) $YS < ZT$ (d) $YS = ZT$

Q.563. In $\triangle ABC$, $AB = 9$ cm and $AC = 13$ cm, then the length of BC could be:

- (a) 4 cm (b) 2 cm (c) 22 cm (d) 20 cm

Q.564. In the following figure, $\angle A = \angle P$ and $AC = PR$.



Which of the following options needs to be satisfied for $\triangle PQR$ and $\triangle ABC$ to be congruent ?

- (a) $BC = QR$ by ASS criteria
(b) $AB = PQ$ by SAS criteria
(c) $BC = QR$ by SSA criteria
(d) $AB = PQ$ by SSA criteria

Q.565. Let ABC and PQR be two congruent triangles, such that $\angle A = \angle P = 90^\circ$. If $BC = 29$ cm, $PR = 21$ cm, find the value (in cm) of AB .

- (a) 23 (b) 19 (c) 20 (d) 22

Q.566. In $\triangle ABC$, $\angle A = 85^\circ$ and $\angle C = 58^\circ$. If $\triangle PQR$ is similar to $\triangle ABC$, and in correspondence, then find $\angle Q$.

- (a) 37° (b) 85° (c) 58° (d) 73°

Q.567. In $\triangle XYZ$, if $XY = 5$ cm, $XZ = 7$ cm and Q is a point on YZ such that XQ bisects $\angle X$, then $YQ : QZ$ is:

- (a) 5 : 7 (b) 7 : 12 (c) 7 : 5 (d) 5 : 12

Q.568. The measures of two angles of a triangle are in the ratio 3 : 7. If the sum of these two measures is equal to the measure of the third angle, then find the smallest angle.

- (a) 27° (b) 3° (c) 9° (d) 36°

Q.569. An equilateral triangle ABC surmounts a square $BCDE$. The value of $\angle EAB + 3\angle AEB$ is :

- (a) 60° (b) 80° (c) 280° (d) 240°

Q.570. One chord of a circle is given as 20.5 cm. Then the radius (r) of the circle must be:

- (a) $r = 10$ cm (b) $r < 10.25$

- (c) $r > 10.25$ (d) $r \geq 10.25$

Q.571. If the hypotenuse of a right-angled triangle is 29 cm and the sum of the other two sides is 41 cm, then the difference between the other two sides is:

- (a) 2 cm (b) 5 cm (c) 10 cm (d) 1 cm

Q.572. For a triangle ABC , D and E are two points on AB and AC such that $AD = \frac{1}{6} AB$, $AE = \frac{1}{6} AC$. If $BC = 22$ cm, then DE is _____ (Consider up to two decimals)

- (a) 3.67 cm (b) 1.67 cm
(c) 1.33 cm (d) 3.33 cm

Q.573. A sector of a circle has a central angle of 45° and an arc length of 22 cm. Find the radius of the circle.

(Use $\pi = \frac{22}{7}$)

- (a) 28 cm (b) 35 cm (c) 36 cm (d) 32 cm

Q.574. In two circles centered at O and O' , the distance between the centers of both circles is 17 cm. The points of contact of a direct common tangent between the circles are P and Q . If the radii of both circles are 7 cm and 15 cm, respectively, then the length of PQ is equal to:

- (a) 15 cm (b) 22 cm (c) 10 cm (d) 17 cm

Q.575. PQR is a triangle. The bisectors of the internal angle $\angle Q$ and external angle $\angle R$ intersect at S . If $\angle QSR = 40^\circ$, then $\angle P$ is:

- (a) 80° (b) 60° (c) 40° (d) 30°

Q.576. E, F, G , and H are four points lying on the circumference of a circle to make a cyclic quadrilateral. If $\angle FGH = 57^\circ$, then what will be the measure of the $\angle HEF$?

- (a) 33° (b) 143° (c) 123° (d) 93°

Q.577. In a $\triangle PQR$, $\angle P : \angle Q : \angle R = 3 : 4 : 8$. The shortest side and the longest side of the triangle, respectively, are:

- (a) QR and PR (b) PQ and PR
(c) PQ and QR (d) QR and PQ

Q.578. In a triangle ABC , if $\angle B = 90^\circ$, $\angle C = 45^\circ$ and $AC = 4$ cm, then the value of BC is :

- (a) 4 cm (b) $2\sqrt{2}$ cm
(c) $4\sqrt{2}$ cm (d) $\sqrt{2}$ cm

Q.579. Find the area of a triangle whose length of two sides are 4 cm and 5 cm and the angle between them is 45° .

- (a) $7\sqrt{2} \text{ cm}^2$ (b) $4\sqrt{2} \text{ cm}^2$
(c) $6\sqrt{2} \text{ cm}^2$ (d) $5\sqrt{2} \text{ cm}^2$

Q.580. Let ABC, PQR be two congruent triangles such that $\angle A = \angle P = 90^\circ$. If $BC = 17$ cm, $PR = 8$ cm, find AB (in cm).

(a) 9 (b) 14 (c) 12 (d) 15

Q.581. Out of two concentric circles, the radius of the outer circle is 6 cm and the chord PQ of the length 10 cm is a tangent to the inner circle. Find the radius (in cm) of the inner circle.

(a) $\sqrt{11}$ (b) $\sqrt{7}$ (c) $\sqrt{13}$ (d) 4

Q.582. The chord of a circle is equal to its radius. Find the difference between the angle subtended by this chord at the minor arc and major arc of the circle.

(a) 30° (b) 120° (c) 150° (d) 60°

Q.583. In $\triangle ABC$ with sides 8 cm, 9 cm and 12 cm, the angle bisector of the largest angle divides the opposite sides into two segments. What is the length of the shorter segment?

(a) $5\frac{11}{17}$ cm (b) $6\frac{13}{17}$ cm(c) $4\frac{11}{17}$ cm (d) $3\frac{9}{17}$ cm

Q.584. A tangent AB at point A of a circle of radius 6 cm meets a line through the centre O at point B. If OB = 10 cm, then the length of AB (in cm) is equal to:

(a) 5 (b) 6 (c) 4 (d) 8

Q.585. Which of the following sets of angles forms a triangle?

(a) 60°, 70°, 80° (b) 60°, 70°, 30°

(c) 60°, 40°, 80° (d) 60°, 50°, 80°

Q.586. Two triangles MNO and XYZ are given similar with the ratio of their side as 9 : 4. If the area of the larger triangle is 243 sq .cm, then the area of smaller triangle will be:

(a) 162 sq .cm (b) 28 sq .cm

(c) 48 sq .cm (d) 16 sq .cm

Q.587. In a circle of radius 14 cm, an arc subtends an angle of 90° at the centre. The length of arc (in cm) is equal to:

Take $(\pi = \frac{22}{7})$

(a) 22 (b) 18 (c) 20 (d) 24

Q.588. AB is a chord of the circle and AOC is its diameter such that angle ACB = 65°. If AT is the tangent to the circle at the point A, then angle BAT is equal to:

(a) 65° (b) 75° (c) 55° (d) 40°

Q.589. $\triangle PQR$ and $\triangle SQR$ are both isosceles triangles on a common base QR such that P and S lie on the same side of QR. If $\angle QSR = 60^\circ$ and $\angle QPR = 100^\circ$, then find $\angle SRP$.

(a) 80° (b) 60° (c) 100° (d) 20°

Q.590. Which option with given sides forms a Triangle?

(a) 8, 6, 9 (b) 100, 200, 300

(c) 10, 25, 11 (d) 8, 2, 6

Q.591. In a triangle PQR, QR is produced to S. If $\angle PRS = (9x - 15)^\circ$, $\angle RPQ = (2x)^\circ$ and $\angle PQR = (4x + 15)^\circ$, what is the value of x?

(a) 55° (b) 20° (c) 10° (d) 75°

Q.592. If in $\triangle PQR$ and $\triangle DEF$, $\angle P = 52^\circ$, $\angle Q = 74^\circ$, $\angle R = 54^\circ$, $\angle D = 54^\circ$, $\angle E = 74^\circ$ and $\angle F = 52^\circ$, then which of the following is correct?

(a) $\triangle PQR \sim \triangle FED$ (b) $\triangle RQP \sim \triangle FED$ (c) $\triangle PRQ \sim \triangle FED$ (d) $\triangle PQR \sim \triangle DEF$

Q.593. Minor arc BC subtends angles BAC and BDC at points A and D, respectively, on the circumference in the major sector of the circle with centre O. What is the value (in degrees) of $(\angle ABC + \angle ACB)$, if $\angle BDC = 73^\circ$?

(a) 117° (b) 107° (c) 103° (d) 113°

Q.594. In $\triangle ABC$, the internal bisectors of $\angle ABC$ and $\angle ACB$ meet at X and $\angle BAC = 30^\circ$. The measure of $\angle BXC$ is:

(a) 120° (b) 115° (c) 105° (d) 150°

Q.595. In triangle RST, M and N are two points on RS and RT such that MN is parallel to the base ST of the triangle

RST. If $RM = \frac{1}{3} MS$, and $ST = 5.6$ cm,

what is the ratio of

 $\frac{\text{Area of Triangle RMN}}{\text{Area of Trapezium MNST}}$?(a) $\frac{14}{15}$ (b) $\frac{15}{16}$ (c) $\frac{1}{15}$ (d) $\frac{1}{16}$

Q.596. If two tangents inclined at an angle 120° are drawn to a circle of radius 6 cm, then what is the length (in cm) of each tangent?

(a) $3\sqrt{3}$ (b) $2\sqrt{3}$ (c) $4\sqrt{3}$ (d) $\sqrt{3}$

Q.597. PS and PT are two tangents from a point P outside the circle with centre O. If S and T are points on the circle such that $\angle SPT = 130^\circ$, then the degree measure of $\angle OST$ is equal to:

(a) 25° (b) 55° (c) 65° (d) 35°

Q.598. In a rhombus STUV, S and U are joined $\angle SUV = 44^\circ$, $\angle STU = 92^\circ$, what is the degree measure of $4\angle SVU - 3\angle TSU$?

(a) 451° (b) 360° (c) 169° (d) 236°

Q.599. MN is the diameter of a circle with centre O. P and S are two points on the circumference of the circle on either side of MN, such that $\angle PMN = 50^\circ$ and $\angle MNS = 35^\circ$. What is the degree measure of the difference of $\angle PMS$ and $\angle PNS$?

(a) 25° (b) 55° (c) 65° (d) 30°

Q.600. $\triangle ABC \sim \triangle PQR$, AB = 12 cm, PQ = 18 cm and the perimeter of $\triangle ABC$ is 45

cm. Find the perimeter of $\triangle PQR$.

(a) 60 cm (b) 30 cm

(c) 67.5 cm (d) 70 cm

Q.601. The sides of similar triangle $\triangle ABC$ and $\triangle DEF$ are in the ratio of $\frac{\sqrt{3}}{\sqrt{5}}$. If the

area of $\triangle ABC$ is 90 cm², then the area of $\triangle DEF$ (in cm²) is:

(a) 150 (b) 152 (c) 154 (d) 156

Q.602. The radius of a circle is 21 cm. What will be the length (in cm) of an arc of the circle that subtends a 22.5° angle

at the center? (user $\pi = \frac{22}{7}$)

(a) 8.25 (b) 7.75 (c) 8.50 (d) 7.50

Q.603. One angle of an isosceles obtuse triangle is $28\frac{1}{2}^\circ$. Find the measure of its obtuse angle in degrees.

(a) 132° (b) 112° (c) 121° (d) 123°

Q.604. In $\triangle ABC$ and $\triangle PQR$, B = Q, C = R and AB = 2PQ, then the two triangles are

_____.
(a) congruent as well as similar
(b) neither similar nor congruent
(c) similar but not congruent
(d) congruent but not similar

Q.605. If the angles P, Q and R of $\triangle PQR$ satisfy the relation $2R - P = Q - R$, then find the measure of $\angle R$.

(a) 45° (b) 60° (c) 50° (d) 55°

Q.606. If the radius of a circle is 8 cm, then find the length of its largest chord.
(a) 16 cm (b) 64 cm (c) 8 cm (d) 4 cm

Q.607. In a square ABCD, E is a point inside the square such that $\triangle DEC$ is an equilateral triangle. If E is joined to vertices A and B of the square, what is the degree measure of $\angle AEB$?

(a) 135° (b) 150° (c) 210° (d) 225°

Q.608. At one end of a diameter PQ of a circle of radius 8 cm a tangent APB is drawn to the circle. Find the length of the chord RS, which is parallel to AB and at a distance of 12 cm from P.

(a) $6\sqrt{3}$ cm (b) $4\sqrt{3}$ cm
(c) $2\sqrt{3}$ cm (d) $8\sqrt{3}$ cm

Q.609. The perpendicular distance from the centre of a circle to the chord is 20 cm. Calculate the chord's length in centimetres if the circle's diameter is 58 cm.

(a) 42 (b) 21 (c) 56 (d) 28

Q.610. The diameter PQ of a circle with centre O is perpendicular to the chord RS. PQ intersects RS at T. If RS = 16 cm



and $QT = 4$ cm, what is the length (in cm) of the diameter of the circle?

- (a) 20 (b) 48 (c) 24 (d) 10

Q.611. Half of the length of the chord of a circle is 12 cm and the perpendicular distance between the centre and the chord is 5 cm. The radius of the circle is:

- (a) 12 cm (b) 13 cm (c) 10 cm (d) 24 cm

Q.612. $\triangle LON$ and $\triangle LMN$ are two right-angled triangles with common hypotenuse LN such that $\angle LON = 90^\circ$ and $\angle LMN = 90^\circ$. LN is the bisector of $\angle OLM$. If $LN = 29$ cm and $ON = 20$ cm, then what is the perimeter (in cm) of $\triangle LMN$?

- (a) 67 (b) 62 (c) 65 (d) 70

Q.613. In $\triangle ABC$, $\angle A = 5x - 60^\circ$, $\angle B = 2x + 40^\circ$, $\angle C = 3x - 80^\circ$. find $m\angle A$.

- (a) 75° (b) 90° (c) 80° (d) 60°

Q.614. If the two interior opposite angles of an exterior angle of a triangle are 35° and 65° , then the measurement of the exterior angle is:

- (a) 135° (b) 125° (c) 100° (d) 90°

Q.615. In a triangle ABC , AP , the bisector of $\angle A$, is perpendicular to BC at point P . The measures of BP and PC are x and $3y$, respectively. The measures of AB and AC are $4x$ and $(5y + 21)$, what is the value of $(x + y)$?

- (a) 21 (b) 15 (c) 12 (d) 18

Q.616. The largest chord of the circle is:

- (a) point (b) tangent
(c) diameter (d) radius

Q.617. Chords $\overline{AB}$ and $\overline{CD}$ of a circle intersect inside the circle at point F . If $m(\overline{AF}) = 2.4$ cm, $m(\overline{BF}) = 1.8$ cm, and $m(\overline{CD}) = 5.7$ cm, what is the length (in cm) of the longer of the two line segments, $\overline{CF}$ and $\overline{DF}$?

- (a) 4.5 (b) 4.8 (c) 5.4 (d) 3.6

Q.618. Let ABC and PQR be two congruent right-angled triangles such that $\angle A = \angle P = 90^\circ$. If $BC = 13$ cm and $PR = 12$ cm, then find the length of AB .

- (a) 25 cm (b) 20 cm (c) 10 cm (d) 5 cm

Q.619. $\triangle ABC$ is an equilateral triangle. The side BC is produced to point D . If A joins D and $BC = CD$, then the degree measure of angle CAD is equal to:

- (a) 30° (b) 15° (c) 45° (d) 18°

Q.620. The lengths of two parallel chords of a circle are 10 cm and 24 cm lie on the opposite sides of the centre. If the smaller chord is 12 cm from the centre, what is the distance (in cm) between the two chords?

- (a) 13 (b) 5 (c) 17 (d) 12

Q.621. In $\triangle ABC$, P and Q are the middle points of the sides AB and AC , respectively. R is point on the segment PQ such that $PR : RQ = 1 : 3$. If $PR = 6$ cm, then BC :

- (a) 46 cm (b) 48 cm (c) 44 cm (d) 50 cm

Q.622. In a circle of radius 8 units, a chord of length 10 units is drawn. What is the perpendicular distance of the chord from the centre of the circle in units?

- (a) $\sqrt{39}$ Unit (b) $\sqrt{35}$ Unit
(c) $\sqrt{30}$ Unit (d) $\sqrt{33}$ Unit

Q.623. In a circle, an arc subtends an angle of 84° at the centre. If the length of the arc is 22 cm, then the radius of the

circle (in cm) is equal to: Take $\pi = \frac{22}{7}$

- (a) 19 (b) 15 (c) 17 (d) 13

Q.624. In triangle STU , V is a point on the side ST and W is a point on the side SU such that $VWUT$ is a trapezium. Given that $VW : TU = 2 : 7$, what is the ratio of area of trapezium $VWUT$ to the area of the triangle STU ?

- (a) 4 : 49 (b) 4 : 45
(c) 45 : 49 (d) 49 : 81

Q.625. The sides of a triangle are 20 cm, 21 cm and 29 cm. The area of the triangle formed by joining the mid points of the sides of triangle will be:

- (a) $67\frac{2}{3} \text{ cm}^2$ (b) $52\frac{1}{2} \text{ cm}^2$
(c) $47\frac{1}{2} \text{ cm}^2$ (d) $58\frac{1}{3} \text{ cm}^2$

Q.626. If the triangles ABC and PQR are similar and if $\angle A = 35^\circ$, $\angle B = 65^\circ$ then $\angle R =$ _____.

- (a) 80° (b) 90° (c) 35° (d) 65°

Q.627. If one diagonal of a rhombus is equal to its side, then the diagonals of the rhombus are in the ratio of:

- (a) $\sqrt{3} : \sqrt{6}$ (b) $\sqrt{2} : 1$ (c) $\sqrt{2} : \sqrt{3}$ (d) $1 : \sqrt{3}$

Q.628. In a triangle ABC , $8\angle A = 6\angle B = 3\angle C$. What are the degree measures of $\angle A$, $\angle B$ and $\angle C$?

- (a) $48^\circ, 96^\circ, 36^\circ$ (b) $36^\circ, 96^\circ, 48^\circ$
(c) $36^\circ, 48^\circ, 96^\circ$ (d) $96^\circ, 48^\circ, 36^\circ$

Q.629. Two similar triangles are given i.e. $\triangle LMN \sim \triangle PQR$, with measurement of angle and side as angle $L = 40^\circ$, angle $N = 80^\circ$, $LM = 6$ cm, $LN = 8$ cm and $PQ = 7.5$ cm. Find the value of angle Q and side PR , respectively.

- (a) $60^\circ, 20$ cm (b) $50^\circ, 6.5$ cm
(c) $40^\circ, 10$ cm (d) $60^\circ, 10$ cm

Q.630. In $\triangle ABC$, $\overline{BD} \perp \overline{AC}$, intersecting $\overline{AC}$ at D . Also, $BD = 12$ cm. If $m(\overline{AD}) = 6$ cm and $m(\overline{CD}) = 4$ cm, find the area (in cm^2) of $\triangle ABC$.

- (a) 45 (b) 50 (c) 60 (d) 75

Q.631. It is given that $\triangle PQR \cong \triangle MNY$ and $PQ = 8$ cm, $\angle Q = 55^\circ$ and $\angle P = 72^\circ$. Which of the following is true?

- (a) $NY = 8$ cm, $\angle Y = 72^\circ$
(b) $NM = 8$ cm, $\angle M = 53^\circ$
(c) $NM = 8$ cm, $\angle Y = 53^\circ$
(d) $NY = 8$ cm, $\angle N = 55^\circ$

Q.632. The lengths of the two sides adjacent to the right angle of a right-angled triangle are 1.6 cm and 6.3 cm. Find the length of the hypotenuse.

- (a) 6.7 cm (b) 7.5 cm (c) 6.5 cm (d) 7 cm

Q.633. AB and BC are two chords of a circle with centre O . Both chords are on either side of the centre O . Point A and point C are connected to the centre O , such that $\angle BAO = 36^\circ$ and $\angle BCO = 48^\circ$. What is the degree measure of the angle subtended by the minor arc AC at the centre O ?

- (a) 136° (b) 144° (c) 120° (d) 168°

Q.634. In $\triangle XYZ$, L and M are the middle points of the sides XY and XZ , respectively. N is a point on the segment LM such that $LN : NM = 1 : 2$. If $LN = 5$ cm, then YZ is equal to:

- (a) 30 cm (b) 24 cm (c) 28 cm (d) 26 cm

Q.635. Two equal circles of radius 6 cm intersect each other such that each pass through the centre of the other. The length (in cm) of the common chord is:

- (a) $5\sqrt{3}$ (b) $6\sqrt{3}$ (c) $4\sqrt{3}$ (d) $3\sqrt{3}$

Answer Key :-

1.(b)	2.(a)	3.(b)	4.(a)
5.(b)	6.(c)	7.(c)	8.(b)
9.(d)	10.(b)	11.(c)	12.(a)
13.(c)	14.(c)	15.(b)	16.(d)
17.(a)	18.(b)	19.(b)	20.(c)
21.(b)	22.(a)	23.(d)	24.(c)
25.(a)	26.(d)	27.(b)	28.(d)
29.(b)	30.(b)	31.(d)	32.(a)
33.(b)	34.(d)	35.(c)	36.(c)
37.(c)	38.(c)	39.(a)	40.(d)
41.(d)	42.(d)	43.(d)	44.(c)
45.(c)	46.(c)	47.(b)	48.(a)
49.(c)	50.(c)	51.(a)	52.(a)
53.(d)	54.(a)	55.(d)	56.(a)



57.(c)	58.(c)	59.(c)	60.(c)
61.(d)	62.(c)	63.(c)	64.(a)
65.(a)	66.(d)	67.(c)	68.(a)
69.(a)	70.(d)	71.(a)	72.(d)
73.(a)	74.(a)	75.(d)	76.(a)
77.(b)	78.(d)	79.(d)	80.(d)
81.(c)	82.(d)	83.(c)	84.(b)
85.(d)	86.(c)	87.(b)	88.(b)
89.(c)	90.(c)	91.(b)	92.(b)
93.(c)	94.(d)	95.(b)	96.(b)
97.(d)	98.(c)	99.(c)	100.(b)
101.(b)	102.(c)	103.(d)	104.(b)
105.(d)	106.(c)	107.(b)	108.(a)
109.(a)	110.(d)	111.(a)	112.(c)
113.(d)	114.(b)	115.(c)	116.(d)
117.(a)	118.(d)	119.(b)	120.(a)
121.(d)	122.(d)	123.(d)	124.(a)
125.(c)	126.(a)	127.(a)	128.(b)
129.(a)	130.(d)	131.(a)	132.(a)
133.(d)	134.(d)	135.(b)	136.(c)
137.(b)	138.(d)	139.(a)	140.(b)
141.(a)	142.(b)	143.(c)	144.(b)
145.(a)	146.(b)	147.(a)	148.(b)
149.(c)	150.(c)	151.(a)	152.(a)
153.(b)	154.(a)	155.(d)	156.(b)
157.(a)	158.(d)	159.(b)	160.(a)
161.(d)	162.(a)	163.(d)	164.(c)
165.(d)	166.(a)	167.(a)	168.(a)
169.(a)	170.(c)	171.(a)	172.(a)
173.(b)	174.(a)	175.(b)	176.(b)
177.(c)	178.(c)	179.(a)	180.(c)
181.(c)	182.(c)	183.(a)	184.(c)
185.(c)	186.(a)	187.(c)	188.(b)
189.(a)	190.(b)	191.(b)	192.(c)
193.(b)	194.(c)	195.(a)	196.(a)
197.(b)	198.(b)	199.(c)	200.(a)
201.(d)	202.(a)	203.(b)	204.(c)
205.(a)	206.(c)	207.(d)	208.(a)
209.(c)	210.(b)	211.(b)	212.(c)
213.(d)	214.(a)	215.(c)	216.(b)
217.(b)	218.(c)	219.(d)	220.(a)
221.(b)	222.(c)	223.(b)	224.(b)
225.(c)	226.(b)	227.(a)	228.(c)
229.(d)	230.(b)	231.(a)	232.(a)
233.(b)	234.(a)	235.(c)	236.(b)
237.(a)	238.(c)	239.(d)	240.(c)

241.(b)	242.(b)	243.(c)	244.(a)
245.(c)	246.(d)	247.(a)	248.(b)
249.(a)	250.(b)	251.(c)	252.(c)
253.(a)	254.(c)	255.(a)	256.(a)
257.(c)	258.(a)	259.(a)	260.(a)
261.(d)	262.(b)	263.(b)	264.(d)
265.(c)	266.(a)	267.(a)	268.(a)
269.(b)	270.(c)	271.(c)	272.(c)
273.(c)	274.(c)	275.(a)	276.(d)
277.(c)	278.(a)	279.(b)	280.(d)
281.(c)	282.(c)	283.(b)	284.(a)
285.(b)	286.(d)	287.(b)	288.(d)
289.(b)	290.(a)	291.(a)	292.(c)
293.(d)	294.(b)	295.(d)	296.(a)
297.(a)	298.(d)	299.(c)	300.(d)
301.(c)	302.(c)	303.(b)	304.(d)
305.(b)	306.(b)	307.(d)	308.(b)
309.(a)	310.(d)	311.(c)	312.(a)
313.(a)	314.(b)	315.(c)	316.(a)
317.(b)	318.(d)	319.(c)	320.(b)
321.(b)	322.(c)	323.(b)	324.(c)
325.(b)	326.(d)	327.(d)	328.(c)
329.(b)	330.(a)	331.(b)	332.(a)
333.(b)	334.(a)	335.(b)	336.(c)
337.(c)	338.(b)	339.(c)	340.(d)
341.(b)	342.(b)	343.(d)	344.(c)
345.(d)	346.(c)	347.(b)	348.(a)
349.(c)	350.(b)	351.(a)	352.(c)
353.(c)	354.(a)	355.(b)	356.(b)
357.(d)	358.(b)	359.(a)	360.(c)
361.(a)	362.(b)	363.(a)	364.(c)
365.(a)	366.(d)	367.(b)	368.(b)
369.(d)	370.(a)	371.(b)	372.(a)
373.(a)	374.(d)	375.(b)	376.(a)
377.(a)	378.(d)	379.(d)	380.(d)
381.(c)	382.(d)	383.(c)	384.(c)
385.(a)	386.(d)	387.(a)	388.(a)
389.(d)	390.(d)	391.(b)	392.(a)
393.(b)	394.(c)	395.(c)	396.(b)
397.(a)	398.(a)	399.(d)	400.(a)
401.(c)	402.(c)	403.(d)	404.(d)
405.(a)	406.(d)	407.(d)	408.(c)
409.(d)	410.(d)	411.(b)	412.(d)
413.(a)	414.(d)	415.(b)	416.(d)
417.(c)	418.(d)	419.(d)	420.(c)
421.(b)	422.(b)	423.(a)	424.(a)

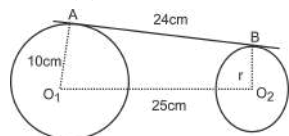
425.(c)	426.(c)	427.(a)	428.(c)
429.(b)	430.(b)	431.(d)	432.(c)
433.(c)	434.(d)	435.(d)	436.(c)
437.(d)	438.(d)	439.(a)	440.(b)
441.(b)	442.(b)	443.(d)	444.(a)
445.(a)	446.(a)	447.(d)	448.(d)
449.(a)	450.(c)	451.(a)	452.(c)
453.(a)	454.(a)	455.(d)	456.(b)
457.(d)	458.(a)	459.(c)	460.(a)
461.(a)	462.(c)	463.(a)	464.(c)
465.(b)	466.(d)	467.(b)	468.(a)
469.(a)	470.(d)	471.(d)	472.(a)
473.(b)	474.(a)	475.(c)	476.(a)
477.(a)	478.(b)	479.(b)	480.(a)
481.(c)	482.(d)	483.(b)	484.(d)
485.(a)	486.(b)	487.(c)	488.(c)
489.(c)	490.(d)	491.(a)	492.(b)
493.(b)	494.(b)	495.(b)	496.(a)
497.(d)	498.(b)	499.(a)	500.(d)
501.(b)	502.(b)	503.(c)	504.(a)
505.(b)	506.(c)	507.(a)	508.(c)
509.(a)	510.(d)	511.(a)	512.(a)
513.(d)	514.(b)	515.(c)	516.(b)
517.(c)	518.(d)	519.(d)	520.(c)
521.(a)	522.(a)	523.(b)	524.(a)
525.(d)	526.(d)	527.(d)	528.(b)
529.(a)	530.(c)	531.(d)	532.(a)
533.(c)	534.(a)	535.(d)	536.(d)
537.(c)	538.(a)	539.(d)	540.(b)
541.(b)	542.(b)	543.(b)	544.(a)
545.(a)	546.(d)	547.(d)	548.(d)
549.(c)	550.(b)	551.(d)	552.(c)
553.(c)	554.(c)	555.(a)	556.(b)
557.(d)	558.(b)	559.(a)	560.(d)
561.(b)	562.(d)	563.(d)	564.(b)
565.(c)	566.(a)	567.(a)	568.(a)
569.(a)	570.(d)	571.(d)	572.(a)
573.(a)	574.(a)	575.(a)	576.(c)
577.(d)	578.(b)	579.(d)	580.(d)
581.(a)	582.(b)	583.(a)	584.(d)
585.(c)	586.(c)	587.(a)	588.(a)
589.(d)	590.(a)	591.(c)	592.(a)
593.(b)	594.(c)	595.(c)	596.(b)
597.(c)	598.(d)	599.(d)	600.(c)
601.(a)	602.(a)	603.(d)	604.(c)
605.(a)	606.(a)	607.(b)	608.(d)



609.(a)	610.(a)	611.(b)	612.(d)
613.(c)	614.(c)	615.(c)	616.(c)
617.(b)	618.(d)	619.(a)	620.(c)
621.(b)	622.(a)	623.(b)	624.(c)
625.(b)	626.(a)	627.(d)	628.(c)
629.(d)	630.(c)	631.(c)	632.(c)
633.(d)	634.(a)	635.(b)	

Solutions :-

Sol.1.(b)



Length of the direct tangent (l)

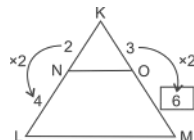
$$= \sqrt{d^2 - (r_1 - r_2)^2}$$

$$24 = \sqrt{25^2 - (10 - r)^2}$$

$$\Rightarrow 576 = 625 - (10 - r)^2$$

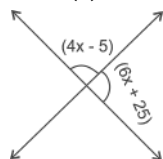
$$\Rightarrow (10 - r)^2 = 49 \Rightarrow r = 3 \text{ cm}$$

Sol.2.(a) Since $NO \parallel LM$, $\Delta KLM \sim \Delta KNO$ (both triangles are proportional)



Sol.3.(b) In a right-angled triangle, the circumcenter lies at the midpoint of the hypotenuse (it is the center of the circle passing through all three vertices). Hence, the Circumcenter is at the midpoint of the hypotenuse.

Sol.4.(a)



Supplementary Equation:

$$(4x - 5) + (6x + 25) = 180^\circ$$

$$10x = 160 \Rightarrow x = 16$$

$$\text{Now, Angle} = 4x - 5 = 4(16) - 5 = 59^\circ$$

Vertical opposite angles are equal, so the angle vertically opposite to the $4x - 5$ degree angle is also 59°

Sol.5.(b)

Sum of two complementary angles = 90°

Ratio $2 : 7 \Rightarrow$ Total parts = 9

$$1^{\text{st}} \text{ angle} = \frac{90}{9} \times 2 = 20^\circ$$

$$2^{\text{nd}} \text{ angle} = \frac{90}{9} \times 7 = 70^\circ$$

$$\text{Sol.6.(c) Radians} = \text{Degrees} \times \frac{\pi}{180^\circ}$$

$$= 135^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{4} \text{ radians}$$

Sol.7.(c) From an external point, the two tangents drawn to a circle are always equal in length because the tangents from a common external point to a circle are congruent.

Sol.8.(b) Sum of length of opposite sides are equal

$$PQ + RS = QR + SP \dots (i)$$

$$\text{Given, } PQ = 12 \text{ cm, } QR = 14 \text{ cm}$$

$$A/Q, PQ + QR + RS + SP = 42$$

$$26 + RS + SP = 42$$

$$RS + SP = 16 \dots (ii)$$

From eq. (i)

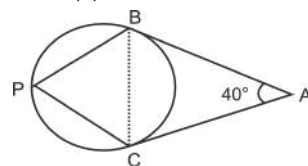
$$QR - PQ = RS - SP$$

$$2 = RS - SP \dots (iii)$$

Add eq. (ii) and (iii),

$$2RS = 18 \text{ cm} \Rightarrow RS = 9 \text{ cm}$$

Sol.9.(d)



$AB = AC \dots (\text{tangent to the circle})$

$$\angle ABC = \angle ACB$$

$$\angle ABC + \angle ACB + \angle BAC = 180^\circ$$

$$2\angle ABC + 40^\circ = 180^\circ \Rightarrow \angle ABC = 70^\circ$$

It is given that $CP \parallel BA$ and BC is a transversal. Alternate interior angles are equal, therefore: $\angle BCP = \angle ABC = 70^\circ$

The angle formed between a tangent and a chord is equal to the angle formed in the alternate segment of the circle.

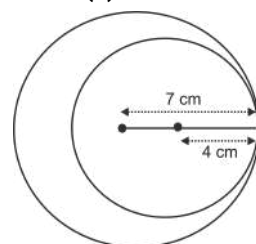
The angle between the tangent AB and the chord BC is $\angle ABC$. This angle is equal to the angle $\angle BPC$ formed by the chord BC in the remaining part of the circle.

$$\angle CBP + \angle BCP + \angle BPC = 180^\circ$$

$$\angle CBP + 70^\circ + 70^\circ = 180^\circ$$

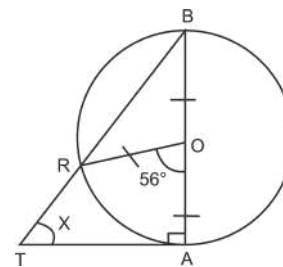
$$\angle CBP = 40^\circ$$

Sol.10.(b)



So, Required difference = $7 - 4 = 3 \text{ cm}$

Sol.11.(c)



Straight angle AOB ,

$$\Rightarrow \angle ROB = 180^\circ - 56^\circ = 124^\circ$$

In ΔROB ,

$$\Rightarrow \angle R + \angle B + 124^\circ = 180^\circ$$

$$\Rightarrow 2\angle B = 56^\circ \quad [\because \angle R = \angle B]$$

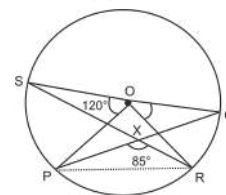
$$\Rightarrow \angle B = 28^\circ$$

In ΔTAB ,

$$\Rightarrow x + 28^\circ + 90^\circ = 180^\circ \Rightarrow x = 62^\circ$$

So, the angle $3x = 186^\circ$

Sol.12.(a)



We know that,

$$\Rightarrow \angle SRP = \frac{\angle SOP}{2} = \frac{120^\circ}{2} = 60^\circ = \angle R$$

In ΔPXR ,

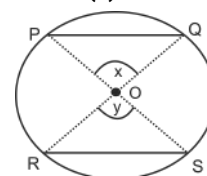
$$\Rightarrow \angle P + \angle X + \angle R = 180^\circ$$

$$\Rightarrow \angle P + 85^\circ + 60^\circ = 180^\circ$$

$$\Rightarrow \angle P = 35^\circ = \angle QPR$$

$$\text{Now, } \angle QOR = 2 \times \angle QPR = 2 \times 35^\circ = 70^\circ$$

Sol.13.(c)



In ΔPOQ , and ΔROS

$PQ = RS \dots (\text{Given})$

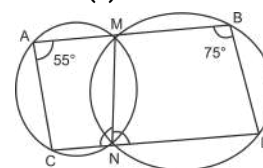
$OR = OQ \dots (\text{Radius})$

$OS = OP \dots (\text{Radius})$

$$\therefore \Delta POQ \cong \Delta ROS$$

Hence $\angle x = \angle y$

Sol.14.(c)



CAMN is a cyclic quadrilateral

$$\angle CAM = \angle MND \dots (i)$$

MNBD is a cyclic quadrilateral

$$\angle DBM = \angle MNC \dots (ii)$$

Adding the eq (i) & (ii)

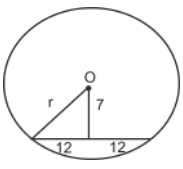
$$\angle CAM + \angle DBM = \angle MND + \angle MNC$$

$$55^\circ + 75^\circ = \angle MND + \angle MNC$$

$$\angle MND + \angle MNC = 130^\circ$$

Sol.15.(b)





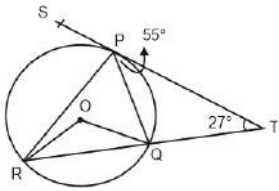
By using pythagoras theorem

$$r^2 = 12^2 + 7^2$$

$$r = \sqrt{144 + 49} \Rightarrow r = \sqrt{193} \text{ cm}$$

Sol.16.(d) The chord that passes through the center of the circle is the longest chord, and it is called the Diameter.

Sol.17.(a)



In ΔPTQ

$$\angle PQR = 55^\circ + 27^\circ = 82^\circ \dots \text{(exterior angle property)}$$

$$\angle RPS = \angle PQR$$

$$\angle RPS = 82^\circ$$

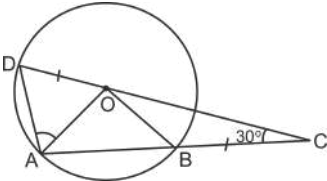
$$\text{Now, } \angle RPS + \angle RPQ + \angle QPT = 180^\circ \dots \text{(linear pair)}$$

$$82^\circ + \angle RPQ + 55^\circ = 180^\circ$$

$$\angle RPQ = 180^\circ - 137^\circ = 43^\circ$$

$$\text{Now, } \angle ROQ = 2\angle RPQ = 2 \times 43^\circ = 86^\circ$$

Sol.18.(b)



Given: $BC = OD$

then, $OD = OB = OA = \text{radii of the circle}$

In ΔOBC ,

$$\angle OCB = \angle COB = 30^\circ$$

$$\angle OBC = 180^\circ - (30^\circ + 30^\circ) = 120^\circ$$

In ΔOAB ,

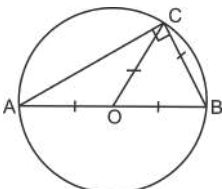
$$\angle OBA = 180^\circ - 120^\circ = 60^\circ = \angle OAB$$

$$\angle AOB = 180^\circ - (60^\circ + 60^\circ) = 60^\circ$$

Now,

$$\angle AOD = 180^\circ - 60^\circ - 30^\circ = 90^\circ$$

Sol.19.(b)



$$\angle ABC = x \angle BAC$$

$$\frac{\angle ABC}{\angle BAC} = x \dots \dots (i)$$

In ΔOCB all sides are equal to each other

then all angles to equal to each other,

$$\angle ABC = \angle BCO = \angle COB = 60^\circ$$

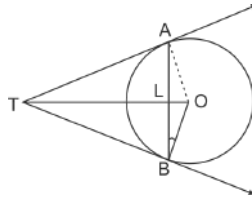
then,

$$\angle ACO = \angle ACB - \angle BCO = 90^\circ - 60^\circ = 30^\circ$$

$$\angle CAO = \angle ACO = 30^\circ \text{ (AO = OC)}$$

$$\text{Now, } x = \frac{60^\circ}{30^\circ} = 2$$

Sol.20.(c) Given, $OB = 20 \text{ cm}$ and $AB = 32 \text{ cm}$, So $AL = LB = 16 \text{ cm}$



Now, $TA = TB$ (length of tangents to a circle from external point are equal)

In ΔOLA

$$LO = \sqrt{400 - 256} = \sqrt{144} = 12 \text{ cm}$$

$$\tan \angle AOL = \frac{AL}{OL} = \frac{16}{12}$$

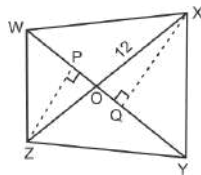
Similarly in ΔOAT

$$\tan \angle AOT = \frac{AT}{OA} = \frac{AT}{20}$$

$$\Rightarrow \frac{16}{12} = \frac{AT}{20} \text{ (}\angle AOL = \angle AOT = \text{common)}$$

$$\Rightarrow AT = \frac{16 \times 20}{12} = 26.67$$

Sol.21.(b)



In ΔZOP and ΔXOQ

$$XQ = ZP \dots \dots (\text{Given})$$

$$\angle P = \angle Q \dots \text{(each } 90^\circ)$$

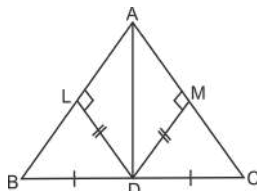
$$\angle POZ = \angle XOQ \dots \text{(vertically opposite angle)}$$

$$\therefore \Delta ZOP \cong \Delta XOQ \text{ (AAS Congruence)}$$

$$\text{So, } OX = ZO = 12 \text{ cm (CPCT)}$$

$$\text{Hence, } ZX = XO + ZO = 12 + 12 = 24 \text{ cm}$$

Sol.22.(a)



In ΔBLD and ΔCMD

$$BD = DC \text{ (given)}$$

$$LD = DM \text{ (given)}$$

$$\angle BLD = \angle CMD = 90^\circ$$

$$\Delta BLD \cong \Delta CMD \text{ [by SAS]}$$

$$\text{So, } BL = MC \dots \dots (i)$$

Now, in the triangle ΔALD and ΔAMD

$$LD = DM \text{ (given)}$$

$$AD = AD \text{ (common)}$$

$$\angle ALD = \angle AMD = 90^\circ$$

$$\Delta ALD \cong \Delta AMD \text{ [by SAS]}$$

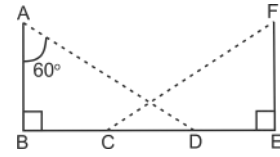
$$\text{then, } AL = AM \dots \dots (ii)$$

$$\text{From equation (i) and (ii)}$$

$$AB = AC$$

Hence, triangle ABC is an isosceles triangle

Sol.23.(d) Given, In ΔABD and ΔFEC , $\angle BAD = 60^\circ$, $I(BD) = I(EC)$, $\angle ABD = \angle FEC = 90^\circ$, and $I(AB) = I(FE)$



Now in ΔABD ,

$$\angle A + \angle B + \angle D = 180^\circ$$

$$60^\circ + 90^\circ + \angle D = 180^\circ \Rightarrow \angle D = 30^\circ$$

Now, According to question,

In ΔABD and ΔFEC

$$AB = FE \text{ (Given)}$$

$$\angle B = \angle E \text{ (Each } 90^\circ)$$

$$BD = CE \text{ (Given)}$$

$$\text{By SAS, } \Delta ABD \cong \Delta FEC$$

$$\therefore \angle D = \angle C = 30^\circ$$

$$\text{Now, ratio of } \frac{\angle BAD}{\angle FCE} = 60^\circ : 30^\circ = 2 : 1$$

Sol.24.(c) Length of common external

$$\text{tangent} = \sqrt{d^2 - (R - r)^2}$$

$$\Rightarrow d^2 - (R - r)^2 = 81$$

$$\Rightarrow (R - r)^2 = d^2 - 81 \dots \dots (i)$$

Length of common internal tangent =

$$\sqrt{d^2 - (R + r)^2}$$

$$\Rightarrow d^2 - (R + r)^2 = 25$$

$$\Rightarrow (R + r)^2 = d^2 - 25 \dots \dots (ii)$$

From equation (i) and (ii)

$$4Rr = 56 \Rightarrow Rr = 14$$

Short trick:-

$$(\text{External tangent})^2 - (\text{internal tangent})^2 = 4Rr$$

$$\Rightarrow (9)^2 - (5)^2 = 4 \times Rr$$

$$\Rightarrow 81 - 25 = 4 \times Rr$$

$$\Rightarrow 56 = 4 \times Rr \Rightarrow Rr = 14$$

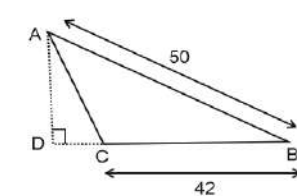
Sol.25.(a) As, $XY + YZ = XZ$ ($11 + 13 = 24$)

implies that X, Y, Z are collinear points.

So, the number of circles that travel

through all three points X, Y, Z is 0.

Sol.26.(d)



Let the height of ΔABC is AD

$$\text{Area of } \Delta ABC = \frac{1}{2} \times BC \times AD$$

$$294 = \frac{1}{2} \times 42 \times AD$$

$$14 = AD \dots \dots (i)$$

Now in ΔADB

$$AB^2 = AD^2 + BD^2$$

$$50^2 = 14^2 + BD^2 \dots [\text{from (i)}]$$

$$BD^2 = 2500 - 196 = 2304$$

$$BD = 48 \text{ cm}$$

$$DC = BD - BC$$

$$= 48 - 42 = 6 \text{ cm}$$

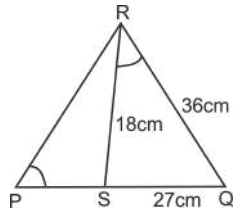
In $\triangle ADC$

$$AC^2 = AD^2 + DC^2$$

$$= 14^2 + 6^2 = 196 + 36 = 232 \text{ cm}$$

$$AC = 2\sqrt{58} \text{ cm}$$

Sol.27.(b)



In $\triangle QPR$ and $\triangle QRS$

$$\angle QPR = \angle QRS \dots (\text{given})$$

$$\angle PQR = \angle SQR \dots (\text{common})$$

So, $\triangle QPR \sim \triangle QRS$ by AA similarity

$$\text{Hence, } \frac{QP}{QR} = \frac{QR}{QS} \Rightarrow QP = \frac{36 \times 36}{27} = 48 \text{ cm}$$

$$\text{and } \frac{PR}{RS} = \frac{QR}{QS} \Rightarrow PR = \frac{36 \times 18}{27} = 24 \text{ cm}$$

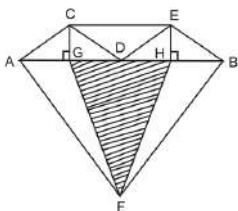
$$\Rightarrow PS = PQ - SQ = 48 - 27 = 21 \text{ cm}$$

In $\triangle PRS$ and $\triangle QSR$,

$$\frac{\text{Perimeter of } (\triangle PRS)}{\text{Perimeter of } (\triangle QSR)} = \frac{PR + RS + PS}{QS + SR + QR}$$

$$= \frac{24 + 18 + 21}{27 + 18 + 36} = \frac{63}{81} = \frac{7}{9} \text{ or } 7 : 9$$

Sol.28.(d)



I) For, $\triangle ACG \cong \triangle BEH$

In the $\triangle AGC$ and $\triangle EHB$

$$\angle AGC = \angle EHB = 90^\circ$$

$$\text{Side (AG)} = \text{side (HB)} \quad [\text{given}]$$

And, $CG = EH$ [lengths between two parallel lines]

So, by SAS theorem,

$$\triangle ACG \cong \triangle BEH \text{ (true)}$$

II) Now, for $\triangle CDE \cong \triangle DEB \cong \triangle DCA$

In the $\triangle CDE$ and $\triangle DEB$, $CE \parallel AB$ (given)

$$DE = DE \quad [\text{common}]$$

$$\angle CED = \angle BDE \quad [\text{alternate interior angle}]$$

$$CE = DB$$

So, $\triangle CDE \cong \triangle DEB$ by SAS theorem,

Similarly,

In $\triangle CDE$ and $\triangle DCA$, $CE \parallel AB$ (given)

$$CD = CD \quad [\text{common}]$$

$$\angle ECD = \angle ADC \quad [\text{alternate interior angle}]$$

$$CE = AD$$

So, $\triangle CDE \cong \triangle DCA$ by SAS theorem,

Hence, $\triangle CDE \cong \triangle DEB \cong \triangle DCA$ [true]

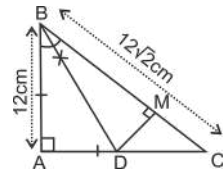
III) For, $\triangle HFB \cong \triangle AFG$ [false]

Only $AG = HB$ [given]

$\therefore$ No other side or angles are equal

Therefore, option (d) is the correct answer.

Sol.29.(b)



In $\triangle DAB$ and $\triangle DMB$,

$$\angle ABD = \angle MBD$$

$$\angle BAD = \angle BMD = 90^\circ$$

$$BD = BD \quad (\text{common})$$

So, $\triangle DAB \cong \triangle DMB$ by ASA

$$\Rightarrow AB = BM = 12 \text{ cm}$$

$$\text{Then, } MC = BC - BM = (12\sqrt{2} - 12) \text{ cm}$$

Now,

$$\angle MDC = 180^\circ - (90^\circ + 45^\circ) = 45^\circ$$

$$\text{So, } MC = DM = (12\sqrt{2} - 12) \text{ cm}$$

$$\text{As, } DM = AD \quad (\text{by congruence})$$

$$\text{Hence, } AD = (12\sqrt{2} - 12) = 12(\sqrt{2} - 1) \text{ cm}$$

Alternate method:-

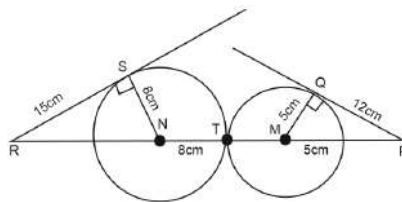
In this type of question we use direct property,

The property is $AB + AD = BC$

$$\Rightarrow AD = BC - AB$$

$$= 12\sqrt{2} - 12 = 12(\sqrt{2} - 1) \text{ cm}$$

Sol.30.(b)



(8, 15, 17) are pythagorean triplet

$$RN = 17 \text{ cm}$$

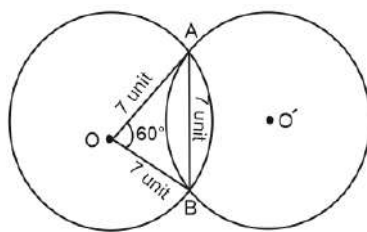
(5, 12, 13) are pythagorean triplet

$$MP = 13 \text{ cm}$$

$$\text{Now, } PR = RN + NT + TM + MP$$

$$= 17 + 8 + 5 + 13 = 43 \text{ cm}$$

Sol.31.(d)



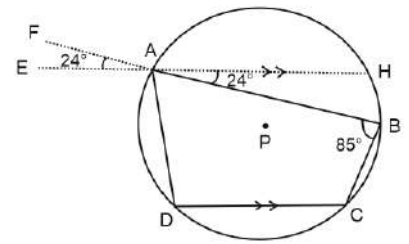
Common area = $2 \times (\text{area of sector OABO} - \text{Area of equilateral } \triangle OAB)$

$$= 2 \times \left(\frac{60}{360} \pi \times (7)^2 - \frac{\sqrt{3}}{4} \times (7)^2 \right)$$

$$= 2 \times (7)^2 \left(\frac{1}{6} \pi - \frac{\sqrt{3}}{4} \right) = 98 \left(\frac{1}{6} \pi - \frac{\sqrt{3}}{4} \right)$$

$$\text{Hence, required area} = 98 \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right)$$

Sol.32.(a)



In cyclic quadrilateral ABCD

$$\angle B + \angle D = 180^\circ$$

$$\angle D = 180^\circ - 85^\circ = 95^\circ$$

Now $EA \parallel DC$

So,

$$\angle DAH + \angle D = 180^\circ \dots (\text{co-interior angle})$$

$$\angle DAB + \angle BAH + \angle D = 180^\circ$$

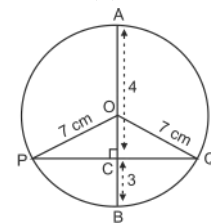
$$\angle DAB = 180^\circ - 95^\circ - 24^\circ = 61^\circ$$

Now

$$\angle C + \angle DAB = 180^\circ$$

$$\angle C = 180^\circ - 61^\circ = 119^\circ$$

Sol.33.(b)



$$AB = 2 \times \text{radius} = 14 \text{ cm}$$

$$AC = 14 \times \frac{4}{7} = 8$$

$$BC = 14 - 8 = 6$$

$$PC = QC \dots (\text{diameter } \perp \text{ chord})$$

Now,

$$PC \times QC = AC \times BC$$

$$PC^2 = 8 \times 6 = 48$$

$$PC = \sqrt{48}$$

$$PQ = 2\sqrt{48} = 8\sqrt{3} \text{ cm}$$

Sol.34.(d) Let the angles are $x, 2x, 3x$

Sum of angles of triangle = 180°

$$x + 2x + 3x = 180^\circ \Rightarrow x = 30^\circ$$

Hence angles are ; $30^\circ, 60^\circ, 90^\circ$

On Using sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{a}{\sin 30^\circ} = \frac{b}{\sin 60^\circ} = \frac{c}{\sin 90^\circ}$$

$$\frac{a}{\frac{1}{2}} = \frac{b}{\frac{\sqrt{3}}{2}} = \frac{c}{1} = k$$

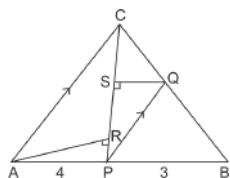
$$2a = \frac{2}{\sqrt{3}} b = \frac{c}{1} = k$$

$$a = \frac{1}{2}, b = \frac{\sqrt{3}}{2}, c = 1$$

$$a : b : c = 1 : \sqrt{3} : 2$$

Sol.35.(c)





As, $PQ \parallel AC$, $\triangle BQP \sim \triangle BCA$ (by AAA rule)

$$\text{Then, } \frac{BP}{AB} = \frac{QP}{AC} = \frac{BQ}{BC} = \frac{3}{7}$$

Now, In $\triangle SPQ$ & $\triangle RCA$

$$\angle QSP = \angle ARC = 90^\circ$$

$$\angle QPS = \angle ACR \text{ (alternate angles)}$$

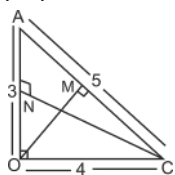
So, $\triangle SPQ \sim \triangle RCA$ (by AAA rule)

$$\text{Then, } \frac{QS}{AR} = \frac{QP}{AC} = \frac{3}{7}$$

$$\Rightarrow \frac{9}{AR} = \frac{3}{7} \Rightarrow AR = \frac{9 \times 7}{3} = 21 \text{ cm}$$

Sol.36.(c) Concept used :

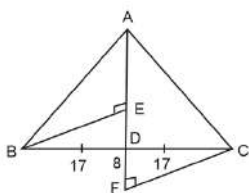
Altitudes of the triangle are inversely proportional to the side of that triangle.



Sides $\rightarrow 3 : 4 : 5$

$$\text{Altitude} \rightarrow \frac{1}{3} : \frac{1}{4} : \frac{1}{5} = 20 : 15 : 12$$

Sol.37.(c) According to the question,



$$CF = \sqrt{17^2 - 8^2} = 15 \text{ cm}$$

In $\triangle BED$ and $\triangle CFD$,

$$\angle DEB = \angle CFD = 90^\circ$$

$$\angle EDB = \angle FDC \text{ (vertical angle)}$$

BD = DC (given)

So, $\triangle BED \cong \triangle CFD$ (By AAS rule)

Then, BE = CF = 15 cm

Sol.38.(c) The minute hand and hour hand coincide after every 65 minutes.

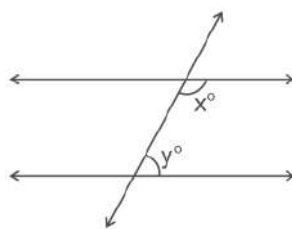
Let us take the time when they meet after 12

12 : 00, 01 : 05, 02 : 10, 03 : 15, 04 : 20, 05 : 25, 06 : 30, 07 : 35, 08 : 40, 09 : 45, 10 : 50, 11 : 55

Hence, in half a day hour hand and minute hand coincide 11 times.

Sol.39.(a) Let the interior angles on the same side are x° and y°

Where $(x > y)$



We know that the sum of interior angles is 180° .

$$\therefore x + y = 180^\circ \text{(i)}$$

$$\text{And } x - y = 40^\circ \text{(ii)}$$

From solving the equations (i) and (ii) we get,

$$y = 70^\circ$$

Hence, small interior angle $(y) = 70^\circ$

Sol.40.(d)

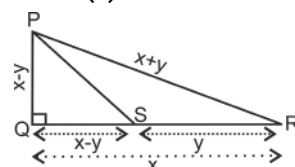
By checking all options one by one we find that option d is incorrect.

Sides of triangle = $2a, 2a, 3a$

$$\text{Height of triangle from base with side } 3a = \frac{\sqrt{7}a}{2}$$

$$\text{Area of triangle} = \frac{1}{2} \times 3a \times \frac{\sqrt{7}a}{2} = \frac{3\sqrt{7}a^2}{4}$$

Sol.41.(d)

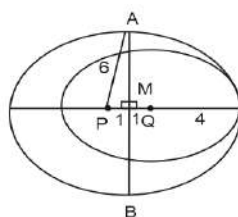


(3, 4, 5) is a pythagorean triple.

Let $x = 4, y = 1$

So the ratio of QS : SR = 3 : 1

Sol.42.(d)

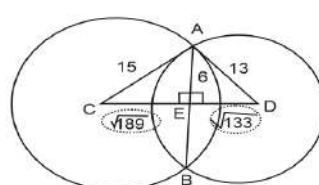


$$\text{So the length of AM} = \sqrt{36 - 1} = \sqrt{35}$$

Now,

$$\text{Length of AB} = 2 \times \sqrt{35} = 2\sqrt{35} \text{ cm}$$

Sol.43.(d)



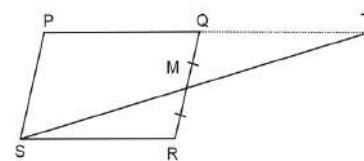
Given, length of AB = 12 cm, then AE = 6 cm

$$\text{In } \triangle ACE, CE = \sqrt{225 - 36} = \sqrt{189}$$

$$\text{In } \triangle ADE, DE = \sqrt{169 - 36} = \sqrt{133}$$

So, the distance between the two centers = $\sqrt{189} + \sqrt{133}$

Sol.44.(c)



SR = PQ(PQRS is a parallelogram)

$$\triangle SMR \cong \triangle QMT$$

$\angle RMS = \angle QMT$ (vertically opposite)

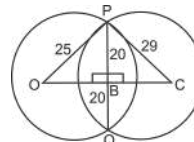
$\angle MRS = \angle MQT$ (alternate angle)

$$\text{So, } \frac{RM}{MQ} = \frac{SR}{QT} \Rightarrow \frac{1}{1} = \frac{SR}{QT}$$

$$\Rightarrow SR = PQ = QT \Rightarrow PT = 2PQ = 2SR$$

Sol.45.(c) Pythagorean triplets :- (15, 20, 25) and (20, 21, 29)

$$AP = BP = 20 \text{ cm}$$



So, distance between the centres (OC) = 15 + 21 = 36

Sol.46.(c)

Using tangent - secant theorem :-

$$RX^2 = RQ \times RP$$

$$144 = (x - 2) \times (2x - 2)$$

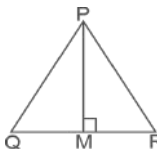
$$144 = 2x^2 - 6x + 4$$

$$2x^2 - 6x - 140 = 0$$

$$x^2 - 3x - 70 = 0 \Rightarrow x = 10, -7$$

Length can't be -ve, so $x = 10 \text{ cm}$

Sol.47.(b)



In an equilateral triangle, both altitude and median are the same.

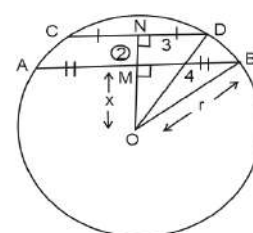
$$PQ = QR = RP = 2QM = 2MR$$

$$\text{In } \triangle PMQ, PQ^2 = PM^2 + QM^2$$

$$PQ^2 = PM^2 + \left(\frac{PQ}{2}\right)^2$$

$$PM^2 = \frac{3PQ^2}{4} \Rightarrow 3PQ^2 = 4PM^2$$

Sol.48.(a)



Equating the value of radius,

$$\Rightarrow (3)^2 + (x + 2)^2 = (4)^2 + (x)^2$$

$$\Rightarrow 9 + 4 + 4x = 16 \Rightarrow x = \frac{3}{4}$$

$$\text{Now, radius (OB=OD)} = \sqrt{(4)^2 + (x)^2}$$

$$= \sqrt{(4)^2 + \left(\frac{3}{4}\right)^2} = \frac{\sqrt{265}}{4}$$

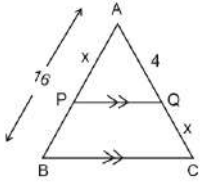


Sol.49.(c)

Length of transverse common tangent

$$= \sqrt{d^2 - (r_1 + r_2)^2} = \sqrt{12^2 - (5 + 4)^2}$$

$$= \sqrt{144 - (81)} = \sqrt{63}$$

Sol.50.(c)

By using Thales theorem

$$\frac{AP}{AB} = \frac{AQ}{AC} \Rightarrow \frac{x}{16} = \frac{4}{x+4}$$

$$= x^2 + 4x - 64 = 0$$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4 \times 1 \times (-64)}}{2 \times 1}$$

$$x = \frac{-4 \pm \sqrt{272}}{2 \times 1} = (2\sqrt{17} - 2)$$

Sol.51.(a)

Value of third side will lie between :-

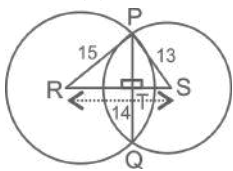
$$(24 - 9) < S < (24 + 9)$$

$$(15) < S < (33)$$

Sol.52.(a) According to the option ,

Option, 'a' is the false statement .

Correct statement is :- ratio of same sides in congruent triangles always be the same.

Sol.53.(d)In ΔPRS

$$S(\text{semi perimeter}) = \frac{a+b+c}{2}$$

$$= \frac{15+13+14}{2} = \frac{42}{2} = 21$$

Area of ΔPRS

$$= \sqrt{21(21-15)(21-13)(21-14)}$$

$$= \sqrt{21 \times 6 \times 8 \times 7} = 84$$

$$\text{Now, area of } \Delta PRS = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\Rightarrow \frac{1}{2} \times RS \times PT = 84 \Rightarrow \frac{1}{2} \times 14 \times PT = 84$$

$$\Rightarrow 7 PT = 84 \Rightarrow PT = 12$$

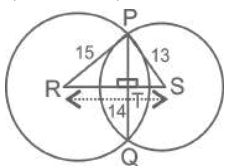
So, the length of the common chord PQ

$$= 2 \times 12 = 24 \text{ cm}$$

Short trick :-

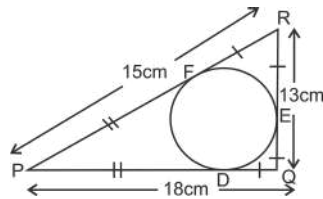
Pythagoras triplets :- (5, 12, 13) and

$$(9, 12, 15)$$



So, length of common chord (PQ)

$$= 2 \times 12 = 24 \text{ cm}$$

Sol.54.(a)Let, $PF = PD = x$, $RF = RE = y$ and

$$QE = QD = z$$

According to the question,

$$2(x + y + z) = 13 + 15 + 18 = 46$$

$$x + y + z = 23 \text{ cm}$$

Now, length of PD (x) = (x + y + z) -

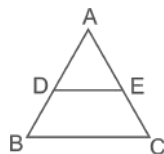
$$(y + z) = 23 - 13 = 10 \text{ cm}$$

Short trick :-

$$PD = \left(\frac{PQ + QR + RP}{2} \right) - RQ$$

$$PD = \left(\frac{18 + 13 + 15}{2} \right) - 13$$

$$PD = 23 - 13 = 10 \text{ cm}$$

Sol.55.(d)As $DE \parallel BC$ DE divides the area of ΔABC into two equal parts

Hence, D and E is the midpoint of AB and AC

As $\Delta ADE \sim \Delta ABC$

$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta ADE} = \left(\frac{AB}{AD} \right)^2$$

$$\Rightarrow \left(\frac{AB}{AD} \right)^2 = \frac{2}{1} \Rightarrow AB = \sqrt{2} AD$$

$$\Rightarrow AB = \sqrt{2} (AB - BD) \Rightarrow \sqrt{2} BD = (\sqrt{2} - 1) AB$$

$$\Rightarrow \frac{BD}{AB} = \frac{(\sqrt{2} - 1)}{\sqrt{2}}$$

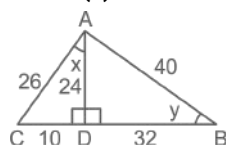
$$\text{Sol.56.(a)} \pi^c = 180^\circ \Rightarrow 1^c = \left(\frac{180 \times 7}{22} \right)^\circ$$

$$\theta \text{ (in radian)} = \frac{\text{length of arc}}{\text{radius}} = \frac{3x}{x} = 3$$

$$\theta \text{ (in degree)} = \frac{3 \times 180 \times 7}{22}$$

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \frac{22}{7} \times r^2$$

$$= \frac{3 \times 180 \times 7}{22 \times 360} \times \frac{22}{7} \times r^2 = 1.5 r^2$$

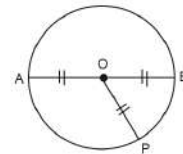
Sol.57.(c)

$$\frac{6}{\cos x} - \frac{5}{\cos y} + 8 \tan y$$

$$= \frac{6}{\frac{24}{26}} - \frac{5}{\frac{32}{40}} + 8 \times \frac{24}{32}$$

$$= \frac{26}{4} - \frac{5 \times 5}{4} + 8 \times \frac{3}{4}$$

$$= \frac{26 - 25 + 24}{4} = \frac{25}{4} \text{ units}$$

Sol.58.(c)

Given that ,

$$\angle AOP = 95^\circ$$

$$\angle AOP + \angle BOP = 180^\circ \Rightarrow \angle BOP = 85^\circ$$

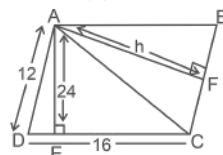
OB = OP = radius

So, $\angle OBP = \angle OPB$ Now, In ΔBOP

$$\angle POB + \angle OBP + \angle BPO = 180^\circ$$

$$2 \times \angle OBP = 180^\circ - 85^\circ$$

$$\angle OBP = \frac{95^\circ}{2} = 47.5^\circ$$

Sol.59.(c)

Let ABCD is a parallelogram,

 $DC \parallel AB$ and $DA \parallel CB$

We know that,

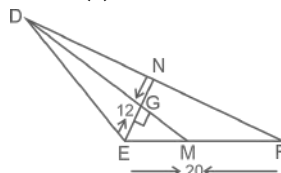
$$DC = AB = 16 \text{ and } DA = CB = 12$$

According to the question,

$$\Rightarrow \text{Area of } \Delta ADC = \text{Area of } \Delta ABC$$

$$\Rightarrow \frac{1}{2} \times 16 \times 24 = \frac{1}{2} \times 12 \times h$$

$$\Rightarrow h = \frac{16 \times 24}{12} = 32 \text{ cm}$$

Sol.60.(c)

The centroid divides each median in the ratio 2 : 1 .

$$\text{So the value of } EG = 12 \times \frac{2}{3} = 8 \text{ cm}$$

EM = 10 cm (median divides side equally)

 ΔEGM follow, tripates = (6, 8, 10)

So the value of GM = 6 cm

(GM) 1 unit = 6 cm

Then (DM) 3 units = 18 cm

Sol.61.(d) Possible no. of values of AC =

$$BC - AB < AC < BC + AB \Rightarrow 5 < AC < 17$$

Total possible values = 11

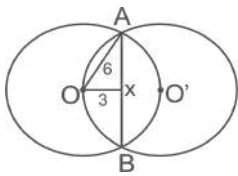
Short trick :-

Total no. of possible values of third side

$$(AC) = 2 \times (\text{smallest side}) - 1$$

$$= 2 \times (6) - 1 = 11$$

Sol.62.(c)



Let common chord AB perpendicularly bisect the distance between the centres at x

$$\text{So, } OX = \frac{OO'}{2} = \frac{6}{2} = 3 \text{ cm}$$

Now, as per figure

$$(AO)^2 = (AX)^2 + (OX)^2 \Rightarrow (AX)^2 = 36 - 9$$

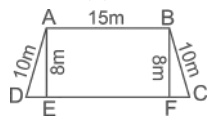
$$(AX) = \sqrt{27} = 3\sqrt{3}$$

$$AB = 2 \times AX = 6\sqrt{3} \text{ cm}$$

Short Trick :-

$$AB = 2 \times \left(\frac{\sqrt{3}}{2} \times 6\right) = 6\sqrt{3} \text{ cm}$$

Sol.63.(c)



$$EF = 15$$

$$\angle AED = \angle BFC = 90^\circ$$

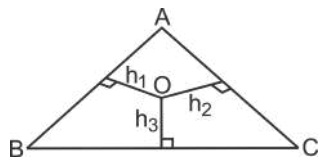
$$AD^2 = DE^2 + AE^2, DE = 6 \text{ m}$$

Similarly, FC = 6 m

Perimeter of Trapezium ABCD

$$= 15 + 10 + 10 + 27 = 62 \text{ m}$$

Sol.64.(a)

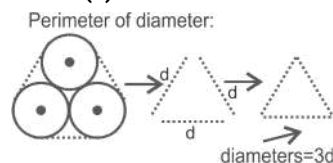


$$\text{Side (a)} = \frac{2}{\sqrt{3}} (h_1 + h_2 + h_3)$$

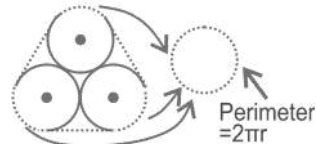
$$= \frac{2}{\sqrt{3}} (\sqrt{3} + 2\sqrt{3} + 5\sqrt{3}) = 16 \text{ cm}$$

$$\text{Perimeter of equilateral triangle} = 3 \times a = 48 \text{ cm}$$

Sol.65.(a)



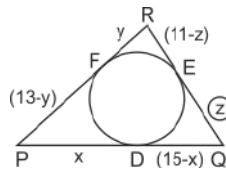
Perimeter of circumference:



Now, the length of string = $3d + 2\pi r$

$$\text{Where } d = \text{diameter of the circles} \\ \text{the length of string} = 3 \times 12 + 2\pi \times 6 = 36 + 12\pi$$

Sol.66.(d) Length of tangents drawn from a common external point to the circle is always equal.



$$\text{Perimeter} = PQ + QR + RP = 2x + 2y + 2z$$

$$x + y + z = \frac{39}{2} \text{ eq. (1)}$$

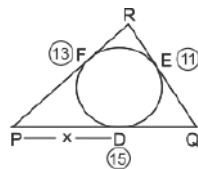
$$RF = RE \text{ (tangent from external point)}$$

$$y + z = 11 \text{ eq. (2)}$$

On solving eq. (1) and eq. (2), we get

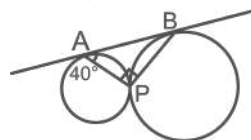
$$\text{the length of PD (x)} = \frac{39}{2} - 11 = \frac{39 - 22}{2} = 8.5 \text{ cm}$$

Short Trick :-



$$PD = \frac{PQ + PR - RQ}{2} = \frac{15 + 13 - 11}{2} = 8.5 \text{ cm}$$

Sol.67.(c)



When two circle touches each other externally at point P and AB is direct common tangent, Then $\angle APB = 90^\circ$

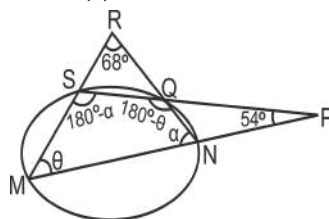
Now, In $\triangle APB$

$$\angle APB + \angle PBA + \angle BAP = 180^\circ$$

$$90^\circ + \angle PBA + 40^\circ = 180^\circ$$

$$\angle PBA = 50^\circ$$

Sol.68.(a)



In $\triangle RMN$

$$\theta + \alpha + 68^\circ = 180^\circ$$

$$\theta + \alpha = 112^\circ \text{ eq 1}$$

Now, In $\triangle PMS$

$$\theta + 180 - \alpha + 54^\circ = 180^\circ$$

$$\theta + 54^\circ = \alpha$$

Putting the value of α in equation 1

$$\theta + \theta + 54^\circ = 112^\circ$$

$$\theta = 29^\circ$$

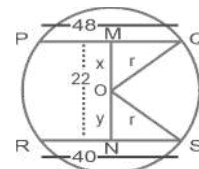
Short trick :-

$$\angle SMN = 90^\circ - \frac{\angle R + \angle P}{2}$$

$$\angle SMN = 90^\circ - \frac{68^\circ + 54^\circ}{2}$$

$$\angle SMN = 29^\circ$$

Sol.69.(a)



$$\text{From the given figure, } MQ = \frac{48}{2} = 24 \text{ cm}$$

$$\text{and } NS = \frac{40}{2} = 20 \text{ cm}$$

$$x + y = 22 \text{ (1)}$$

In $\triangle MOQ$, we have :

$$r^2 = x^2 + 24^2 \text{ (2)}$$

In $\triangle NOS$, we have :

$$r^2 = y^2 + 20^2 \text{ (3)}$$

From equation (2) and (3) we have :

$$x^2 + 24^2 = y^2 + 20^2$$

$$\Rightarrow x^2 + 576 = y^2 + 400$$

$$\Rightarrow x^2 - y^2 = -176 \Rightarrow (x - y)(x + y) = -176$$

$$(x - y)22 = -176 \text{ (from eqn(1))}$$

$$x - y = -8 \text{ (4)}$$

Solving eqn(1) and (4) we have :

$$x = 7 \text{ and } y = 15$$

Now putting $x = 7$ in eqn(2), we get :

$$r = \sqrt{49 + 576} = 25 \text{ cm}$$

Short Trick :-

Pythagoras triplets $\rightarrow$

(15, 20, 25) and (7, 24, 25)

20 and 24 are the given chords of the circle.

$$\text{Distance between the chord} = (15 + 7) = 22 \text{ cm}$$

Given conditions satisfied.

Clearly, radius of the circle = 25 cm

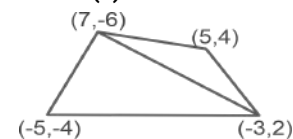
Sol.70.(d) Radius of the circle = $\frac{l}{\theta}$, where l is arc of circle and θ is the angle in radian

$$r = \frac{15}{24 \times \frac{\pi}{180}} = \frac{225}{2\pi} \text{ cm}$$

Circumference of the circle

$$= 2 \times \pi \times \frac{225}{2\pi} = 225 \text{ cm}$$

Sol.71.(a)



Let the vertices of quadrilateral be A(-3, 2), B(5, 4), C(7, -6), D(-5, -4)

Area of the triangle whose vertices are

$$(x_1, y_1) (x_2, y_2) (x_3, y_3) = \frac{1}{2} [x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)]$$

Area of the $\triangle ABC$ =

$$\frac{1}{2} |-3 \times (10) + 5 \times (-8) + 7 \times (-2)|$$

$$= \frac{1}{2} \times 84 = 42 \text{ square units}$$

Area of the $\triangle ACD$

$$= \frac{1}{2} |-3 \times (-2) + 7 \times (-6) - 5 \times (8)|$$



$$= \frac{1}{2} \times 76 = 38 \text{ square units}$$

Area of quadrilateral ABCD = 42 + 38
= 80 square units

Short trick :-

Area of quadrilateral =

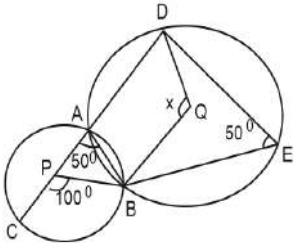
$$\frac{1}{2} \begin{vmatrix} x_1 - x_3 & y_1 - y_3 \\ x_2 - x_4 & y_2 - y_4 \end{vmatrix}$$

$$\frac{1}{2} \begin{vmatrix} -10 & 8 \\ 10 & 8 \end{vmatrix}$$

$$\text{Required area} = \frac{1}{2} \times |(-80) - (80)|$$

= 80 square units

Sol.72.(d)



Construction : we take a point E on the arc such that ABED is a cyclic quadrilateral.

$\angle PAB = \frac{1}{2} \angle BPC$ (angle subtended at arc is half the angle subtended at the centre.)

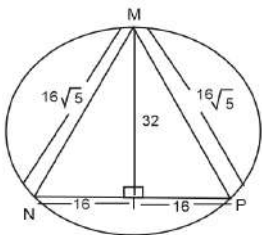
$$\angle PAB = \frac{1}{2} \times 100^\circ = 50^\circ$$

$\angle BED = \angle PAB = 50^\circ$ (external angle property of cyclic quadrilateral)

$\angle BQD = 2\angle BED$ (angle subtended at centre)

$$\angle BQD = 2 \times 50^\circ = 100^\circ$$

Sol.73.(a)



$$\text{Given, } MN = MP = 16\sqrt{5},$$

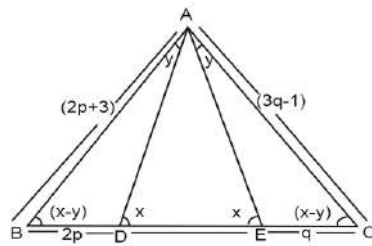
$$\text{In right } \triangle NIM, MI = \sqrt{(16\sqrt{5})^2 - 16^2} = 32 \text{ cm.}$$

$$\text{Circumradius (R)} = \frac{a \times b \times c}{4 \times \text{ar. } \triangle NIM}$$

$$= \frac{16\sqrt{5} \times 16\sqrt{5} \times 32}{4 \times \frac{1}{2} \times 32 \times 32} = 4 \times \sqrt{5} \times \sqrt{5}$$

= 20 cm.

Sol.74.(a)



As, AD = AE, $\angle D = \angle E = x$

As $\angle BAD = \angle CAE = y$ (let)

By exterior angle property, $\angle B = \angle C = (x - y)$

By symmetry, BD = CE, so, $2p = q$,

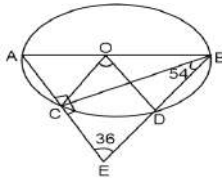
$\Rightarrow$ As, $\angle B = \angle C$, AB = AC

$$\Rightarrow 2p + 3 = 3q - 1 \Rightarrow 2p + 3 = 3 \times 2p - 1$$

$$\Rightarrow 4 = 4p \Rightarrow p = 1.$$

$$\text{So, } q = 2p = 2 \Rightarrow (p + q) = 2 + 1 = 3.$$

Sol.75.(d)



Construction : Join CB.

As AB is diameter we have

$$\angle ACB = \angle BCE = 90^\circ$$

$$\angle EBC = 90^\circ - \angle AEB = 90^\circ - 36^\circ = 54^\circ$$

$$\angle DOC = 2\angle EBC = 2 \times 54 = 108^\circ$$

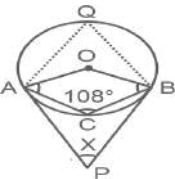
(As $\angle DBC$ is made by arc CD at circumference while $\angle DOC$ is angle made by it on centre.)

Short Trick :-

$$\angle DOC = 180^\circ - 2 \times \angle CED$$

$$= 180^\circ - 2 \times 36^\circ = 108^\circ$$

Sol.76.(a)



Let Q be the point on the major arc, and O be the centre of the circle.

So, $\square APBQ$ becomes cyclic quadrilateral.

So,

$$\angle AQB = 180^\circ - \angle ACB = 180^\circ - 108^\circ = 72^\circ$$

$$\text{and } \angle AOB = 2(\angle AQB) = 2(72^\circ) = 144^\circ$$

In $\square AOBP$, $\angle OAP = \angle OBP = 90^\circ$ ($\because$ radius from centre is perpendicular to the tangent)

So, in $\square AOBP$,

$$144^\circ + 90^\circ + 90^\circ + x = 360^\circ$$

$$x = 360^\circ - 324^\circ = 36^\circ$$

$$\text{So, } \angle APB = x = 36^\circ$$

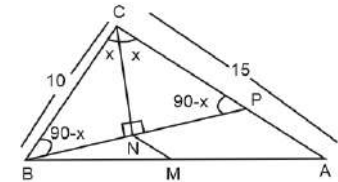
Short Trick :-

$$\angle ACB = 90^\circ + \frac{\angle APB}{2}$$

$$\angle APB = 2 \times (108^\circ - 90^\circ)$$

$$\angle APB = 36^\circ$$

Sol.77.(b)



As $\angle BCN = \angle PCN$ and $\angle BNC = \angle PNC$

so, $\triangle CNP \sim \triangle CNB$

$$\text{Now, } \frac{CN}{BC} = \frac{CN}{CP},$$

clearly, BC = CP = 10 cm.

Similarly, BN = NP,

hence N is the midpoint of BP.

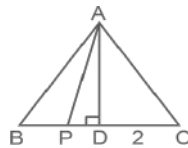
$$AP = AC - CP = 15 - 10 = 5 \text{ cm}$$

In $\triangle ABP$, M is midpoint of AB and N is midpoint of BP

So, by midpoint theorem,

$$NM = \frac{1}{2} AP = \frac{5}{2} = 2.5 \text{ cm}$$

Sol.78.(d)



$$PC = \frac{3\sqrt{7}}{3} \times 2 = 2\sqrt{7}$$

AD is the altitude and median of triangle

$$DC = \frac{3\sqrt{7}}{2}$$

$$PD = 2\sqrt{7} - \frac{3\sqrt{7}}{2} = \frac{\sqrt{7}}{2}$$

$$AD = \frac{\sqrt{3}a}{2} = \frac{\sqrt{3} \cdot 3 \cdot \sqrt{7}}{2} = \frac{3\sqrt{21}}{2}$$

$$AP = \sqrt{AD^2 + DP^2} = 7$$

Short Trick :-

$$BP = 3\sqrt{7} \times \frac{1}{3} = \sqrt{7}$$

$$PC = 3\sqrt{7} \times \frac{2}{3} = 2\sqrt{7}$$

$$(AP)^2 = (\text{Side})^2 - BP \times PC$$

$$AP = \sqrt{(63) - 14} = 7 \text{ cm}$$

Sol.79.(d) There are two possibilities to solve this question

First possibility :

if the sum of equal sides = 16

Then any value of the third side is possible from 1 to 15, so total 15 triangles

Second possibility : if the sum of two unequal sides = 16

Then the possible cases are = (1, 15, 15), (2, 14, 14), (3, 13, 13), (4, 12, 12), (5, 11, 11), (6, 10, 10), (6, 10, 6), (7, 9, 9) and (7, 9, 7) so total 9 triangles

Total triangles = 15 + 9 = 24

Short Trick :-

No. of possible isosceles triangles, when sum of two of the sides are given :-

$$\frac{5 \times (\text{sum})}{3} - 3$$



$$= \frac{5 \times (16)}{3} - 3 = 23.66 \approx 24$$

Sol.80.(d) Possible value of a and b are (1, 24); (2, 12); (3, 8); (4, 6)

Sides of the triangle are a, b, c.

$$\text{If } a = 1, b = 24 \Rightarrow a - b < c < a + b$$

$$23 < c < 25$$

Possible value of c = 24

$$\text{If } a = 2, b = 12 \Rightarrow a - b < c < a + b$$

$$10 < c < 14$$

Possible value of c = 11, 12, 13

$$\text{If } a = 3, b = 8 \Rightarrow a - b < c < a + b$$

$$5 < c < 11$$

Possible value of c = 6, 7, 8, 9, 10

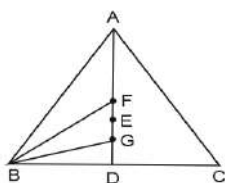
$$\text{If } a = 4, b = 6 \Rightarrow a - b < c < a + b$$

$$2 < c < 10$$

Possible value of c = 3, 4, 5, 6, 7, 8, 9

Number of possible value of c = 16

Sol.81.(c)



As Median divides the triangle into two equal areas.

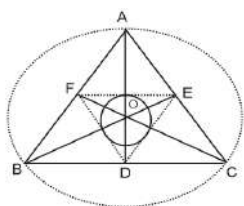
And in $\triangle ADB$,

Area of $\triangle ADB = 2 \times$ Area of $\triangle BGF$ (base(AD = 2GF) is half and height is same)

So, Area of $\triangle ABC = 4 \times$ Area of $\triangle BGF$

$$\Rightarrow \text{Area of } \triangle BGF = \frac{1}{4} \times \text{Area of } \triangle ABC$$

Sol.82.(d) Concept : triangle formed by the joining the altitude has its incenter at the same point as the circumcenter of the bigger triangle.



From the diagram

OE is the bisector of the $\angle FED$

So, $\angle FEO = \angle DEO = 24^\circ$

In Triangle BFE

$$\angle FBE = 180^\circ - (110^\circ + 24^\circ)$$

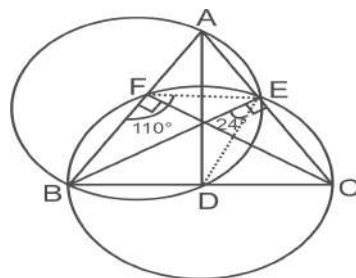
$$\angle FBE = 180^\circ - 134^\circ = 46^\circ$$

Short trick :-

Draw a circle of Diameter AB which

passed through D as $\angle ADB = \angle AEB = 90^\circ$

Draw another circle of diameter BC which passed through E as $\angle BEC = \angle BFC = 90^\circ$



$$\angle EFC = 110^\circ - 90^\circ = 20^\circ$$

Now considering EC a chord,

$$\angle EBC = \angle EFC = 20^\circ$$

Now considering BD a chord,

$$\angle BED = \angle BAD = 24^\circ$$

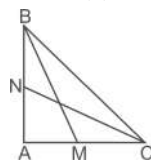
So, in $\triangle ABD$,

$$\angle BAD + \angle ADB + \angle ABD = 180^\circ$$

$$\Rightarrow 24^\circ + 90^\circ + \angle ABE + 20^\circ = 180^\circ$$

$$\angle ABE = 46^\circ, \angle EBF = 46^\circ$$

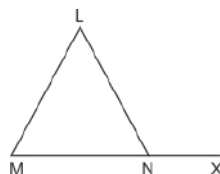
Sol.83.(c) Short-Trick:



$$\text{We know, } 4(BM^2 + CN^2) = 5BC^2$$

$$\frac{BM^2 + CN^2}{BC^2} = \frac{5}{4}$$

Sol.84.(b)



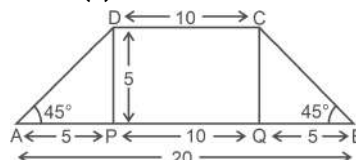
$$\angle M = x^\circ \text{ then } \angle L = 2x$$

$$2x + x = 117^\circ \text{ (Exterior angle)}$$

$$\Rightarrow 3x = 117^\circ \Rightarrow x = 39^\circ$$

$$\Rightarrow \angle L = 2 \times 39^\circ = 78^\circ$$

Sol.85.(d)



In trapezium ABCD,

$$CD = PQ = 10 \text{ cm}$$

$$AP = QB = 5 \text{ cm}$$

$$\angle DAB + \angle CBA = 90^\circ$$

$$\text{let } \angle DAB = \angle CBA = 45^\circ$$

In right angle triangle APD,

$$\angle DAP + \angle APD + \angle PDA = 180^\circ$$

$$45^\circ + 90^\circ + \angle PDA = 180^\circ$$

$$\angle PDA = 180^\circ - 135^\circ = 45^\circ$$

$$\text{If } \angle PDA = \angle DAP = 45^\circ$$

$$\text{then } AP = PD = 5 \text{ cm}$$

Area of trapezium

$$= \frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$$

$$= \frac{1}{2} \times 30 \times 5 = 75 \text{ cm}^2$$

Sol.86.(c) $\angle CBD = 80^\circ$ (exterior angle)

In quadrilateral BCDE

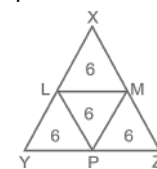
$$\angle CED = 360^\circ - (80^\circ + 45^\circ + 35^\circ)$$

$$\angle CED = 200^\circ \text{ (it is a reflex angle)}$$

$$\text{Obtuse angle } (\angle CED) = 360^\circ - 200^\circ$$

$$= 160^\circ$$

Sol.87.(b) concept - we know that median divides triangle in two equal parts.



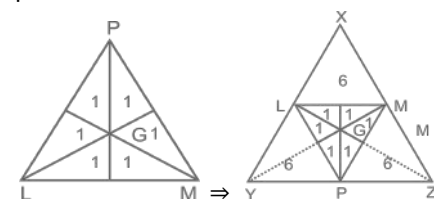
If the mid points of sides of a triangle joined. Then the triangle is divided into four equal parts (areas).

Let area of each triangle $\triangle XLM$

$$= \triangle YLP = \triangle PMZ = \triangle LMP = 6 \text{ units}$$

$$\text{Total area of } \triangle XYZ = 24 \text{ units}$$

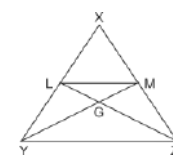
Median divides triangle in two equal parts



Area of $\triangle GLM$: Area of

$$\triangle XYZ = 2 : 24 = 1 : 12$$

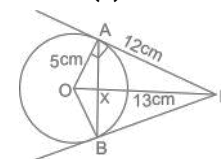
Short Trick :-



We know,

$$\text{Area of } \triangle GLM : \text{Area of } \triangle XYZ = 1 : 12$$

Sol.88.(b)



$$\text{Area of } \triangle PAB, M = \frac{1}{2} \times AB \times PX$$

Let $OX = a$ and $PX = 13 - a$

$$\text{Area of } \triangle PAO = \frac{1}{2} \times AO \times PA$$

$$= \frac{1}{2} \times AX \times PO$$

$$\Rightarrow \frac{1}{2} \times 5 \times 12 = \frac{1}{2} \times AX \times 13$$

$$\Rightarrow AX = \frac{60}{13}$$

$$AB = 2 \times AX = \frac{120}{13}$$

Use pythagoras theorem in $\triangle AXO$;

$$OX^2 = OA^2 - AX^2$$

$$\Rightarrow OX^2 = 5^2 - \left(\frac{60}{13}\right)^2 \Rightarrow OX = \frac{25}{13}$$

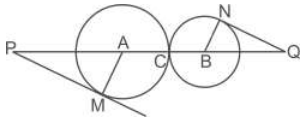
$$PX = 13 - \frac{25}{13} = \frac{169 - 25}{13} = \frac{144}{13}$$

Area of $\triangle PAB$,

$$M = \frac{1}{2} \times \frac{120}{13} \times \frac{144}{13}$$

$$\sqrt{\frac{M}{15}} = \sqrt{\frac{1}{15} \times \frac{1}{2} \times \frac{120}{13} \times \frac{144}{13}} = \frac{24}{13}$$

Sol.89.(c) MP = 15 cm, NQ = 8 cm,
PA = 17 cm and BQ = 10 cm

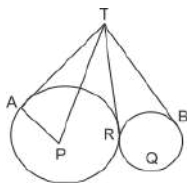


$$AM = \sqrt{17^2 - 15^2} = 8 \text{ cm}$$

$$NB = \sqrt{10^2 - 8^2} = 6 \text{ cm}$$

$$AB = 8 + 6 = 14 \text{ cm}$$

Sol.90.(c) AP, radius of first circle = 6 cm
and PT = 10 cm



Tangent is perpendicular to the radius.

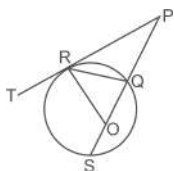
Apply pythagoras theorem in ΔTAP ,

$$TP^2 = AP^2 + AT^2 \Rightarrow AT = 8 \text{ cm}$$

Tangents to a circle from a point outside the circle are equal in length.

Thus, AT = TR = TB = 8 cm

Sol.91.(b) In the given diagram,
 $\angle QRP = 28^\circ$, Join OR.



We get $OR \perp PT$.

$$\text{Then } \angle ORQ = \angle OQR = 90^\circ - 28^\circ = 62^\circ$$

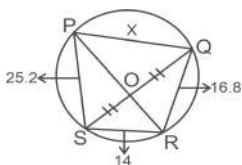
$$\text{Then } \angle ROQ = 180^\circ - 124^\circ = 56^\circ.$$

In ΔORP ,

$$\angle ORP = 90^\circ$$

$$\text{Then } \angle RPO = 90^\circ - 56^\circ = 34^\circ$$

Sol.92.(b) As PR and SQ bisect each other at O.



In ΔPOS and ΔROQ ,

$$\angle POS = \angle ROQ, PO = RO, OS = QO$$

Thus, $\Delta POS \sim \Delta ROQ$

$$\frac{PO}{OR} = \frac{OS}{OQ} = \frac{PS}{QR} = \frac{25.2}{16.8} \Rightarrow \frac{3}{2}$$

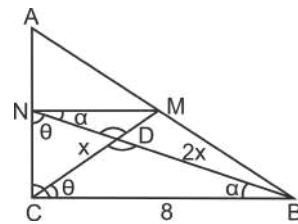
Similarly, $\Delta POQ \sim \Delta ROS$

$$\text{So, } \frac{PO}{OR} = \frac{PQ}{SR} \Rightarrow \frac{3}{2} = \frac{x}{14} \Rightarrow x = 21$$

Short trick :- As PR and SQ bisect each other at O.

$$\text{Then } \frac{PS}{QR} = \frac{PQ}{SR} \Rightarrow \frac{25.2}{16.8} = \frac{x}{14} \Rightarrow x = 21$$

Sol.93.(c)



$$\text{Since } MN \parallel CB, MN = \frac{1}{2}CB$$

ΔMND & ΔCBD are similar

$$\frac{NM}{BC} = \frac{1}{2} = \frac{ND}{DB}$$

$$\text{Let } ND = x \text{ and } DB = 2x$$

In ΔNCB and ΔCDB

$$\angle CDB = \angle NCB = 90^\circ$$

$$\alpha = \alpha, \theta = \theta$$

$$\text{Thus, } \frac{x+2x}{8} = \frac{8}{2x} \Rightarrow 6x^2 = 64$$

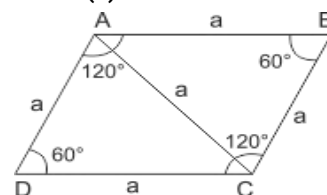
$$x^2 = \frac{64}{6} \Rightarrow x = \frac{8}{\sqrt{6}}$$

$$3x = \frac{8}{\sqrt{6}} \times 3 \Rightarrow x = 4\sqrt{6} \text{ cm}$$

Short trick :-

$$BN = \frac{BC\sqrt{6}}{2} = \frac{8\sqrt{6}}{2} = 4\sqrt{6} \text{ cm}$$

Sol.94.(d)



When two equilateral triangles are joined

then they make a parallelogram

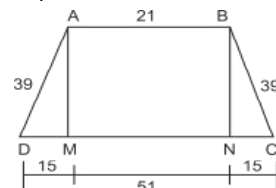
Area of the parallelogram = base $\times$ height

$$a \times H = 2 \times \frac{\sqrt{3}}{4} a^2$$

$$H = \frac{\sqrt{3}}{2} a \quad (\text{Putting } a = 10\sqrt{3})$$

$$H = 15 \text{ cm}$$

Sol.95.(b) As given, the mentioned trapezium is isosceles



In right triangle BNC

$$BN = \sqrt{(39)^2 - (15)^2}$$

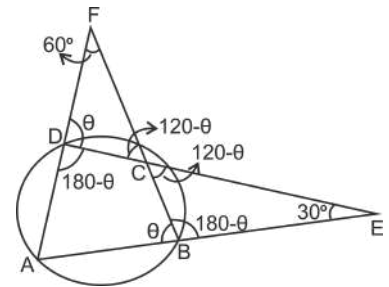
$$= \sqrt{54 \times 24} = 3 \times 2 \times 6 = 36$$

Area of the trapezium

$$= \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$$

$$= \frac{1}{2} \times 72 \times 36 = 1296 \text{ cm}^2$$

Sol.96.(b)



Let $\angle ABC = \theta$ then $\angle ADC = 180 - \theta$ (as sum of opposite angles of cyclic quadrilateral)

$\angle ADC = \angle CBE = 180 - \theta$ (Using alternate segment theorem)

similarly, $\angle ABC = \angle CDF = \theta$

So, in triangle ΔFDC ,

$$\angle FCD = 180 - 60 - \theta = 120 - \theta \quad (\text{Sum of angles of triangle property})$$

Now, $\angle FCD = \angle BCE$ (vertically opposite angles)

$$\text{In } \Delta BCE = 180^\circ$$

$$\Rightarrow 120 - \theta + 180 - \theta + 30 = 180$$

$$\Rightarrow 300 - 2\theta = 150 \Rightarrow \theta = 75^\circ$$

Short Trick :-

$$\angle FAE = 90^\circ - \frac{\angle F + \angle E}{2}$$

$$\angle FAE = 90^\circ - \frac{60^\circ + 30^\circ}{2}$$

$$\angle FAE = 45^\circ$$

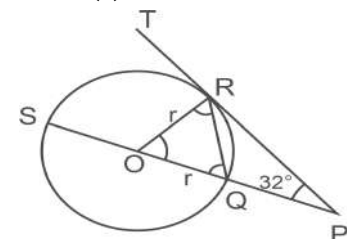
Now, In ΔAFB ,

$$\angle AFB + \angle FBA + \angle ABF = 180^\circ$$

$$\angle FBA = 180^\circ - (60^\circ + 45^\circ)$$

$$\angle FBA = 75^\circ$$

Sol.97.(d)



We know that OR is the radius of the circle that is perpendicular to the tangent PT.

$$\angle ORP = 90^\circ$$

$$\text{In } \Delta ORQ, OR = OQ = r$$

$$\angle ORQ = \angle OQR$$

$$\Rightarrow \text{In } \Delta ORP$$

$$\angle ORP = 90^\circ, \angle RPO = 32^\circ$$

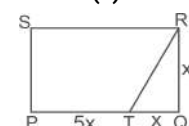
$$\angle ROP = 58^\circ$$

$$\Rightarrow \text{Thus, In } \Delta ORQ$$

$$\angle ORQ = \angle OQR = 61^\circ$$

$$\angle QRP = 90 - 61 = 29^\circ$$

Sol.98.(c)



Since QRT is a right angled isosceles triangle due to $\angle Q = 90^\circ$

Area of $\Delta QRT = 12\sqrt{3}$ (Given)



$$\Rightarrow \frac{1 \times x \times x}{2} = 12\sqrt{3} \Rightarrow x^2 = 24\sqrt{3} \dots (1)$$

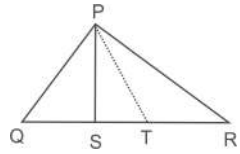
Now $PQ = 6x$, $QR = x$

Area of rectangle = $l \times b = 6x \times x = 6x^2$
 $= 6 \times 24\sqrt{3} \dots (\text{using 1st}) = 144\sqrt{3} \text{ sq cm}$

Sol.99.(c) Given, $\angle P = 120^\circ$, $PS \perp QR$ and $PQ + QS = SR$.

Draw $PT = PQ$. Let $PQ = y$ and $QT = 2x$,
 $\angle PQS = \theta$ and $\angle PRQ = \phi$

In $\triangle PQR$ $\angle PQR + \angle PRQ + \angle QPR = 180$
 $\Rightarrow \theta + \phi = 180 - 120 = 60 \dots (1)$



$\Rightarrow$ PQT is an isosceles triangle and $PS \perp$

$QR \Rightarrow QS = ST = x$

$\Rightarrow \angle PTQ = \theta \dots (PQ = PT)$

And $\angle QPT = 180 - 2\theta$

Now, $PQ + QS = SR \dots (\text{given})$

$\Rightarrow y + x = x + TR \Rightarrow TR = y$

$\Rightarrow PT = TR \dots (\text{Both are equal to } y)$

$\Rightarrow \angle PRT = \angle TPR = \phi \dots (PT = TR)$

According to the question

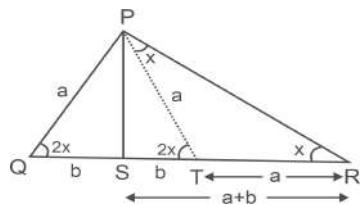
$\angle QPT + \angle TPR = 120$

$\Rightarrow 180 - 2\theta + \phi = 120 \Rightarrow 2\theta - \phi$

$= 60 \dots (2)$

Add eq (1) and (2) $\Rightarrow \theta = 40^\circ$

Short trick :-



Draw PT such that $PT = PQ$,

$PQ + QS = SR$

From external angle property,

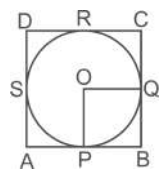
$\angle PQR = \angle PTQ = 2x$

In triangle PQR,

$2x + x + 120^\circ = 180^\circ \Rightarrow x = 20^\circ$

Therefore $\angle Q = 2x = 2 \times 20^\circ = 40^\circ$

Sol.100.(b) Given, $\angle B = 90^\circ$. If $AD = 24$ cm, $AB = 27$ cm and $DR = 6$ cm



$DR = DS = 6$ cm $\dots (\text{tangents})$

$AS = AD - DS = 24 - 6 = 18$ cm

$AS = AP = 18$ cm $\dots (\text{tangents})$

$PB = AB - AP = 27 - 18 = 9$ cm

$OP \perp AB$, $OQ \perp BC \dots (\text{angle made by radius on tangent})$

In quadrilateral OPBQ

$\angle B = 90^\circ$, $\angle OQB = 90^\circ$, $\angle OPB = 90^\circ$

$\Rightarrow \angle POQ = 90^\circ \Rightarrow PB = BQ \dots (\text{tangents})$

$\Rightarrow OPBQ$ is a square

$PB = BQ = OP = OQ = 9$ cm

Perimeter of the circle = $2\pi r$

$= 2 \times \pi \times 9 = 18\pi$

Sol.101.(b) sum of the interior angles of a regular polygon = $(n - 2) \times 180$

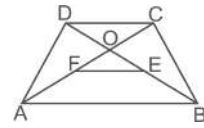
$1260 = (n - 2) \times 180 \Rightarrow n = 9$

$\Rightarrow$ Exterior angle = $\frac{360}{n} = \frac{360}{9} = 40^\circ$

$\Rightarrow$ Interior angle = $\frac{1260}{9} = 140^\circ$

Required difference = $140^\circ - 40^\circ = 100^\circ$

Sol.102.(c)



Given, $DC \parallel AB$, $AB = 12$ cm and $DC = 7.2$ cm

Let E and F be the points joining the mid points of BD and AC.

Short Trick :-

$$EF = \frac{(AB - CD)}{2} = \frac{(12 - 7.2)}{2} = 2.4 \text{ cm}$$

Sol.103.(d) Each Interior angle of a regular polygon = $\frac{(n - 2) \times 180}{n}$

Where n is the number of sides

$$\Rightarrow (128 \frac{4}{7})^\circ = \frac{(n - 2) \times 180}{n}$$

$$\Rightarrow \frac{900}{7} = \frac{(n - 2) \times 180}{n}$$

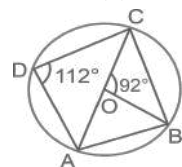
$$\Rightarrow 5n = 7n - 14 \Rightarrow n = 7$$

$$\text{Number of diagonals} = \frac{n(n - 3)}{2}$$

$$= \frac{7(7 - 3)}{2} = 14$$

Desired sum = $7 + 14 = 21$

Sol.104.(b)



$\angle ABC + \angle ADC = 180^\circ$

$\Rightarrow \angle ABC = 180^\circ - 112^\circ = 68^\circ$

In $\triangle OBC$

$\angle OBC = \angle OCB \dots (OB = OC = \text{radius})$

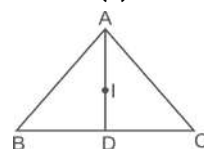
$\angle OBC + \angle OCB + \angle COB = 180^\circ$

$2\angle OBC = 180^\circ - 92^\circ (\angle OBC = \angle OCB)$

$\angle OBC = 44^\circ$

$\angle ABO = \angle ABC - \angle OBC = 68^\circ - 44^\circ = 24^\circ$

Sol.105.(d)



$\Rightarrow AB + AC + BC = 24$ cm

$\Rightarrow AB + AC = 24 - 9 = 15$ cm

$$\Rightarrow \frac{AI}{ID} = \frac{AB + AC}{BC} \Rightarrow \frac{AI}{ID} = \frac{15}{9} = \frac{5}{3}$$

Sol.106.(c) We know that

Circumference of a circle (C) = $2\pi r$

$$\Rightarrow r = \frac{C}{2\pi}$$

Area of the sector (A) = $\frac{\theta}{360} \times \pi r^2$

$$\Rightarrow A = \frac{60}{360} \times \pi \times \frac{C}{2\pi} \times \frac{C}{2\pi} = \frac{C^2}{24\pi}$$

Sol.107.(b) We know that,

Inradius of triangle = $\frac{\text{Area}}{\text{Semiperimeter}}$

$$\Rightarrow r = \frac{A}{\frac{p}{2}} \Rightarrow A : p = r : 2$$

Sol.108.(a) We know that the circumradius of a hexagon is equal to the side of the hexagon.

Let side of the hexagon = radius of the circle = a

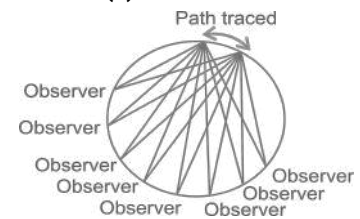
$\Rightarrow$ Area of the circle = πa^2

$$\Rightarrow \text{Area of hexagon} = 6 \times \frac{\sqrt{3}}{4} \times a^2$$

$$\text{Required ratio} = \frac{\pi a^2}{6 \times \frac{\sqrt{3}}{4} \times a^2} = \frac{2\pi}{3\sqrt{3}}$$

$$= 2\pi : 2\pi - 3\sqrt{3}$$

Sol.109.(a)



Whatever be the position of the observer, the angle formed from him will be the half of the angle formed by the coach from the centre.

$$\theta = \frac{1}{2} \times 10^\circ = 5^\circ$$

Sol.110.(d)

Equilateral Triangle (only in equilateral triangle all converge at a point)

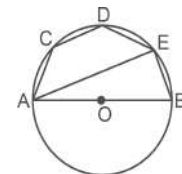
Sol.111.(a)

Required number of triangles = ${}^{11}C_3$

We know that

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{11!}{3!(11-3)!} = \frac{11 \times 10 \times 9 \times 8!}{(3 \times 2 \times 1)8!} = \frac{11 \times 10 \times 9}{6} = 165$$

Sol.112.(c)



Here, ACDE is a cyclic quadrilateral.

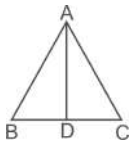
$\angle ACD + \angle DEA = 180^\circ \dots (\text{opposite angles of cyclic quadrilateral})$

$\angle AEB = 90^\circ \dots (\text{angle of semicircle})$

$\angle ACD + \angle DEB \Rightarrow \angle ACD + \angle DEA + \angle AEB$

$$= 180^\circ + 90^\circ = 270^\circ$$

Sol.113.(d)



$$\text{Given, } \angle BAD = \frac{1}{2} \times \angle ADC$$

$$, \angle BAC = 87^\circ \text{ and } \angle C = 42^\circ$$

$$\angle B = 180^\circ - \angle BAC - \angle C$$

$$= 180^\circ - 87^\circ - 42^\circ = 51^\circ$$

$$\angle BAD + \angle B = \angle ADC$$

(external opposite angle)

$$\angle BAD + 51^\circ = 102^\circ$$

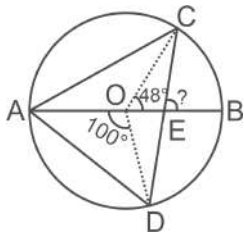
$$\angle BAD = 51^\circ$$

$$\angle ADB = 180^\circ - \angle BAD - \angle ABD$$

$$= 180^\circ - 51^\circ - 51^\circ = 78^\circ$$

Sol.114.(b)

$$\angle BOC = 48^\circ \text{ and } \angle AOD = 100^\circ$$



$$\angle BAC = \frac{1}{2} \times \angle BOC$$

$$\angle BAC = \frac{1}{2} \times 48 = 24^\circ$$

$$\angle ACE = \frac{1}{2} \times \angle AOD$$

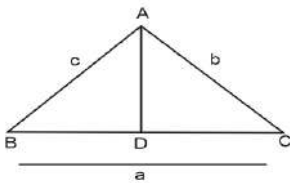
$$\angle ACE = \frac{1}{2} \times 100 = 50^\circ$$

$$\angle CEB = \angle BAC + \angle ACE \text{ (external opposite angle)}$$

$$= 24^\circ + 50^\circ = 74^\circ$$

Sol.115.(c) $BC = a$ cm

$$\text{Let } BD = x, DC = a - x$$



We know that, if AD is bisector of $\angle BAC$ then,

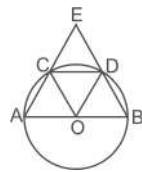
$$\frac{AB}{AC} = \frac{BD}{DC} \Rightarrow \frac{c}{b} = \frac{x}{a-x}$$

$$\Rightarrow ac - cx = bx \Rightarrow x = \frac{ac}{b+c}$$

$$\Rightarrow DC = a - x \Rightarrow DC = a - \frac{ac}{b+c} = \frac{ab}{b+c}$$

$$BD - DC = \frac{ac}{b+c} - \frac{ab}{b+c} = \frac{a(c-b)}{c+b}$$

Sol.116.(d)



$$\text{Given, } \angle CED = 70^\circ$$

In $\triangle AEB$

$$\angle EAB + \angle EBA = 180^\circ - \angle CED$$

$$= 180^\circ - 70^\circ = 110^\circ$$

In $\triangle AOC$

$$\angle OAC = \angle OCA \quad \dots (\text{OC} = \text{OA} = \text{radius})$$

$$\angle COA = 180^\circ - (\angle OAC + \angle OCA)$$

$$= 180^\circ - 2\angle OAC$$

In $\triangle BOD$

$$\angle OBD = \angle ODB \quad \dots (\text{OD} = \text{OB} = \text{radius})$$

$$\angle BOD = 180^\circ - (\angle OBD + \angle ODB)$$

$$= 180^\circ - 2\angle OBD$$

$$\angle AOB = \angle COA + \angle BOD + \angle COD = 180^\circ$$

$$\Rightarrow \angle COA + \angle BOD + \angle COD = 180^\circ$$

$$\Rightarrow 180^\circ - 2\angle OAC + 180^\circ - 2\angle OBD$$

$$+ \angle COD = 180^\circ$$

$$\Rightarrow 180^\circ - 2(\angle OAC + \angle OBD) + \angle COD = 0$$

$$\Rightarrow 180^\circ - 2(110^\circ) + \angle COD = 0$$

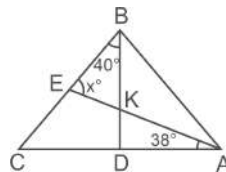
$$\Rightarrow -40^\circ + \angle COD = 0 \Rightarrow \angle COD = 40^\circ$$

Short trick :-

$$\angle COD = 180^\circ - 2 \times \angle CED$$

$$= 180^\circ - 2 \times 70^\circ = 180^\circ - 140^\circ = 40^\circ$$

Sol.117.(a)



$$\text{Given, } BD \perp AC, \angle EAC = 38^\circ \text{ and } \angle EBD = 40^\circ$$

$$\text{In } \triangle ADK, \angle DKA = 180^\circ - \angle KDA - \angle DAK$$

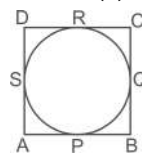
$$= 180^\circ - 90^\circ - 38^\circ = 52^\circ$$

$$\angle DKA = \angle BKE \text{ (Vertically opposite angles)}$$

$$\text{In } \triangle BEK, \angle BEK = 180^\circ - \angle BKE - \angle EBK$$

$$= 180^\circ - 52^\circ - 40^\circ = 88^\circ$$

Sol.118.(d)



$$\text{Given, } AS = 8 \text{ cm, } BC = 11 \text{ cm and } CR = 5 \text{ cm}$$

$$AS = AP = 8 \text{ cm} \quad (\text{Tangent of same circle})$$

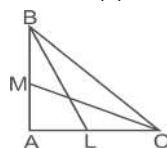
$$CQ = RC = 5 \text{ cm} \quad (\text{Tangent of same circle})$$

$$BQ = BC - QC = 11 - 5 = 6 \text{ cm}$$

$$BP = BQ = 6 \text{ cm} \quad (\text{Tangent of same circle})$$

$$AB = AP + BP = 8 + 6 = 14 \text{ cm}$$

Sol.119.(b)



In $\triangle ABL$

$$BL^2 = AB^2 + AL^2 \Rightarrow BL^2 = AB^2 + \left(\frac{AC}{2}\right)^2$$

$$\Rightarrow 4BL^2 = 4AB^2 + (AC)^2 \dots\dots(1)$$

In $\triangle CAM$

$$CM^2 = AC^2 + AM^2 \Rightarrow CM^2 = AC^2 + \left(\frac{AB}{2}\right)^2$$

$$\Rightarrow 4CM^2 = 4AC^2 + (AB)^2 \dots\dots(2)$$

Adding (1) and (2)

$$\Rightarrow 4BL^2 + 4CM^2 = 4AB^2 + (AC)^2 + 4$$

$$AC^2 + (AB)^2$$

$$\Rightarrow 4BL^2 + 4CM^2 = 5AB^2 + 5AC^2$$

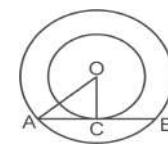
$$\Rightarrow 4(BL^2 + CM^2) = 5(AB^2 + AC^2)$$

$$\Rightarrow 4(BL^2 + CM^2) = 5(BC^2)$$

Short trick :-

$$\text{Its direct relation, } 4(BL^2 + CM^2) = 5(BC^2)$$

Sol.120.(a) Let OA be the radius of a bigger circle and OC be the radius of a smaller circle. And AB be the chord of a bigger circle.

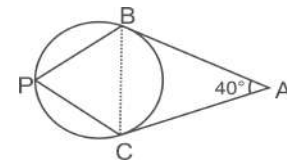


$$OA^2 = OC^2 + AC^2$$

$$AC = \sqrt{15^2 - 9^2} = 12 \text{ cm}$$

$$AB = 2 \times AC = 2 \times 12 = 24 \text{ cm}$$

Sol.121.(d)



In $\triangle ABC$

$$AB = AC \quad \dots (\text{tangent to the circle})$$

$$\angle ABC = \angle ACB \quad \dots (\text{AB} = \text{AC})$$

$$\angle ABC + \angle ACB + \angle BAC = 180^\circ$$

$$2\angle ABC + 40^\circ = 180^\circ \Rightarrow \angle ABC = 70^\circ$$

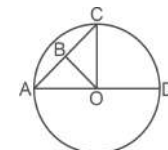
$$\angle BPC = 90^\circ - \frac{\angle BAC}{2} = 90^\circ - \frac{40}{2} = 70^\circ$$

$$\angle ABP + \angle BPC = 180^\circ \quad \dots (\text{CP} \parallel \text{AB})$$

$$\Rightarrow \angle ABP = 180^\circ - 70^\circ = 110^\circ$$

$$\Rightarrow \angle CBP = \angle ABP - \angle ABC = 110^\circ - 70^\circ = 40^\circ$$

Sol.122.(d)



$$\angle OCA = \angle OAC \quad (\text{OA} = \text{OC} = \text{radius})$$

$$\angle DOC = 60^\circ$$

$$\angle OCA + \angle OAC = \angle DOC$$

(external opposite angle)

$$2\angle OCA = 60^\circ \Rightarrow \angle OCA = 30^\circ$$

In $\triangle OBC$

$$\angle OCB = 30^\circ$$

$$\angle OBC = 180^\circ - \angle OBA$$

$$= 180^\circ - 60^\circ \quad (\angle OBA = 60^\circ \text{ given})$$

$$= 120^\circ$$

$$\angle BOC = 180^\circ - \angle OCB - \angle OBC$$

$$= 180^\circ - 30^\circ - 120^\circ = 30^\circ$$



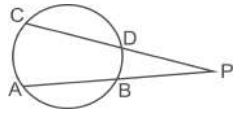
$\Rightarrow OB = BC = 5 \text{ cm} \dots (\angle OCB = \angle BOC)$

Short trick :-

Remember, in such type of questions $OB = BC = (\text{given value})$

Sol.123.(d) $CD = 3$, $PD = 5$ and $AB = 6$ given

$\Rightarrow PC = PD + CD = 5 + 3 = 8$



Let $PB = x$, Then, $PA = x + 6$

We know that

$PB \times PA = PD \times PC$

$x(x + 6) = 5 \times 8 \Rightarrow x(x + 6) = 40$

Put $x = 4$

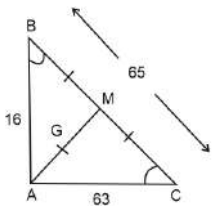
$x(x + 6) = 40 \Rightarrow 4(4 + 6) = 40$

The Condition satisfied clearly $x = 4 \text{ cm}$

So, $PB = 4 \text{ cm}$

Sol.124.(a)

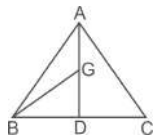
Pythagoras triplets :- (16, 63, 65)



Here, AM is the median of the right angle triangle ABC. We know that,

$BM = MC = AM = \frac{BC}{2} = 32.5 \text{ cm}$

Sol.125.(c)



Given, $AG : GD = 2 : 1$

Let $AG = 2 \text{ unit}$ And $GD = 1 \text{ unit}$

Height of triangles ABG and BDG will be the same as both the triangles are formed on the same base line and have a common vertex B. Let the height is h.

$$\frac{\text{ar}\triangle ABG}{\text{ar}\triangle BDG} = \frac{\frac{1}{2} \times h \times AG}{\frac{1}{2} \times h \times DG}$$

$$\Rightarrow \frac{\frac{1}{2} \times h \times 2}{\frac{1}{2} \times h \times 1} = \frac{2}{1}$$

Area of $\triangle ABD = \text{ar}\triangle ABG + \text{ar}\triangle BDG$

$= 2 + 1 = 3 \text{ unit}$

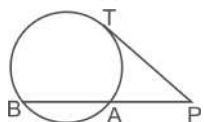
Area of $\triangle ABC = 2 \times \text{Area of } \triangle ABD$

(AD is median)

$= 2 \times 3 = 6 \text{ unit}$

$\text{ar}(\triangle BDG) : (\triangle ABC) = 1 : 6$

Sol.126.(a)



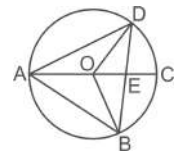
We know that

$PT^2 = PA \times PB$

$$5^2 = 4 \times (4 + x) \Rightarrow 25 - 16 = 4x$$

$$x = \frac{9}{4} = 2.25$$

Sol.127.(a)



Given, $\angle BOC = 50^\circ$

We, know that, $2\angle BAC = \angle BOC = 50^\circ$

$\angle BAC = 25^\circ$

Similarly, $2\angle ABD = \angle AOD = 110^\circ$

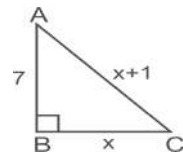
$\angle ABD = 55^\circ$

In triangle ABE,

$\angle BAE + \angle ABE = \angle BEC$ (exterior angle property)

$25^\circ + 55^\circ = \angle BEC \Rightarrow \angle BEC = 80^\circ$

Sol.128.(b)



$$\Rightarrow 7^2 + x^2 = (x + 1)^2$$

$$\Rightarrow 49 + x^2 = (x^2 + 1 + 2x) \Rightarrow x = 24$$

$$\sec C = \frac{AC}{BC} = \frac{25}{24} \Rightarrow \cot A = \frac{AB}{BC} = \frac{7}{24}$$

$$(\sec C + \cot A) = \frac{25}{24} + \frac{7}{24} = \frac{32}{24} = \frac{4}{3}$$

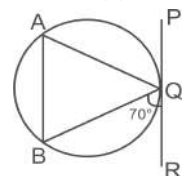
Short trick :-

Triplet 7, 24, 25 satisfies the given condition.

So, $BC = 24 \text{ cm}$

$$(\sec C + \cot A) = \frac{25}{24} + \frac{7}{24} = \frac{32}{24} = \frac{4}{3}$$

Sol.129.(a)



$\angle BQR = 70^\circ$ (given)

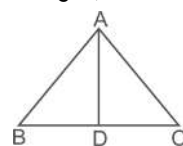
$\Rightarrow \angle BQR = \angle ABQ = 70^\circ$ (Alternate Angle)

$\Rightarrow \angle BQR = \angle BAQ = 70^\circ$

...(Alternate segment theorem)

$\Rightarrow \angle AQB = 180^\circ - 70^\circ - 70^\circ = 40^\circ$

Sol.130.(d) Let $\triangle ABC$ be the given triangle, $AD = 12 \text{ cm}$ and $BC = 10 \text{ cm}$



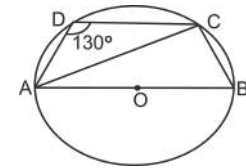
We know that the altitude of an isosceles triangle divides the base into two equal parts.

$$\Rightarrow BD = DC = \frac{BC}{2} = 5 \text{ cm}$$

In $\triangle ABD \Rightarrow AD^2 + BD^2 = AB^2$

$$\Rightarrow AB = \sqrt{12^2 + 5^2} = 13 \text{ cm}$$

Sol.131.(a)



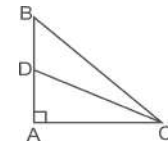
$\angle ADC + \angle ABC = 180^\circ$ (cyclic quadrilateral)

$\Rightarrow \angle ABC = 180^\circ - 130^\circ = 50^\circ$

$\Rightarrow \angle ACB = 90^\circ$ (Angle of semicircle)

$\angle BAC = 180^\circ - \angle ABC - \angle ACB$
 $= 180^\circ - 90^\circ - 50^\circ = 40^\circ$

Sol.132.(a)



In $\triangle ABC$

$AB^2 + AC^2 = BC^2$ (Pythagoras Theorem)

$$\Rightarrow AB^2 + 6^2 = 10^2$$

$$\Rightarrow AB = \sqrt{10^2 - 6^2} = 8$$

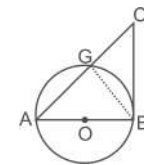
$AD = AB - BD = 8 - 4 = 4 \text{ cm}$

In $\triangle ADC$

$AD^2 + AC^2 = DC^2$

$$\Rightarrow 4^2 + 6^2 = DC^2 \Rightarrow DC = \sqrt{52} = 2\sqrt{13}$$

Sol.133.(d)



In $\triangle AGB$

$AG = 8 \text{ cm}$ and $AB = 12 \text{ cm}$ (given)

$\angle AGB = 90^\circ$ (Angle made by Diameter)

$\Rightarrow BG = \sqrt{AB^2 - AG^2}$ (Pythagoras Theorem)

$$\Rightarrow \sqrt{12^2 - 8^2} = 4\sqrt{5}$$

In $\triangle ABG$ and $\triangle ABC$

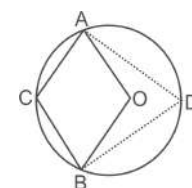
$\angle BAG = \angle BAC$ (same angle)

$\angle BGA = \angle ABC$ (Both 90°)

$$\Rightarrow \triangle AGB \sim \triangle ABC \Rightarrow \frac{BG}{BC} = \frac{AG}{AB}$$

$$\Rightarrow \frac{4\sqrt{5}}{BC} = \frac{8}{12} \Rightarrow BC = 6\sqrt{5} \text{ cm}$$

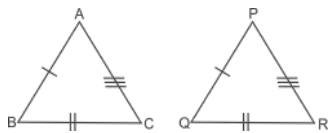
Sol.134.(d) Let $\angle ACB = x$ and $\angle CBO = y$
 $\Rightarrow \angle ADB = 180^\circ - x \Rightarrow \angle AOB = 2(180^\circ - x)$



$x + y = 180^\circ$ (AC || OB)

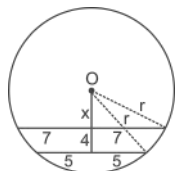
$\Rightarrow y = 180^\circ - x$



Sol.152.(a)

When all three sides of one triangle are equal to the three sides of another triangle, the triangles are congruent by Side-Side-Side (SSS) rule.

Sol.153.(b) In a cyclic quadrilateral, opposite angles are supplementary.
So, $\angle P + \angle R = 180^\circ \Rightarrow 110^\circ + \angle R = 180^\circ$
 $\Rightarrow \angle R = 70^\circ$.

Sol.154.(a)

Let dist. of 14 cm chord from centre = x cm

Then dist. of 10 cm chord = x + 4 cm

Now, $7^2 + x^2 = r^2$

$5^2 + (x + 4)^2 = r^2$

$\Rightarrow 49 + x^2 = 25 + x^2 + 8x + 16$

$\Rightarrow 49 = 41 + 8x \Rightarrow x = 1$ cm

$r^2 = 49 + 1 = 50 \Rightarrow r = \sqrt{50} = 7.07$ cm

Radius = 7.07 cm

Sol.155.(d) 1 radian = 57.3°

$3.5 \times 57.3 = 200.55^\circ$

Sol.156.(b) We know that in similar triangles ratio of their area = (ratio of corresponding altitudes) $^2 = 9 : 16$

Sol.157.(a) In any circle, the longest chord is the one that passes through the center of the circle, and such a chord is called the diameter of the circle.
Therefore, the maximum length of the chord = 15 cm.

Sol.158.(d) D = 14 cm and radius = 7 cm

For an equilateral triangle with inradius r

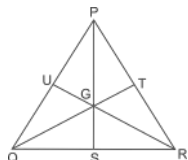
$a = 2\sqrt{3}r$

$a = 2 \times 1.73 \times 7 = 24.22$ cm

Area of equilateral triangle = $\frac{\sqrt{3}}{4} \times 24.22$
 $\times 24.22 \approx 254.00$ cm 2

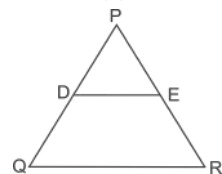
Area of circle = $\pi \times 7 \times 7 = \frac{22}{7} \times 7 \times 7$
 $= 154$ cm 2

Remaining portion = $254.00 - 154$
 $= 100.00$ cm 2

Sol.159.(b)

These three medians split triangle PQR into 6 smaller triangles of equal area. Triangle GPQ is made of 2 of these triangles.

$$\text{Reqd. ratio} = \frac{\text{ar. of } \triangle GPQ}{\text{ar. of } \triangle PQR} = \frac{2}{6} = \frac{1}{3}$$

Sol.160.(a)

DE $\parallel$ QR

Area($\triangle PDE$) : Area of trapezium DERQ
 $= 4 : 21$

Total area of $\triangle PQR$ = Area of $\triangle PDE$ +
Area of trapezium DERQ

Area ($\triangle PDE$) : Area ($\triangle PQR$) = $4 : (4 + 21)$
 $= 4 : 25$

We know that, DE $\parallel$ QR

$$\frac{4}{25} = \left(\frac{PD}{PQ}\right)^2 \Rightarrow \left(\frac{2}{5}\right)^2 = \left(\frac{PD}{PQ}\right)^2$$

$$\frac{2}{5} = \left(\frac{PD}{PQ}\right)$$

PQ = PD + DQ

$5 = 2 + DQ \Rightarrow DQ = 3$

$$\frac{PD}{DQ} = \frac{2}{3}$$

Sol.161.(d) Direct common tangent

$$= \sqrt{d^2 - (r_1 - r_2)^2}$$

$$24 = \sqrt{d^2 - (12 - 5)^2}$$

$$576 + 49 = d^2$$

$$d = \sqrt{625} = 25$$
 cm

Sol.162.(a) Direct common tangent (P)

$$= \sqrt{D^2 - (r_1 - r_2)^2}$$

Transverse common tangent (Q)

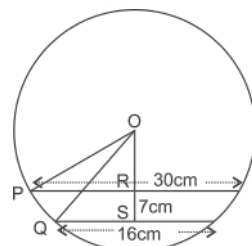
$$= \sqrt{D^2 - (r_1 + r_2)^2}$$

$$P^2 + Q^2 = 2D^2 - [(r_1 - r_2)^2 + (r_1 + r_2)^2]$$

$$400 = 2D^2 - 2(r_1^2 + r_2^2)$$

$$400 + 2(200) = 2D^2$$

$$400 = D^2 \Rightarrow D = 20$$
 cm

Sol.163.(d)

Let the length of OR = x cm

In $\triangle OPR$

$$OP^2 = PR^2 + OR^2$$

$$OP^2 = 15^2 + x^2$$

In $\triangle OQS$

$$OQ^2 = QS^2 + OS^2$$

$$OQ^2 = 8^2 + (x + 7)^2$$

Since both OP and OQ are radius then they both should be equal

$$225 + x^2 = 64 + x^2 + 49 + 14x$$

$$112 = 14x \Rightarrow x = 8$$
 cm

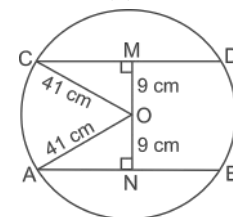
In $\triangle OPR$

$$OP^2 = 225 + 64 = 289$$

$$OP = 17$$
 cm = Radius

$$\text{Diameter} = 34$$
 cm

Sol.164.(c) the circles are touching externally, so 3 common tangent exists

Sol.165.(d)

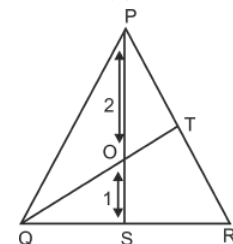
In $\triangle OAN$,

$$\Rightarrow AN^2 = AO^2 - ON^2$$

$$\Rightarrow AN = \sqrt{41^2 - 9^2}$$

$$= \sqrt{1681 - 81} = 40$$
 cm

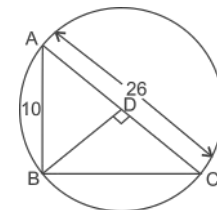
So, the length of the chord AB = 2×40
 $= 80$ cm

Sol.166.(a)

A/Q, PS (3 unit) $\rightarrow$ 15 cm

$$PO (2 \text{ unit}) \rightarrow \frac{15 \times 2}{3} = 10$$
 cm

Sol.167.(a) If one side and hypotenuses is the same of two triangles, the triangle must be congruent by RHS rule.

Sol.168.(a)

We know that, $AB^2 = AD \times AC$

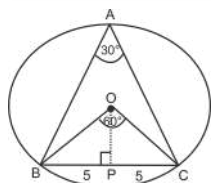
$$\Rightarrow 100 = AD \times 26$$

$$\Rightarrow AD = \frac{100}{26} = 3.8$$
 cm

Sol.169.(a) There are three common tangents to the circles

Sol.170.(c) We know in this case,
Length of Direct common tangent (D)
 $= 2\sqrt{Rr}$

Sol.171.(a)



In $\triangle OPB$, $\angle P = 90^\circ$, $\angle O = 30^\circ$,
 $\therefore \angle B = 180^\circ - (90^\circ + 30^\circ) = 60^\circ$

Now, $\tan 60^\circ = \frac{OP}{PB}$

$$\Rightarrow \sqrt{3} = \frac{OP}{5} \Rightarrow OP = 5\sqrt{3} \text{ cm}$$

Sol. 172.(a) Area of sector = $\frac{\theta}{360} \times \pi r^2$

$$= \frac{60}{360} \times \pi \times 4 \times 4 = \frac{8\pi}{3}$$

Area of equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

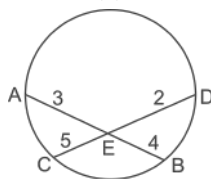
$$= \frac{\sqrt{3}}{4} \times 4 \times 4 = 4\sqrt{3}$$

Area of the segment formed by a chord

$$= \frac{8\pi}{3} - 4\sqrt{3}$$

Sol. 173.(b) Each pair of disjoint circles has 4 common tangents. Number of pairs = ${}^4C_2 = 6$. If no tangent is common to more than two circles, total $6 \times 4 = 24$

Sol. 174.(a)



Chord-intersection theorem:

$$AE \times EB = CE \times ED$$

Given

$$AE : EB = 3 : 4 \Rightarrow AE = 3k, EB = 4k$$

$$\Rightarrow AE \times EB = 12k^2.$$

$$CE : ED = 5 : 2 \Rightarrow CE = 5m, ED = 2m$$

$$\Rightarrow CE \times ED = 10m^2.$$

$$\text{So, } 12k^2 = 10m^2 \Rightarrow \frac{k^2}{m^2} = \frac{10}{12} = \frac{5}{6}$$

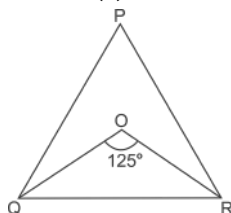
$$\text{Now } AB = AE + EB = 7k$$

$$\text{and } CD = CE + ED = 7m$$

$$\frac{x}{y} = \frac{AB}{CD} = \frac{7k}{7m} = \frac{k}{m} = \sqrt{\frac{5}{6}}$$

$$\text{Or } \frac{x^2}{y^2} = \frac{5}{6}$$

Sol. 175.(b) Given that $\angle QOR = 125^\circ$



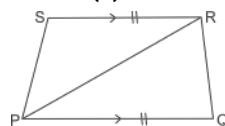
Using the angle bisector theorem and the relation:

$$\angle QOR = 90^\circ + \frac{1}{2} \angle QPR$$

$$\Rightarrow 125^\circ = 90^\circ + \frac{1}{2} \angle QPR$$

$$\Rightarrow \frac{1}{2} \angle QPR = 35^\circ \Rightarrow \angle QPR = 70^\circ$$

Sol. 176.(b)



In $\triangle PQR$ and $\triangle RSP$

PQ = RS (Given)

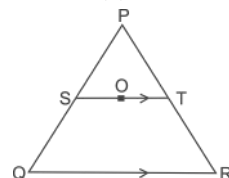
PR = PR (Common side)

$\angle QPR = \angle SRP$ (Alternate interior angles,

Since PQ $\parallel$ RS)

Hence, $\triangle PQR \cong \triangle RSP$ by SAS

Sol. 177.(c)



The centroid O divides each median in the ratio 2 : 1.

Since QR $\parallel$ ST and passes through the centroid, $\triangle PST$ and $\triangle PQR$ are similar triangles with side ratio 2 : 3.

$$\text{Area ratio} = \left(\frac{2}{3}\right)^2 = \frac{4}{9} \quad (\triangle PST \text{ to } \triangle PQR)$$

Therefore, $\text{STRQ} = \text{PQR} - \text{PST}$

$$= 1 - \frac{4}{9} = \frac{5}{9}$$

Therefore, the area ratio PST : STRQ

$$= \frac{4}{9} : \frac{5}{9} = 4 : 5$$

Sol. 178.(c) Given :- $r_1 = 9 \text{ cm}$ & $r_2 = 5 \text{ cm}$

Distance between centres(d) = 15 cm

Formula for internal common tangent:

$$\begin{aligned} L &= \sqrt{d^2 - (r_1 + r_2)^2} \\ &= \sqrt{15^2 - (9 + 5)^2} \\ &= \sqrt{225 - 196} = \sqrt{29} \text{ cm} \end{aligned}$$

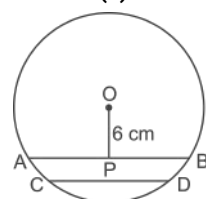
Sol. 179.(a) Given:

$$r_1 = 14 \text{ cm}, r_2 = 6 \text{ cm}, d = 40 \text{ cm}$$

For the transverse common tangent, the formula is:

$$\begin{aligned} L &= \sqrt{d^2 - (r_1 + r_2)^2} \\ &= \sqrt{40^2 - (14 + 6)^2} \\ &\Rightarrow \sqrt{1600 - 400} = \sqrt{1200} \text{ cm} \end{aligned}$$

Sol. 180.(c)



Let the radius of the circle = r.

Since the radius is the same for both chords, we can set r^2 equal to -

$$\Rightarrow \left(\frac{10}{2}\right)^2 + (6)^2 = \left(\frac{8}{2}\right)^2 + (d_2)^2$$

$$\Rightarrow 25 + 36 = 16 + (d_2)^2$$

$$\Rightarrow (d_2)^2 = 45 \Rightarrow d_2 = \sqrt{45} = 3\sqrt{5} \text{ cm}$$

Hence, the distance of an 8 cm chord from the center is greater than 3 cm.

Sol. 181.(c) Given: $\theta = 150^\circ$ and $R = r$

According to the question,

$$\Rightarrow \text{side} = 2r \sin \theta \Rightarrow a = 2R \sin(150^\circ)$$

$$\Rightarrow a = 2R \sin(180^\circ - 30^\circ)$$

$$\Rightarrow a = 2R \left(\frac{1}{2}\right) \Rightarrow a = R$$

Sol. 182.(c)

Only one right or one obtuse angle

Sol. 183.(a) Given: Area of $\triangle PQR = 8 \text{ cm}^2$

Side of $\triangle DEF = 2 \times (\text{Side of } \triangle PQR)$

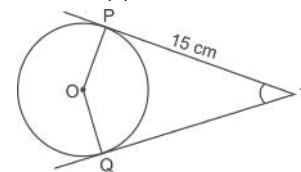
We know that,

$$\Rightarrow \frac{\text{Area of } \triangle PQR}{\text{Area of } \triangle DEF} = \frac{PQ^2}{DE^2}$$

$$\Rightarrow \frac{8}{\text{Area of } \triangle DEF} = \frac{1^2}{2^2}$$

$$\Rightarrow \text{Area of } \triangle DEF = 32 \text{ cm}^2$$

Sol. 184.(c)



The lengths of the two tangents drawn from an external point to a circle are equal.

$$\Rightarrow TP = TQ$$

$$\text{So, } TQ = 15 \text{ cm}$$

Sol. 185.(c)

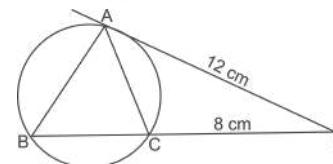
(12, 16, 20) is a Pythagorean triplet.

So, $a = 12$, $b = 16$, $c = 20$

$$r = \frac{a + b - c}{2}$$

$$= \frac{12 + 16 - 20}{2} = \frac{8}{2} = 4 \text{ cm}$$

Sol. 186.(a)



$$AD^2 = CD \times BD$$

$$(12)^2 = 8 \times (8 + BC)$$

$$\frac{144}{8} = 8 + BC$$

$$18 - 8 = BC \Rightarrow BC = 10 \text{ cm}$$

Sol. 187.(c) Direct common tangent

$$= \sqrt{d^2 - (r_1 - r_2)^2}$$

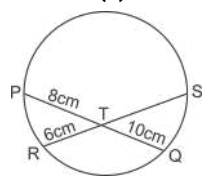


Direct common tangent

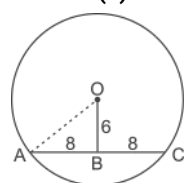
$$= \sqrt{36x^2 - (2x)^2} = \sqrt{32}x = 4\sqrt{2}x$$

Sol.188.(b) $r_1 = 5\text{cm}$, $r_2 = 3\text{cm}$ and $d = 8\text{cm}$ $r_1 + r_2 = 8\text{cm}$ and $r_1 - r_2 = 2\text{cm}$

We know that if $d = r_1 + r_2$ then the circles touch externally and there will be 3 tangents. (2 direct and 1 transverse).

Sol.189.(a)

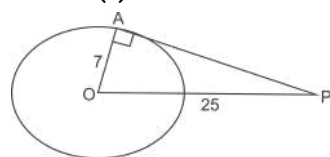
A/Q,
 $PT \times TQ = RT \times TS$
 $8 \times 10 = 6 \times TS$
 $\frac{40}{3} \text{ cm} = TS$

Sol.190.(b)

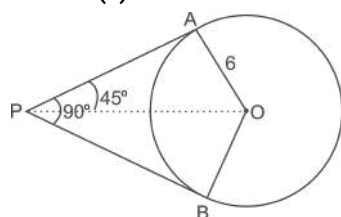
In $\triangle OAB$
 $OA^2 = AB^2 + OB^2$
 $OA^2 = 64 + 36$
 $OA^2 = 100 \Rightarrow OA = 10 \text{ cm}$

Sol.191.(b)Sum of interior angles $= (n - 2) \times 180^\circ$ Given: $n = 8$

$$\Rightarrow (8 - 2) \times 180^\circ = 1080^\circ$$

Sol.192.(c)

In right $\triangle OAP$,
 $(25)^2 = (7)^2 + (PA)^2$
 $\Rightarrow PA = 24 \text{ cm}$

Sol.193.(b)

Now, in quadrilateral AOBP,
 $\angle AOP = 360^\circ - (90^\circ + 90^\circ + 90^\circ) = 90^\circ$
Hence, quadrilateral AOBP is square.
We know that diagonal of square
 $= \text{side} \sqrt{2}$
 $\therefore OP = 6\sqrt{2} \text{ cm}$

Sol.194.(c) According to the question,

$$\text{Side of the hexagon} = \frac{24}{6} = 4 \text{ cm}$$

$$\text{Side of the octagon} = \frac{3}{4} \times 4 = 3 \text{ cm}$$

$$\text{Perimeter of the octagon} = 8 \times 3 = 24 \text{ cm}$$

$$\text{Sol.195.(a)} \quad D = \frac{n(n-3)}{2}$$

For decagon, $n = 10$

$$D = \frac{10(10-3)}{2} = \frac{10 \times 7}{2} = 35$$

Sol.196.(a) Internal tangent

$$= \sqrt{d^2 - (R_1 + R_2)^2}$$

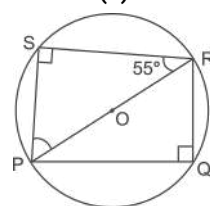
$$12 = \sqrt{20^2 - (10 + R_2)^2}$$

Squaring on both sides

$$144 = 400 - (10 + R_2)^2$$

$$(10 + R_2)^2 = 256 \Rightarrow (10 + R_2) = \sqrt{256}$$

$$R_2 = 16 - 10 = 6 \text{ cm}$$

Sol.197.(b)

Since PR is the diameter, the angle subtended by it at any point on the circle is 90° .

In $\triangle PRS$,

$$\angle PSR = 90^\circ, \angle PRS = 55^\circ$$

$$\text{So, } \angle RPS = 180^\circ - 90^\circ - 55^\circ = 35^\circ$$

Sol.198.(b) Each interior angle

$$= \frac{(n-2) \times 180^\circ}{n} = \frac{(12-2) \times 180^\circ}{12} = 150^\circ$$

Sol.199.(c) Let parallel sides $= 2x, 3x$

Area of trapezoid

$$= \frac{1}{2}(\text{sum of parallel sides}) \times \text{height}$$

$$360 = \frac{1}{2}(2x + 3x) \times 12$$

$$360 = 30x \Rightarrow x = 12$$

$$\text{So, sides} = 24, 36$$

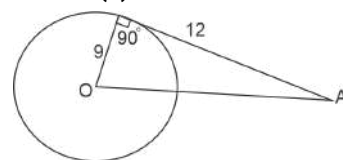
Sol.200.(a)

Since $\triangle MNO$ is congruent to $\triangle PQR$, their corresponding sides will also be equal.

$$PQ = MN = 8, QR = NO = 9$$

$$RP = OM = 10$$

$$\text{Perimeter} = MN + NO + OM = 8 + 9 + 10 = 27$$

Sol.201.(d)

The tangent line forms a right angle with the radius at the point where it meets the circle.

Using the Pythagorean theorem:

$$AO^2 = 12^2 + 9^2$$

$$AO^2 = 144 + 81$$

$$AO^2 = 225 \Rightarrow AO = 15$$

Sol.202.(a) The lengths of the two tangent segments drawn from an external point to a circle are equal. Therefore, the length of the second tangent will be 15.

Sol.203.(b)Sum of interior angles $= (n - 2) \times 180^\circ$ For a nonagon ($n = 9$)

$$(9 - 2) \times 180^\circ = 7 \times 180^\circ = 1260^\circ$$

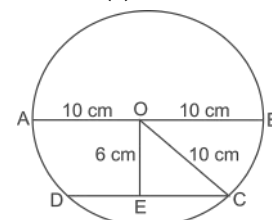
Sol.204.(c) Given: $L = 4$, $D = 5$ Let radius $= 2r$, r

For direct common tangent:

$$L = \sqrt{D^2 - (R - r)^2}$$

$$4 = \sqrt{5^2 - (2r - r)^2}$$

$$16 = 25 - r^2 \Rightarrow r = 3 \text{ cm}$$

Sol.205.(a)

$$\text{Radius } (r) = \frac{20}{2} = 10 \text{ cm}$$

In $\triangle OEC$,

Using Pythagoras theorem

$$(OC)^2 = (OE)^2 + (EC)^2$$

$$(10)^2 = (6)^2 + (EC)^2 \Rightarrow EC = 8 \text{ cm}$$

length of the chord (DC) $= 8 + 8 = 16 \text{ cm}$

Sol.206.(c) Diagonals bisect each other in a parallelogram. That is, each diagonal cuts the other into two equal parts.

Sol.207.(d) As $8^2 + 15^2 = 17^2$, the set (8, 15, 17) is a Pythagorean triplet

$$\text{Area}_1 = \frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$$

Since the second triangle is similar and its hypotenuse is 51 cm, then the scale factor

$$k = \frac{51}{17} = 3$$

$$\text{Area}_2 = \text{Area}_1 \times 3^2 = 60 \times 9 = 540 \text{ cm}^2$$

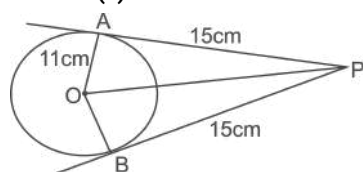
Sol.208.(a) In triangle ABC and PQR

$$AB = PQ$$

$$AC = PR$$

$$BC = QR$$

Since all three sides are equal, the SSS congruence rule applies.

Sol.209.(c)

By the Pythagorean theorem in triangle APO

$$\text{Hypotenuse}^2 = \text{Base}^2 + \text{Perpendicular}^2$$

$$\text{Hypotenuse} = \sqrt{15^2 + 11^2}$$

$$= \sqrt{346} = 18.6 \approx 19 \text{ cm}$$

Sol.210.(b) By the Pythagorean theorem

$$\text{Hypotenuse} = \sqrt{9^2 + 12^2} = \sqrt{225} = 15 \text{ cm}$$

Radius of the incircle

$$= \frac{\text{Base} + \text{Perpendicular} - \text{Hypotenuse}}{2}$$

$$r = \frac{9 + 12 - 15}{2} = 3$$

Sol.211.(b) Number of sides in a polygon

$$= \frac{360^\circ}{\text{Exterior angle}} \Rightarrow \frac{360^\circ}{72^\circ} = 5$$

Sol.212.(c) we know that,

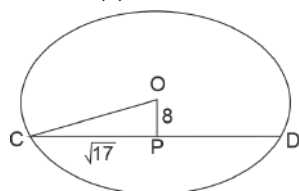
$$\text{Distance between the centres} =$$

$$\sqrt{(\text{transverse common tangent})^2 - (r_1 + r_2)^2}$$

According to question,

$$d = \sqrt{(d)^2 - (r_1 + r_2)^2}$$

$$d^2 = (d)^2 - (r_1 + r_2)^2 \Rightarrow (r_1 + r_2) = 0$$

Sol.213.(d)

$$\text{Length of radius(OC)} = \sqrt{17 + 64} = 9$$

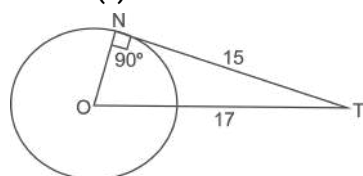
Sol.214.(a)

Let the number of sides of polygon = n

$$\text{each interior angle} = \frac{(n-2) \times 180^\circ}{n}$$

$$120^\circ = \frac{(n-2) \times 180^\circ}{n}$$

$$2n = 3n - 6 \Rightarrow n = 6$$

Sol.215.(c)

Radius of circle = NO

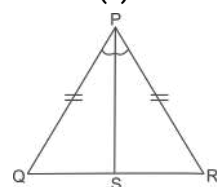
$$OT^2 = NO^2 + NT^2$$

$$17^2 = NO^2 + 15^2$$

$$NO^2 = 289 - 225 = 64$$

$$NO = 8$$

$$\text{Area of circle} = \pi r^2 \Rightarrow 64\pi \text{ cm}^2$$

Sol.216.(b)

PQ = PR, $\triangle PQR$ is an isosceles triangle

Therefore, $\angle PQR = \angle PRQ$

PS is the bisector of $\angle QPR$

So,

$$\angle QPS = \angle SPR$$

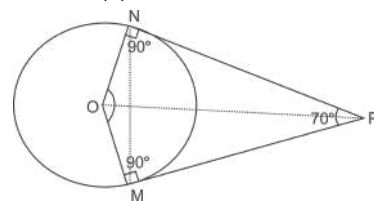
$\triangle PQS$ and $\triangle PRS$

PQ = PR,

$$\angle QPS = \angle SPR$$

PS is the common side

Therefore : Side - Angle - Side (SAS)

Sol.217.(b)

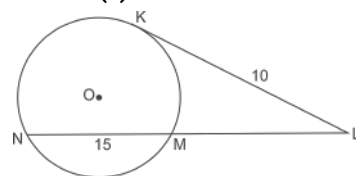
If two tangent segments are drawn to a circle from the same external point, then those segments are equal in length.

So, $\angle PNO = \angle PMO = 90^\circ$

$$\angle MON + \angle MPN = 180^\circ$$

$$\angle MON + 70^\circ = 180^\circ$$

$$\angle MON = 110^\circ$$

Sol.218.(c)

Let LM = y

$$LK^2 = LN \times LM$$

$$10^2 = (15 + y) y$$

$$100 = 15y + y^2$$

$$y^2 + 15y - 100 = 0$$

$$y = -20, 5 \Rightarrow y = 5 \text{ cm}$$

Sol.219.(d) Corresponding angles are equal when a transversal intersects parallel lines

First corresponding angle = 135°

Second corresponding angle = 135°

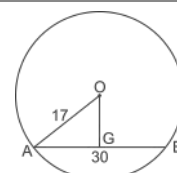
Sol.220.(a) $\triangle PQR \sim \triangle MNO$

So the ratio of corresponding sides is

$$\text{constant} = \frac{MN}{PQ} = \frac{16}{8} = 2$$

Therefore, QR corresponds to NO

$$NO = QR \times 2 = 10 \times 2 = 20 \text{ cm}$$

Sol.221.(b)

From Pythagoras' theorem in triangle $\triangle OAG$.

$$OA^2 = OG^2 + AG^2$$

$$17^2 = OG^2 + 15^2$$

$$OG = \sqrt{289 - 225} = 8 \text{ cm}$$

Sol.222.(c)

Corresponding angles are equal when a transversal intersects parallel lines

First corresponding angle = 100°

Second corresponding angle = 100°

Sol.223.(b) $\triangle PQR \sim \triangle MNO$

$P \leftrightarrow M$

$Q \leftrightarrow N$

$R \leftrightarrow O$

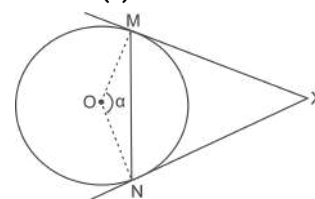
$$\angle P = 70^\circ \Rightarrow \angle M = 70^\circ$$

$$\angle Q = 80^\circ \Rightarrow \angle N = 80^\circ$$

$$\therefore \angle N = 80^\circ$$

Sol.224.(b) All chords of equal length are at the same distance from the center

$\therefore$ distance from center of second chord = 2 cm

Sol.225.(c)

Radii OM and ON are perpendicular to the tangents at points M and N. Therefore, the angle between the tangents ($\angle MXN$) and the angle subtended by the chord of contact at the center ($\angle MON = \alpha$) are supplementary. Hence $\angle MXN = 180^\circ - \alpha$

$$\text{Sol.226.(b) exterior angle} = \frac{360^\circ}{n}$$

$$\Rightarrow 30^\circ = \frac{360^\circ}{n} \Rightarrow n = \frac{360^\circ}{30} \Rightarrow n = 12$$

Sol.227.(a)

$$\text{Area of } \triangle PQS = \frac{1}{2} \times QS \times \text{height}$$

$$\text{Area of } \triangle PRS = \frac{1}{2} \times SR \times \text{height}$$

Since, QS = SR

Area of $\triangle PQS$ = Area of $\triangle PRS$

$$\text{the ratio of the areas is} = \frac{\Delta PQS}{\Delta PRS} = \frac{1}{1}$$

Sol.228.(c) Given that in $\triangle ABC$ and $\triangle DEF$,

$$\angle A = \angle D,$$

$$AB = DE, \text{ and } \angle B = \angle E$$

In $\triangle ABC$, there are $\angle A$ and $\angle B$ and the side AB between them. Similarly, in $\triangle DEF$, there are $\angle D$ and $\angle E$ and the side DE between them.

When two angles and the included side of one triangle are equal to the corresponding two angles and included side of another triangle, then the triangles are congruent by the ASA (Angle-Side-Angle) rule.

Therefore,
 $\triangle ABC \cong \triangle DEF$ (By ASA Rule)

Sol.229.(d) Given: Radius = 17 cm
 Distance from center to midpoint of chord (a) = 8 cm

Half the length of the chord (b)

The equation is: $8^2 + b^2 = 17^2$

for half the chord length = $64 + b^2 = 289$

$b^2 = 225 \Rightarrow b = \sqrt{225} = 15$ cm

full length of the chord is $2b$

$= 2 \times 15 = 30$ cm

Sol.230.(b) Given :-

A chord of a circle subtends an angle of 80° at the center.

The angle subtended at the center of a circle is twice the angle subtended at any point on the circle in the same segment.

Angle at center

$= 2 \times \text{Angle at circumference}$

Now,

$80^\circ = 2 \times \text{Angle at circumference}$

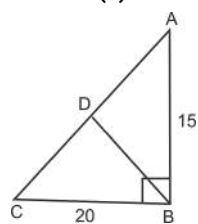
Angle at circumference = $\frac{80^\circ}{2} = 40^\circ$

Sol.231.(a) Sum of the interior angles

$= (n - 2) \times 180^\circ$

$= (18 - 2) \times 180^\circ = 16 \times 180^\circ = 2880^\circ$

Sol.232.(a)



(15, 20, 25) is a pythagorean triplet

So, The length of $BD = \frac{20 \times 15}{25} = 12$ cm

Sol.233.(b) The exterior angle at C is 112° which is the sum of interior angles A and B.

Let $\angle A = 3x$, $\angle B = 4x$

Then $3x + 4x = 112^\circ$

$7x = 112^\circ \Rightarrow x = 16^\circ$

Now $\angle A = 48^\circ$, $\angle B = 64^\circ$

$\angle C = 180^\circ - \text{Exterior angle at C}$

$\angle C = 180^\circ - 112^\circ = 68^\circ$

Sol.234.(a) $\triangle ABC$ is congruent to $\triangle DEF$ ($\triangle ABC \cong \triangle DEF$),
 their corresponding angles are equal.

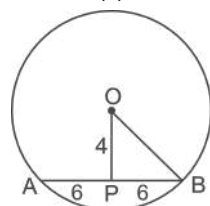
$\angle A$ corresponds to the $\angle D$.

Therefore, the measure of $\angle D$ is equal to the measure of $\angle A$.

Hence,

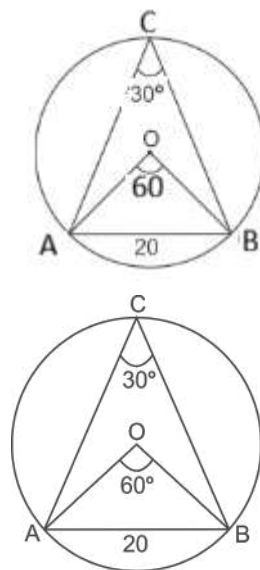
The measure of $\angle D$ is 40° .

Sol.235.(c)



$OB = \sqrt{4^2 + 6^2} = \sqrt{16 + 36} = \sqrt{52}$
 $= 7.2 \approx 7$ cm

Sol.236.(b)



In $\triangle BOC$, $OB = OC = \text{radius}$

$\angle O = 60^\circ$

$\therefore \triangle BOC$ is an equilateral triangle.

$OB = OC = BC = 20$ cm

Sol.237.(a) According to question,

$A + B = 180^\circ$ ----- (i)

$A = 2B - 30^\circ$ ----- (ii)

Putting the value of 'A' in equ.(i):-

$(2B - 30^\circ) + B = 180^\circ$

$3B - 30^\circ = 180^\circ$

$3B = 210 \Rightarrow B = 70^\circ$

Then, putting the value of 'B' in equ.(i)

$A + 70^\circ = 180^\circ$

$A = 110^\circ$

hence, larger angle = 110°

Sol.238.(c) According to the question,

$\angle BEC = 90^\circ - \frac{1}{2} \angle A$

$\Rightarrow 50^\circ = 90^\circ - \frac{1}{2} \angle A$

$\Rightarrow \angle A = 40^\circ \times 2 = 80^\circ$

Sol.239.(d) According to the question,

$LM = LO + OM$

$12 = 4 + OM$

$OM = 8$ cm

In $\triangle LMN$

OP is parallel to MN

By Thale's Theorem

$$\frac{LO}{OM} = \frac{LP}{PN} \Rightarrow \frac{4}{8} = \frac{5}{PN}$$

$PN = 10$ cm

Sol.240.(c) According to question,

$\Rightarrow \text{Sum of 9 interior angle} = 180^\circ \times (9 - 2)$
 $= 1260^\circ$

Now,

$\Rightarrow 9\text{th interior angle} = 1260^\circ - 1120^\circ$

$= 140^\circ$

We know that,

$\Rightarrow \text{Exterior angle} + \text{interior angle} = 180^\circ$

$\Rightarrow \text{Exterior angle} = 180^\circ - 140^\circ = 40^\circ$

Sol.241.(b) Given:

$AB = XY$, $AC = XZ$ and $\angle BAC = \angle YXZ$

Two sides and one angle are equal.

Rule of congruence : SAS (Side - Angle - Side)

Sol.242.(b) Given that,

the angle subtended by the chord at the center is 140°

Angle at circumference = $\frac{140}{2} = 70^\circ$

Sol.243.(c) A regular hexagon has 6 equal sides, so the center splits the figure into 6 equal central angles

Angle = $\frac{360^\circ}{\text{no. of side}}$

Angle = $\frac{360^\circ}{6} = 60^\circ$

Sol.244.(a) Given :

$r_1 = 7$ cm, $r_2 = 4$ cm

We know that

distance between centers d = sum of radii:

$d = r_1 + r_2 = 7 + 4 = 11$ cm

Sol.245.(c) We know: Angle at the centre = $2 \times$ (angle subtended by the same chord at the circumference).

So angle at the circumference = $\frac{1}{2} \times$ angle at center.

$\angle ACB = \frac{1}{2} \times \angle AOB$

$\angle ACB = \frac{1}{2} \times 100^\circ = 50^\circ$

Sol.246.(d) For two chords intersecting at right angles:

$r^2 = (\text{distance from O to AB})^2 + (\text{distance from O to CD})^2$

$r^2 = 6^2 + 8^2 = 36 + 64 = 100$

$r^2 = 100 \Rightarrow r = 10$

Sol.247.(a) Given, the angle subtended by a chord at the center is 120°

Angle at center = $2 \times$ Angle at the circumference

$120^\circ = 2 \times \text{Angle at circumference}$



$$\text{Angle at circumference} = \frac{120^\circ}{2} = 60^\circ$$

Sol.248.(b) We know

Area of a triangle in terms of its radius T
 $= r \times s$

Where $s = \frac{p+q+r}{2}$ is the semi-perimeter.

$$T = r \times s \Rightarrow r = \frac{T}{s}$$

$$\Rightarrow r = \frac{T}{\frac{p+q+r}{2}} = \frac{2T}{p+q+r}$$

Sol.249.(a) Let each interior angle $= x^\circ$

exterior angle $= 180^\circ - x^\circ$.

According to the question,

$$x - (180^\circ - x) = 60^\circ$$

$$\Rightarrow 2x - 180^\circ = 60^\circ \Rightarrow 2x = 240^\circ \Rightarrow x = 120^\circ$$

For a regular polygon,

$$\text{Exterior angle} = 180^\circ - 120^\circ = 60^\circ$$

$$\text{No. of sides (n)} = \frac{360^\circ}{60^\circ} = 6$$

Sol.250.(b)

Convert 0.6 radians to degrees

We know

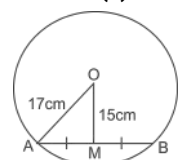
$$\text{Degree} = \text{radian} \times \frac{180}{\pi}$$

$$= 0.6 \times \frac{180}{\pi} = \frac{108}{\pi} \quad (\text{where } \pi = 3.14)$$

$$= \frac{108}{3.14} = 34.39^\circ$$

Sol.251.(c) In an acute-angled triangle, the orthocenter lies inside the triangle.

Sol.252.(c)



In $\triangle OAM$,

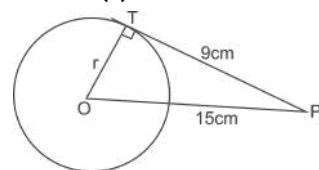
By using Pythagorean triplet (8, 15, 17)

Then, $AM = 8$ cm

$AM = BM = 8$ cm

So, $AB = 16$ cm

Sol.253.(a)



In $\triangle PTO$,

By Pythagorean triplet (9, 12, 15)

So, $r = 12$ cm

Sol.254.(c)

Length of direct common tangent

$$= \sqrt{d^2 - (r_1 - r_2)^2}$$

$$\Rightarrow \sqrt{30^2 - (16 - 6)^2}$$

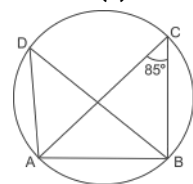
$$\Rightarrow \sqrt{900 - 100} = \sqrt{800} = 20\sqrt{2} \text{ cm.}$$

Sol.255.(a) A line from the center perpendicular to a chord always bisects the chord due to symmetry of the circle.

Sol.256.(a) The angle at the center is twice the angle at the circumference on the same arc.

$$\text{So, central angle} = 2 \times 90^\circ = 180^\circ.$$

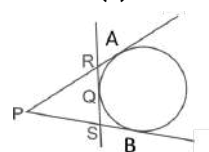
Sol.257.(c)



Angles in the same segment of a circle are equal.

$$\therefore \angle ADB = \angle ACB = 85^\circ$$

Sol.258.(a)



Tangents from P are equal, so

$$PA = PB = 20.$$

Let $RA = x$, and $SB = y$

then, $PR = 20 - x$, $PS = 20 - y$,

$RS = RQ + QS = x + y$,

$$\text{so perimeter} = (20 - x) + (20 - y) + (x + y) = 40 \text{ cm.}$$

Shortcut Method:-

$$\text{Given } PA = 20 \text{ cm, the perimeter of } \triangle PRS = 2 \times PA = 2 \times 20 \text{ cm} = 40 \text{ cm.}$$

Sol.259.(a) we know that:

$$\text{Each interior angle} = \frac{(n-2) \times 180^\circ}{n}$$

Interior angle of regular pentagon

$$= \frac{(5-2) \times 180^\circ}{5} = \frac{3 \times 180^\circ}{5} = 108^\circ$$

Interior angle of regular decagon

$$= \frac{(10-2) \times 180^\circ}{10} = 8 \times 18 = 144^\circ$$

$$\text{Ratio} \rightarrow 108^\circ : 144^\circ = 3 : 4$$

Sol.260.(a) Let angle P, Q, R = $2x$, $3x$, $4x$

We know that:

The sum of the angles of a quadrilateral is 360 degrees.

$$2x + 3x + 4x + 90^\circ = 360^\circ$$

$$9x = 270^\circ \Rightarrow x = 30^\circ$$

$$\text{So, } \angle Q = 3 \times 30^\circ = 90^\circ$$

$$\text{Sol.261.(d) Given: } \frac{5\pi}{6} \text{ radians}$$

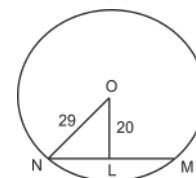
$$\text{Degrees} = \text{Radians} \times \frac{180^\circ}{\pi}$$

$$\text{Degrees} = \frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ$$

Sol.262.(b) one altitude is a side itself (because the two legs are mutually

perpendicular; the second altitude is dropped from the vertex of the right angle onto the hypotenuse. Therefore, the point of intersection of the altitudes (the orthocenter) is at the vertex containing the right angle itself.

Sol.263.(b)



$$ON^2 = OL^2 + LN^2$$

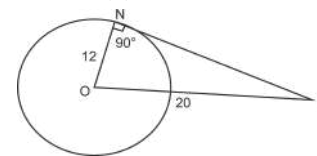
$$29^2 = 20^2 + LN^2$$

$$LN^2 = 841 - 400 = 441$$

$$LN = 21$$

$$\text{So, } MN = 2 \times 21 = 42 \text{ cm}$$

Sol.264.(d)



By the Pythagorean Theorem

$$OP^2 = NP^2 + NO^2$$

$$20^2 = NP^2 + 12^2$$

$$NP^2 = 400 - 144 = 256$$

$$NP = 16 \text{ cm}$$

Sol.265.(c) According to the question

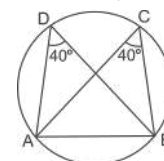
$$L = \sqrt{a^2 - (r_1 + r_2)^2}$$

$$L = \sqrt{(12)^2 - (8 + 3)^2} = \sqrt{144 - 121}$$

$$L = \sqrt{23} \text{ cm}$$

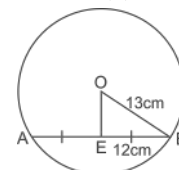
Sol.266.(a) Angles in the same segment of a circle are equal.

Therefore,



$$\angle ACB = \angle ADB = 40^\circ$$

Sol.267.(a)

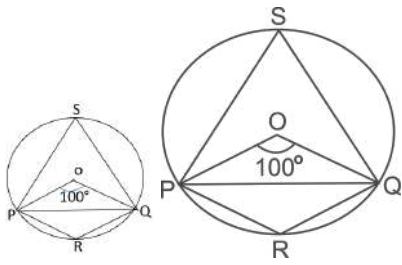


Distance of the chord from the center of

$$\text{the circle (OE)} = \sqrt{(13)^2 - (12)^2}$$

$$= \sqrt{169 - 144} = 5 \text{ cm}$$

Sol.268.(a)



The angle subtended by a chord at the circumference of a circle is half the angle subtended by the same chord at the centre.

$$\angle PSQ = 50^\circ$$

In the cyclic quadrilateral PRQS

$$\angle PSQ + \angle PRQ = 180^\circ$$

$$50^\circ + \angle PRQ = 180^\circ$$

$$\angle PRQ = 130^\circ$$

Sol.269.(b)

$$\begin{aligned} \text{Length of tangent} &= \sqrt{15^2 - 12^2} \\ &= \sqrt{225 - 144} = \sqrt{81} = 9 \text{ cm} \end{aligned}$$

Sol.270.(c) Let each exterior angle = x

Then each interior angle = $180 - x$

According to question,

Half the interior angle = $k \times$ exterior angle

$$\Rightarrow \frac{1}{2} (180 - x) = kx$$

$$\Rightarrow 180 = x(2k + 1) \Rightarrow x = \frac{180}{2k + 1}$$

Now, number of sides(n)

$$= \frac{360}{\frac{180}{2k + 1}} = 2(2k + 1) = 4k + 2$$

Sol.271.(c)

$$\frac{5\pi}{4} \text{ radians} = \frac{5\pi}{4} \times \frac{180^\circ}{\pi} = 225^\circ$$

Sol.272.(c) In a right-angled triangle, the altitudes from the two perpendicular sides meet at the right-angled vertex itself.

Hence, the orthocenter lies at the vertex of the right angle.

Sol.273.(c) In a circle, equal chords are equidistant from the center.

Hence, the distance from O to CD = distance from O to AB = 16 cm.

Sol.274.(c) A tangent to a circle is perpendicular to the radius (or diameter) at the point of contact.

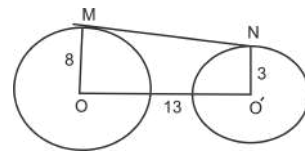
Therefore, the angle between the tangent and the diameter = 90° .

Sol.275.(a) Angle made in full circle = 2π

$$\text{Required \%} = \frac{\frac{2\pi}{3}}{2\pi} \times 100$$

$$= \frac{100}{3} = 33.33\%$$

Sol.276.(d)



Length of direct common tangent

$$\begin{aligned} &= \sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{13^2 - (8 - 3)^2} \\ &= \sqrt{169 - 25} = 12 \text{ cm} \end{aligned}$$

Sol.277.(c)

The angle subtended by the arc at any point in the major segment will be half the angle subtended by it at the center.

Angle subtended at a point in the major

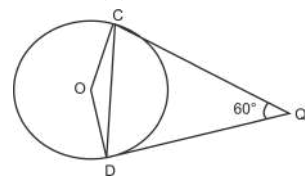
$$\text{segment} = \frac{100^\circ}{2} = 50^\circ$$

Sol.278.(a) If two chords, PQ and RS, in a circle are of equal length, then they subtend equal arcs and equal angles.

Therefore, the angle subtended by RS

$$= 75^\circ$$

Sol.279.(b)



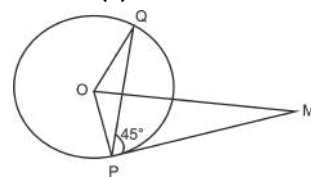
$$\angle COD = 180^\circ - \angle CQD = 180^\circ - 60^\circ = 120^\circ$$

The angle subtended by a chord on the circumference is half the angle subtended by it at the center.

Therefore, the angle subtended by the

$$\text{chord CD} = \frac{120^\circ}{2} = 60^\circ$$

Sol.280.(d) Given: PQ = 10 cm



$$\angle OPM = 90^\circ$$

$$\angle OPQ = \angle OPM - \angle QPM = 90^\circ - 45^\circ = 45^\circ$$

$$OP = OQ \dots \dots \{ \text{radius of circle} \}$$

$$\text{So, } \angle OPQ = \angle OQP = 45^\circ$$

$$\angle POQ = 90^\circ$$

By Pythagoras theorem

$$OP^2 + OQ^2 = PQ^2$$

$$r^2 + r^2 = 10^2 \Rightarrow r = 5\sqrt{2} \text{ cm}$$

Sol.281.(c) Radius of the middle circle

$$= \frac{5 + 15}{2} = 10 \text{ cm}$$

$$\text{Sol.282.(c) Degrees = Radians} \times \frac{180}{\pi}$$

$$= 5 \times \frac{180}{\pi} = \frac{900}{3.14} = 286.62^\circ$$

Sol.283.(b) Sum of interior angles of a pentagon = $(5 - 2) \times 180 = 540^\circ$

$$\text{According to question, } y + (y + 5) + (y + 10) + (y + 15) + (y + 20) = 540$$

$$\Rightarrow 5y + 50 = 540 \Rightarrow y = 98$$

$$\therefore \text{Smallest angle} = y = 98^\circ$$

Sol.284.(a) We know that,

$$\angle \text{Circumference angle} = \frac{\angle \text{Central angle}}{2}$$

$$= \frac{100}{2} = 50^\circ$$

Sol.285.(b) $\theta = 3$ radians

$$\Rightarrow 1^\circ = 57.2958^\circ$$

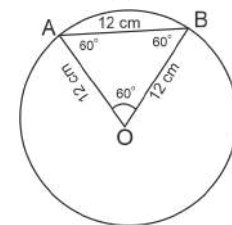
$$= 3 \times 57.2958^\circ = 171.89^\circ$$

percentage of the full circle

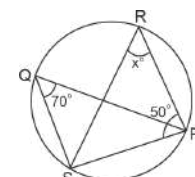
$$= \frac{171.89^\circ}{360^\circ} \times 100 = 47.74\%$$

Sol.286.(d) If $(d = r_1 + r_2) \Rightarrow 3 + 5 = 8 \rightarrow 3$, then there will be 3 tangents (two external and one at the point where they touch externally).

Sol.287.(b)

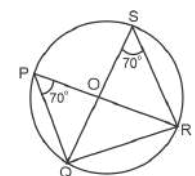


Sol.288.(d) If two or more angles are subtended by the same arc in a circle, then all those angles are equal at the circumference.

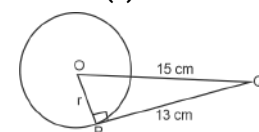


$$\angle PQS = 70^\circ = \angle PRS \Rightarrow x^\circ = 70^\circ$$

Sol.289.(b) If two or more angles are subtended by the same arc in a circle, then all those angles are equal at the circumference.



Sol.290.(a)



$$15^2 = r^2 + 13^2$$

$$\Rightarrow r^2 = 225 - 169 = 56 \Rightarrow r = \sqrt{56}$$

Sol.291.(a) Area of sector = $\frac{\theta}{360} \times \pi r^2$

$$38.5 = \frac{\theta}{360} \times \frac{22}{7} \times (7)^2$$

$$\theta = \frac{38.5 \times 360}{154} = 90^\circ$$

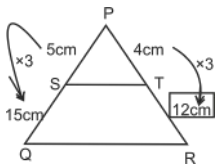
Sol.292.(c) Degrees = Radians $\times \frac{180}{\pi}$

$$= 3.5 \times \frac{180}{3.14} = 200.63^\circ$$

Sol.293.(d) In right triangle,
One acute angle = 60° , so the other = 30° .
Exterior angle = $180^\circ - 90^\circ = 90^\circ$

Sol.294.(b) A triangle has one angle = 105° (i.e., an obtuse triangle).
In an obtuse-angled triangle, the orthocenter lies outside the triangle.

Sol.295.(d) Since $ST \parallel QR$, $\Delta PST \sim \Delta PQR$



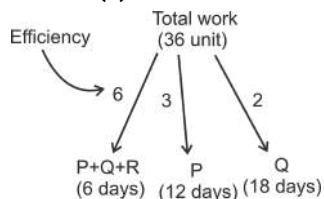
Sol.296.(a)

$$\begin{aligned} \text{Reflex Angle} &= 360^\circ - \text{Angle at center} \\ &= 360^\circ - 105^\circ = 255^\circ \end{aligned}$$

Sol.297.(a) Radius(r) = $\frac{d}{2}$

$$\text{Length of an arc} = r \times \theta = \frac{d\theta}{2}$$

Sol.298.(d)



$$\text{Efficiency of R's} = 6 - 3 - 2 = 1$$

$$\text{Time} = \frac{36}{1} = 36 \text{ days.}$$

Sol.299.(c) According to question ,

$$\Rightarrow (n-2) \times 180^\circ = 5 \times 360^\circ$$

$$\Rightarrow n-2 = 5 \times 2 \Rightarrow n = 12$$

Number of diagonal in the polygon

$$= \frac{n(n-3)}{2} = \frac{12 \times 9}{2} = 54$$

Sol.300.(d)

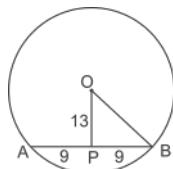
$$\text{Semiperimeter} = \frac{8 + 15 + 17}{2} = 20\text{cm}$$

$$\text{Inradius} = 2\text{cm}$$

$$\text{Area} = \text{inradius} \times \text{semiperimeter}$$

$$= 2 \times 20 = 40\text{cm}^2$$

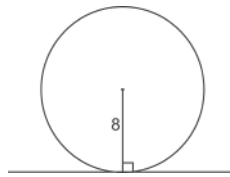
Sol.301.(c)



In ΔOPB

$$\begin{aligned} OB &= \sqrt{13^2 + 9^2} = \sqrt{169 + 81} \\ &= \sqrt{250} = 15.81 \approx 16\text{cm} \end{aligned}$$

Sol.302.(c)



Required angle would be 90°

Sol.303.(b) According to the question ,

$$\frac{\text{arc length}}{\text{circumference}} = \frac{\text{central angle}}{\text{full circle angle}}$$

$$\frac{11}{2\pi r} = \frac{36^\circ}{360^\circ} \Rightarrow 2\pi r = 110^\circ$$

$$r = \frac{110^\circ \times 7}{22 \times 2} = 17.5\text{ cm}$$

Sol.304.(d) Radians = Degrees $\times \frac{\pi}{180}$

$$= 6 \times \frac{\pi}{180} = \frac{\pi}{30}$$

Sol.305.(b) Vertically opposite angles are always equal.

So, if one angle = 55° , the other angle = 55° .

Sol.306.(b)

Using the Basic Proportionality Theorem (Thales theorem),

$$\Rightarrow \frac{PD}{DQ} = \frac{PE}{ER} \Rightarrow \frac{1}{2} = \frac{6}{ER} \Rightarrow ER = 12\text{ cm}$$

Sol.307.(d) Since equal chords subtend equal angles at the center,

$$x = 35^\circ$$

Sol.308.(b) According to question

$$\text{Ratio} \rightarrow a : b$$

$$a : b$$

$$\dots\dots\dots 100 : 144$$

$$\Rightarrow \frac{a^2}{b^2} = \frac{100}{144} \Rightarrow \left(\frac{a}{b}\right)^2 = \left(\frac{10}{12}\right)^2$$

$$\Rightarrow \frac{a}{b} = \frac{10}{12}$$

$$\text{Reqd. \%} = \frac{2}{10} \times 100 = 20\%$$

Sol.309.(a) Let angle A, B, C = $5x, 4x, 5x$

We know that:

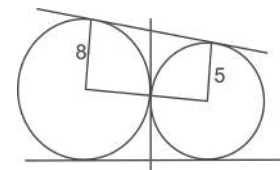
The sum of the angles of a quadrilateral is 360 degrees.

$$5x + 4x + 5x + 80^\circ = 360^\circ$$

$$14x = 280^\circ \Rightarrow x = 20^\circ$$

$$\text{So, } \angle B = 4 \times 20^\circ = 80^\circ$$

Sol.310.(d)



We can see in the figure that the common tangent is three.

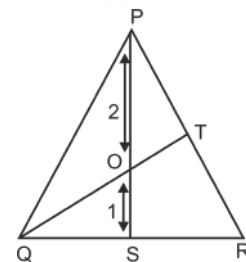
Sol.311.(c)

$$\frac{3\pi}{2} \text{ radians} = \frac{3\pi}{2} \times \frac{180^\circ}{\pi} = 270^\circ$$

Sol.312.(a) If two chords, EF and GH, in a circle are of equal length, then they subtend equal arcs and equal angles.

Therefore, the angle subtended by GH = 95°

Sol.313.(a)

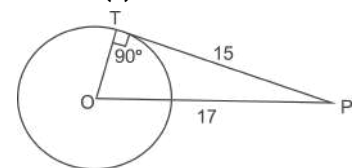


$$\text{A/Q, PS (3 unit)} \rightarrow 21\text{ cm}$$

$$\text{PO (2 unit)} \rightarrow \frac{21 \times 2}{3} = 14\text{ cm}$$

Sol.314.(b) A decagon has 10 sides. Connecting its center to all vertices gives 10 triangles. Each triangle has two equal sides, so all are isosceles.

Sol.315.(c)



Radius of circle = TO

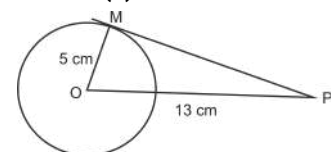
$$OP^2 = TO^2 + TP^2 \Rightarrow 17^2 = TO^2 + 15^2$$

$$TO^2 = 289 - 225 = 64 \Rightarrow TO = 8$$

$$\text{Perimeter of circle} = 2\pi r \Rightarrow 16\pi\text{ cm}$$

Sol.316.(a) (5, 12, 13) is a Pythagorean triple ($5^2 + 12^2 = 13^2$), therefore the triangle will be a right-angled triangle .

Sol.317.(b)



Pythagorean theorem in ΔOPM

$$(OP)^2 = (OM)^2 + (PM)^2$$

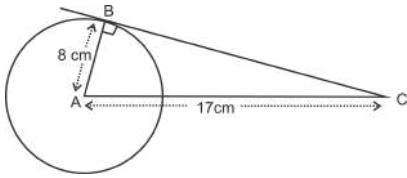
$$(13)^2 = (5)^2 + (PM)^2$$

$$(PM) = \sqrt{169 - 25} = 12\text{ cm}$$

Sol.318.(d) All three angles are different, so all three sides are different. Therefore, the triangle is a scalene triangle.

Sol.319.(c) There is exactly **one** tangent to a circle at a specific point on it.

Sol.320.(b)



In $\triangle ABC$,

$$\Rightarrow BC^2 = AC^2 - AB^2 \Rightarrow BC^2 = 17^2 - 8^2$$

$$\Rightarrow BC = \sqrt{289 - 64} = \sqrt{225}$$

$$\Rightarrow BC = 15 \text{ cm}$$

Sol.321.(b) The sum of all interior angles of a triangle is 180°

Given angles = 35° and 65°

$$\text{Third angle} = 180^\circ - (35^\circ + 65^\circ) = 80^\circ$$

Sol.322.(c) Chord

Chord:- A line segment with endpoints on the circle.

Radius:- Distance from the center to the circle boundary.

Tangent:- A line that touches the circle at exactly one point.

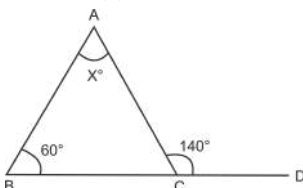
Secant:- A line that extends through the circle, intersecting it at two points.

Sol.323.(b) Sum of interior angles

$$\rightarrow (n - 2) \times 180^\circ = 1440^\circ$$

$$(n - 2) = 8 \Rightarrow n = 10$$

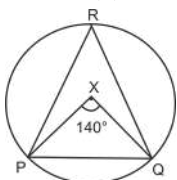
Sol.324.(c)



Exterior angle is equal to sum to its interior opposite angle:-

$$140^\circ = 60^\circ + x^\circ \Rightarrow x^\circ = 80^\circ$$

Sol.325.(b)



In a circle, the angle formed on the circumference by an arc is half of the angle formed at the center by the same arc.

$$\angle PRQ = \frac{1}{2} \angle PXQ = \frac{1}{2} \times 140^\circ = 70^\circ$$

Sol.326.(d) Let the angle = $3x$, $1x$, $2x$
sum of three angles of a triangle = 180°

$$3x + 1x + 2x = 180^\circ$$

$$6x = 180^\circ \Rightarrow x = 30^\circ$$

$$\text{Angle} = 3 \times 30^\circ, 1 \times 30^\circ, 2 \times 30^\circ = 90^\circ, 30^\circ, 60^\circ$$

Sol.327.(d) Sum of all three angles of a triangle = 180°

$$\text{Third angle} = 180^\circ - (55^\circ + 75^\circ) = 50^\circ$$

$$\text{Sol.328.(c)} \angle R = 180^\circ - (60^\circ + 50^\circ) = 70^\circ$$

Sol.329.(b)

Number of rotations of a wheel:

$$\text{Rotations} = \frac{\text{Distance travelled}}{\text{Circumference of wheel}}$$

$$= \frac{2\pi R}{2\pi r} = \frac{R}{r}$$

$$\frac{200}{0.6} = \frac{220}{r_2} \Rightarrow r_2 = \frac{22 \times 0.3}{10} = 0.66 \text{ m}$$

or 66 cm

$$\text{Sol.330.(a)} \text{Diagonals} = \frac{n(n-3)}{2}$$

$$= \frac{10(10-3)}{2} = \frac{70}{2} = 35$$

Sol.331.(b) Becomes half

Sol.332.(a)

Both Assertion (A) and Reason (R) are true, and the Reason correctly explains the Assertion.

Sol.333.(b) The sum of opposite angles of a cyclic quadrilateral = 180°

$$\angle P + \angle R = 180^\circ$$

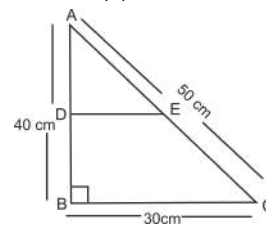
$$60^\circ + 120^\circ = 180^\circ$$

Since $\angle Q$ and $\angle S$ are also opposite angles,

$$\therefore \angle Q + \angle S = 180^\circ$$

Sol.334.(a) The angle formed in a semicircle (i.e., angle subtended by the diameter at the circumference) is always 90° .

Sol.335.(b)



Since D is the midpoint of AB:

$$AD = \frac{AB}{2} = \frac{40}{2} = 20 \text{ cm}$$

In $\triangle AED$ and $\triangle ACB$:

$\angle A$ is common

$$\angle AED = 90^\circ = \angle ABC$$

$$\therefore \triangle AED \sim \triangle ACB \text{ (AA similarity)}$$

$$\frac{\text{Area}(\triangle AED)}{\text{Area}(\triangle ACB)} = \left(\frac{AD}{AC}\right)^2 = \left(\frac{20}{50}\right)^2 = \frac{4}{25}$$

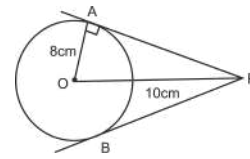
$$\text{Area}(\triangle ABC) = \frac{1}{2} \times 40 \times 30 = 600 \text{ cm}^2$$

$$\text{Area}(\triangle AED) = 600 \times \frac{4}{25} = 96 \text{ cm}^2$$

Sol.336.(c) In a right-angled triangle, the midpoint of the hypotenuse is the circumcenter of the triangle.

Sol.337.(c) If the corresponding sides of two triangles are proportional, it is called the (Side-Side-Side) similarity.

Sol.338.(b)



As we know,

$$\angle OAP = 90^\circ \text{ [tangent]}$$

$$OP^2 = AO^2 + AP^2$$

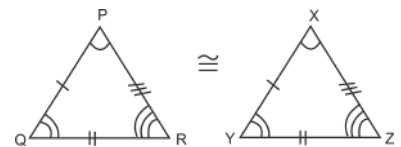
$$(10)^2 = (8)^2 + AP^2$$

$$AP^2 = (10)^2 - (8)^2 = 100 - 64 = 36$$

$$AP = \sqrt{36} = 6 \text{ cm}$$

Sol.339.(c)

When $\triangle PQR \cong \triangle XYZ$, the order shows the correspondence:



So the matching sides are:

$$PQ = XY$$

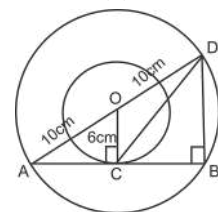
$$QR = YZ$$

$$RP = XZ$$

Sol.340.(d) If chord = radius, the triangle formed at the center is equilateral (all sides = radius).

So each angle = 60° , hence the chord subtends 60° at the center.

Sol.341.(b)



According to question, $AO = OD = 10 \text{ cm}$

If a line divides any two sides in the

same ratio ($\frac{AO}{OD} = \frac{AC}{CB}$) then the line is

parallel to the third line.

$$DB \parallel OC$$

$$\therefore OC = 6 \text{ cm}, BD = 12$$

In $\triangle AOC$,

$$AC = \sqrt{(10)^2 - (6)^2} = 8 \text{ cm}$$

$$\Rightarrow AC = BC = 8 \text{ cm}$$

In $\triangle BCD$,

$$CD = \sqrt{(12)^2 + (8)^2} = \sqrt{208} \text{ cm}$$

$$\therefore \text{The length of the CD is } \sqrt{208} \text{ cm.}$$



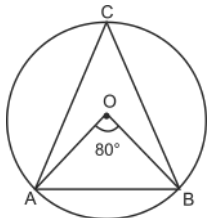
Sol.342.(b) Length of Transverse

$$\text{common tangent} = \sqrt{d^2 - (r_1 + r_2)^2}$$

$$= \sqrt{(58)^2 - (28 + 14)^2}$$

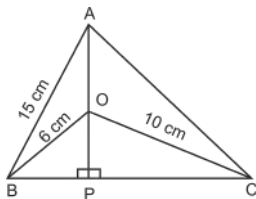
$$= \sqrt{3364 - 1764} = 40 \text{ cm}$$

Sol.343.(d) SSA does NOT guarantee similarity because two sides and a non-included angle can form more than one possible triangle (ambiguous case). Other conditions (AAA, SAS, SSS) always give a unique similarity between triangles.

Sol.344.(c)

We know that,

$$\Rightarrow \angle C = \frac{\angle AOB}{2} = \frac{80^\circ}{2} = 40^\circ$$

Sol.345.(d)

In $\triangle OPB$ and $\triangle OPC$,

$$\Rightarrow OP^2 = 36 - BP^2 \dots (i)$$

$$\Rightarrow OP^2 = 100 - CP^2 \dots (ii)$$

From eq. (i) and (ii),

$$\Rightarrow CP^2 - BP^2 = 100 - 36 = 64 \dots (iii)$$

In $\triangle APB$ and $\triangle APC$,

$$\Rightarrow AP^2 = 225 - BP^2 \dots (iv)$$

$$\Rightarrow AP^2 = AC^2 - CP^2 \dots (v)$$

From eq. (iv) and (v),

$$\Rightarrow CP^2 - BP^2 = AC^2 - 225 \dots (vi)$$

Now, From eq. (iii) and (vi),

$$\Rightarrow 64 = AC^2 - 225$$

$$\Rightarrow AC^2 = 289 \Rightarrow AC = 17 \text{ cm}$$

Sol.346.(c) According to question,

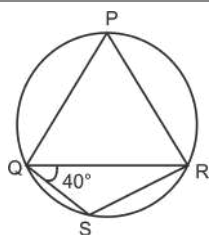
$$\Rightarrow \text{Internal angle} + \text{External angle} = 180^\circ$$

$$\Rightarrow 4x + x = 180^\circ$$

$$\Rightarrow x = \frac{180^\circ}{5} = 36^\circ$$

$$\text{So, Side of the polygon} = \frac{360^\circ}{\text{exterior angle}}$$

$$= \frac{360^\circ}{36^\circ} = 10$$

Sol.347.(b)

$\triangle PQR$ is equilateral, so

$$\angle PQR = \angle PRQ = \angle QPR = 60^\circ$$

We know that, $\angle QPR + \angle QSR = 180^\circ$

$$\Rightarrow 60^\circ + \angle QSR = 180^\circ$$

$$\Rightarrow \angle QSR = 180^\circ - 60^\circ = 120^\circ$$

In $\triangle QRS$,

$$\Rightarrow \angle RQS + \angle QSR + \angle QRS = 180^\circ$$

$$\Rightarrow 40^\circ + 120^\circ + \angle QRS = 180^\circ$$

$$\Rightarrow \angle QRS = 180^\circ - 40^\circ - 120^\circ = 20^\circ$$

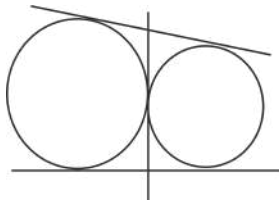
Sol.348.(a) By Basic Proportionality Theorem,

Since, $EF \parallel QR$, triangle PEF is similar to triangle PQR

$$\Rightarrow \frac{PE}{PQ} = \frac{PF}{PR} \Rightarrow \frac{16}{30} = \frac{PF}{90} \Rightarrow PF = 48 \text{ cm}$$

Sol.349.(c)

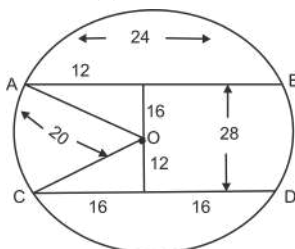
When all sides of a parallelogram touch a circle, it becomes a tangential parallelogram. Such a parallelogram can only be a rhombus. In a rhombus, the opposite angles are equal.

Sol.350.(b) A-4, B-2, C-3, D-1**Sol.351.(a)**

Three common tangents can be drawn.

Sol.352.(c) Incenter

The incenter always lies inside the triangle, regardless of its type.

Sol.353.(c) 32

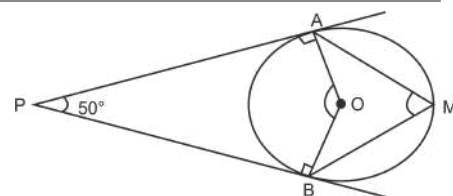
Length of $CD = 16 + 16 = 32 \text{ cm}$

Sol.354.(a) Let the two complementary angle be x and $90 - x$

$$A/Q, 90^\circ - x - x = 24^\circ$$

$$90^\circ - 24^\circ = 2x \Rightarrow 66^\circ = 2x \Rightarrow x = 33^\circ$$

$$\text{And other angle} = 90^\circ - 33^\circ = 57^\circ$$

Sol.355.(b)

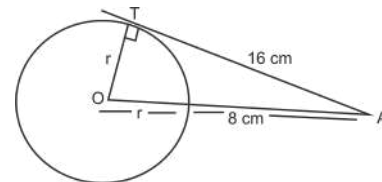
In the quadrilateral $OAPB$, the sum of angles = 360°

$$\Rightarrow \angle AOB + \angle OBP + \angle APB + \angle OAP = 360^\circ$$

$$\Rightarrow \angle AOB + 90^\circ + 50^\circ + 90^\circ = 360^\circ$$

$$\Rightarrow \angle AOB = 130^\circ$$

$$\text{Now, } \angle AMB = \frac{\angle AOB}{2} = \frac{130^\circ}{2} = 65^\circ$$

Sol.356.(b)

In $\triangle OTA$

$$OA^2 = OT^2 + TA^2$$

$$(r + 8)^2 = r^2 + (16)^2$$

$$r^2 + 64 + 16r = r^2 + 256$$

$$16r = 192 \Rightarrow r = 12 \text{ cm}$$

Sol.357.(d) Let the angle be x

$$\text{Complement angle} = 2x + 9^\circ$$

We know that,

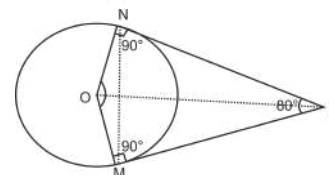
$$\Rightarrow x + 2x + 9^\circ = 90^\circ \Rightarrow 3x = 81^\circ \Rightarrow x = 27^\circ$$

Sol.358.(b) Let angle = x°

$$A/Q, 2(90^\circ - x) = x^\circ$$

$$\Rightarrow 180^\circ - 2x = x^\circ \Rightarrow 3x^\circ = 180^\circ \Rightarrow x = 60^\circ$$

$$\text{Reqd. angle} = 60^\circ$$

Sol.359.(a)

If two tangent segments are drawn to a circle from the same external point, then those segments are equal in length.

$$\angle MON + \angle MPN = 180^\circ$$

$$\angle MON + 80^\circ = 180^\circ \Rightarrow \angle MON = 100^\circ$$

Sol.360.(c)

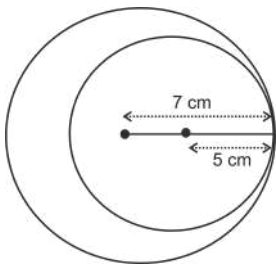
Length of direct common tangent

$$= \sqrt{d^2 - (r_1 - r_2)^2}$$

$$\Rightarrow \sqrt{25^2 - (10 - 5)^2}$$

$$\Rightarrow \sqrt{625 - 25} = \sqrt{600} = 10\sqrt{6} \text{ cm.}$$

Sol.361.(a)



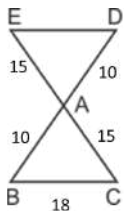
So, Required difference = $7 - 5 = 2$ cm

Sol.362.(b) Let the angle be x and other angle be $90^\circ - x$

$$A/Q, x - (90^\circ - x) = 24^\circ$$

$$2x = 114^\circ \Rightarrow x = 57^\circ$$

Sol.363.(a)



In $\triangle EAD$ and $\triangle CAB$

$AE = AC$ (Given)

$AD = AB$ (Given)

$\angle EAD = \angle CAB$ (Vertically opposite angles are equal.)

$\triangle EAD \cong \triangle CAB$ (From SAS)

$ED = BC = 18$ cm

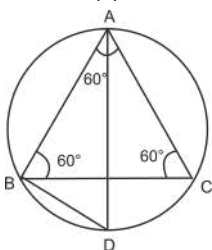
Sol.364.(c) Length = $\sqrt{25^2 - (8 + 7)^2}$
 Length = $\sqrt{625 - 225} = \sqrt{400} = 20$ cm

Sol.365.(a) A/Q,

$$90 - y = y + 20 \Rightarrow 70 = 2y \Rightarrow y = 35^\circ$$

Sol.366.(d) For incentre we have to find the either area or side length. It cannot be determined.

Sol.367.(b)



We know that angles made by the same chord at circumference are the same.

So, $\angle ACB$ and $\angle ADB$ are same

$$\angle ADB = 60^\circ$$

$$\frac{1}{2} \angle ADB = 30^\circ$$

Sol.368.(b) Sum of the opposite sides of parallelogram are equal

$$PQ + RS = QR + PS$$

$$5 + 7 = 6 + PS \Rightarrow PS = 6 \text{ cm}$$

Sol.369.(d)

$$\text{Area of each sector} = \frac{360^\circ}{4} = 90^\circ$$

Sol.370.(a) Let the angle be x° and its complementary angle be $90^\circ - x$

$$\text{Given, } 90^\circ - x = \frac{x}{5} \Rightarrow 90^\circ = \frac{6}{5}x \Rightarrow x = 75^\circ$$

Sol.371.(b) Inradius = $\frac{a+b-c}{2}$, when given right angle triangle

$$r = \frac{9+40-41}{2} \Rightarrow r = \frac{8}{2} = 4 \text{ cm}$$

Sol.372.(a) We know that,

$$AO \times OB = CO \times OD$$

$$\Rightarrow 16 \times 7 = 8 \times OD \Rightarrow OD = 14 \text{ cm}$$

Sol.373.(a)

Length of the arc = radius $\times$ angle

$$\text{Length of arc} = 12 \times 3 = 36 \text{ cm}$$

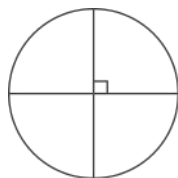
Sol.374.(d) We know that, Area of

$$\text{triangle} = \frac{\text{Product of sides}}{4 \times \text{circumradius}}$$

According to question,

$$\begin{aligned} \text{Product of three sides} &= 84 \times 4 \times 21 \\ &= 7056 \text{ cm}^3 \end{aligned}$$

Sol.375.(b)



Central angle of one part = 90°

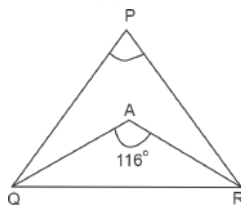
$$\text{Area} = \frac{1}{4} \times 300 = 75 \text{ m}^2$$

Sol.376.(a) Area of sector = $\frac{\theta}{360^\circ} \times \pi r^2$

$$\text{Area of sector} = \frac{90^\circ}{360^\circ} \times \pi (8^2)$$

$$\text{Area of sector} = \frac{1}{4} \times \pi \times 64 = 16\pi \text{ cm}^2$$

Sol.377.(a)



$$\angle QAR = 90^\circ + \frac{\angle P}{2}$$

$$\Rightarrow 116^\circ = 90^\circ + \frac{\angle P}{2}$$

$$\Rightarrow 116^\circ = \frac{180^\circ + \angle P}{2}$$

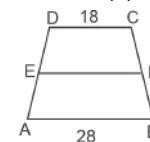
$$\angle P = 232^\circ - 180^\circ = 52^\circ$$

Sol.378.(d) Angle corresponding to commerce postgraduates

$$= \frac{40}{180} \times 360^\circ = 80^\circ$$

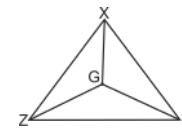
Sol.379.(d) A quadrilateral having equal diagonals and equal opposite sides is always a rectangle.

Sol.380.(d)



$$EF = \frac{AB + DC}{2} = \frac{28 + 18}{2} = \frac{46}{2} = 23 \text{ cm}$$

Sol.381.(c)

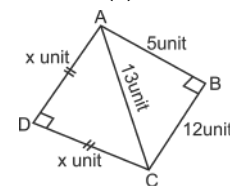


The centroid divides the triangle into 3 smaller ($\triangle GYZ$, $\triangle GXY$ and $\triangle GXZ$) triangles of equal area

$$\text{Area of } \triangle GYZ = \frac{\text{area of } \triangle XYZ}{3}$$

$$= \frac{45}{3} = 15 \text{ cm}^2$$

Sol.382.(d)



$$\text{Area of } \triangle ABC = \frac{1}{2} \times 12 \times 5 = 30 \text{ unit}^2$$

In the $\triangle ADC$ applying pythagoras theorem

$$x^2 + x^2 = (13)^2$$

$$2x^2 = 169 \Rightarrow x = \frac{13}{\sqrt{2}}$$

$$\text{Now, area of } \triangle ADC = \frac{1}{2} \times \frac{13}{\sqrt{2}} \times \frac{13}{\sqrt{2}}$$

$$= \frac{169}{4} = 42.25 \text{ unit}^2$$

$$\begin{aligned} \text{Hence, area of quadrilateral ABCD} \\ = 30 + 42.25 = 72.25 \text{ unit}^2 \end{aligned}$$

Sol.383.(c) Perimeter of equilateral

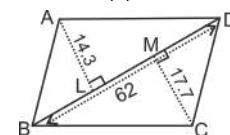
$$\text{triangle } (3a) = 24 \text{ cm}$$

$$\text{Each side } (a) = 8 \text{ cm}$$

$$\text{Length of median} = \frac{\sqrt{3}a}{2} = \frac{\sqrt{3} \times 8}{2}$$

$$= 4\sqrt{3} \text{ cm}$$

Sol.384.(c)



Area of ABCD = area of $\triangle ABD$ + area of $\triangle CDB$

$$= \frac{1}{2} \times BD \times AL + \frac{1}{2} \times BD \times CM$$

$$= \frac{1}{2} \times 62 (14.3 + 17.7)$$

$$= \frac{1}{2} \times 62 \times 32 = 992 \text{ cm}^2$$

Sol.385.(a) If two sides of both triangles



are equal, and the perimeter is the same, the third side must also be equal because the perimeter is the sum of all sides. Therefore, all three sides of both triangles would be equal, satisfying the SSS (Side-Side-Side) congruence condition. So, Option (a) is sufficient to conclude that the two triangles are congruent.

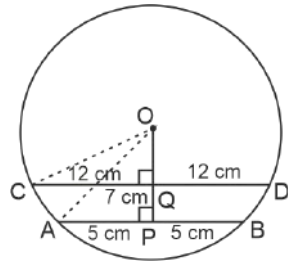
Sol.386.(d) Let the side of an equilateral triangle be x

According to the question,

$$2x = 16 \Rightarrow x = 8 \text{ cm}$$

We know that an equilateral triangle all side equal to each other so the third side will be 8 cm.

Sol.387.(a)



In ΔOQC ,

By using pythagoras theorem,

$$OQ^2 + CQ^2 = OC^2$$

$$OQ^2 + 12^2 = OC^2 \dots\dots(i)$$

In ΔOPA ,

$$OP^2 + AP^2 = OA^2$$

$$(OQ + 7)^2 + 5^2 = OA^2 \dots\dots(ii)$$

Equating the equation (i) and (ii) because $OC = OA$ (radii of the circle)

$$OQ^2 + 12^2 = (OQ + 7)^2 + 5^2$$

$$OQ^2 + 144 = OQ^2 + 49 + 14OQ + 25$$

$$144 - (49 + 25) = 14OQ$$

$$70 = 14OQ$$

$$OQ = 5 \text{ cm}$$

Put the value of OQ in equation (i)

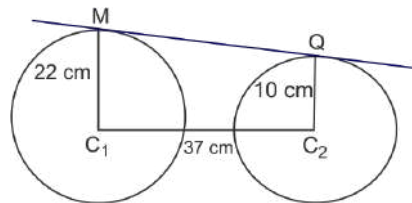
$$5^2 + 12^2 = OC^2$$

$$OC^2 = 25 + 144 = 169, OC = 13$$

So, the diameter of the circle

$$= 2 \times 13 = 26 \text{ cm}$$

Sol.388.(a)



Length of the common tangent(MQ) =

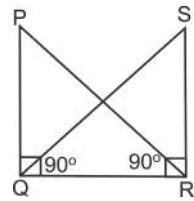
$$\sqrt{(\text{distance between centre of two circle})^2 - (c_1 - c_2)^2}$$

$$MQ = \sqrt{(37)^2 - (22 - 10)^2}$$

$$= \sqrt{(37)^2 - (12)^2}$$

$$MQ = \sqrt{1369 - 144} = \sqrt{1225} = 35 \text{ cm}$$

Sol.389.(d)



In the ΔPQR and ΔSRQ

$$PQ = SR \text{ \{given\}}$$

$$\angle Q = \angle R \text{ \{both are } 90^\circ\}$$

$$QR = QR \text{ \{common side\}}$$

Hence, $\Delta PQR \cong \Delta SRQ$

\{by SAS congruence\}

Sol.390.(d)

After checking all the options one by one only option (d) satisfies.

$$1) \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}$$

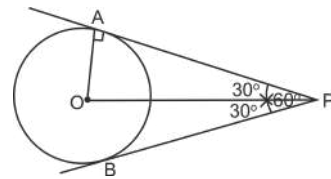
Multiplying by $\frac{180^\circ}{\pi}$ to change in degree

$$\Rightarrow 30^\circ, 60^\circ, 90^\circ$$

$$\Rightarrow \frac{30^\circ + 90^\circ}{2} = 60^\circ \times \frac{\pi}{180} \text{ (change in}$$

radian to multiply by $\frac{\pi}{180}) = \frac{\pi}{3}$ (is average of two other angles)

Sol.391.(b)



In the ΔPOA

$\angle OPA = 30^\circ$ [when two tangents are drawn to a circle from an external point, the line drawn from that external point to the center of the circle bisects the angle formed between the two tangents.]

$\angle OAP = 90^\circ$ [radius $\perp$ tangent]

$$\text{Hence, } \angle POA = 180^\circ - (30 + 90) = 60^\circ$$

Sol.392.(a) Concept :- congruent triangles always have the same area.

$$\text{Hence, ar}\Delta HIJ = \text{ar}\Delta EFG = 124 \text{ cm}^2$$

Sol.393.(b)

Applying angle-angle-angle(AAA) criteria, two triangles can never be proved congruent.

Sol.394.(c) Distance between centres (d)

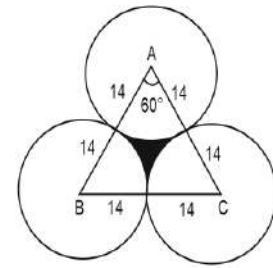
$$= 18 + 12 = 30 \text{ cm}$$

Direct common tangent

$$= \sqrt{d^2 - (R - r)^2} = \sqrt{30^2 - (18 - 12)^2}$$

$$= \sqrt{900 - 36} = \sqrt{864} = 12\sqrt{6} \text{ cm}$$

Sol.395.(c)



Area of 3 sectors =

$$3 \times \left(\frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 14^2 \right) = 308 \text{ cm}^2$$

$$\text{Area of } \Delta ABC = \frac{\sqrt{3}}{4} \times 28^2 = 339.08 \text{ cm}^2$$

Then, area of shaded region

$$= 339.08 - 308 = 31.08 \text{ cm}^2$$

Sol.396.(b) Given: $\Delta EFG \sim \Delta HIJ$

$$\text{Then, } \frac{EF}{HI} = \frac{FG}{IJ} = \frac{GE}{HJ}$$

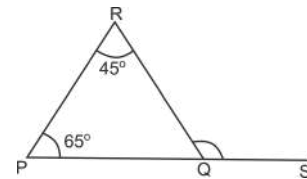
According to the question,

$$\Rightarrow 3EF = HI \Rightarrow \frac{EF}{HI} = \frac{1}{3}$$

Now,

$$\Rightarrow \frac{EF}{HI} = \frac{FG}{IJ} \Rightarrow \frac{1}{3} = \frac{9}{IJ} \Rightarrow IJ = 27 \text{ cm}$$

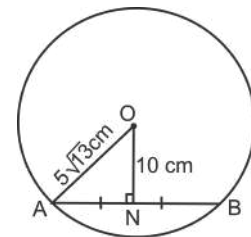
Sol.397.(a)



Using Exterior angle property

$$\angle RQS = (\angle P + \angle R) = 65^\circ + 45^\circ = 110^\circ$$

Sol.398.(a)



In ΔAON ,

By using pythagoras theorem,

$$AN^2 + ON^2 = AO^2$$

$$(10)^2 + AN^2 = (5\sqrt{13})^2$$

$$100 + AN^2 = 325$$

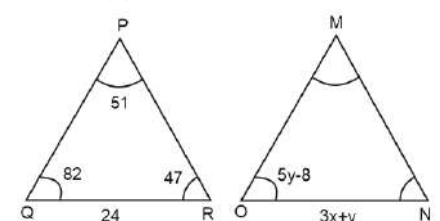
$$AN = \sqrt{325 - 100} = \sqrt{225}$$

$$AN = 15 \text{ cm}$$

$$AN = NB = 15 \text{ cm}$$

$$\text{So, } AB = 30 \text{ cm}$$

Sol.399.(d)



$$\Delta PQR \cong \Delta MON$$



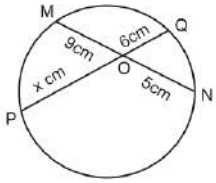
$\Rightarrow \angle P = \angle M, \angle Q = \angle O, \angle R = \angle N$ and
 $QR = ON$

$$\Rightarrow 5y - 8 = 82 \Rightarrow y = 18$$

$$\text{And } 3x + y = 24 \Rightarrow x = 2$$

$$\text{Hence, } x + y = 18 + 2 = 20$$

Sol.400.(a)



From the figure,

$$MO \times ON = OQ \times OP$$

$$9 \times 5 = 6 \times x \Rightarrow x = 7.5 \text{ cm}$$

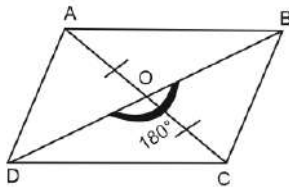
Sol.401.(c) Since, the triangle is right angled at C, we know that $A + B = 90^\circ$

$$\Rightarrow \cot(A + B) = 90^\circ = 0$$

$$\text{Hence, } \sin A \cos B \cot(A + B)$$

$$= \sin A \cos B \times 0 = 0$$

Sol.402.(c)

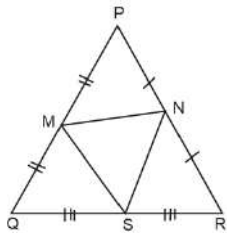


As per figure,

Since, DOB is straight line

$$\text{So, } \frac{5}{9} \text{ of } \angle DOB = \frac{5}{9} \times 180^\circ = 100^\circ$$

Sol.403.(d)



$$\text{Area of each triangle} = \frac{46}{4} = 11.5 \text{ cm}^2$$

$$\text{Area of quadrilateral MNRQ} = 11.5 \times 3$$

$$= 34.5 \text{ cm}^2$$

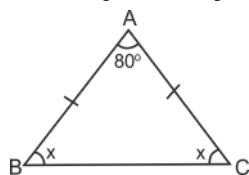
$$\text{Sol.404.(d)} \sin A = \frac{1}{2}$$

$$\sin A = \sin 30^\circ \Rightarrow A = 30^\circ$$

$$\text{Then, } \angle C = 90^\circ - 30^\circ = 60^\circ$$

$$\cos C = \cos 60^\circ = \frac{1}{2}$$

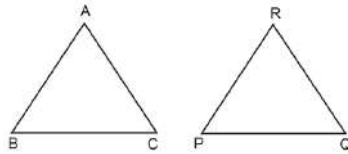
Sol.405.(a) According to question, we have given triangle;



$$80^\circ + 2x = 180^\circ$$

$$2x = 180^\circ - 80^\circ = 100^\circ \Rightarrow x = 50^\circ$$

Sol.406.(d)



$$\triangle ABC \cong \triangle RPQ \text{ (given)}$$

$$\text{Then, } BC = PQ$$

Sol.407.(d)

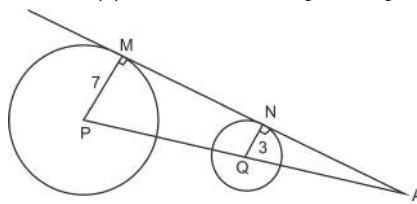
$$\angle A + \angle B + \angle C = 180^\circ \text{ ----- (1)}$$

$$\frac{1}{2} \angle A + \angle B + \angle C = 120^\circ \text{ ----- (2)}$$

Solving eqn(1) - eqn(2) we have;

$$\frac{1}{2} \angle A = 60^\circ \Rightarrow \frac{1}{4} \angle A = 30^\circ$$

Sol.408.(c) ATQ, we have given figure;

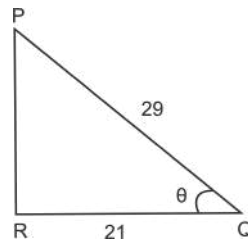


$$\triangle APM \sim \triangle AQN \text{ (BY AAA rule)}$$

$$\frac{AP}{AQ} = \frac{7}{3}$$

Clearly, we can see that A divides PQ externally in the ratio 7 : 3.

Sol.409.(d)



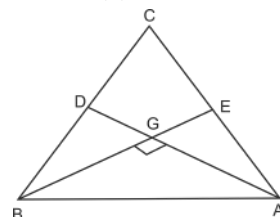
$$PR = \sqrt{29^2 - 21^2} = \sqrt{400} = 20 \text{ units}$$

$$\cos \theta = \frac{21}{29}, \sin \theta = \frac{20}{29}$$

$$\text{Then, } \cos^2 \theta - \sin^2 \theta = \left(\frac{21}{29}\right)^2 - \left(\frac{20}{29}\right)^2$$

$$= \frac{441 - 400}{841} = \frac{41}{841}$$

Sol.410.(d)



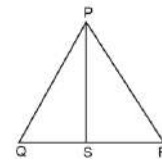
As we know, centroid (G) divides the median in the ratio 2 : 1. So, we have;

$$AG = \frac{2}{3} AD = \frac{2}{3} \times 12 = 8 \text{ cm}$$

$$BG = \frac{2}{3} BE = \frac{2}{3} \times 9 = 6 \text{ cm}$$

$$\text{Now, In } \triangle BGA, AB = \sqrt{8^2 + 6^2} = \sqrt{100} = 10 \text{ cm}$$

Sol.411.(b)



In a triangle PQR, S is a point on the side QR such that $PS \perp QR$

$$PR^2 + QS^2 = PQ^2 + SR^2$$

Always satisfies the condition.

Sol.412.(d) The triangle is congruent by SAS if two sides and their included angle in one triangle are exactly equal to the corresponding two sides and their included angle in another triangle.

Hence for congruency for triangles ABC and DEF

$$AB = FD \text{ and } \angle A = \angle D \Rightarrow AC = DE$$

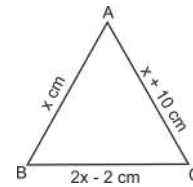
$$\text{Sol.413.(a)} 17x + 13x + 15x = 180^\circ$$

$$\Rightarrow 45x = 180^\circ \Rightarrow x = 4^\circ$$

Now,

$$17x - 13x = 4x = 4 \times 4^\circ = 16^\circ$$

Sol.414.(d)



Perimeter of triangle ABC

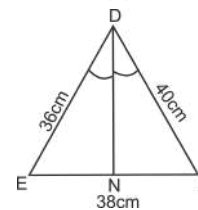
$$= x + 2x - 2 + x + 10 \Rightarrow 4x + 8 = 40$$

$$\Rightarrow 4x = 32 \Rightarrow x = 8$$

Hence,

The smallest side = 8 cm

Sol.415.(b)



The angle bisector theorem states that when an angle in a triangle is bisected, the opposite side is divided into two segments that are proportional to the other two sides of the triangle.

Here,

$$DE : DF = 36 : 40 = 9 : 10$$

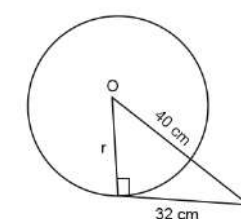
$$9x + 10x = 38 \Rightarrow x = 2$$

Hence,

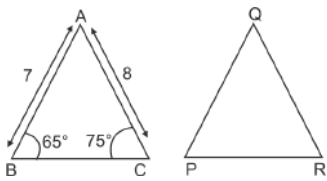
$$EN = 9 \times 2 = 18$$

$$\text{And } NF = 10 \times 2 = 20$$

Sol.416.(d)

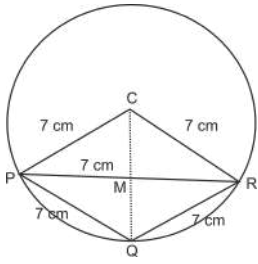


Let, r = radius



∴ PQ = 7 cm and ∠R = 75° is true.

Sol.431.(d)



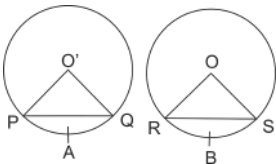
CR = PC = CQ = radii of the circle
 $\triangle CRQ$ and $\triangle PCQ$ are an equilateral triangle
 So,

$$RM = \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{2} \times 7 = \frac{7\sqrt{3}}{2}$$

Now,

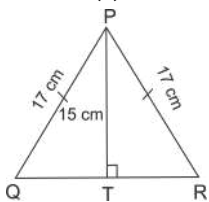
$$PR = 2 \times RM = 2 \times \frac{7\sqrt{3}}{2} = 7\sqrt{3} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{21}{\sqrt{3}} \text{ units}$$

Sol.432.(c)



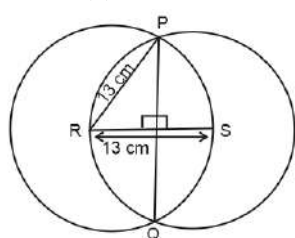
Arcs PAQ and RBS are congruent, then
 PQ = RS
 So, Ratio of PQ and RS = 1 : 1

Sol.433.(c)



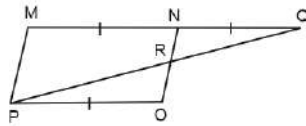
In $\triangle PTR$,
 By using pythagoras theorem,
 $\Rightarrow PR^2 = PT^2 + TR^2$
 $\Rightarrow TR^2 = PR^2 - PT^2$
 $= 17^2 - 15^2 = 289 - 225 = 64 \Rightarrow TR = 8$
 $PT \perp QR$, then
 $TR = QT = 8 \text{ cm}$
 So, $QR = TR + QT = 8 + 8 = 16 \text{ cm}$

Sol.434.(d)



Length of common chord
 $= \sqrt{3} \times \text{radius} = 13\sqrt{3} \text{ cm}$

Sol.435.(d) MNOP is a parallelogram.

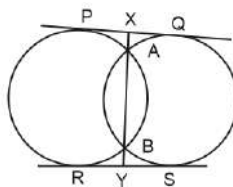


In $\triangle QNR$ and $\triangle POR$
 $\angle QNR = \angle POR$... (alternate interior angle)
 $QN = PO$... (MN = OP = QN)
 $\angle RPO = \angle NQR$... (alternate interior angle)
 Hence,
 $\triangle QNR \cong \triangle POR$ by SAS congruency
 Therefore,
 $OR = NR$ by C.P.C.T.
 So, $OR : NR = 1 : 1$

Sol.436.(c) If we extend a chord infinitely on both sides, then the chord becomes a secant.

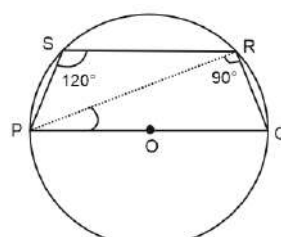
Sol.437.(d) Here,
 $P - Q = 20^\circ \Rightarrow \angle P = Q + 20^\circ$ and $Q - R = 26^\circ$
 $\Rightarrow \angle R = Q - 26^\circ$
 $\Rightarrow \angle P + \angle Q + \angle R = 180^\circ$
 $\Rightarrow \angle Q + 20^\circ + \angle Q + \angle Q - 26^\circ = 180^\circ$
 $\Rightarrow 3\angle Q = 186^\circ \Rightarrow \angle Q = 62^\circ$
 $\therefore \angle P = 62^\circ + 20^\circ = 82^\circ$

Sol.438.(d)



Given, $AB = 3 \text{ cm}$, and $XY = 5 \text{ cm}$
 We know that $XA = YB$
 $XY = XA + AB + BY \Rightarrow 5 = 2XA + 3$
 $2XA = 2$, $XA = 1 \Rightarrow XB = 1 + 3 = 4$
 We know that,
 $PX^2 = XA \times XB$
 $PX^2 = 1 \times 4 \Rightarrow PX = 2$
 Similarly $XQ = 2 \text{ cm}$
 $PQ = PX + XQ = 2 + 2 = 4$
Short method:-
 Given, $AB = 3 \text{ cm}$, $XY = 5 \text{ cm}$
 We know,
 $PQ^2 = XY^2 - AB^2 = 5^2 - 3^2 = 25 - 9 = 16$
 $PQ = 4 \text{ cm}$

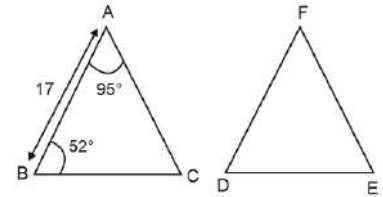
Sol.439.(a)



We know that, $\angle PRQ = 90^\circ$

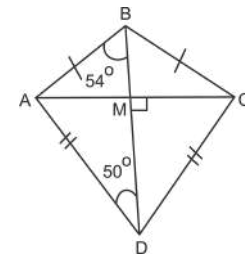
Since, PQRS is cyclic quadrilateral, so
 $\angle PSR + \angle RQP = 180^\circ$
 $\Rightarrow \angle RQP = 180^\circ - 120^\circ = 60^\circ$
 In $\triangle PRQ$
 $\angle P + \angle Q + \angle R = 180^\circ$
 $\angle P = 180^\circ - 90^\circ - 60^\circ \Rightarrow \angle P = 30^\circ$
 So, $\angle QPR = 30^\circ$

Sol.440.(b) $\triangle ABC \cong \triangle FDE$ ∴ there corresponding angle and side are equal



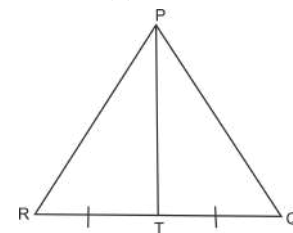
∴ $\angle C = 180^\circ - 95^\circ - 52^\circ = 33^\circ$
 Hence, Option (b) $DF = 17 \text{ cm}$, $\angle E = 33^\circ$ is true.

Sol.441.(b)



In $\triangle AMB$,
 $\angle ABM + \angle BAM + \angle BMA = 180^\circ$
 $54^\circ + 90^\circ + \angle BAM = 180^\circ$
 $\angle B = 180^\circ - 144^\circ = 36^\circ$
 In $\triangle AMD$,
 $\angle MAD + \angle AMD + \angle MDA = 180^\circ$
 $\angle MAD + 90^\circ + 50^\circ = 180^\circ$
 $\angle MAD = 180^\circ - 140^\circ = 40^\circ$
 Now,
 $\angle BAD = \angle BAM + \angle DAM = 36^\circ + 40^\circ = 76^\circ$
 Hence,
 $\angle BCD = \angle BAD = 76^\circ$

Sol.442.(b)



In $\triangle PTR$,
 By using pythagoras theorem,
 $PR^2 = PT^2 + RT^2$ (i)
 In $\triangle PTQ$,
 By using pythagoras theorem,
 $PQ^2 = PT^2 + TQ^2$ (ii)
 Add equation (i) and (ii)
 $PR^2 + PQ^2 = PT^2 + RT^2 + PT^2 + TQ^2$
 $PR^2 + PQ^2 = 2PT^2 + (RT^2 + TQ^2)$
 $PR^2 + PQ^2 = 2PT^2 + 2TQ^2$ (∵ $RT = TQ$)
 $PR^2 + PQ^2 = 2(PT^2 + TQ^2)$

Short trick:

This is apollonius theorem, hence you should remember it .

Sol.443.(d)

We know that in a right angled triangle the square of the hypotenuse is equal to the sum of the squares of the other two sides.

After checking all options one by one, only option (d) satisfies

(5, 12, 13)

$$\Rightarrow 5^2 + 12^2 = 13^2$$

$$\Rightarrow 25 + 144 = 169 \Rightarrow \text{LHS} = \text{RHS}$$

Sol.444.(a)

We know that a scalene triangle has three angles of different measurements.

So, 60° , 30° and 90°

Only option (a) is the correct answer because in this option all angles are different .

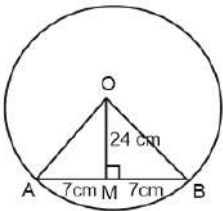
Sol.445.(a)

We know the longest chord of the circle is also called the diameter of the circle.

So, the distance between the longest chord and centre of the circle = 0 unit

Sol.446.(a) If the medians of a triangle are equal, the triangle will be **equilateral**.

This is because in any triangle, the medians are proportional to the sides. If all three medians are of equal length, it implies that all three sides of the triangle must also be equal. Hence, the triangle must be equilateral.

Sol.447.(d)

In $\triangle OMB$,

$$OM^2 + MB^2 = OB^2$$

$$OB^2 = 24^2 + 7^2$$

$$OB^2 = 576 + 49 = 625$$

$$OB = \sqrt{625} = 25 \text{ cm}$$

Sol.448.(d)

$$\text{Altitude of an equilateral} = \frac{\sqrt{3}}{2}a$$

$$= \frac{\sqrt{3}}{2} \times 37\sqrt{3} = \frac{37 \times 3}{2} = 55.5$$

Sol.449.(a) Let $4\angle P = 5\angle Q = 20\angle R$

$$\Rightarrow \angle P : \angle Q : \angle R = \frac{k}{4} : \frac{k}{5} : \frac{k}{20}$$

$$\Rightarrow \angle P : \angle Q : \angle R = 5 : 4 : 1$$

Now,

$$10 \text{ units} = 180^\circ \Rightarrow 4 \text{ units} = 72^\circ$$

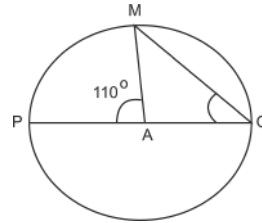
Sol.450.(c) In $\triangle XYZ$ is an isosceles triangle with side $XY = XZ$.

$$\therefore \angle Y \text{ and } \angle Z = 80^\circ$$

In $\triangle XYZ$,

$$\Rightarrow \angle X + \angle Y + \angle Z = 180^\circ$$

$$\Rightarrow \angle X = 180^\circ - 80^\circ - 80^\circ \Rightarrow \angle X = 20^\circ$$

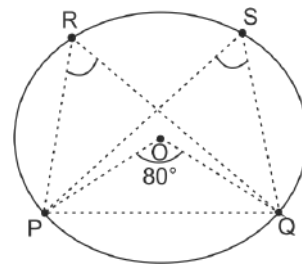
Sol.451.(a)

Here, $\angle MAQ = 180^\circ - 110^\circ = 70^\circ$

Now in $\triangle MAQ$, $\angle M = \angle Q = x$ ($AM = AQ$ = Radius)

$$\Rightarrow \angle A + \angle M + \angle Q = 180^\circ$$

$$\Rightarrow 70^\circ + 2x = 180^\circ \Rightarrow x = 55^\circ$$

Sol.452.(c)

We know that the angle made by the chord on the circumference is half to the angle made by the on the center of the circle .

$$\therefore \angle PRQ = \frac{1}{2} \angle POQ = \angle PSQ$$

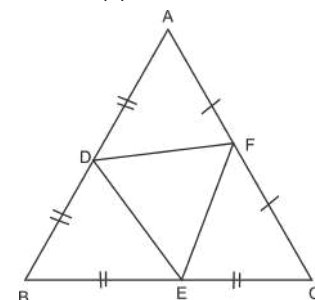
$$\text{So } \angle PRQ = \angle PSQ = 40^\circ$$

Sol.453.(a) If $\triangle DEF$ is congruent to $\triangle XYZ$ their corresponding sides and angle will be equal to each other i.e $DE = XY$, $EF = YZ$, $DF = XZ$

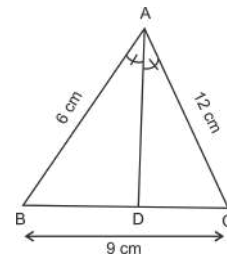
$$\text{In } \triangle DEF, DF = \sqrt{12^2 + 15^2} = \sqrt{369}$$

$$= 3\sqrt{41} \text{ cm, Hence } DF = XZ = 3\sqrt{41} \text{ cm}$$

Sol.454.(a) When two lines coincide with each other then there are infinitely many solution or points of intersection.

Sol.455.(d)

$\triangle DEF$ is not congruent to $\triangle ABC$ because their size are different ,
Hence, $\triangle DEF \not\cong \triangle ABC$.

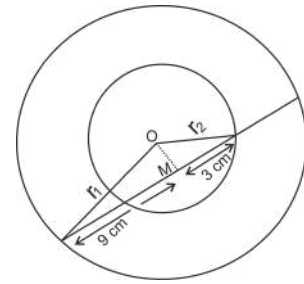
Sol.456.(b)

$$\frac{AB}{BD} = \frac{AC}{DC} \dots (\text{angle bisector theorem})$$

$$\frac{6}{BD} = \frac{12}{DC} \Rightarrow \frac{DC}{BD} = \frac{4}{2}$$

$$BC (6 \text{ unit}) = 9 \text{ cm}$$

$$DC (4 \text{ unit}) = \frac{9}{6} \times 4 = 6 \text{ cm}$$

Sol.457.(d)

OM is perpendicular to chord

Hence, by pythagoras theorem

$$OM^2 = r_1^2 + (9)^2 \dots (i)$$

$$OM^2 = r_2^2 + (3)^2 \dots (ii)$$

From (i) and (ii)

$$r_2^2 + (3)^2 = r_1^2 + (9)^2$$

$$r_2^2 - r_1^2 = 81 - 9 = 72 \text{ cm}$$

Sol.458.(a) It is always possible to draw a circle passing through 3 non-collinear points in a plane.

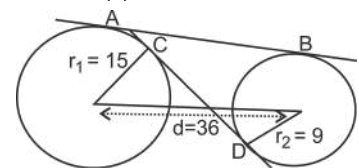
Sol.459.(c) In triangle,

Third side < 18 ($12 + 6$)

Third side > 6 ($12 - 6$)

For isosceles triangle side must be equal to 12 or 6 cm but third side should be greater than 6 cm.

Hence, third side = 12 cm

Sol.460.(a)

$$CD = \sqrt{d^2 - (r_1 + r_2)^2}$$

$$= \sqrt{36^2 - (15 + 9)^2} = \sqrt{1296 - 576}$$

$$= \sqrt{720} = 12\sqrt{5}$$

$$AB = \sqrt{d^2 - (r_1 - r_2)^2}$$

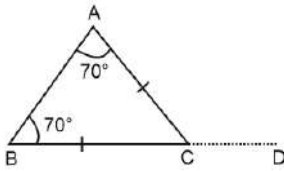
$$= \sqrt{36^2 - (15 - 9)^2} = \sqrt{1296 - 36}$$

$$= \sqrt{1260} = 6\sqrt{35}$$

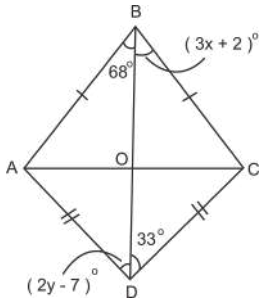
$$\text{Sum } (CD + AB) = 12\sqrt{5} + 6\sqrt{35}$$

$$= 6\sqrt{5} (2 + \sqrt{7})$$



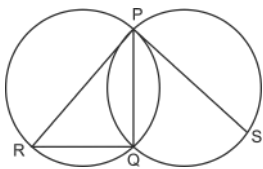
Sol.461.(a)

$\angle BAC = \angle ABC = 70^\circ$... (angle opposite to equal side)
 $\angle ACD = \angle ABC + \angle BAC$... (exterior angle property)
 $\angle ACD - \angle ABC = \angle BAC$
 $\angle ACD - \angle ABC = 70^\circ$

Sol.462.(c)

In $\triangle ABD$ and $\triangle CBD$
 $AB = BC$... (given)
 $BD = BD$... (common)
 $AD = DC$... (given)
 $\triangle ABD \cong \triangle CBD$ by SSS congruency.
Hence,
 $\angle ABD = \angle CBD$ by C.P.C.T.
 $68^\circ = (3x + 2)^\circ$
 $x = \frac{66}{3} = 22^\circ$
and $\angle ADB = \angle CDB$ by C.P.C.T.
 $(2y - 7)^\circ = 33^\circ \Rightarrow y = \frac{40}{2} = 20^\circ$

Now,
 $2x + 3y = 2 \times 22^\circ + 3 \times 20^\circ$
 $= 44^\circ + 60^\circ = 104^\circ$

Sol.463.(a)

As PR is the diameter of the circle then
 $\angle PQR = 90^\circ$

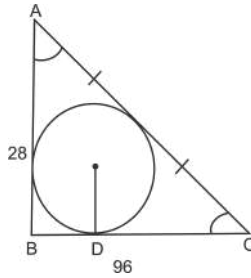
Sol.464.(c) Perimeter of sector
= length of arc + $2 \times$ radius
 $= \frac{\theta}{360^\circ} \times 2\pi r + 2r$
 $= \frac{1}{8} \times 2 \times \frac{22}{7} \times 14 + 2 \times 14$
 $= 11 + 28 = 39$

Sol.465.(b) $\angle QPS = \frac{1}{2} \times \angle QOS = 10x^\circ$
Now,
 $\angle QPS + \angle QRS = 180^\circ$... (opposite angles of cyclic quadrilateral)

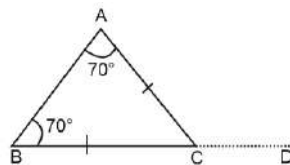
$$10x^\circ + 26x^\circ = 180^\circ \Rightarrow x = \frac{180^\circ}{36} = 5$$

Sol.466.(d) both (a) and (c) are correct.
A triangle can have at most one obtuse angle and sum of all interior angles is always 180° .

Sol.467.(b) Ratio of angle = $7 : 8 : 3$
Sum of angle = 18 unit
18 unit = 180°
Largest angle (8 unit) = $\frac{180^\circ}{18} \times 8 = 80^\circ$

Sol.468.(a)

$AC = \sqrt{AB^2 + BC^2} = \sqrt{28^2 + 96^2}$
 $= \sqrt{784 + 9216} = 100$
Circumradius = $\frac{\text{Hypotenuse}}{2} = \frac{100}{2}$
 $= 50$ cm
Area of circumcircle = $\pi \times 50 \times 50$
 $= 3.14 \times 2500 = 7850 \text{ cm}^2$

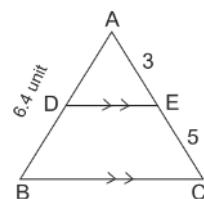
Sol.469.(a)

In $\triangle ABC$
 $\angle A + \angle B + \angle C = 180^\circ$
 $\angle ACD = \angle A + \angle B$... (exterior angle property)
 $\angle ACD = 140^\circ$
now,
 $2.5 \times \angle ACD - 1.5 \angle ABC$
 $2.5 \times 140^\circ - 1.5 \times 70^\circ = 350^\circ - 105^\circ = 245^\circ$

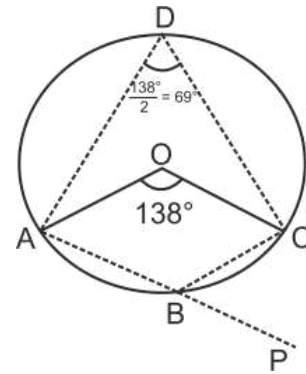
Sol.470.(d)

$$5AE = 3EC$$

$$\frac{AE}{EC} = \frac{3}{5}$$



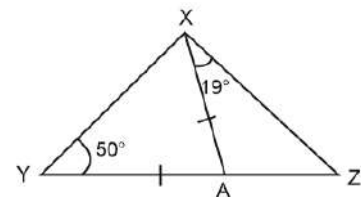
$DE \parallel BC$
Hence by BPT
 $\frac{AD}{AB} = \frac{AE}{AC} \Rightarrow \frac{AD}{6.4} = \frac{3}{8}$
 $AD = 2.4$
 $DB = 6.4 - 2.4 = 4$ unit

Sol.471.(d)

$\angle ADC = \frac{1}{2} \times 138^\circ = 69^\circ$
Now,
 $\angle ABC = 180^\circ - 69^\circ = 111^\circ$
 $\angle ABC + \angle CBP = 180^\circ$... (linear pair)
 $\angle CBP = 180^\circ - 111^\circ = 69^\circ$

Sol.472.(a) Isosceles triangle
If 2 altitudes of a triangle are equal then the triangle formed is an isosceles triangle.

Sol.473.(b) $\angle P + \angle Q + \angle R = 180^\circ$
 $\angle R = 180^\circ - 84^\circ = 96^\circ$

Sol.474.(a)

In $\triangle XYA$
 $\angle YXA = \angle AYX = 50^\circ$... (angle opposite to equal sides)
Now
 $\angle XAZ = \angle YXA + \angle AYX = 50^\circ + 50^\circ$
 $= 100^\circ$
In $\triangle AXZ$
 $\angle XZA + \angle XAZ + \angle AXZ = 180^\circ$
 $\angle XZA = 180^\circ - 100^\circ - 19^\circ = 61^\circ$

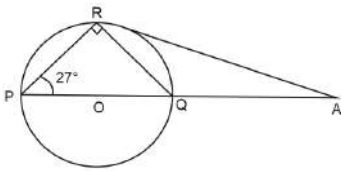
Sol.475.(c)

$\frac{\text{perimeter of } \triangle ABC}{\text{perimeter of } \triangle DEF} = \frac{24}{9} = \frac{AB}{DE} = \frac{BC}{EF}$
 $\frac{24}{9} = \frac{AB}{3}$, $AB = 8$ cm
 $\frac{\text{area of } \triangle ABC}{\text{area of } \triangle DEF} = \left(\frac{AB}{DE}\right)^2$
 $\frac{\text{area of } \triangle ABC}{\text{area of } \triangle DEF} = \left(\frac{8}{3}\right)^2$
Required ratio = $64 : 9$

Sol.476.(a)

Only option (a) is the correct option.
If two circles touch each other, the point of contact lies on the line joining the two centres.

Sol.477.(a)



In ΔPQR

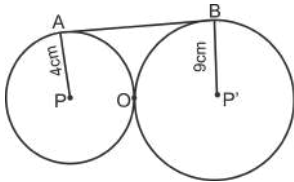
$\angle PRQ = 90^\circ$ [angle made by diameter at the circumference of the circle]

$\angle RPQ = 27^\circ$ [given]

$90^\circ + 27^\circ + \angle RQP = 180^\circ$

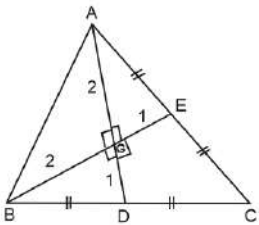
So, $\angle RQP = 180^\circ - 117^\circ = 63^\circ$

Sol.478.(b)



Length of tangent $AB = 2\sqrt{r_1 r_2} = 2\sqrt{4 \times 9}$
 $= 12$ cm

Sol.479.(b)



Centroid intersect the medians into 2 : 1

$AD = 18$ cm

AD (3 unit) = 18 cm $\Rightarrow$ 1 unit = 6 cm

Then, AG (2unit) = 12 cm, GD (1cm) = 6 cm

Now, $BE = 12$ cm

BE (3 unit) = 12 cm $\Rightarrow$ 1 unit = 4 cm

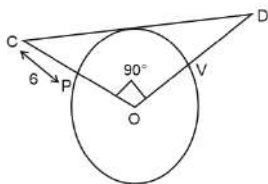
BG (2 unit) = 8 cm, GE (1 unit) = 4 cm

Now, in right angle ΔBGD

$$BD^2 = BG^2 + GD^2 \Rightarrow BD^2 = 8^2 + 6^2 = 100$$

$BD = 10$ cm

Sol.480.(a)



Circumference of the circle = $2\pi r$

$$88 = 2 \times \frac{22}{7} \times r \Rightarrow r = 14$$
 cm

Length of PO and $OV = 14$ cm

So the length of $CO = 6 + 14 = 20$ cm

And the length of $OD = 20 + 1 = 21$

$$(CD)^2 = (CO)^2 + (OD)^2$$

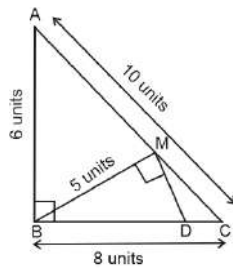
$$(CD)^2 = (20)^2 + (21)^2$$

$$(CD)^2 = 400 + 441 = 841$$

$CD = 29$ cm

$$\text{Half of the } CD = \frac{29}{2} = 14.5$$
 cm

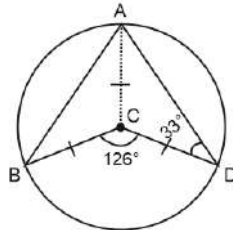
Sol.481.(c)



According to the question,

$$\frac{BM}{AB} = \frac{BD}{AC} \Rightarrow \frac{5}{6} = \frac{BD}{10} \Rightarrow BD = \frac{25}{3}$$
 units

Sol.482.(d)



Construction : join A and C

($\because$ The angle subtended by an arc at the center is twice the angle subtended at the circumference of the circle)

So, $\angle BCD = 2\angle BAD$

$$126^\circ = 2\angle BAD \Rightarrow \angle BAD = 63^\circ$$

in ΔACD

$\angle CAD = \angle CDA = 33^\circ$... {angle opposite to equal side}

Now, $\angle BAC = \angle BAD - \angle CAD$

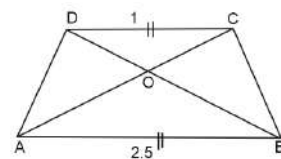
$$\angle BAC = 63^\circ - 33^\circ = 30^\circ$$

in ΔACB

$\angle CAB = \angle ABC = 30^\circ$... {angle opposite to equal side}

Hence, $\angle ABC = 30^\circ$

Sol.483.(b)



In the trapezium ABCD

$$\frac{\text{ar } \Delta AOB}{\text{ar } \Delta DOC} = \frac{AB^2}{DC^2}$$

$$\frac{\text{ar } \Delta AOB}{\text{ar } \Delta DOC} = \frac{(2.5)^2}{(1)^2} = \frac{(\frac{5}{2})^2}{(1)^2} = \frac{25}{4}$$

Hence, required ratio = 25 : 4

Sol.484.(d) $\Delta PQR \sim \Delta XYZ$

The ratio of the area of two similar triangles is equal to the square of the ratio of any pair of the corresponding sides of the similar triangles.

$$\text{Then, } \frac{\text{ar } \Delta PQR}{\text{ar } \Delta XYZ} = \left(\frac{QR}{YZ}\right)^2$$

$$\frac{12.96}{635.04} = \left(\frac{2.9}{YZ}\right)^2 \Rightarrow \frac{1.44}{70.56} = \left(\frac{2.9}{YZ}\right)^2$$

$$\sqrt{\frac{1.44}{70.56}} = \frac{2.9}{YZ} \Rightarrow \frac{1.2}{8.4} = \frac{2.9}{YZ}$$

$$\frac{1}{7} = \frac{2.9}{YZ} \Rightarrow YZ = 20.3$$

Sol.485.(a)

ΔABC and ΔPQR are similar triangles

$$\text{Then, } \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$$

Now, according to the question,

$$\frac{AB}{PQ} = \frac{BC}{QR} \Rightarrow \frac{8}{12} = \frac{BC}{18}$$

$BC = 12$ cm

Now,

$$\frac{BC}{QR} = \frac{AC}{PR} \Rightarrow \frac{12}{18} = \frac{AC}{24}$$

$AC = 16$ cm

So, perimeter of $\Delta ABC = (AB + BC + AC)$

$$= (8 + 12 + 16) = 36$$
 cm

Sol.486.(b)

Length of the common tangent

$$= \sqrt{d^2 - (R - r)^2}$$

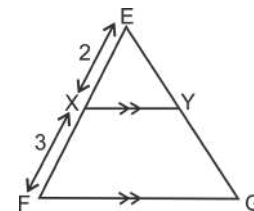
$$= \sqrt{(13)^2 - (10 - 8)^2} = \sqrt{169 - 4}$$

$$= \sqrt{165}$$
 cm

$$\text{Sol.487.(c)} \quad \frac{\text{area of } \Delta ABC}{\text{area of } \Delta QRP} = \left(\frac{BC}{RP}\right)^2$$

$$\sqrt{\frac{9}{4}} = \frac{BC}{RP} \Rightarrow \frac{15}{RP} = \frac{3}{2} \Rightarrow RP = 10$$
 cm

Sol.488.(c)



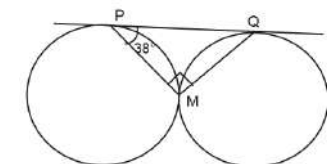
$$\frac{\text{area of } \Delta EXY}{\text{area of } \Delta EFG} = \left(\frac{2}{2+3}\right)^2 = \frac{4}{25}$$

Area of quadrilateral XFGY = 25 - 4 = 21 units

21 units = 44 m²

$$(\text{area of } \Delta EXY) 2 \text{ units} = \frac{44}{21} \times 4 = 8.38$$
 m²

Sol.489.(c)



Concept used :- if two circle touch each other externally at some point and a direct common tangent is drawn to both circle, the angle subtended by the direct common tangent at the point where two circles touch each other is 90°

According to the concept, $\angle PMQ = 90^\circ$

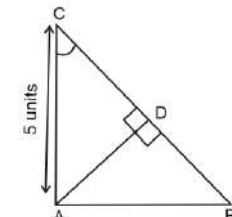
Hence the $\angle PQM = 90^\circ - 38^\circ = 52^\circ$

Sol.490.(d)

Concept used :- if two circle touch each other externally at some point and a direct common tangent is drawn to both circle, the angle subtended by the direct common tangent at the point where two circles touch each other is 90°

According to the concept, $\angle PMQ = 90^\circ$

Hence the $\angle PQM = 90^\circ - 38^\circ = 52^\circ$



$$\frac{\text{area of } \triangle ACB}{\text{area of } \triangle DCA} = \left(\frac{BC}{AC}\right)^2 [\because \triangle ACB \sim \triangle DCA]$$

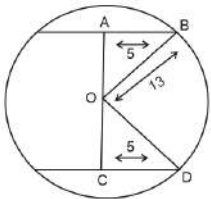
$$\sqrt{\frac{63}{7}} = \frac{BC}{5} \Rightarrow \sqrt{9} = \frac{BC}{5} \Rightarrow BC = 15 \text{ cm}$$

Sol.491.(a) If AB = 11 cm, BC = 13 cm, and AC = 24 cm these three point A, B and C lie on a straight line

Since the points are collinear, the number of circles that can pass through all three points is 0.

No circle can be drawn through three collinear points

Sol.492.(b)



By pythagorean formula :-

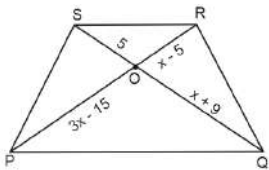
$$AO = \sqrt{(OB)^2 - (AB)^2}$$

$$AO = \sqrt{(13)^2 - (5)^2} \Rightarrow AO = \sqrt{144}$$

$$AO = 12 \text{ cm}$$

$$\text{So the distance between AC} = 12 + 12 = 24 \text{ cm}$$

Sol.493.(b) According to the question,



$$\text{Concept used :- } \frac{PO}{OR} = \frac{QO}{OS}$$

$$\frac{3x-15}{x-5} = \frac{x+9}{5}$$

$$15x - 75 = x^2 + 9x - 5x - 45$$

$$x^2 - 11x + 30 = 0$$

$$x^2 - 6x - 5x + 30 = 0$$

$$x(x-6) - 5(x-6) = 0$$

$$(x-5)(x-6) = 0$$

$$x = 5, x = 6$$

$$\text{Now, the value of } (x_1^2 - x_2^2) = (6^2 - 5^2)$$

$$= 36 - 25 = 11$$

Sol.494.(b)

concept :- in similar triangles the ratio of their area is equal to square of their corresponding sides or perimeters

Perimeter ratio of similar triangles = 4 : 7

So their area ratio = 16 : 49

Sum of areas = (16 + 49) sq . unit

= 65 sq . unit

According to the question,

$$65 \text{ sq . unit} = 195 \text{ sq . cm}$$

$$1 \text{ sq . unit} = 3 \text{ sq . cm}$$

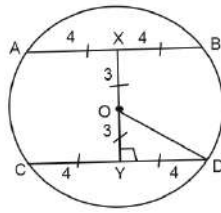
$$\text{Now, difference of area} = (49 - 16) \text{ unit}^2$$

$$= 33 \text{ unit}^2 = 33 \times 3 = 99 \text{ sq . cm}$$

$$\text{So, required difference} = \frac{1}{3} \times 99$$

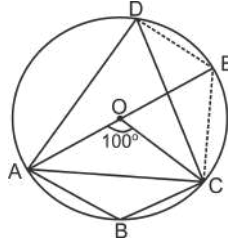
$$= 33 \text{ sq . cm}$$

Sol.495.(b)



$$\text{Radius of circle (OD)} = \sqrt{(OY)^2 + (YD)^2} \\ = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ unit}$$

Sol.496.(a)



Concept used :- The angle made by an arc at the center is twice the angle made by the same arc at the circumference.

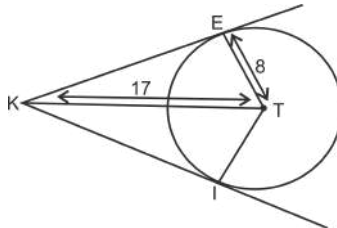
$$\text{So, } \angle CEA = \angle ADC = \frac{100}{2} = 50^\circ$$

$\angle ADE = 90^\circ$ (In a semicircle, any angle formed by a diameter is always 90°)

$$\angle CDE = \angle ADE - \angle ADC = 90^\circ - 50^\circ = 40^\circ$$

$$\angle CDE + \angle CEA = 40^\circ + 50^\circ = 90^\circ$$

Sol.497.(d)



the angle between the radius and the tangent will always be exactly 90°

$$(KE)^2 = (17)^2 - (8)^2 \Rightarrow (KE)^2 = 289 - 64$$

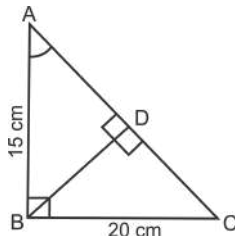
$$(KE)^2 = 225 \Rightarrow KE = 15 \text{ cm}$$

$$\text{Area of } \triangle KET = \frac{1}{2} \times 15 \times 8 = 60 \text{ sq . cm}$$

Area of $\triangle KIT$ = area of $\triangle KET$ (both triangle are congruent)

$$\text{So the Area of quadrilateral TEKI} = 60 \times 2 = 120 \text{ sq . cm}$$

Sol.498.(b)



By pythagorean :-

$$AC = \sqrt{(15)^2 + (20)^2}$$

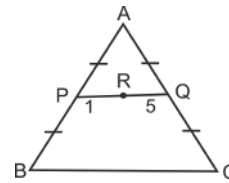
$$AC = \sqrt{225 + 400} \Rightarrow AC = 25 \text{ cm}$$

Property of right angle Δ :-

$$AB \times BC = BD \times AC$$

$$15 \times 20 = BD \times 25 \Rightarrow BD = 12 \text{ cm}$$

Sol.499.(a)



According to the question,

(PR) 1 unit = 6 cm

(PQ) 6 units = 36 cm

$$\text{Hence, } BC = 36 \times 2 = 72 \text{ cm } (\because \frac{BC}{2} = PQ)$$

Sol.500.(d) Let the measure of smallest and largest angle be x° and y°

And the 3rd angle be z°

According to the question,

$$x^\circ + 42^\circ = y^\circ \dots\dots (i)$$

$$x^\circ + 24^\circ = z^\circ \dots\dots (ii)$$

On adding the e.q . (i) and (ii) we get,

$$x^\circ + 42^\circ + x^\circ + 24^\circ = y^\circ + z^\circ$$

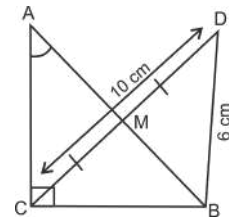
$$2x^\circ + 66^\circ = y^\circ + z^\circ$$

Then, sum of all angle

$$x^\circ + 2x^\circ + 66^\circ = 180^\circ (\because x^\circ + y^\circ + z^\circ = 180^\circ)$$

$$3x^\circ = 180^\circ - 66^\circ \Rightarrow x^\circ = \frac{114}{3} = 38^\circ$$

Sol.501.(b)



$$CM = \frac{10}{2} = 5 \text{ cm}$$

Sol.502.(b)

Formula:- Lengths of the transverse

$$\text{common tangent} = \sqrt{d^2 - (R + r)^2}$$

Lengths of the direct common tangent

$$= \sqrt{d^2 - (R - r)^2}$$

$$\text{Required ratio} = \sqrt{(80)^2 - (35 + 25)^2} :$$

$$\sqrt{(80)^2 - (35 - 25)^2}$$

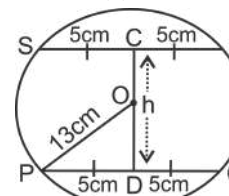
$$= \sqrt{(80)^2 - (60)^2} : \sqrt{(80)^2 - (10)^2}$$

$$= \sqrt{140 \times 20} : \sqrt{90 \times 70}$$

$$= \sqrt{7 \times 20 \times 20} : \sqrt{7 \times 9 \times 100}$$

$$= 20\sqrt{7} : 30\sqrt{7} = 2 : 3$$

Sol.503.(c)



In the right angle $\triangle ODP$

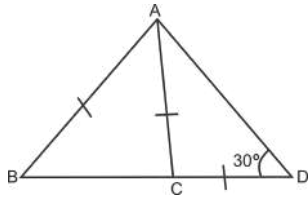
$$OD = \sqrt{(OP)^2 - (PD)^2} = \sqrt{(13)^2 - (5)^2}$$

$$= \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm}$$

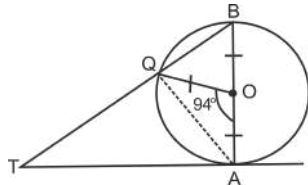
Now, $CD = 2OD = 24 \text{ cm}$

Hence, $h = 24 \text{ cm}$

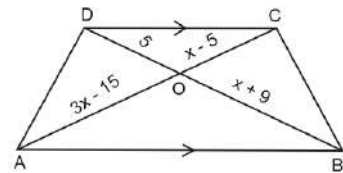


Sol.504.(a)

$\angle CAD = 30^\circ$ ($\because AC = CD$)
 So, $\angle ACD = 180^\circ - (30^\circ + 30^\circ) = 120^\circ$
 Then, $\angle ACB = 180^\circ - 120^\circ = 60^\circ$
 $\angle ABC = 60^\circ$ ($\because AC = AB$)
 Hence, the angles of the triangle are $60^\circ, 60^\circ, 60^\circ$

Sol.505.(b)

$\angle AQB = 90^\circ$ (angle subtended by a diameter on any point of a circle is always 90°)
 So, $\angle TQA = 90^\circ$
 $\angle OAQ = \frac{180^\circ - 94^\circ}{2} = 43^\circ$
 $\angle QAT = 90^\circ - 43^\circ = 47^\circ$
 In $\triangle TQA$,
 $\angle QTA = 180^\circ - (90^\circ + 47^\circ)$
 $\angle QTA = 180^\circ - 137^\circ = 43^\circ$
 Required angle ($\angle ATQ$) = 43°

Sol.506.(c)

In trapezium ABCD,
 The diagonals of a trapezium divide each other proportionally,

$$\Rightarrow \frac{AO}{OC} = \frac{OB}{OD} \Rightarrow \frac{3x-15}{x-5} = \frac{x+9}{x-5}$$

$$\Rightarrow 15x - 75 = x^2 + 9x - 5x - 45$$

$$\Rightarrow x^2 + 4x - 15x + 75 - 45 = 0$$

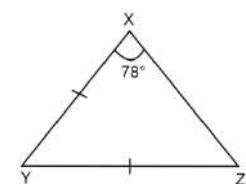
$$\Rightarrow x^2 - 11x + 30 = 0 \dots (i)$$

On solving the eqn. (i), we get

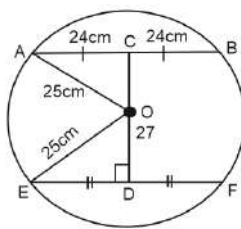
$$\Rightarrow x = 5 \text{ or } 6$$

Now,

$$(x_1^2 + x_2^2) = (5)^2 + (6)^2 = 25 + 36 = 61$$

Sol.507.(a)

In $\triangle XYZ$,
 $\Rightarrow \angle X + \angle Y + \angle Z = 180^\circ$
 $\Rightarrow 78^\circ + \angle Y + 78^\circ = 180^\circ$ ($\angle X = \angle Z$)
 $\Rightarrow \angle Y = 180 - 156^\circ \Rightarrow \angle Y = 24^\circ$

Sol.508.(c)

Applying pythagorean theorem in the right angle $\triangle ACO$

$$OC^2 = AO^2 - AC^2$$

$$OC^2 = (25)^2 - (24)^2 = 49 \quad OC = 7 \text{ cm}$$

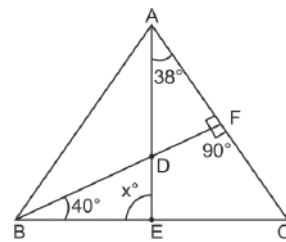
$$\text{Now, } OD = CD - OC = 27 - 7 = 20 \text{ cm}$$

Applying pythagorean theorem in the right angle $\triangle ODE$

$$ED^2 = OE^2 - OD^2$$

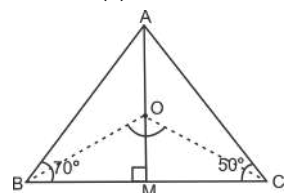
$$ED^2 = (25)^2 - (20)^2 = 225 \Rightarrow ED = 15 \text{ cm}$$

So, the length of chord (EF) = $15 \times 2 = 30 \text{ cm}$

Sol.509.(a)

In the right angle $\triangle AFD$,
 $38^\circ + 90^\circ + \angle ADF = 180^\circ \Rightarrow \angle ADF = 52^\circ$
 Now, $\angle ADF = \angle BDE = 52^\circ$ [vertically opposite angle]
 In the $\triangle BDE$
 $40^\circ + 52^\circ + \angle BED = 180^\circ \Rightarrow \angle BED = 88^\circ$

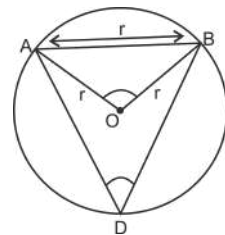
Sol.510.(d) Length of arc = $\frac{\theta}{360} \times 2\pi r$
 $= \frac{120}{360} \times 2 \times \frac{22}{7} \times 7 = \frac{44}{3} \text{ cm}$

Sol.511.(a)

$$\angle BAC = 180^\circ - \angle B - \angle C$$

$$= 180^\circ - 120^\circ = 60^\circ$$

$$\angle BOC = 90^\circ + \frac{\angle BAC}{2} = 90^\circ + \frac{60^\circ}{2} = 120^\circ$$

Sol.512.(a)

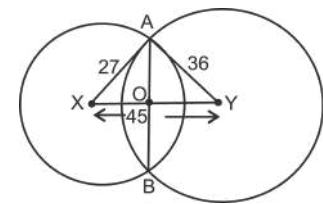
$\triangle AOB$ is an equilateral triangle.

So, $\angle AOB = 60^\circ$ ($\because$ all angle of equilateral triangle is 60°)

And $\angle ADB = 30^\circ$ ($\because$ The angle subtended by an arc at the center is

twice the angle subtended at the circumference of the circle)

Hence, the required angle will be 60° and 30°

Sol.513.(d)

$\triangle XOY$ is right angled as [tangent (AY) $\perp$ radius (XA)]

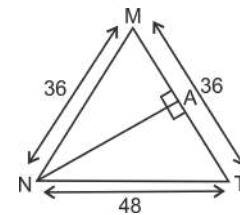
Now, in right angled $\triangle XOY$

$$XA \times AY = OA \times XY$$

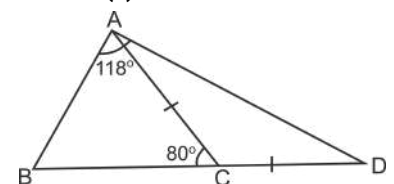
$$27 \times 36 = OA \times 45 \Rightarrow OA = 21.6 \text{ cm}$$

$$AB = 2 \times OA = 2 \times 21.6 = 43.2 \text{ cm}$$

Hence, length of common chord (AB) = 43.2 cm

Sol.514.(b)

Area of isosceles triangle = $\frac{a}{4} \sqrt{4b^2 - a^2}$
 $= \frac{1}{2} \times \text{base} \times \text{height (NA)}$
 $\Rightarrow \frac{48}{4} \sqrt{4(36)^2 - (48)^2} = \frac{1}{2} \times 36 \times \text{height}$
 $\Rightarrow 12\sqrt{5184 - 2304} = 18 \times \text{height}$
 $\Rightarrow 12 \times 24\sqrt{5} = 18 \times \text{height}$
 Height = $\frac{12 \times 24\sqrt{5}}{18} = 16\sqrt{5} \text{ cm}$

Sol.515.(c)

$\angle ACD = 180 - 80 = 100^\circ$ [sum of angles at one side of linear line is 180°]

Side, $AC = CD$

Then, $\angle CAD = \angle ADC$

In $\triangle CAD$

$$2\angle CAD + 100^\circ = 180^\circ$$

$$\angle CAD = 40^\circ$$

$$\text{Now, } \angle BAC = 118 - 40 = 78^\circ$$

In $\triangle ABC$

$$\angle BAC + \angle ACB + \angle CBA = 180^\circ$$

$$78^\circ + 80^\circ + \angle ABC = 180^\circ \Rightarrow \angle ABC = 22^\circ$$

Sol.516.(b) Concept :- ratio of the area of similar triangles is equal to the square of corresponding sides.

Now, smallest side of $\triangle ABC$ ($3x$) = 15 unit
 $x = 5 \text{ unit}$

So, other sides of $\triangle ABC$

$$4x = 20 \text{ unit}, 5x = 25 \text{ unit}$$

As, (15, 20, 25) are the pythagorean triplet
So, ΔABC is right angle triangle

$$\text{Area of } \Delta ABC = \frac{1}{2} \times 15 \times 20$$

$$= 150 \text{ square unit}$$

According to question,

$$\text{Area of } \Delta DEF = \frac{1}{2} \times 150 = 75 \text{ square unit}$$

$$\text{So, } \frac{\text{Ar}(\Delta DEF)}{\text{Ar}(\Delta ABC)} = \frac{(x)^2}{(25)^2}$$

$$\frac{75}{150} = \frac{(x)^2}{(25)^2} \Rightarrow x^2 = \frac{(25)^2}{2} \Rightarrow x = \frac{25}{\sqrt{2}}$$

$$\text{Hence, largest side of } \Delta DEF (x) = \frac{25}{\sqrt{2}} \text{ unit}$$

Sol.517.(c) Smallest side = x

Then, hypotenuse = $2x + 6$

And, third side = $2x + 4$

Take $x = 10$ m

Then, sides of right angle triangle

= 10, 24 and 26 m (pythagorean triplet)

$$\text{So, area of triangle} = \frac{1}{2} \times 10 \times 24 = 120 \text{ m}$$

Sol.518.(d) Let angles of the triangles be

$2x, 4x$ and $6x$

$$2x + 4x + 6x = 180^\circ$$

$$12x = 180^\circ \Rightarrow x = 15^\circ$$

Angles of triangle = $30^\circ, 60^\circ, 90^\circ$

Ratio of sides = $1 : \sqrt{3} : 2$

Since, one angle is 90° so, triangle is right angled

Now, circumradius of right angle triangle

$$= \frac{\text{hypotenuse}}{2}$$

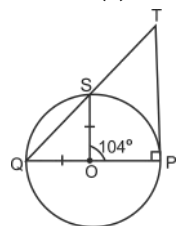
$$\frac{\text{hypotenuse}}{2} = 12 \Rightarrow \text{hypotenuse} = 24 \text{ cm}$$

Now, 2 unit = 24 cm, 1 unit = 12 cm

Other sides be 12 cm, $12\sqrt{3}$ cm

Hence, sides are 12 cm, $12\sqrt{3}$ cm, 24 cm

Sol.519.(d)



In ΔSOQ

$\angle QSO + \angle SQO = 104^\circ$ [exterior angle is equal to sum of opposite interior angle]

$2\angle SQO = 104^\circ$ [opposite angle of same side (OS = OQ = Radius)]

$$\angle SQO = 52^\circ$$

Now, in right angle ΔTPQ

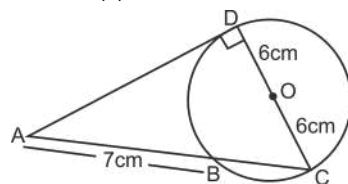
$$52^\circ + 90^\circ + \angle PTQ = 180^\circ \Rightarrow \angle PTQ = 38^\circ$$

Sol.520.(c) $\Delta XYZ \sim \Delta PQR$

$$\frac{\text{perimeter of } \Delta XYZ}{\text{perimeter of } \Delta PQR} = \frac{XY}{PQ}$$

$$\frac{XY}{3.6} = \frac{16}{9} \Rightarrow XY = \frac{64}{10} = 6\frac{2}{5}$$

Sol.521.(a)



Let $BC = x$

In ΔADC

$$AC^2 = AD^2 + CD^2 \quad (\text{pythagoras theorem})$$

$$(7 + x)^2 = AD^2 + (12)^2$$

$$AD^2 = (7 + x)^2 - (12)^2 \quad \text{---(i)}$$

$$AD^2 = 7(7 + x) \quad \text{---(ii)} \quad [\because AD^2 = AB \times AC]$$

Now

$$(7 + x)^2 - (12)^2 = 7(7 + x)$$

$$x^2 + 7x - 144 = 0$$

$$x^2 + 16x - 9x - 144 = 0$$

$$x(x + 16) - 9(x + 16) = 0$$

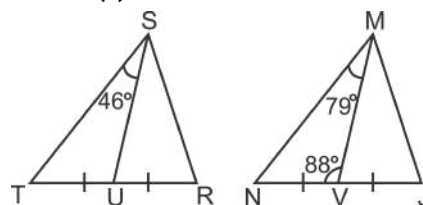
$$(x - 9)(x + 16) = 0$$

$$x = 9, -16 \quad (\because \text{side cannot be negative})$$

$$x = 9$$

$$AC = AB + BC = 7 + 9 = 16 \text{ cm}$$

Sol.522.(a)



$$\angle STU = 180^\circ - (46^\circ + 88^\circ) = 46^\circ$$

$$\angle SRT = 180^\circ - (46^\circ + 79^\circ) = 55^\circ$$

$$\Rightarrow \frac{7}{11} \angle SRT = \frac{7}{11} \times 55^\circ = 35^\circ$$

Sol.523.(b) Concept :- if any triangle has

three sides x, y and z

Then, $(x + y) > z > (x - y)$

According to question,

$$(25 - 19) < y < (19 + 25)$$

$$6 < y < 44$$

Sol.524.(a)

(6, 8, 10) are the pythagorean triplet

$$\text{So, area of triangle} = \frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$$

Sol.525.(d) Let initial angles are $3x + 17^\circ$,

$4x + 17^\circ$ and $7x + 17^\circ$

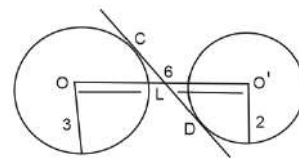
$$\text{Then, } 14x + 51^\circ = 180^\circ$$

$$14x = 129^\circ \Rightarrow x = \frac{129^\circ}{14}$$

$$\text{Largest angle } (7x + 17) = 7 \times \frac{129^\circ}{14} + 17$$

$$= \frac{163^\circ}{2} \Rightarrow \frac{163^\circ}{2} \times \frac{\pi}{180} = \frac{163\pi}{360} \text{ radian.}$$

Sol.526.(d) Concept used : For maximum distance between the centres of the circle, we need to draw the transverse common tangent. For the minimum distance between the centres of the circle, we need to draw the direct common tangent.



Let the maximum distance between the centres be 'L'

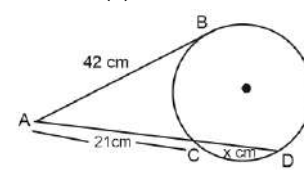
Transverse common tangent

$$= \sqrt{L^2 - (R + r)^2}$$

$$6 = \sqrt{L^2 - (3 + 2)^2} = \sqrt{L^2 - 5^2}$$

$$36 = L^2 - 25 \Rightarrow 61 = L^2 \Rightarrow L = \sqrt{61} \text{ Units}$$

Sol.527.(d)



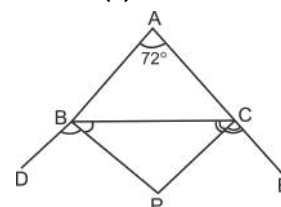
$$(AB)^2 = AC \times AD$$

$$42 \times 42 = 21 \times (21 + x)$$

$$84 = 21 + x \Rightarrow x = 63$$

$$\text{Required ratio} = 42 : 63 = 2 : 3$$

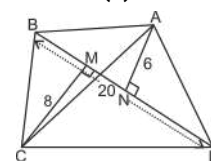
Sol.528.(b)



We know that,

$$\angle CPB = 90^\circ - \frac{72^\circ}{2} = 90^\circ - 36^\circ = 54^\circ$$

Sol.529.(a)



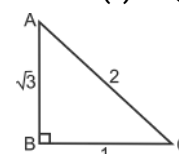
Area of quadrilateral ABCD

$$= \text{ar}(\Delta ABD) + \text{ar}(\Delta BCD)$$

$$= \frac{1}{2} (\text{CM} + \text{AN}) \times \text{BD}$$

$$= \frac{1}{2} (8 + 6) \times 20 = 140 \text{ cm}^2$$

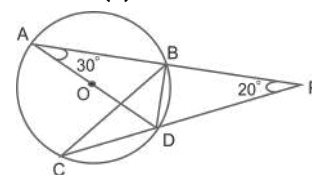
Sol.530.(c) ATQ, we get given triangle ;



$$\tan \angle ACB = \frac{\sqrt{3}}{1}$$

$$\tan \angle ACB = \tan 60^\circ \Rightarrow \angle ACB = 60^\circ$$

Sol.531.(d)



$\angle DAB = \angle DCB = 30^\circ$ (angles in same segment are equal)

$\angle ABD = 90^\circ$ (angle in semicircle)

Then, $\angle DBP = 90^\circ$

In $\triangle BDP$, $\angle PDB = 180^\circ - (90^\circ + 20^\circ) = 70^\circ$

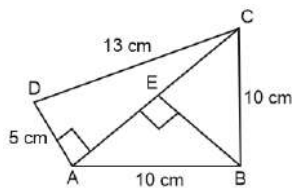
Now, in $\triangle BCD$, $\angle BDP$ is exterior angle. So, we have;

$\angle DCB + \angle DBC = \angle BDP$

$30^\circ + \angle DBC = 70^\circ$

$\angle DBC = 70^\circ - 30^\circ = 40^\circ$

Sol.532.(a)



$\triangle DAC$ is right triangle. Then, AC

$$= \sqrt{13^2 - 5^2} = \sqrt{144} = 12 \text{ cm}$$

AE = EC = 6 cm ($\triangle BCE \cong \triangle BAE$)

Now, $\triangle BEC$ is right triangle. Then, BE

$$= \sqrt{10^2 - 6^2} = \sqrt{64} = 8 \text{ cm}$$

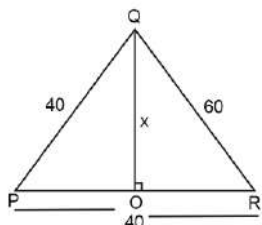
Area of quadrilateral = area of ($\triangle ABC$) + area of ($\triangle ADC$)

$$= \frac{1}{2} \times BE \times AC + \frac{1}{2} \times AD \times AC$$

$$= \frac{1}{2} \times AC(BE + AD) = \frac{1}{2} \times 12(8 + 5)$$

$$= 78 \text{ cm}^2$$

Sol.533.(c)



$$\text{Semiperimeter}(s) = \frac{40 + 40 + 60}{2} = \frac{140}{2}$$

$$= 70 \text{ cm}$$

Using Heron's formula, Area of triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{70(70-40)(70-40)(70-60)}$$

$$= \sqrt{70 \times 30 \times 30 \times 10} = \sqrt{90000 \times 7} = 300\sqrt{7}$$

$$\text{Also, area of triangle} = \frac{1}{2} \times x \times 40 = 20x$$

$$\text{Now, } 20x = 300\sqrt{7} \Rightarrow x = 15\sqrt{7} \text{ cm}$$

Sol.534.(a) Option (a) is not true as "There is no circle passing through any three points in any plane".

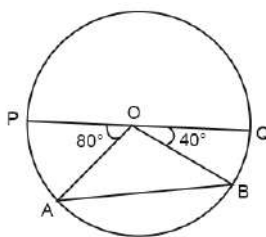
Sol.535.(d) According to the question,

$$\frac{\text{area of } \triangle ABC}{\text{area of } \triangle EDF} = \frac{(BC)^2}{(DF)^2}$$

$$\frac{1}{4} = \frac{8^2}{(DF)^2} \Rightarrow (DF)^2 = 64 \times 4 = 256$$

$$DF = 16 \text{ cm}$$

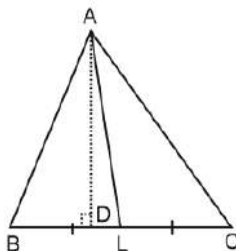
Sol.536.(d)



$$\angle POA + \angle AOB + \angle QOB = 180^\circ$$

$$80^\circ + \angle AOB + 40^\circ = 180^\circ \Rightarrow \angle AOB = 60^\circ$$

Sol.537.(c) Let AD be the height of $\triangle ABC$



$$\text{Area of } \triangle ABL = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$15 = \frac{1}{2} \times BL \times AD$$

$$30 = BL \times AD \quad \dots (i)$$

$$\text{Area of } \triangle ACL = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$= \frac{1}{2} \times LC \times AD$$

$$= \frac{1}{2} \times BL \times AD \quad \dots (L \text{ is the midpoint of } BC \because BL = LC)$$

$$= \frac{1}{2} \times 30 \quad \dots [\text{from (i)}]$$

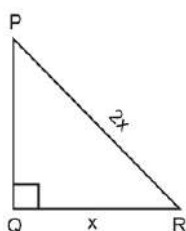
$$= 15 \text{ cm}^2$$

$$\text{Area of } \triangle ABC$$

$$= \text{Area of } \triangle ABL + \text{Area of } \triangle ACL$$

$$= 15 + 15 = 30$$

Sol.538.(a)



Using pythagoras theorem

$$PQ^2 = PR^2 - QR^2 = 4x^2 - x^2 = 3x^2$$

$$PQ = x\sqrt{3}$$

$$\text{Area of triangle} = 600$$

$$\frac{1}{2} \times QR \times PQ = 600$$

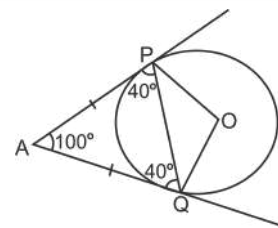
$$\frac{1}{2} \times x \times x\sqrt{3} = 600 \Rightarrow x^2 = \frac{1200}{\sqrt{3}}$$

$$x^2 = 400\sqrt{3} \Rightarrow x^2 = 400 \times 1.73$$

$$x^2 = 692 \Rightarrow x = 26.30$$

$$\text{Difference between hypotenuse and base} \\ (2x - x) = x = 26.30$$

Sol.539.(d) Theorem : Tangents drawn from an external point to a circle are equal.



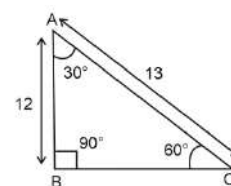
As we know, $\angle PAQ + \angle POQ = 180^\circ$

$$\angle POQ = 180^\circ - 100^\circ = 80^\circ$$

Sol.540.(b) Length of the arc of circle

$$= \frac{72^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 35 = 44 \text{ cm}$$

Sol.541.(b)



It forms a triplet of 5, 12, 13

Hence, BC = 5 cm

Sol.542.(b) Since FG is the mid-point of equilateral triangle,

So, FG will be median

Median of equilateral triangle

$$= \frac{\sqrt{3} \times \text{side}}{2} = \frac{\sqrt{3} \times 12}{2} = 6\sqrt{3} \text{ cm}$$

Sol.543.(b) B and D

A triangle can have two acute angles and a triangle can have one obtuse angle also.

Sol.544.(a) An obtuse triangle is a triangle with one angle greater than 90° .

Sol.545.(a) Concept : In any triangle the sum of any two sides is always greater than the third side.

Case 1 - taking 2 cm, 5 cm, 7 cm, as side $2 + 5 = 7$ which is equal to the third side. Hence, no triangle can be formed.

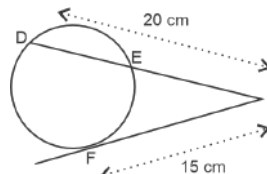
Case 2 - taking 2 cm, 5 cm, 8 cm, as side $2 + 5 < 8$. Hence, no triangle can be formed.

Case 3 - taking 5 cm, 7 cm, 8 cm, as side $5 + 7 = 12 > 8$. Hence, triangles can be formed.

Case 4 - taking 2 cm, 7 cm, 8 cm, as side $2 + 7 = 9 > 8$. Hence, triangles can be formed.

Hence, triangle can be formed in case 3 and 4 only.

Sol.546.(d)



$$CF^2 = CD \times CE$$

$$15^2 = 20 \times CE$$

$$CE = \frac{225}{20} = 11.25$$

$$\text{Hence, } DE = 20 - 11.25 = 8.75 \text{ cm}$$

Sol.547.(d) $AB = 3.8 \text{ cm}$

$$BC = 6 \text{ cm}$$

$$CA < BC - AB$$

$$< 6 - 3.8$$

$$< 2.2 \text{ cm}$$

On checking the option we can see that only option (d) cannot be the side of the scalene triangle.

Sol.548.(d) We know that:

An isosceles triangle has two equal angles.

And sum of three angles of a triangle = 180°

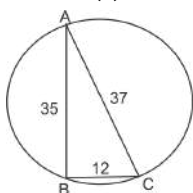
Let each of the other two angles = x° .

$$\Rightarrow x^\circ + x^\circ + 42^\circ = 180^\circ$$

$$\Rightarrow 2x^\circ = 180^\circ - 42^\circ$$

$$\Rightarrow 2x^\circ = 138^\circ \Rightarrow x^\circ = 69^\circ$$

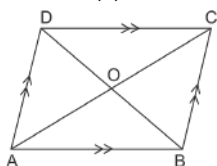
Sol.549.(c)



Circumradius of the triangle

$$= \frac{\text{hypotenuse}}{2} = \frac{37}{2} = 18.5 \text{ cm}$$

Sol.550.(b)



Option (a) $\triangle AOB \cong \triangle COD$

In $\triangle AOB$ and $\triangle COD$, $AB \parallel CD$

$\angle BAO = \angle DCO$ (alternate interior angle)

$\angle ABO = \angle CDO$ (alternate interior angle)

$\angle AOB = \angle COD$ (vertically opposite angle)

So, $\triangle AOB \cong \triangle COD$ (true)

Similarly,

Option (c) $\triangle ABC \cong \triangle ADC$ (true)

Option (d) $\triangle BOC \cong \triangle AOD$ (true)

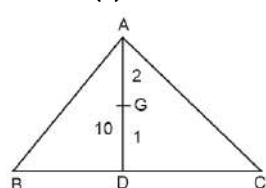
But,

Option (b) $\triangle AOD \cong \triangle COD$ (false)

Both triangles angles are not equal to each other because AD does not parallel to CD.

Therefore, the correct answer is option (b)

Sol.551.(d)



We know

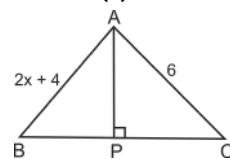
Centroid divides the median in 2 : 1.

$$GD = 1 \rightarrow 10$$

$$AG = 10 \times 2 = 20$$

$$AD = AG + GD = 20 + 10 = 30 \text{ cm}$$

Sol.552.(c)



Given P is the mid point of BC

$$\Rightarrow BP = PC$$

We know

In an equilateral triangle, $AP \perp BC$ if and only if P is the mid point of BC.

Hence, ABC is an equilateral triangle.

$$\Rightarrow AB = BC = CA$$

$$\Rightarrow 2x + 4 = 6 \Rightarrow 2x = 2 \Rightarrow x = 1$$

Sol.553.(c) Given,

$$AC = BD \Rightarrow BC = AD$$

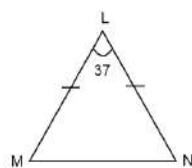
AB is common.

Hence, by SSS rule

Both triangles $\triangle ABC$ and $\triangle BAD$ are congruent.

$$\Rightarrow \triangle ABC \cong \triangle BAD$$

Sol.554.(c)



Since $LM = LN$

$$\Rightarrow \angle LMN = \angle LNM = x \text{ (say)}$$

$$\text{And } \angle L + \angle M + \angle N = 180^\circ$$

$$\Rightarrow 37 + x + x = 180^\circ$$

$$\Rightarrow 2x = 180^\circ - 37^\circ = 143^\circ$$

$$\Rightarrow x = \frac{143^\circ}{2} = 71.5^\circ$$

Sol.555.(a) As we know, $P^2 + B^2 = H^2$

$$(x^2 - 1)^2 + (2x)^2 = (x^2 + 1)^2$$

$$x^4 + 1 - 2x^2 + 4x^2 = (x^2 + 1)^2$$

$$x^4 + 2x^2 + 1 = (x^2 + 1)^2$$

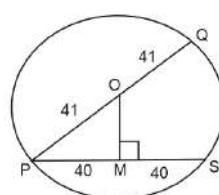
$$(x^2 + 1)^2 = (x^2 + 1)^2$$

$$\text{LHS} = \text{RHS}$$

$$\text{So, hypotenuse} = x^2 + 1$$

Sol.556.(b) Concept used :

The perpendicular distance from centre of circle to the chord bisects the chord.



In $\triangle OPM$, we have ;

$$OM = \sqrt{41^2 - 40^2} = \sqrt{1681 - 1600}$$

$$= \sqrt{81} = 9 \text{ cm}$$

Sol.557.(d) As we know, sum of the angle of triangle is 180°

$$9 + 11 + 16 = 36 \text{ unit} \rightarrow 180^\circ$$

$$\text{Then, } (16 - 9) = 7 \text{ unit} \rightarrow \frac{180^\circ}{36} \times 7 = 35^\circ$$

Sol.558.(b) In $\triangle ABC$ and $\triangle PQR$,

$$AB = PQ = 7 \text{ cm}$$

$$BC = QR = 8 \text{ cm}$$

$$AC = PR = 9 \text{ cm}$$

Then, $\triangle ABC \cong \triangle PQR$ (by SSS rule)

Sol.559.(a)

In the triangle the sum of two sides is always greater than the third side.

Sol.560.(d) Area of two congruent triangles are always equal.

Hence,

$$\text{Area of } \triangle PQR = \text{Area of } \triangle ABC = 225 \text{ cm}^2$$

Sol.561.(b) As we know, Sum of the angle of triangle = 180°

$$(18x + 6^\circ) + (10x + 30^\circ) + (15x + 15^\circ) = 180^\circ$$

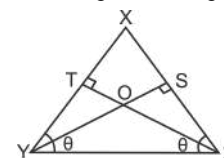
$$43x + 51^\circ = 180^\circ$$

$$43x = 129 \Rightarrow x = 3$$

$$\text{So, } 18x + 6^\circ = 18 \times 3 + 6 = 60^\circ, 10x + 30^\circ = 10 \times 3 + 30 = 60^\circ, 15x + 15^\circ = 15 \times 3 + 15 = 60^\circ$$

Therefore, the triangle is equilateral.

Sol.562.(d) According to the question, we have given triangle



$\angle XYZ = \angle XZY = \theta$ (angles opposite to equal side)

$$\text{In } \triangle TYZ, \angle TZY = 90^\circ - \theta$$

$$\text{In } \triangle SYZ, \angle SYZ = 90^\circ - \theta$$

$$\Rightarrow \angle TZA = \angle SYZ$$

Now, In $\triangle TYZ$ and $\triangle SYZ$, we have ;

$$\angle TYZ = \angle SYZ$$

$$YZ = YZ \text{ (common)}$$

$$\angle TZA = \angle SYZ$$

So, $\triangle TYZ \cong \triangle SYZ$ (By ASA rule)

$$\text{Then, } ZT = YS$$

Sol.563.(d) In the $\triangle ABC$,

As we know,

$$AC + AB > BC > AC - AB$$

$$13 + 9 > BC > 13 - 9$$

$$22 > BC > 4$$

From the above option, we get $BC = 20 \text{ cm}$

Sol.564.(b) From the given option, option (b) is the correct one.

As in $\triangle PQR$ and $\triangle ABC$,

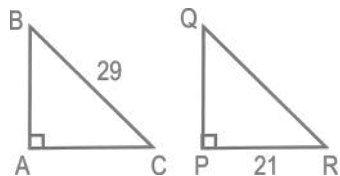
$$\angle A = \angle P \text{ (given)}$$

$$AC = PR \text{ (given)}$$

$$AB = PQ$$

Then, $\triangle PQR \cong \triangle ABC$ (by SAS rule)



Sol.565.(c)

These two triangles are congruent. So length of corresponding sides is equal
 $AB = PQ$, $AC = PR$, $BC = QR$

$$AB = \sqrt{BC^2 - AC^2}$$

$$AB = \sqrt{29^2 - 21^2} = 20 \text{ cm}$$

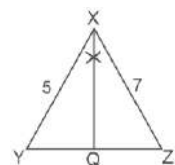
Sol.566.(a)

Given that $\triangle ABC$ is similar to $\triangle PQR$
 So, corresponding sides and angles are equal

$$\angle A = \angle P, \angle B = \angle Q \text{ and } \angle C = \angle R$$

$$\therefore \angle B = 180^\circ - (85^\circ + 58^\circ) = 37^\circ$$

$$\therefore \angle Q = 37^\circ$$

Sol.567.(a)

According to angle bisector theorem,

$$YQ : QZ = XY : XZ$$

$$\therefore YQ : QZ = 5 : 7$$

Sol.568.(a) Let the measure of first two angles = $3x$ and $7x$

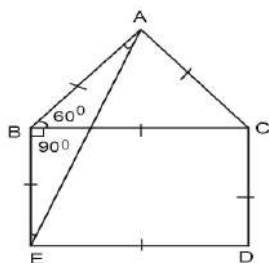
$$\therefore \text{Measure of third angle} = 10x$$

$$\text{Sum of all three angles of a triangle} = 180^\circ$$

$$3x + 7x + 10x = 180^\circ$$

$$\therefore x = 9^\circ$$

$$\text{Hence smallest angle} = 3x = 27^\circ$$

Sol.569.(a)

In $\triangle ABE$, $\angle BAE = \angle AEB = x^\circ$ (Angles opposite to equal sides are equal)

$$\text{Now, } 150^\circ + 2x = 180^\circ$$

$$2x = 180^\circ - 150^\circ = 30^\circ$$

$$x = 15^\circ$$

$$\text{Then, } \angle EAB + 3\angle AEB = x + 3x = 4x$$

$$= 4 \times 15^\circ = 60^\circ$$

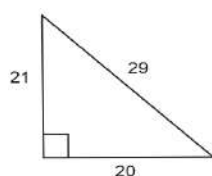
Sol.570.(d) As we know, the longest chord of a circle is its diameter.

$$\text{Diameter} \geq 20.5$$

$$2r \geq 20.5$$

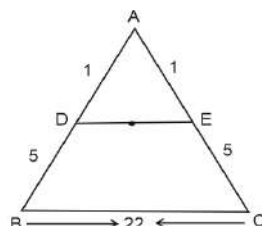
$$r \geq 10.25$$

So, the radius (r) of the circle must be $r \geq 10.25$

Sol.571.(d) triplet : (20, 21, 29)

Given, sum of the other two sides
 $= 41 = (20 + 21)$

Now, difference between the two sides
 $= (21 - 20) = 1 \text{ cm}$

Sol.572.(a)

$$\triangle ABC \sim \triangle ADE$$

Let the length of $DE = x$

$$\frac{AD}{AB} = \frac{x}{22} \Rightarrow \frac{1}{6} = \frac{x}{22}$$

$$(\text{Length of } DE) x = 3.67 \text{ cm}$$

Sol.573.(a) According to the question,
 $45^\circ = 22 \text{ cm}$

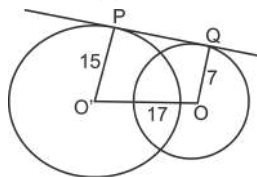
$$(\text{Perimeter of circle}) 360^\circ = \frac{22}{45} \times 360$$

$$= 176 \text{ cm}$$

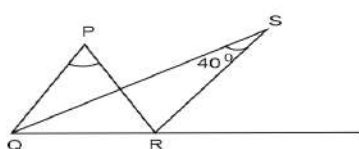
$$\text{Perimeter of circle} = 2\pi r$$

$$2 \times \frac{22}{7} \times r = 176$$

$$(\text{radius}) r = 28 \text{ cm}$$

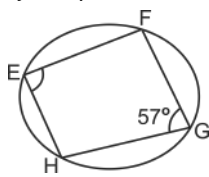
Sol.574.(a)

$$\text{Length of } PQ = \sqrt{17^2 - (15 - 7)^2} = 15 \text{ cm}$$

Sol.575.(a)

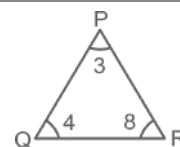
$$\angle P = 2 \times 40 = 80^\circ$$

Sol.576.(c) Sum of opposite angles of cyclic quadrilateral is 180°

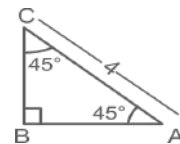


$$\angle HEF = 180 - 57 = 123^\circ$$

Sol.577.(d) The side facing the larger angle will be larger and the side facing the smaller angle will be smaller.



So we can say that QR is the shortest side and PQ is the longest side.

Sol.578.(b)

$$90^\circ, 45^\circ, 45^\circ = \sqrt{2} : 1 : 1$$

$$(\text{Length of } AC) \sqrt{2} \text{ unit} = 4 \text{ cm}$$

$$(\text{Length of } BC) 1 \text{ unit} = \frac{4}{\sqrt{2}} = 2\sqrt{2} \text{ cm}$$

Sol.579.(d) According to the question,

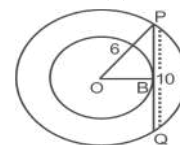
Sine rule = $\frac{1}{2} ab \sin \theta$ (By using the formula)

$$\text{Area of triangle} = \frac{1}{2} \times 4 \times 5 \times \frac{1}{\sqrt{2}} = 5\sqrt{2}$$

Sol.580.(d)

Pythagorean triples (8, 15, 17)

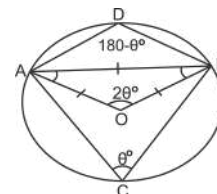
And $\triangle ABC$ and $\triangle PQR$ are the congruent triangles so the length of AB will be 15

Sol.581.(a)

$$\text{Length of } PB = \frac{10}{2} = 5 \text{ cm}$$

length of radius of inner circle

$$= \sqrt{6^2 - 5^2} = \sqrt{11} \text{ cm}$$

Sol.582.(b)

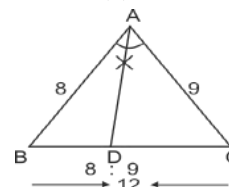
Here $\triangle AOB$ forms an equilateral triangle

Therefore, $2\theta = 60^\circ$

and $\angle C = \theta = 30^\circ$

$\angle D = 180 - \theta = 150^\circ$

$$\text{Required difference} = \angle D - \angle C = 150^\circ - 30^\circ = 120^\circ$$

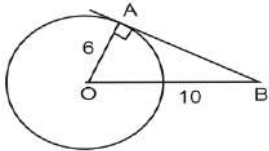
Sol.583.(a)

Angle bisector theorem :- an angle bisector of an angle of a triangle divides the opposite side in two segments that are proportional to the other two sides of the triangle.

Length of shorter segment (BD)

$$= 12 \times \frac{8}{(8+9)} = \frac{96}{17} = 5\frac{11}{17} \text{ cm}$$

Sol.584.(d)



Angle between tangents and radius is always 90° .

Pythagorean triplet :- (6, 8, 10)

So, the length of AB = 8 cm

Sol.585.(c)

Sum of all angles of a triangle = 180°

Only one options (c) is possible i.e. 60° , 40° and 80°

Sol.586.(c) $\Delta MNO \sim \Delta XYZ$

$$\text{So, } \frac{\text{ar. } \Delta MNO}{\text{ar. } \Delta XYZ} = \left(\frac{9}{4}\right)^2$$

$$\frac{243}{\text{ar. } \Delta XYZ} = \frac{81}{16} \Rightarrow \text{ar. } \Delta XYZ = 48 \text{ cm}^2$$

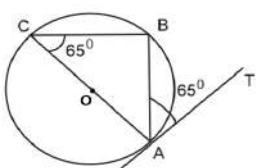
Sol.587.(a)

$$\text{Length of the arc} = \frac{\theta}{360^\circ} \times 2\pi r$$

$$= \frac{90^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 14$$

$$= \frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 22 \text{ cm}$$

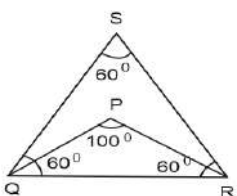
Sol.588.(a)



Alternate segment theorem :-

In any circle, the angle between a chord and a tangent through one of the end points of the chord is equal to the angle made by the same chord in the alternate segment.

Sol.589.(d)



QS = SR

So, $\angle PQR = \angle PRQ = 40^\circ$

PQ = PR

So, $\angle SQR = \angle SRQ = 60^\circ$

Now, $\angle SRP = 60^\circ - 40^\circ = 20^\circ$

Sol.590.(a) Condition to form triangle :-

$$(a - b) < c < (a + b)$$

Only one option is satisfying the condition i.e.

$$8, 6, 9 \Rightarrow (8 - 6) < 9 < (8 + 6)$$

$$\Rightarrow (2) < 9 < (14)$$

Sol.591.(c)

Using exterior angle property :-

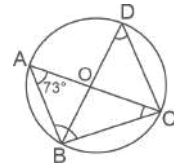
$$2x^\circ + 4x + 15^\circ = 9x - 15^\circ$$

$$3x = 30^\circ \Rightarrow x = 10^\circ$$

Sol.592.(a) $\angle P = \angle F = 52^\circ$,
 $\angle Q = \angle E = 74^\circ$ and $\angle R = \angle D = 54^\circ$

So, $\Delta PQR \sim \Delta FED$

Sol.593.(b)



Angles made by the same arc, anywhere at the circumference are equal.

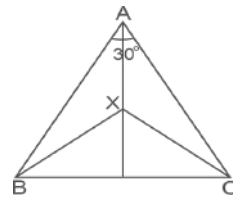
$$\angle BAC = \angle BDC = 73^\circ$$

In ΔABC ,

$$\angle BAC + \angle ACB + \angle ABC = 180^\circ$$

$$\angle ACB + \angle ABC = 180^\circ - 73^\circ = 107^\circ$$

Sol.594.(c)



We know that :-

$$\angle BXC = 90^\circ + \frac{\angle BAC}{2} = 90^\circ + 15^\circ = 105^\circ$$

$$\text{Sol.595.(c)} \frac{\text{ar. } \Delta RMN}{\text{ar. } \Delta RST} = \left(\frac{1}{(1+3)}\right)^2 = \frac{1}{16}$$

$$\text{Now, } \frac{\text{ar. } \Delta RMN}{\text{ar. } \square MNST} = \frac{1}{(16-1)} = \frac{1}{15}$$

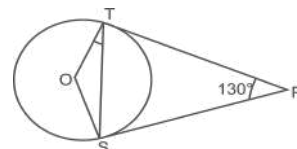
Sol.596.(b) Angle = $30^\circ : 60^\circ : 90^\circ$

$$\text{Side} = 1 : \sqrt{3} : 2$$

$$\text{Here, } \sqrt{3} \text{ units} = 6 \text{ cm}$$

$$\text{So, length of tangents (1 unit)} = 2\sqrt{3}$$

Sol.597.(c)



$$\angle TOS + \angle TPS = 180^\circ$$

$$\Rightarrow \angle TOS = 180^\circ - 130^\circ$$

$$\Rightarrow \angle TOS = 50^\circ$$

Now, In ΔOTS , $OT = OS$ (radius)

$$\angle OTS + \angle TSO + \angle TOS = 180^\circ$$

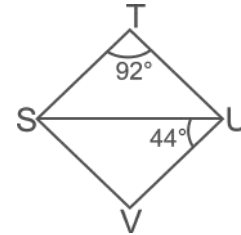
$$\Rightarrow \angle OTS + \angle TSO = 180^\circ - 50^\circ$$

$$\Rightarrow \angle OTS + \angle TSO = 130^\circ$$

$$[\angle OTS = \angle TSO]$$

$$\Rightarrow \angle OTS = \frac{130^\circ}{2} = 65^\circ$$

Sol.598.(d)

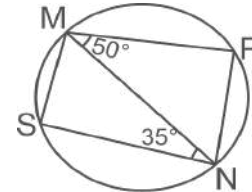


$$\angle STU = \angle SVU = 92^\circ \dots (\text{opposite angle})$$

$$\angle SUV = \angle TSU = 44^\circ \dots (\text{alternate angle})$$

$$\text{So, } 4\angle SVU - 3\angle TSU = 368^\circ - 132^\circ = 236^\circ$$

Sol.599.(d)



$$\angle SMN = 180^\circ - (90^\circ + 35^\circ) = 55^\circ$$

$$\angle MNP = 180^\circ - (90^\circ + 50^\circ) = 40^\circ$$

$$\text{Now, } \angle PMS - \angle PMS$$

$$= (55^\circ + 50^\circ) - (40^\circ + 35^\circ) = 30^\circ$$

Sol.600.(c)

$$\frac{\text{perimeter of } \Delta ABC}{\text{perimeter of } \Delta PQR} = \frac{AB}{PQ}$$

$$\text{Perimeter of } \Delta PQR = 45 \times \frac{18}{12}$$

$$= \frac{135}{2} = 67.5 \text{ cm}$$

$$\text{Sol.601.(a)} \frac{\text{ar. } \Delta ABC}{\text{ar. } \Delta DEF} = \frac{(AB)^2}{(DE)^2}$$

$$\frac{\text{ar. } \Delta ABC}{\text{ar. } \Delta DEF} = \frac{(\sqrt{3})^2}{(\sqrt{5})^2} \Rightarrow \frac{90}{\text{ar. } \Delta DEF} = \frac{3}{5}$$

$$\text{ar. } \Delta DEF = 150 \text{ cm}^2$$

Sol.602.(a)

$$\text{Length of arc} = \frac{22.5}{360} \times 2 \times \frac{22}{7} \times 21$$

$$= \frac{22 \times 21}{8 \times 7} = 8.25 \text{ cm}$$

Sol.603.(d) Required obtuse angle

$$= 180^\circ - (2 \times 28.5^\circ) = 123^\circ$$

Sol.604.(c) $\Delta ABC \sim \Delta PQR$

$$\angle B = \angle Q, \angle C = \angle R$$

$$\text{Thus, } \angle A = \angle P$$

But triangles are not congruent because Sides are not equal.

Sol.605.(a) $P + Q + R = 180^\circ \dots (\text{sum of all angle of triangle})$

$$\text{Here, } 2R - P = Q - R$$

$$3R = P + Q$$

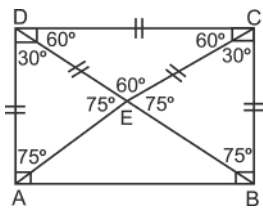
$$3R = 180^\circ - R \Rightarrow 4R = 180^\circ \Rightarrow R = 45^\circ$$

Sol.606.(a) Longest chord of the circle is its diameter.

$$\text{So length of longest chord} = 2 \times 8 = 16 \text{ cm}$$

Sol.607.(b)





$\triangle DEC$ is equilateral so all angles are equal to 60° .

All angles of the square are equal to 90° .

So,

$$\angle ADE = \angle BCE = 30^\circ$$

In $\triangle ADE$, $AD = DE$

$$\text{So, } \angle DAE = \angle DEA = 75^\circ$$

In $\triangle CEB$, $CE = CB$

$$\text{So, } \angle CEB = \angle CBE = 75^\circ$$

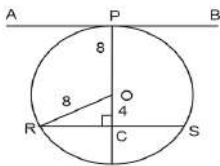
Now,

$$\angle DEC + \angle CEB + \angle AEB + \angle EAD = 360^\circ$$

$$\angle AEB = 360^\circ - (60^\circ + 75^\circ + 75^\circ)$$

$$\angle AEB = 150^\circ$$

Sol.608.(d)



Length of $PC = 12$ cm ... (given)

$PO = OR = OQ = \text{radius} = 8$ cm

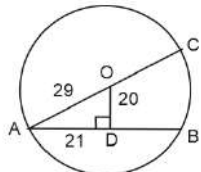
So, $OC = 12 - 8 = 4$ cm

Now, in $\triangle RCO$

$$RC = \sqrt{(8)^2 - (4)^2} = 4\sqrt{3} \text{ cm}$$

$$RS = 2RC = 8\sqrt{3} \text{ cm}$$

Sol.609.(a)



Perpendicular from the centre of the circle to the chord bisect the chord.

Pythagorean triplets :- (20,21,29)

So, length of chord $= 2 \times 21 = 42$ cm

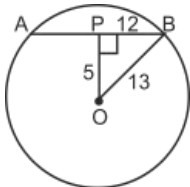
Sol.610.(a)

$$RT \times TS = PT \times TQ$$

$$8 \times 8 = PT \times 4 \Rightarrow PT = 16$$

$$\text{Diameter} = PT + TQ = 16 + 4 = 20 \text{ cm}$$

Sol.611.(b)



Pythagorean triplets :- (5, 12, 13)

So, radius $= 13$ cm

Sol.612.(d)

Pythagorean triplets :- (20, 21, 29)

So, perimeter of triangle $= 20 + 21 + 29 = 70$ cm

Sol.613.(c) $\angle A + \angle B + \angle C = 180^\circ$

$$(5x - 60^\circ) + (2x + 40^\circ) + (3x - 80^\circ) = 180^\circ$$

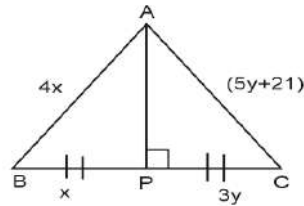
$$(10x) = 280^\circ \Rightarrow x = 28^\circ$$

$$\text{Now, } \angle A = (5x - 60^\circ) = 140^\circ - 60^\circ = 80^\circ$$

Sol.614.(c) Exterior angle = sum of two opposite interior angle of triangle

$$= 35^\circ + 65^\circ = 100^\circ$$

Sol.615.(c)



Using angle bisector theorem :-

$$\frac{4x}{5y+21} = \frac{x}{3y}$$

$$12y = 5y + 21 \Rightarrow y = 3$$

Median and altitude of an isosceles triangle is same.

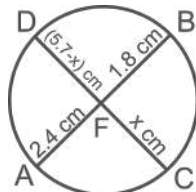
$$\text{So, } BP = PC \Rightarrow x = 3y = 9$$

$$\text{Now, } (x + y) = 9 + 3 = 12$$

Sol.616.(c)

Largest chord of the circle is diameter.

Sol.617.(b)



$$AF \times BF = CF \times DF$$

$$2.4 \times 1.8 = x \times (5.7 - x)$$

$$4.8 \times 0.9 = x \times (5.7 - x)$$

$$0.9 \times (5.7 - 0.9) = x \times (5.7 - x)$$

$$\text{Longest of two line segments} = 4.8 \text{ cm}$$

Sol.618.(d)

Pythagorean triplets :- (5, 12, 13)

$$\triangle ABC \cong \triangle PQR$$

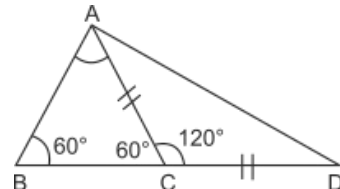
$$\text{So, } \angle A = \angle P = 90^\circ$$

$$\text{and } BC = QR = 13 \text{ cm}$$

$$AC = PR = 12 \text{ cm}$$

$$\text{Then } AB = PQ = 5 \text{ cm}$$

Sol.619.(a)



$$\angle ACB + \angle ACD = 180^\circ \dots (\text{linear pair})$$

$$\Rightarrow \angle ACD = 120^\circ$$

$$\angle CAD = \angle ADC \dots (AC = CD)$$

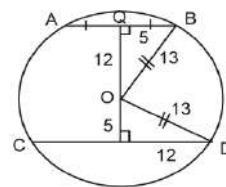
In $\triangle ACD$,

$$\angle CAD + \angle ADC + \angle ACD = 180^\circ$$

$$2 \times \angle CAD = 60 \Rightarrow \angle CAD = 30^\circ$$

Sol.620.(c)

Pythagorean triplets :- (5, 12, 13)



Distance between the chords

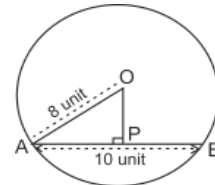
$$= 12 + 5 = 17 \text{ cm}$$

Sol.621.(b) $PQ = (4x) \times 6 = 24$ cm

Using mid-point theorem :-

$$BC = 2PQ = 48 \text{ cm}$$

Sol.622.(a)



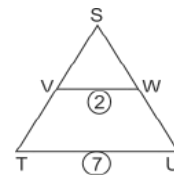
$$\text{Required distance} = \sqrt{(8)^2 - (5)^2} = \sqrt{64 - 25} = \sqrt{39}$$

$$\text{Sol.623.(b)} \quad 84^\circ = \left(\frac{22 \times 84}{7 \times 180} \right)^\circ$$

$$\text{Now, } \theta (\text{in radian}) = \frac{l}{r}$$

$$r = \frac{22 \times 7 \times 180}{22 \times 84} = \frac{180}{12} = 15 \text{ cm}$$

Sol.624.(c)

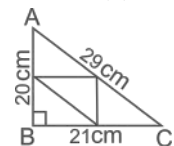


$$\frac{\text{ar. } \triangle STU}{\text{ar. } \triangle VWU} = \frac{(VW)^2}{(TU)^2} = \frac{4}{49}$$

$$\text{So, ar. } \triangle VWU = 49 - 4 = 45$$

$$\text{Required ratio} : 45 : 49$$

Sol.625.(b)



$$\text{Required area} = \frac{1}{4} \times \left(\frac{1}{2} \times 20 \times 21 \right)$$

$$= \frac{210}{4} = 52 \frac{1}{2} \text{ cm}^2$$

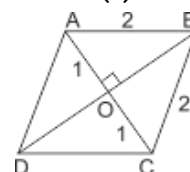
Sol.626.(a)

Given triangles are similar, so

$$\angle A = \angle P, \angle B = \angle Q \text{ and } \angle C = \angle R$$

$$\angle C = \angle R = 180^\circ - (35^\circ + 65^\circ) = 80^\circ$$

Sol.627.(d) Let side of rhombus $= 2$ units



In $\triangle AOB$,

$$OB = \sqrt{(2)^2 - (1)^2} = \sqrt{3}$$

$$\text{Diagonal (DB)} = 2 \times OB = 2\sqrt{3}$$

$$\text{Required ratio} : - AC : DB = 2 : 2\sqrt{3} \\ = 1 : \sqrt{3}$$

$$\text{Sol.628.(c)} \quad 8 \angle A : 6 \angle B : 3 \angle C$$

$$\angle A : \angle B : \angle C = 3 : 4 : 8$$

$$15 \text{ units} = 180^\circ \Rightarrow 1 \text{ unit} = 12^\circ$$

$$\angle A = 36^\circ, \angle B = 48^\circ \text{ and } \angle C = 96^\circ$$

$$\text{Sol.629.(d)}$$

Given triangles are similar, so

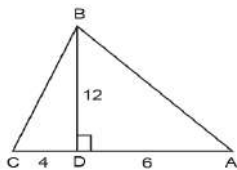
$$\angle L = \angle P, \angle M = \angle Q \text{ and } \angle N = \angle R$$

$$\angle M = \angle Q = 180^\circ - (40^\circ + 80^\circ) = 60^\circ$$

$$\frac{LM}{LN} = \frac{PQ}{PR} \Rightarrow \frac{6}{8} = \frac{7.5}{PR}$$

$$PR = \frac{8 \times 7.5}{6} = 10 \text{ cm}$$

$$\text{Sol.630.(c)}$$



$$\text{Area of } \triangle ABC = \frac{1}{2} \times (4 + 6) \times 12 = 60 \text{ cm}^2$$

$$\text{Sol.631.(c)} \quad \triangle PQR \cong \triangle MNY, \text{ So}$$

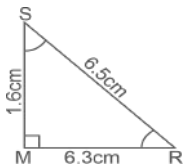
$$\angle P = \angle M = 72^\circ, \angle Q = \angle N = 55^\circ \text{ and}$$

$$\angle R = \angle Y = 180^\circ - (72^\circ + 55^\circ) = 53^\circ$$

$$PQ = MN = 8 \text{ cm}$$

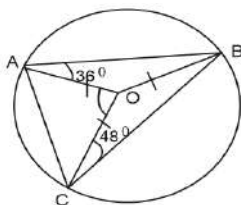
$$\text{Sol.632.(c)}$$

Pythagorean triplets : (16, 63, 65)



$$\text{Hypotenuse} = 6.5 \text{ cm}$$

$$\text{Sol.633.(d)}$$



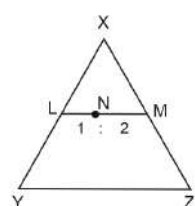
As per figure,

$$\angle BAO = \angle ABO \text{ and } \angle BCO = \angle OBC$$

$$\text{Then, } \angle ABC = 36 + 48 = 84^\circ$$

$$\text{Therefore, } \angle AOC = 2\angle ABC = 168^\circ$$

$$\text{Sol.634.(a)}$$

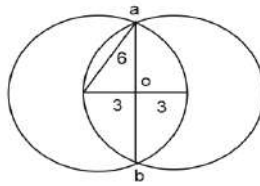


$$\text{LN (1 unit)} = 5 \text{ cm}$$

$$\text{Then (LM) 3 units} = 5 \times 3 = 15 \text{ cm}$$

$$\text{Value of YZ} = 15 \times 2 = 30 \text{ cm}$$

$$\text{Sol.635.(b)}$$



$$\text{Length of } ao = \sqrt{6^2 - 3^2} = 3\sqrt{3}$$

$$\text{So the length of ab (chord)} = 2 \times 3\sqrt{3} \\ = 6\sqrt{3} \text{ cm}$$

